BASICS OF ROTATING INDUSTRIAL EQUIPMENT

An Introduction to Rotating Equipment Maintenance

Objectives

- Define safety needs and lockout procedures.
- Identify rotating equipment.
- List the major components of rotating equipment and explain their function.
- Identify the auxiliary equipment required to maintain rotating equipment operation.
- Define inspection and preventative maintenance techniques.

Equipment

- Compressors
  - Rotating, screw and centrifugal types
- Turbines
  - Gas turbines
- Pumps
  - Basic types and Centrifugal
- Fans, Blowers, and Louvers

Auxiliary and Support Systems

- Lubrication
- Bearing
- Seals
- Alignment
- Vibration Analysis
- Thermal Analysis

TOPICS – Click to view

- General Safety Topics
- Compressors
- Pumps
- Turbines
- Fans and Louvers
- Lubrication Requirements
- Bearings
- Seals
- Alignment
- Vibration Analysis
- Thermal Analysis
- Preventative Maintenance
- Fault Recognition

GENERAL SAFETY TOPICS

Tenets of Maintenance Safety

1. Always operate equipment within design or environment limits.
2. Always work in a safe and controlled condition.
3. Always ensure safety devices are in place and functioning.
4. Always follow safe work practices and procedures.
5. Always meet or exceed customer's requirements.
6. Always maintain integrity of dedicated systems.
7. Always comply with all applicable rules and regulations.

Tenets of Maintenance Safety

5. Always meet or exceed customer's requirements.
6. Always maintain integrity of dedicated systems.
7. Always comply with all applicable rules and regulations.

Safety Meetings

- The primary purpose of safety meetings is to prevent accidents from happening.
- Safety Meetings should discuss recent incidents, accident causes, lessons learned, and hazard awareness.

Accident Causes

- Whenever an accident occurs, someone always asks, “How did it happen?”
- Accidents do not “just happen”—they are caused
- If we are going to eliminate accidents we must have *some* idea of what *causes of accidents can be.*
  - Unsafe Conditions
  - Unsafe Acts

Unsafe Conditions

- Unsafe conditions are those things that can be seen by inspecting and looking for hazards in the work environment.
- Unsafe conditions are usually created by poor housekeeping, improper storage, defective or broken equipment, or removing guards from machinery.
- This is the principle reason that safety inspections should be done on a scheduled basis.

Unsafe Acts

- What are unsafe acts or unsafe practices?
  - Reaching into a running machine
  - Operating a machine without guards
  - Using defective tools or equipment
  - Indulging in horseplay on the job

Hazard Awareness

- The main indicator of an existing hazard is by the posting of signs.
- Other indicators are listed below:
  - Safety Meetings
  - Toolbox Meetings
  - Procedure Warnings and Cautions
  - System and Work Site Familiarity

Rotating Equipment Safety

- All persons working near or around rotating equipment should be familiar with the location and operation of all stopping devices.
- Be alert when in equipment areas, leaning against equipment, and where you put your hands.
- Rotating equipment movements are often sudden and unpredictable.

Rotating Equipment Safety

- Maintain good housekeeping practices.
  - Clear work areas and pathways of debris and obstructions.
  - Properly clean up spilled lubricant and other slippery materials.
- If equipment is down for service, lock out per plant requirements.
  - Always assume equipment can start at any time.

Rotating Equipment Safety

- Beware of and avoid getting too close to machinery where guards have been removed and report such conditions.
- When climbing around or following conveyor paths, be aware of hazards such as sharp edges, protruding objects, and low clearances.

Rotating Equipment Safety

- Do not operate equipment unless authorized to do so.
- Stop-start stations should be clearly marked and located for easy accessibility, do not hesitate to use them when necessary.
- Horseplay, scuffling, or other such actions around equipment is hazardous.

Rotating Equipment Safety

- Promptly report to the proper supervisor all damage or any irregularities in equipment operation.
- In case of injury, take immediate action to obtain aid by competent personnel.

Rotating Equipment Safety

- If potentially dangerous conditions exist, report it to the proper supervisor immediately.
- Do not work around equipment while under the influence of alcohol, drugs, or narcotics.
- Avoid entanglement in rotating equipment by:
  - Removing loose items such as clothing and jewelry
  - Tying back long hair
- Leave repair functions to the properly trained maintenance personnel to perform.

Rotating Equipment Safety

- All personnel performing maintenance or repairs on the equipment shall be qualified and trained in the fundamentals governing proper and safe maintenance and repairs and shall follow the standards for proper lockout energy control procedures.

Rotating Equipment Safety

- Bypassing or jumping safety circuits will cause a hazardous condition and must never be done.
- Do not perform maintenance on a system while it is running unless the nature of the maintenance absolutely requires so.
- Use all recommended safety practices when using mechanical aids, hoists, cables, safety harnesses, and other equipment.

Rotating Equipment Safety

- It may be necessary to bleed lines to any pneumatically or hydraulically powered component of the system to prevent inadvertent operation to prevent injury inherent in stored energy. Lockout any associated electrical interlocked equipment.
- When power needs to remain on for testing electrical components or mechanical functions all operators or personnel involved with the equipment should be made aware of the testing and work being done.

Rotating Equipment Safety

- Be aware of abnormal noises as they often precede mechanical problems and safety hazards. Investigate as soon as possible to protect people and machinery.
- If abnormal noise is due to vibration, check for build-up of foreign material, misalignment, or failed internal rotating components.

Rotating Equipment Safety

- Before restarting a piece of equipment that has been shut down for any reason, insure that all personnel are clear and that everyone at risk within the area is aware that the machine is about to be started. The equipment should be checked to see that all obstructions have been removed which usually requires a walk of the equipment.
- Do not restart the equipment unless all safety devices are working and all guards and fences are in place.

Rotating Equipment Safety

- Before restarting a piece of equipment that has been shut down for any reason, ensure that all personnel are clear and that everyone at risk within the area is aware that the machine is about to be started.

Signs

- The following slides are examples of types of signs that could be used to warn of hazardous areas, materials or conditions. Always refer to your plant safety literature for specific application of signs.

Prohibition Signs





- No Smoking and No Open Flame signs are for posting at entrances to “Open Flame Restricted Areas”

Open Flame Restricted Areas

- Warehouses with easily ignited and flammable materials
- Explosion hazardous areas
- Locations with toxic materials
- Areas where different activities with flammable materials are carried out

Mandatory Signs

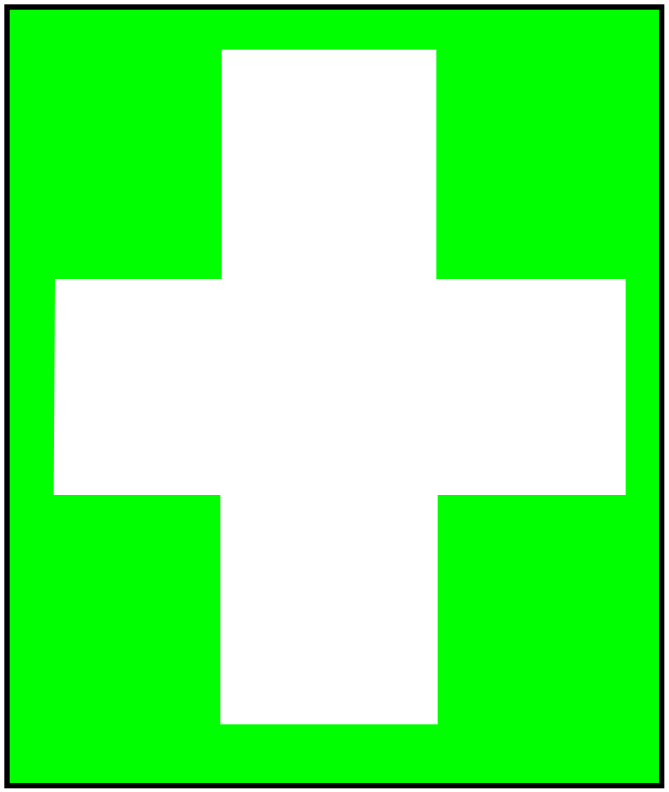
- **Attention, When Entering Facility, Please Advise Operator**
  - Signs are for posting at the entrances to all production facilities

Warning Signs

- Warning signs mean
  - Caution
  - Risk of Danger
  - Hazard ahead
- Warning signs are designated by white background with a black outline of an equilateral triangle, yellow inside the triangle, and black symbol in the triangle.

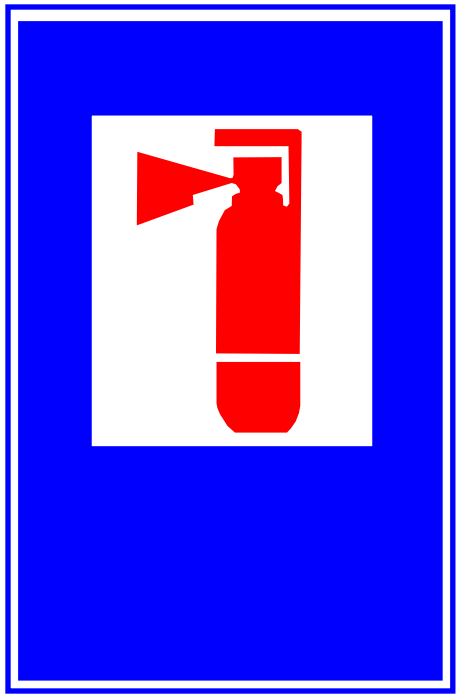
Safety Signs

- **First Aid** signs are for posting at locations having a first aid kit.



Fire Safety Signs

- **Fire Extinguisher** signs are for posting at locations where fire extinguishers of A, B, C and D types are available.



Traffic Signs

- Speed Limit
- It is prohibited to exceed the speed specified on the sign



Traffic Signs

- Pedestrian Crossing



Traffic Signs

- Priority signs shall be posted to establish the passing sequence of road intersection, road crossing or narrow road sections.

Fire Safety

- Obey All Warning and Caution Signs
  - Explosive Hazard Area
  - No Open Flames
- Report Fires and Call for Help
- Report to Muster Area
- Use Appropriate Precautions

Electrical Lock Out

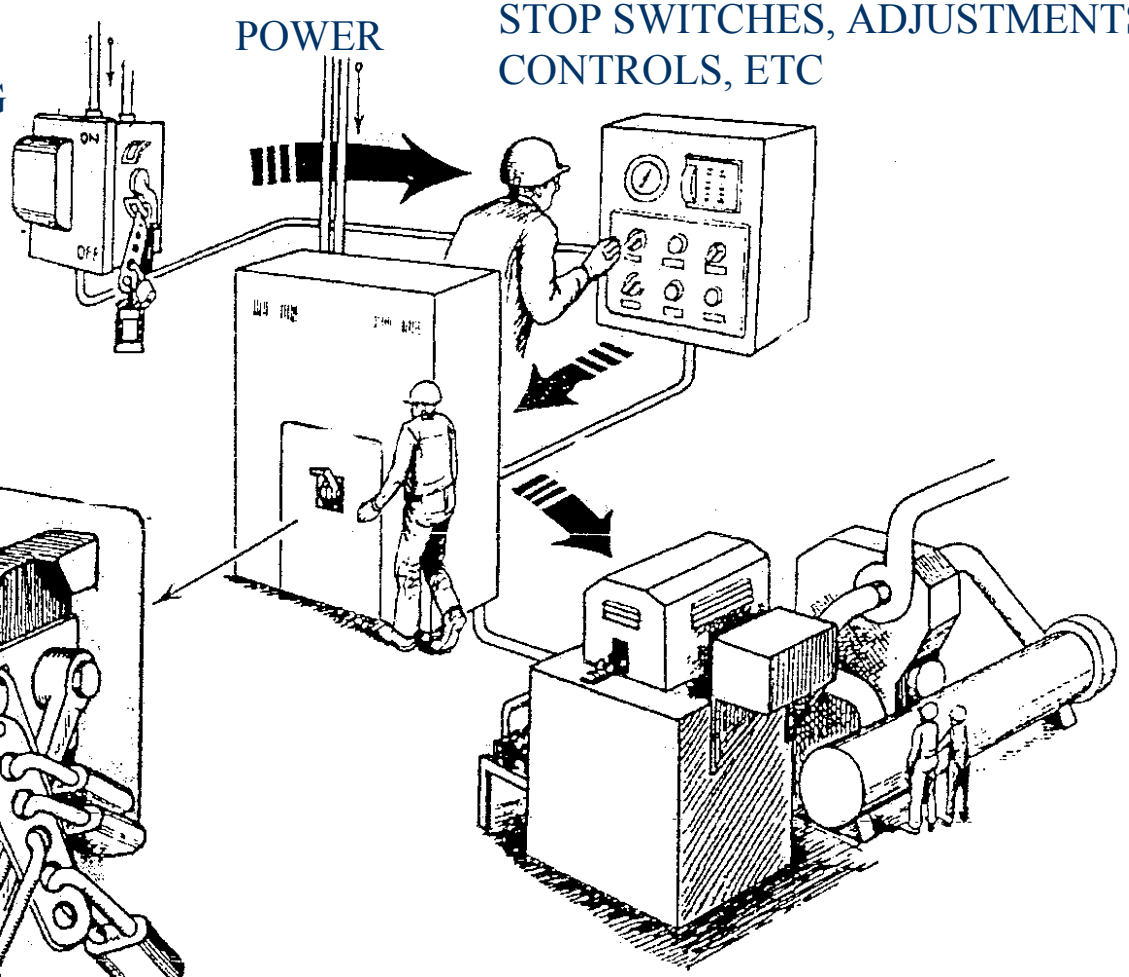
- To protect personnel, equipment that is to be worked on must be deenergized to prevent the accidental release of energy or the inadvertent operation of equipment.
- Lockout is the method of placing a lock on an isolating device to ensure that a piece of equipment cannot be operated.

LOCKOUT

INCOMING POWER

CONTROL PANEL START AND STOP SWITCHES, ADJUSTMENTS, CONTROLS, ETC

DISCONNECT SWITCH LOCKOUT IF WORKING ON CONTROL PANEL OR ON ELECTRICAL CONTROL CIRCUIT



CIRCUIT BREAKER AND MOTOR STARTER LOCKOUT BEFORE WORKING ON MOTOR OR EQUIPMENT SWITCH IN OFF POSITION WITH I.D. TAGS AND TONG AND LOCK SYSTEM WITH EMPLOYEE PADLOCKS

LOCKOUT TERMS

- LOCKOUT
- LOCKOUT DEVICE
- ENERGY SOURCE
- ENERGY ISOLATING DEVICE
- SHALL
- SHOULD

Definitions

- Electric Power Source is the main control panel (i.e., motor control center, circuit breaker, etc.).
- Electrical equipment must be locked out at the power source, not at the start/stop switches.
- Electrical disconnect is the physical removal of electrical leads at the power source (or removal of the fuses), so it is impossible for someone to start the equipment.

Lock Definitions

- Instrumentation/Electrical locks are single-use, disposable locks or locks keyed separately and individually assigned to electricians, maintenance and instrumentation personnel and are used solely for the purpose of locking out equipment that they will be working on.

Tagout Definitions

- Tagout is the installation of “Danger - Do Not Operate” tags on equipment controls to warn workers that the equipment must not be used, or that the position of a valve or isolating device should not be changed.

Summary

- Potential electrical hazards can be minimized when working with electrical equipment by the following.
  - Electrical Regulations
  - Electrical PPE
  - Safety Codes
  - Lock Out
  - Precautions

Personal Protective Equipment

- Personal Protective Equipment must be worn as protection against hazards that cannot be eliminated by other means, or where no other preventive solution is found to be practical.

Definitions

- Personal Protective Equipment
- Impervious Clothing and Gloves
- Safety Equipment

Roles and Responsibilities

- Comply with equipment manufacturer recommendations.
- Visually inspect the PPE daily or before each use.
- Replace torn or damaged PPE.
- Properly clean and store equipment.
- Contact supervisor with questions.

General PPE Requirements

- Make sure that PPE is appropriate to the work condition.
- Using PPE that is not required may get in the way.
  - For example, wearing electrician gloves to calibrate a level indicator would be a hindrance.

General PPE Requirements

- The minimum PPE in plant areas include:
  - Hard Hat
  - Safety Glasses
  - Safety or Sturdy Shoes
  - Mini Filter in some areas

Head Protection

- Hard hats protect the head from impact, and penetration by falling or flying objects and electric shock for insulated hard hats

Eye and Face Protection

- Eye and face protection is required when an employee is exposed to eye or face hazards.

Face Shields

- Face shields must be worn to protect the face and neck.
- Face shields alone do not provide adequate eye protection.

Eye and Face Protection

- Goggles and face shields should be washed with warm soapy water, rinsed thoroughly, and hung to dry before they are stored.
- A soft tissue or soft nonabrasive cloth should be used to clean the lenses.

Hand Protection

- Gloves shall be worn when hands are exposed to hazardous substances, sharp objects, or temperature extremes (hot or cold).
- Impervious gloves must be used when handling hydrocarbons and corrosive chemicals such as acids and caustics.
- Miscellaneous gloves include special-use gloves. The following gloves must be individually assigned: Welding gloves, Fire fighters' gloves, Electrician gloves

Glove Inspection

- Impervious gloves should be checked for pinholes leaks by blowing air into them. They should be replaced when they become cracked or develop holes.

Body Protection

- Appropriate body protection must be worn to keep acidic, corrosive, oily, dirty, or dusty materials off the body. The type of protection required depends upon the nature of the hazard.
- Disposable coveralls and suits are designed to keep dust and dry material off the worker. They provide minimal protection against liquids and oily substances.

Aprons

- Aprons should be worn to keep dirt and material off work clothing when pouring liquids, dumping dry materials, or working with dirty equipment.

Foot Protection

- Employees shall wear safety steel toed footwear when they work in an area where there is danger of foot injury due to falling or rolling objects.
- Areas and jobs, which require safety footwear, shall be determined by the Facility Owner.
- Rubber boots should be worn when it is necessary to protect the feet and shoes from excessive water, oil, mud, muck, or corrosive material.

Definitions

- Air Line Respirator
- Breathing Air Equipment
- Cartridge Respirator
- Face Piece-to-Face Seal
- Hazard Assessment
- Hazardous Atmosphere

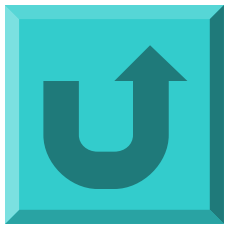
Definitions

- IDLH Atmosphere
- Qualitative Fit Test
- Self Contained Breathing Apparatus (SCBA)
- Single-Use Disposable Dust Respirator
- Tolerance Test

Summary

- Review

CLICK TO RETURN TO TOPICS



COMPRESSORS

Main Topics

- Introduction to compressors
- Centrifugal
- Reciprocating
- Screw

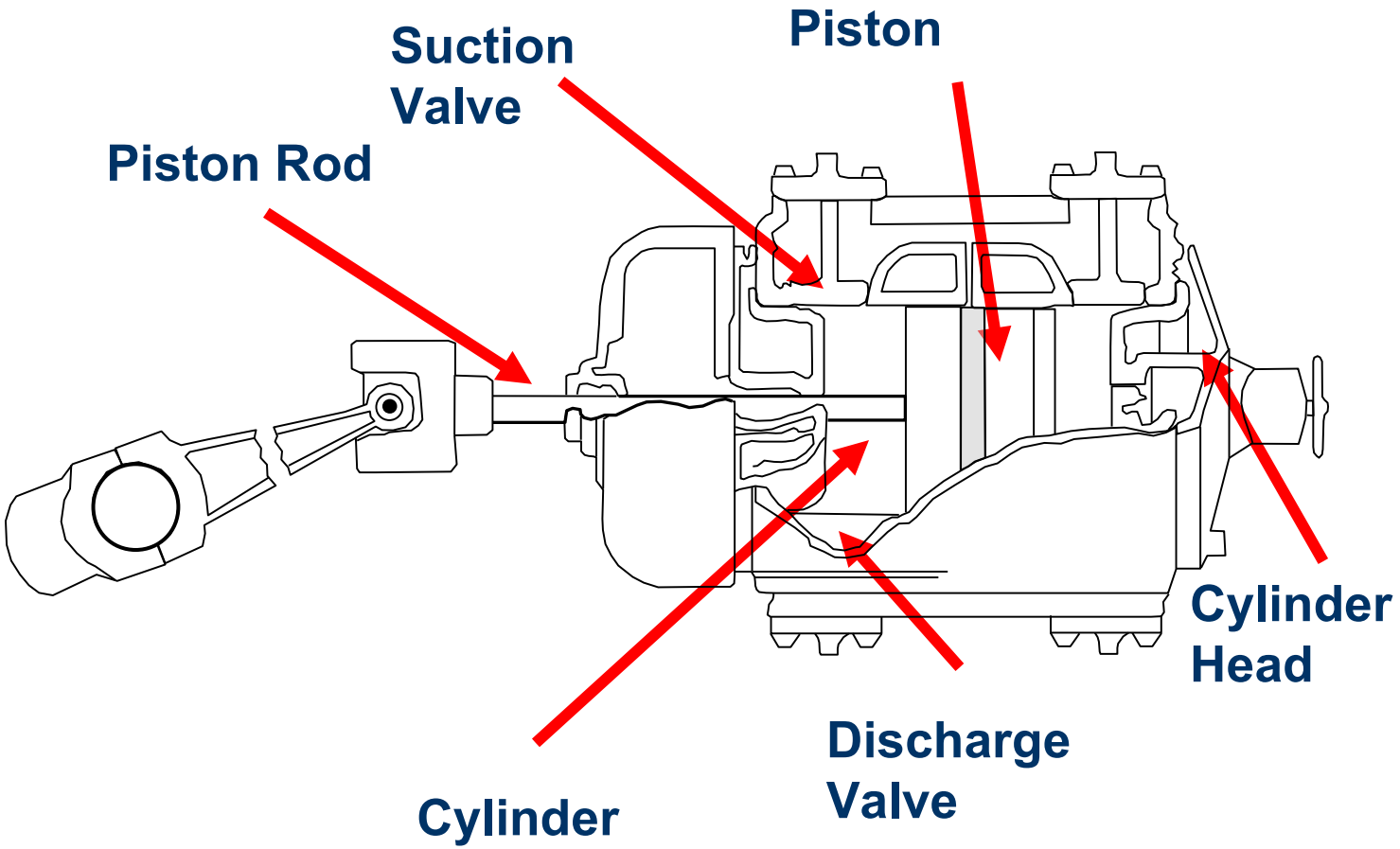
Introduction

- Compression is used in all aspects of gas processing such as:
  - Gas Lift
  - Gas Gathering
  - Helium Recovery
  - Condensate Recovery
  - Transmission
  - Distribution

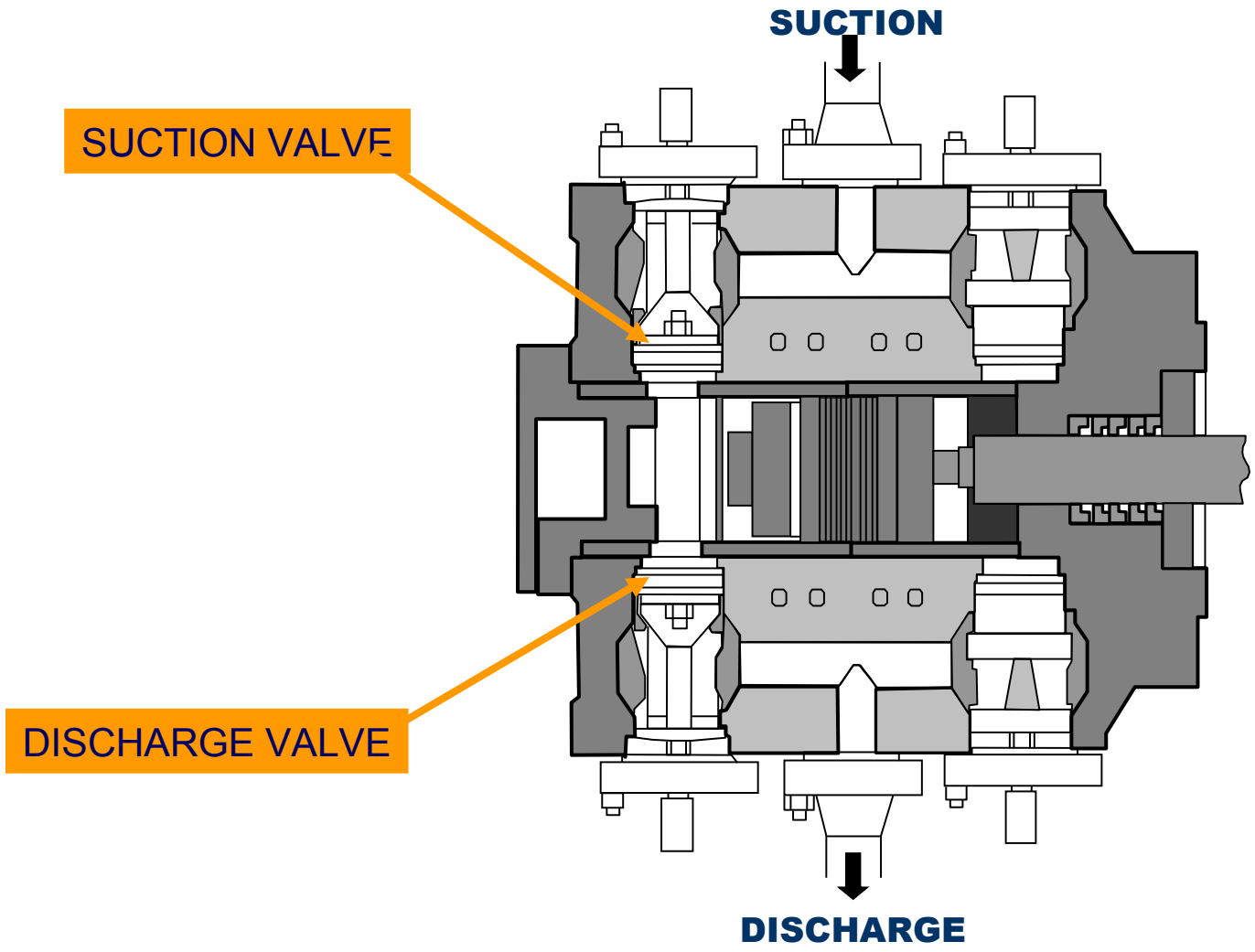
Types

- Reciprocating
- Centrifugal
- Sliding Vane
- Rotary Screw

Reciprocating Compressor



Cylinder Operating Valves

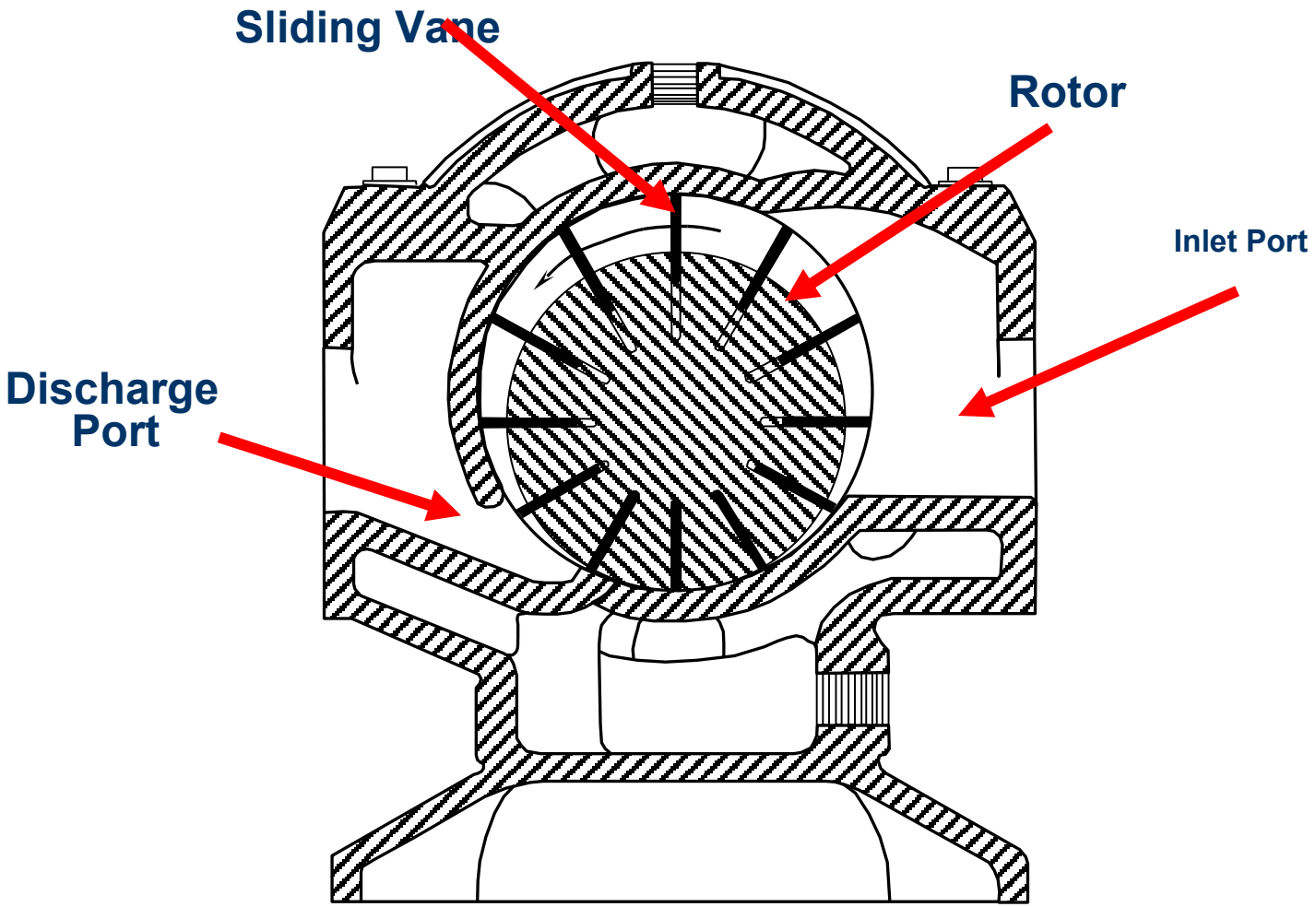


Stages

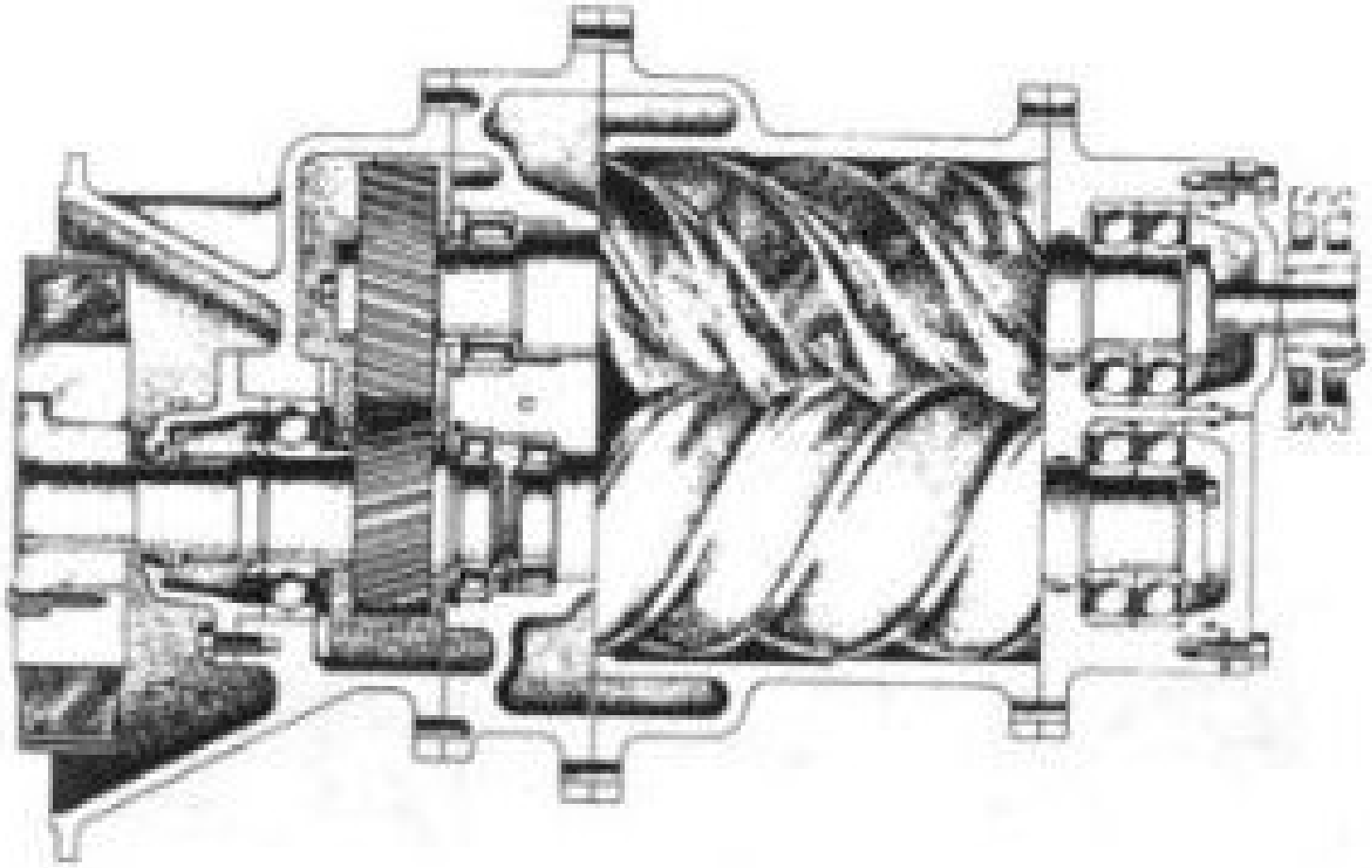
The number of stages is governed by the following factors:

- Allowable discharge temperature.
- Rod loading.
- Existence of a fixed side stream pressure level (where flow is added to or withdrawn from main flow of compressor).
- Allowable working pressure of available cylinders.

Sliding Vane Compressor



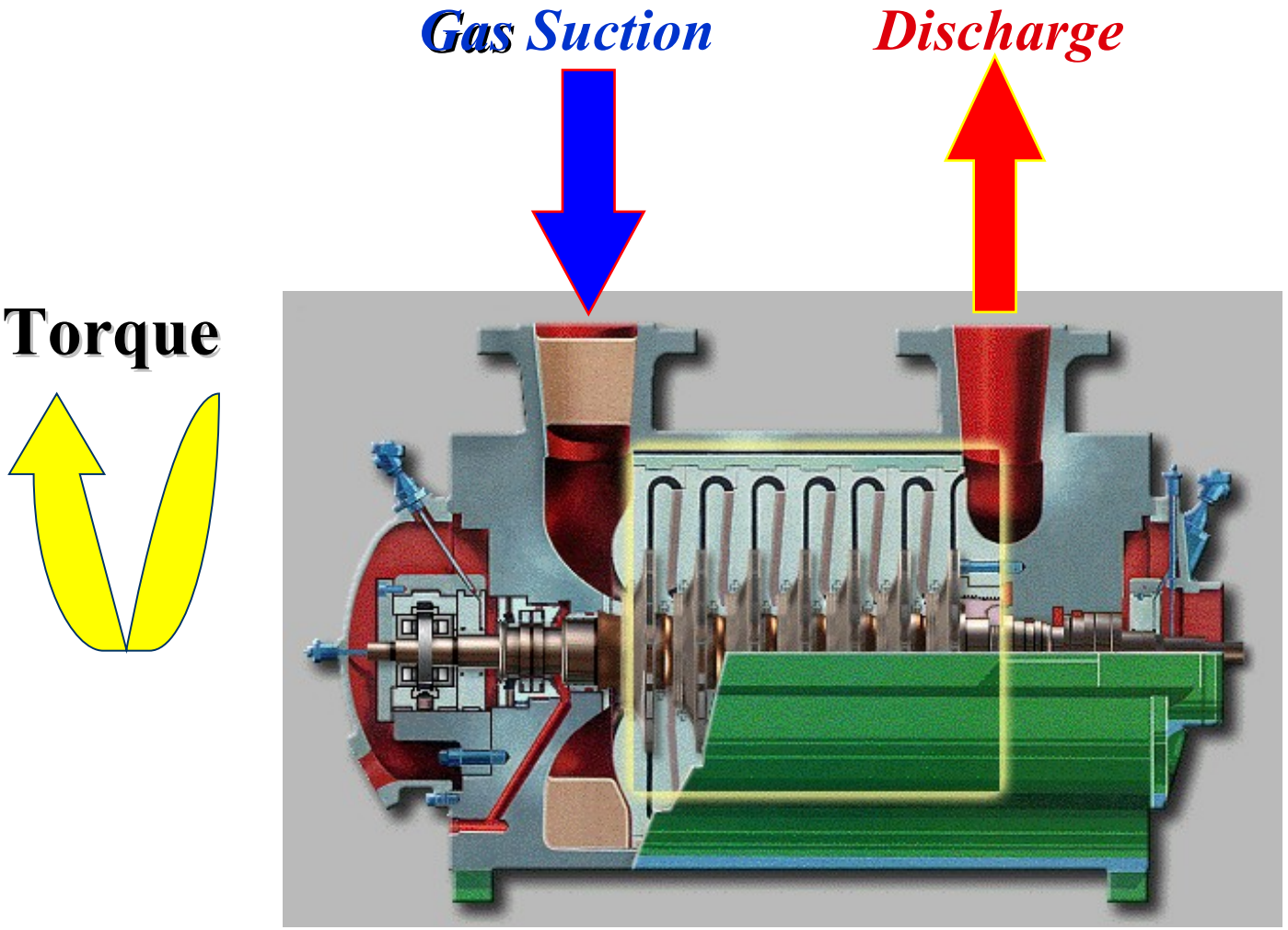
Screw Compressors



Centrifugal Compressor Fundamentals

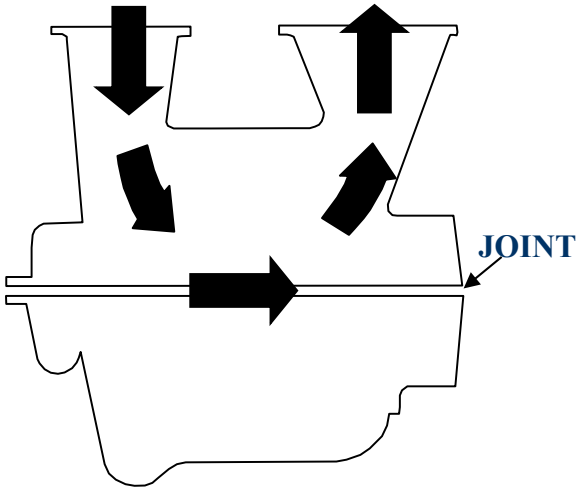
- Gas flow path
- Stage
- Process stage
- Velocity Energy to Pressure

Centrifugal Compressor

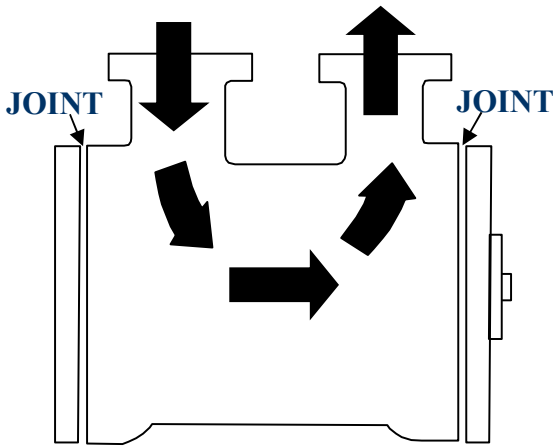


Centrifugal Compressor Types

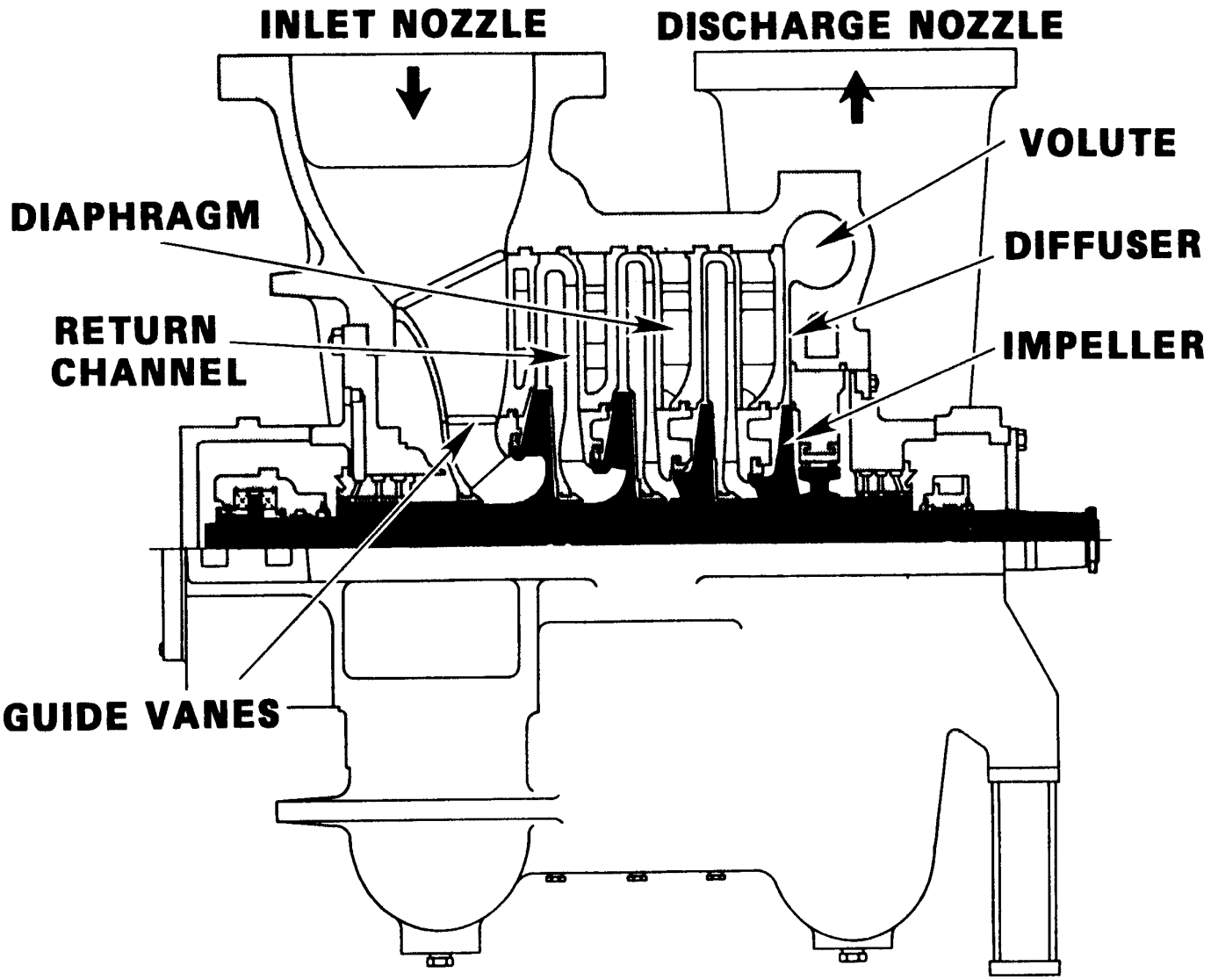
- **Axial**, or horizontally split



- **Radial**, or vertically split



Centrifugal Compressor Stage Components



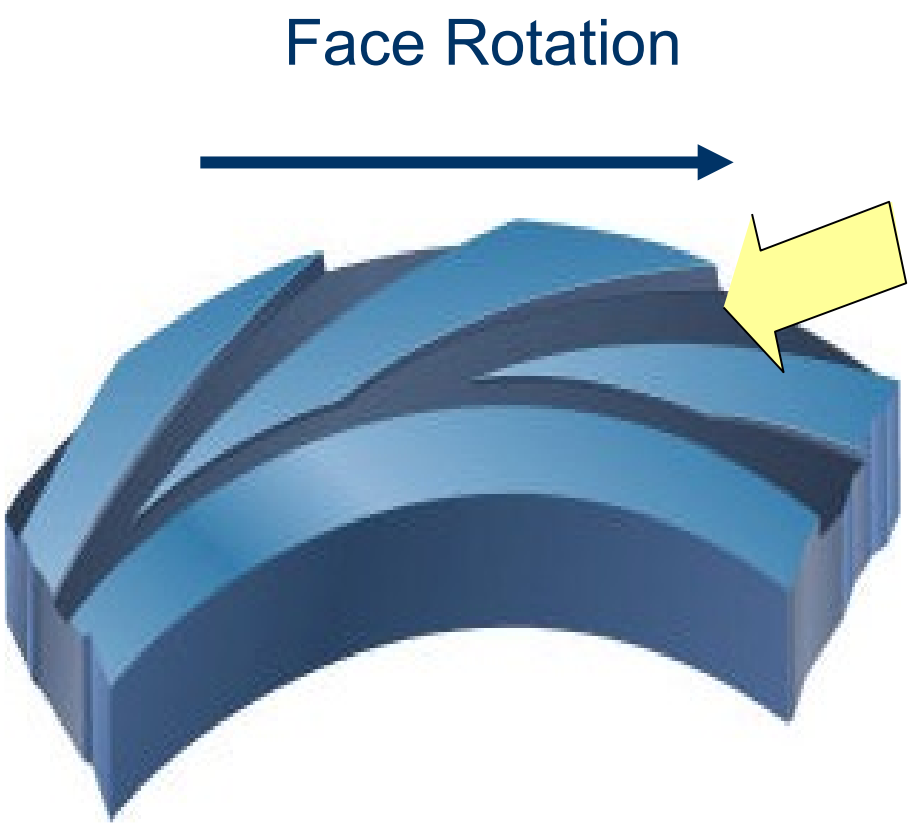
Surge

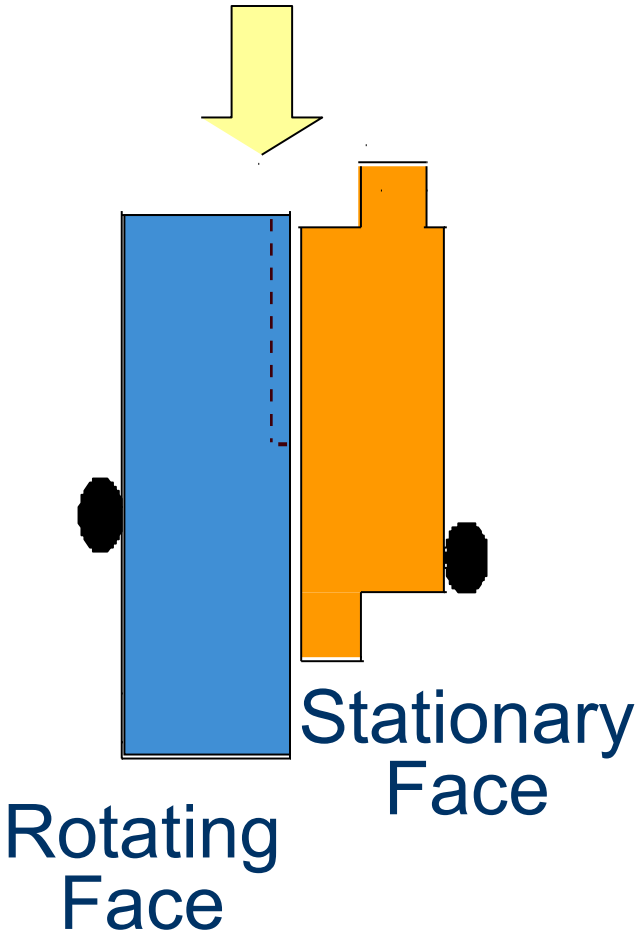
- Surge is caused by unstable flow within compressor which results in flow reversal system pressure fluctuations.
- Frequency of surge

Causes/Effects of Surge

- Restricted suction or discharge such as a plugged strainer.
- Process changes in pressures or gas composition.
- Mis-positioned rotor or internal plugging of flow passages.
- Inadvertent speed change such as from a governor failure.

Dry Gas Seals

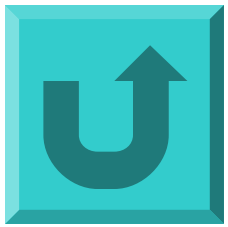




Summary

- Review Question and Answer Session

CLICK TO RETURN TO TOPICS



PUMPS

Course Objectives

- At the completion of this course students will be able to:
  - Identify types of pumps
  - Identify major components for each type of pump
  - Define Characteristics of each type of pump
  - Describe applications in which each type of pump is used

Major Topics

- Pumps – General
- Positive Displacement Pumps
- Centrifugal Pumps

Pumps

- Types
  - Positive Displacement - Overview
    - Screw Pumps
    - Gear Pumps
    - Piston Pumps
    - Plunger Pumps
  - Centrifugal - Overview

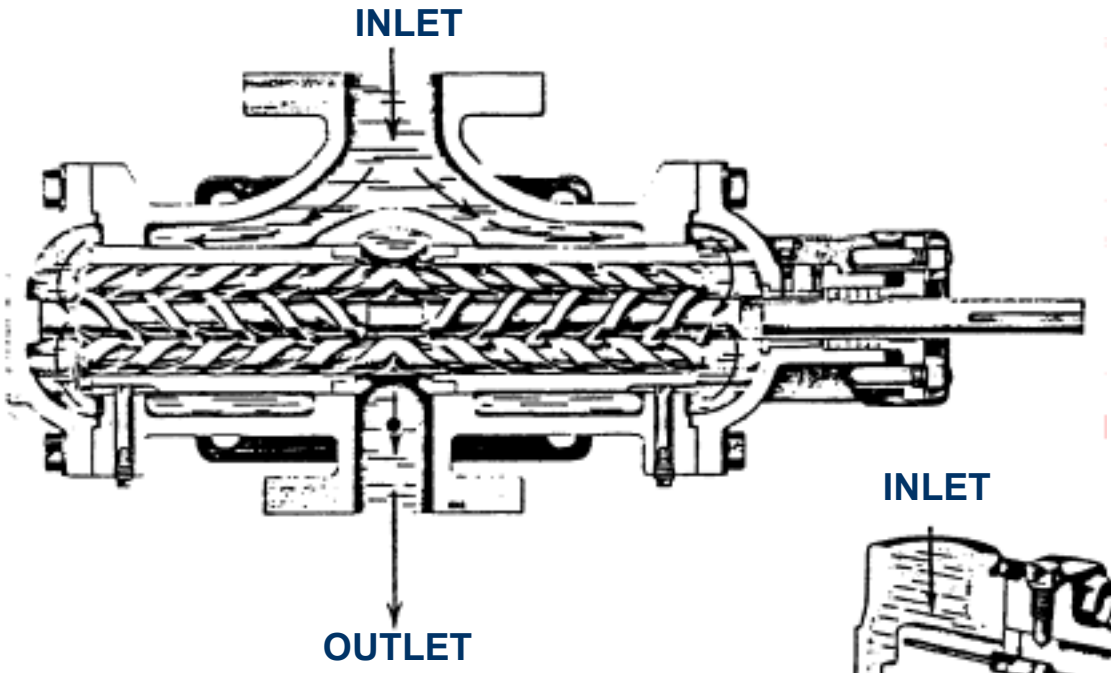
Positive Displacement Pumps

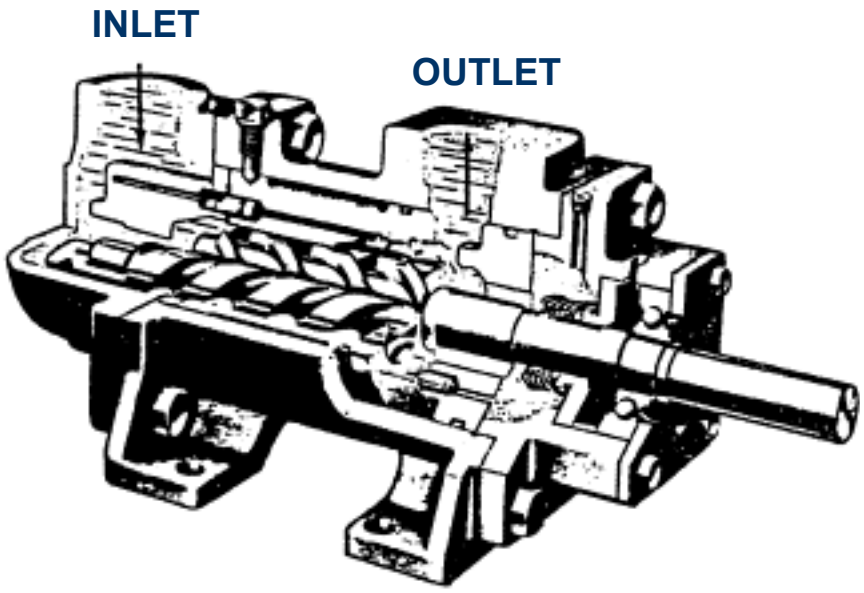
- Screw Pumps
- Gear pumps
- Piston pumps
- Rotating gears
- Centrifugal pumps

Screw Pumps

- Screw pumps are the most common type of rotary pump found in the petroleum industry.
- The three sub-types of screw pumps:
  - three-screw
  - two- screw
  - single-screw

Screw Pumps





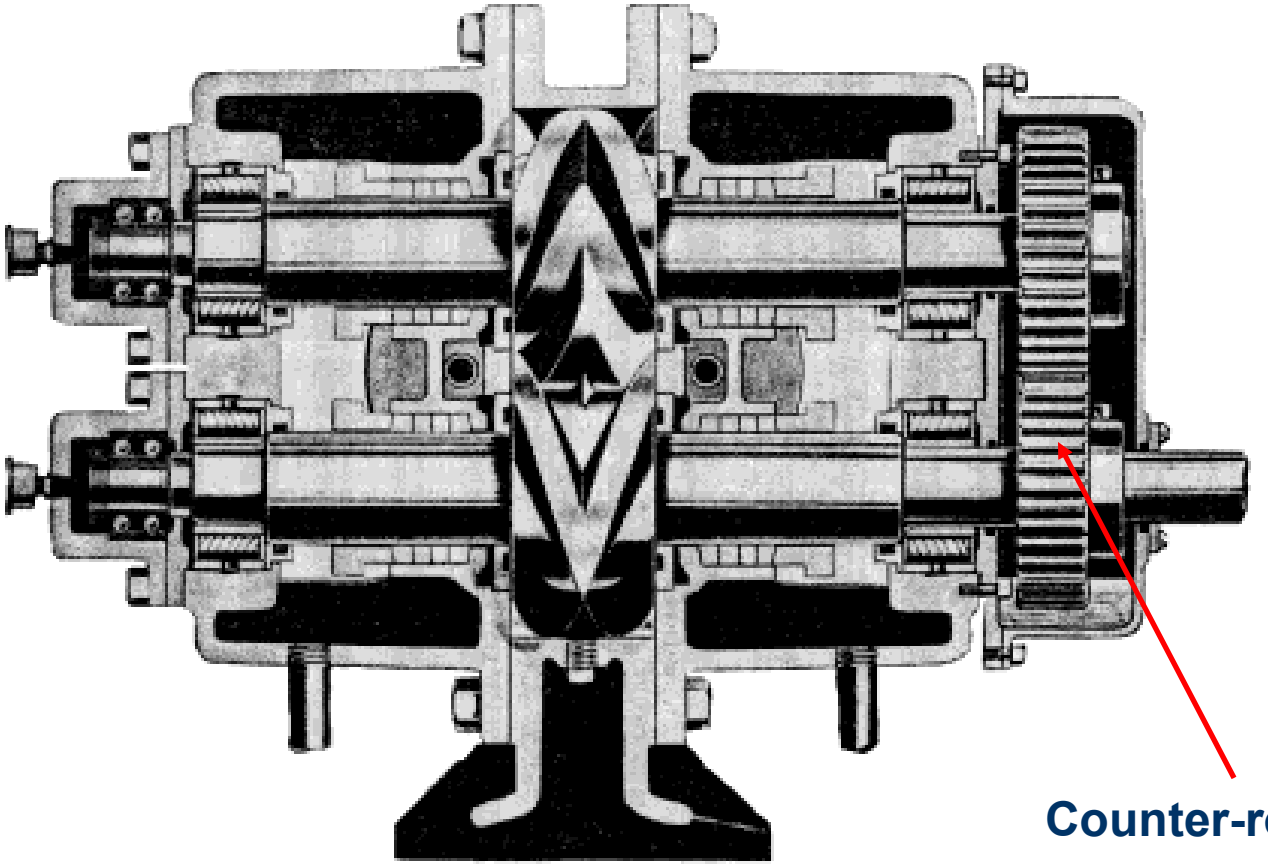
Gear Pumps

Generally less expensive than screw pumps, and used when an inexpensive short-life pump can be tolerated. Also used in intermittent services.

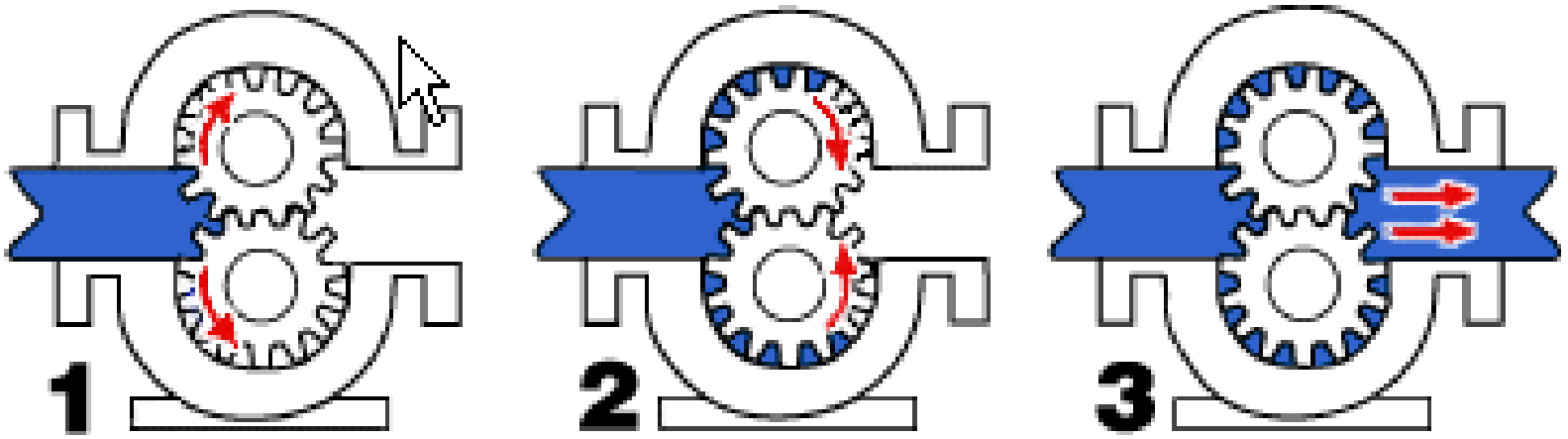
Types:

- External Gear
- Internal Gear
- Lobe

External Gear Pump



External Gear Pumps

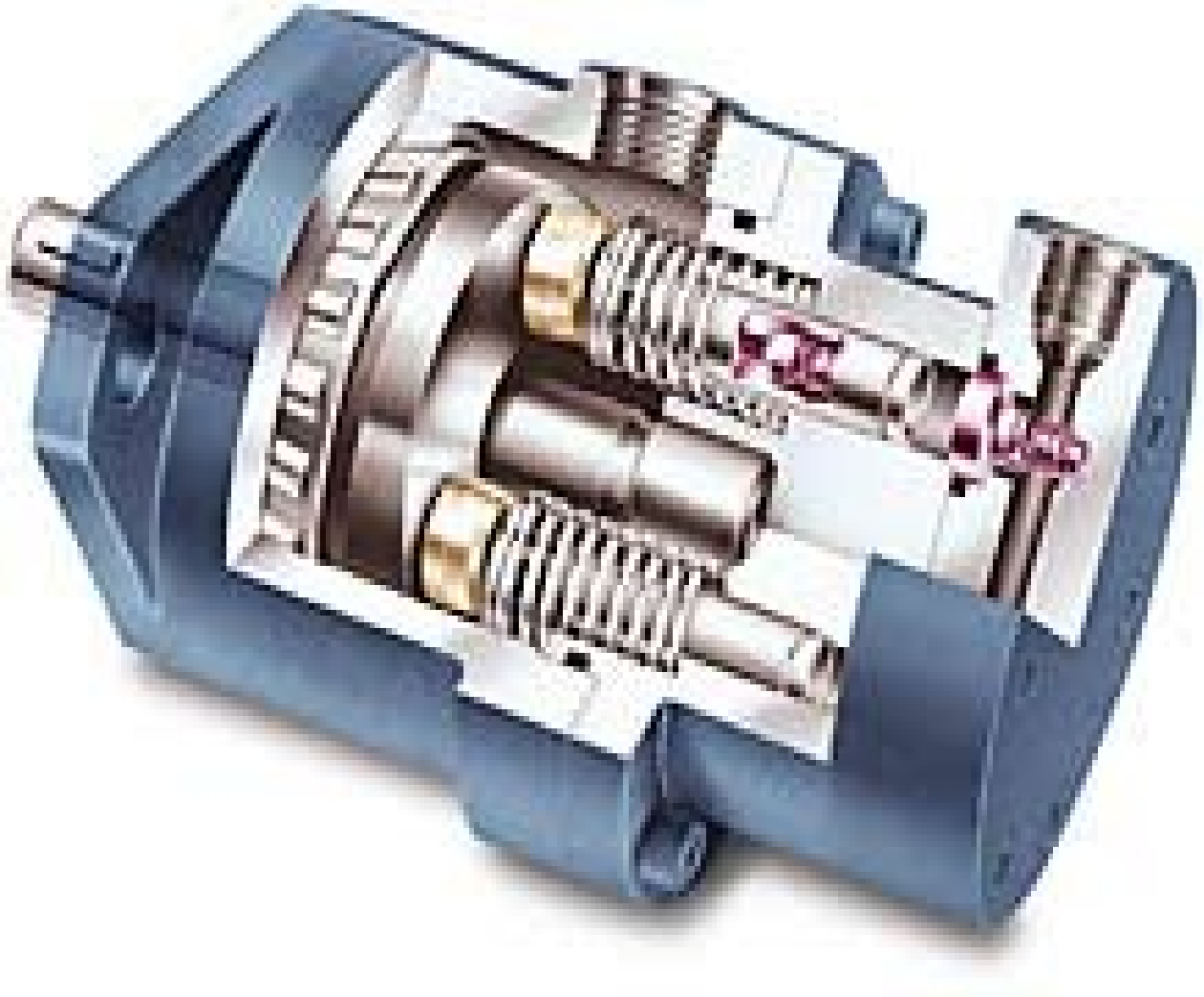


Internal Gear Pump

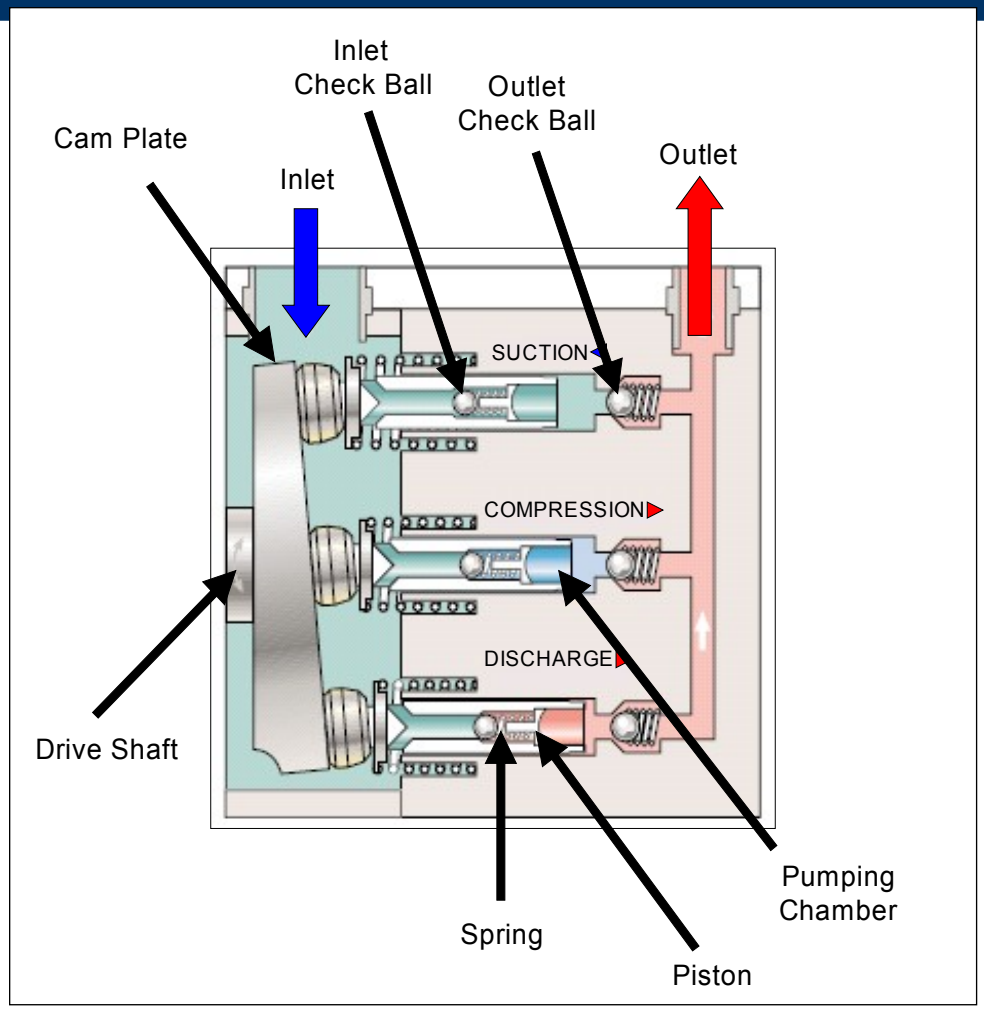
Piston Pumps

- Piston Pump Diagram
- Major Component Review
- Operation and Application
- Maintenance and Troubleshooting

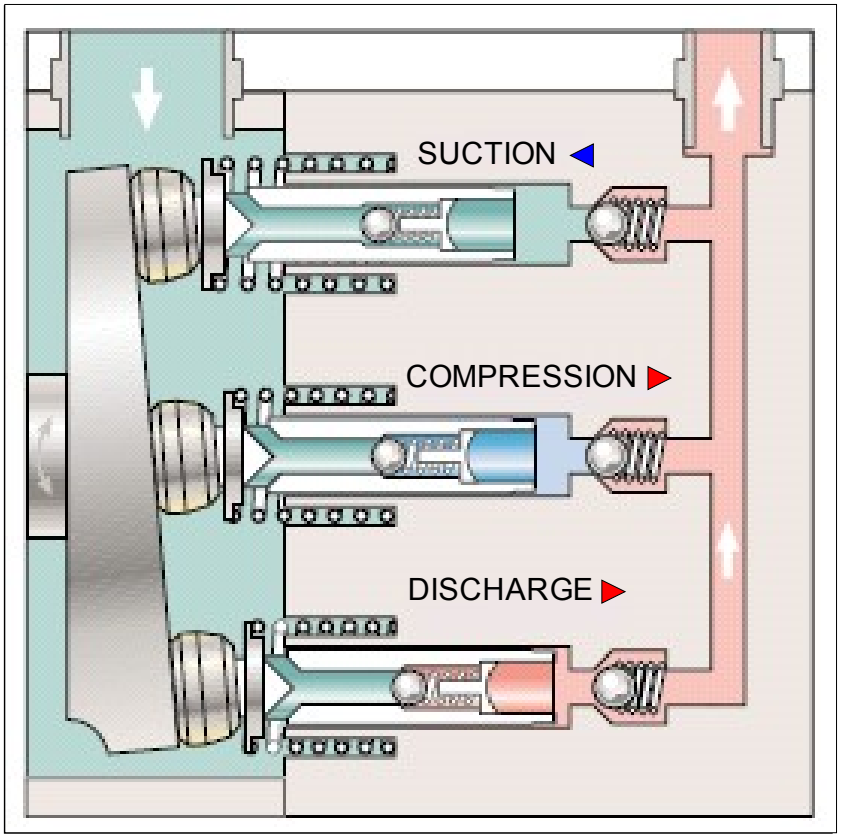
Piston Pump



Major Components



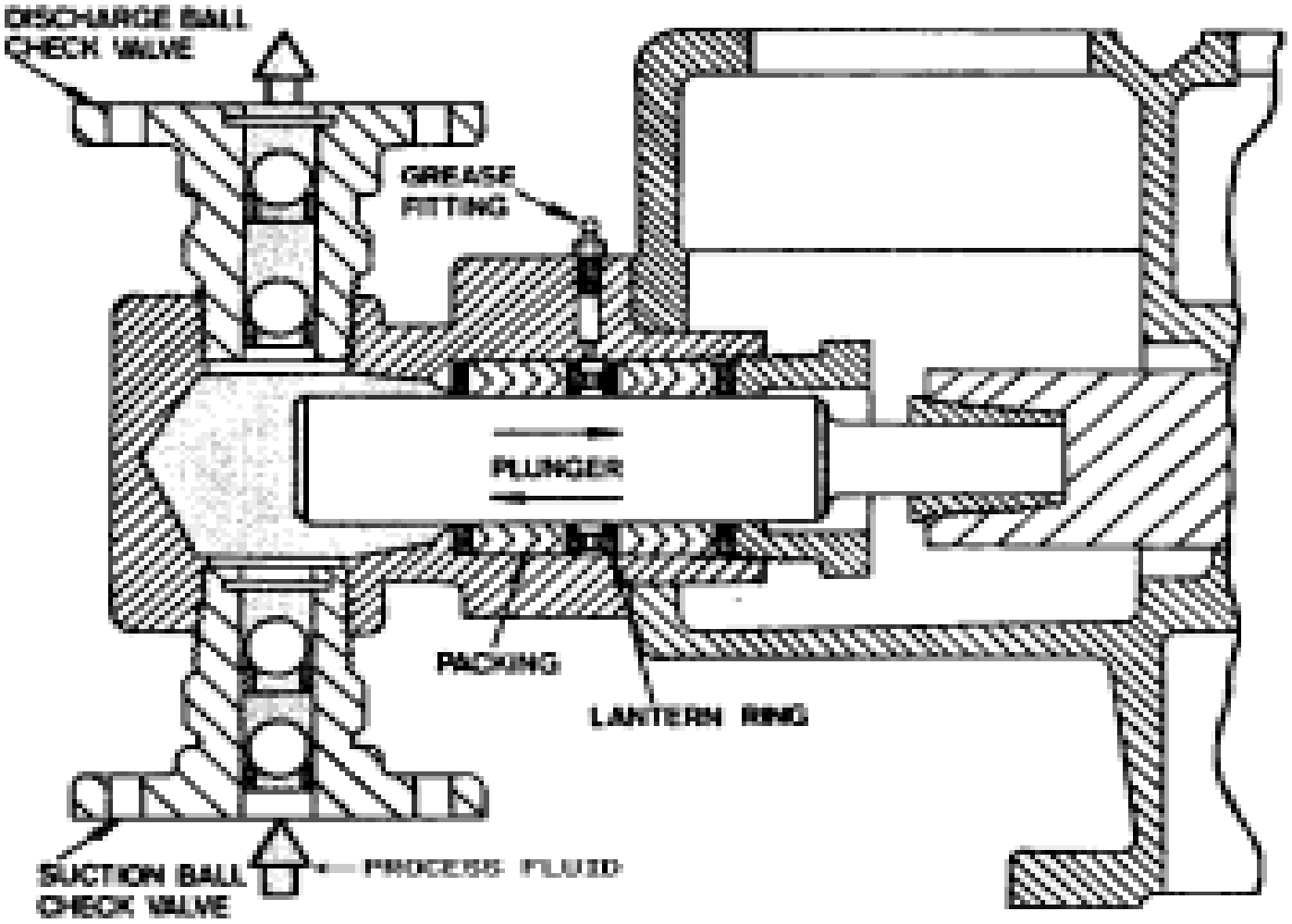
Operation and Application



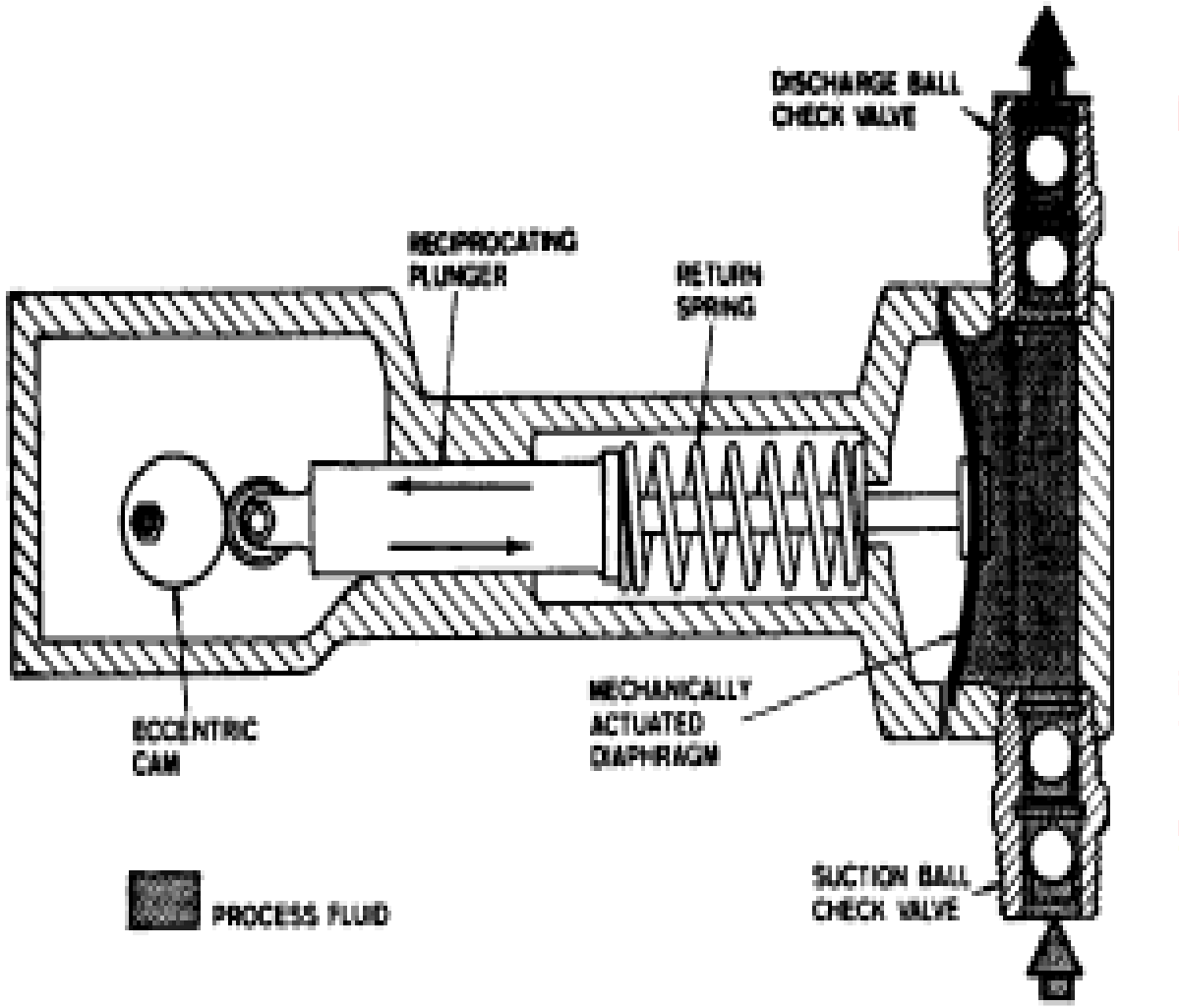
Plunger Pumps

- Plunger Pump Diagram
- Major Component Review
- Operation and Application

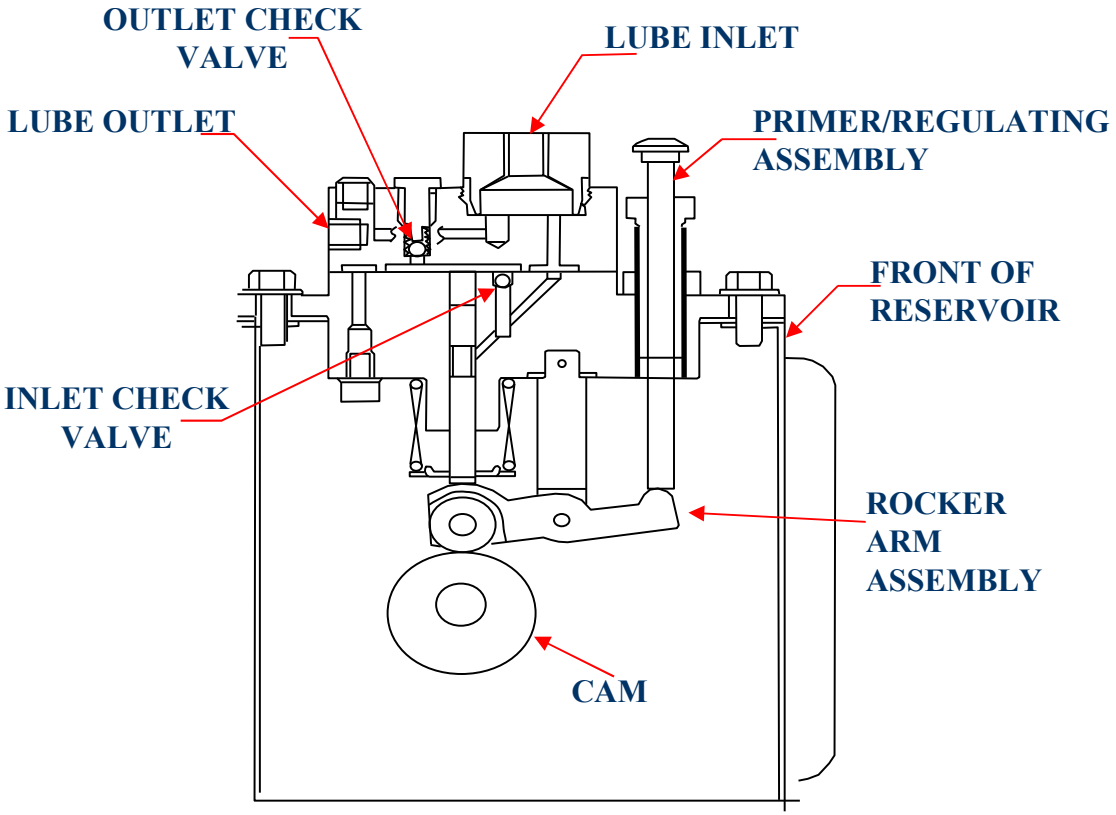
Packed Plunger Pump



Diaphragm Plunger Pump



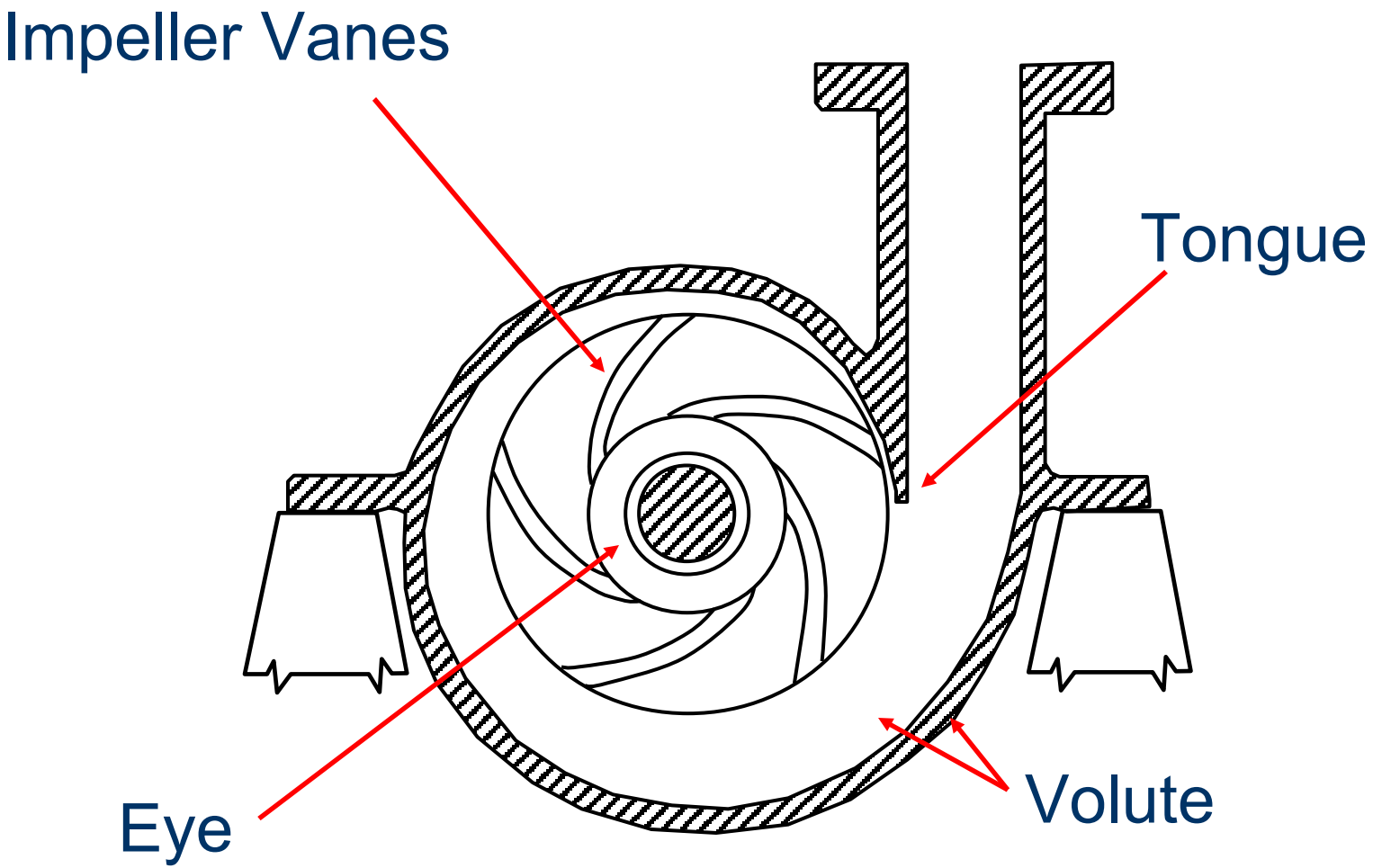
Example Plunger Pump Diagram



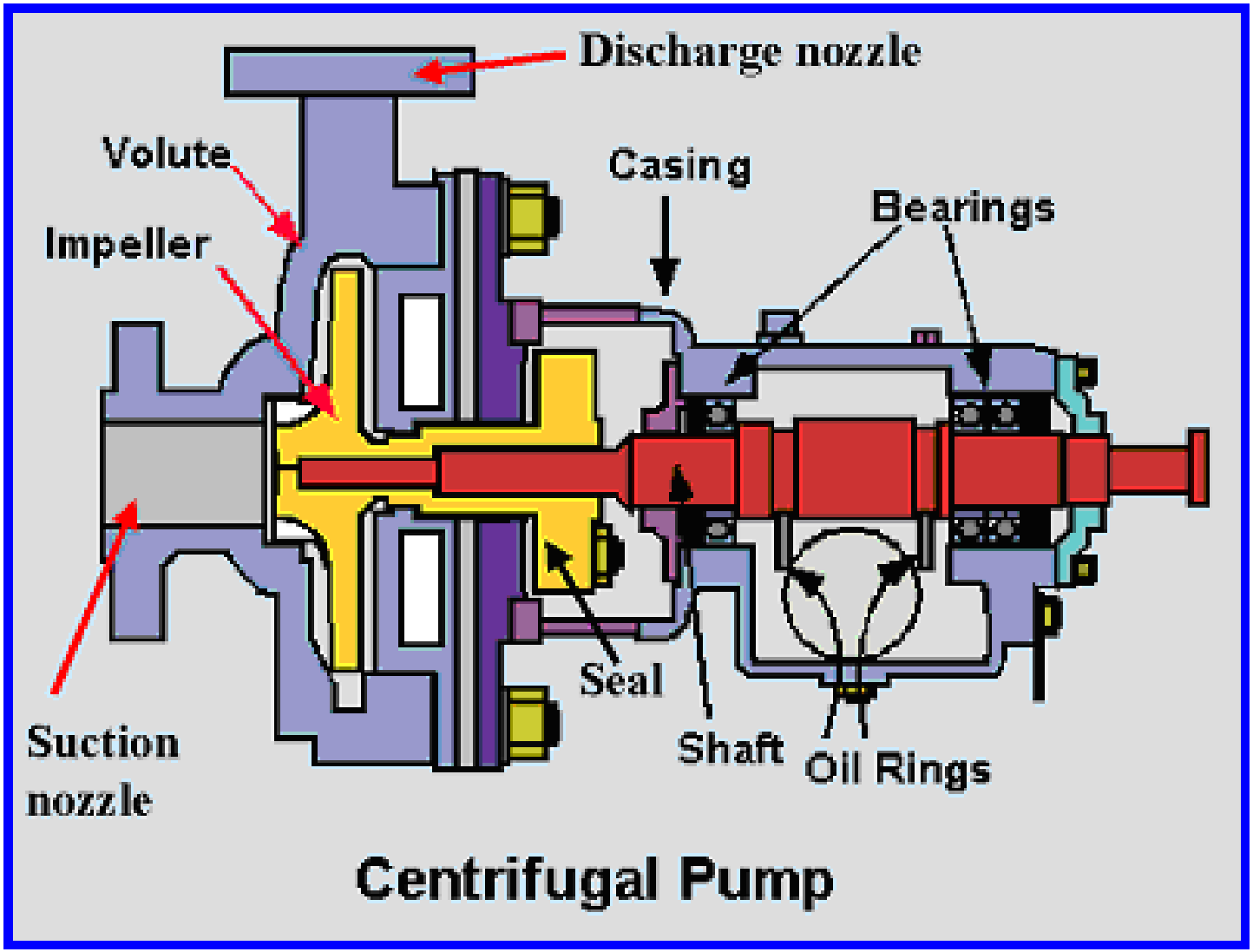
Centrifugal Pumps

- Centrifugal Pump Diagram
- Major Component Review
- Operation and Application
- Pump Laws
- Centrifugal Pumps
- Maintenance and Troubleshooting

Fundamentals



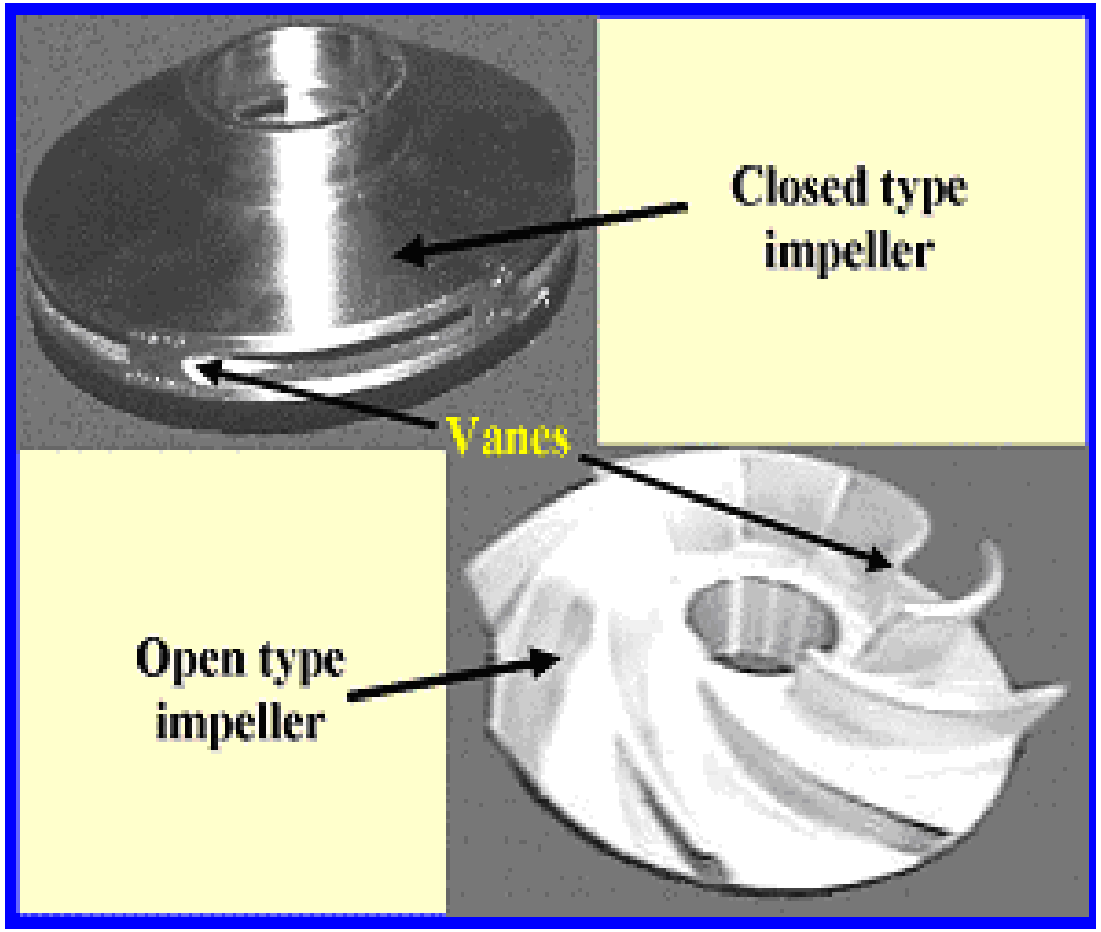
Centrifugal Pump Diagram



Sleeve/Coupling/Bearings

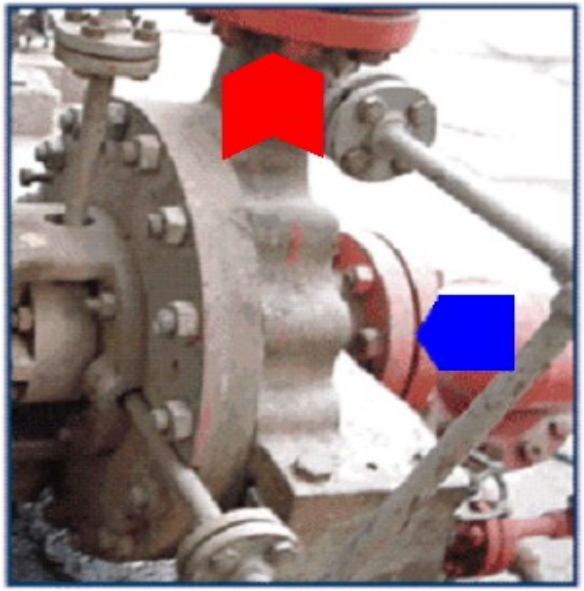
- Shaft Sleeve
- Coupling
  - Elastomeric couplings (having properties that resemble rubber)
  - Non-elastomeric
- Bearings

Impeller Types

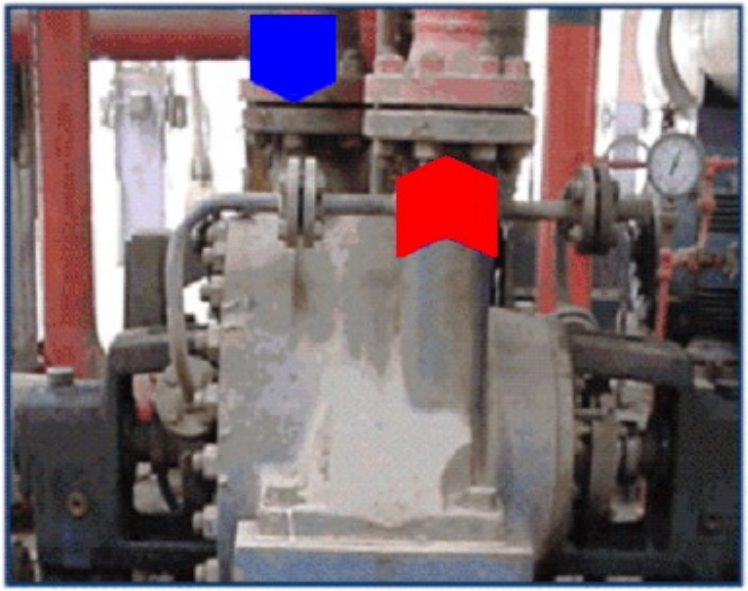


Suction and Discharge

Side Suction
Top Discharge



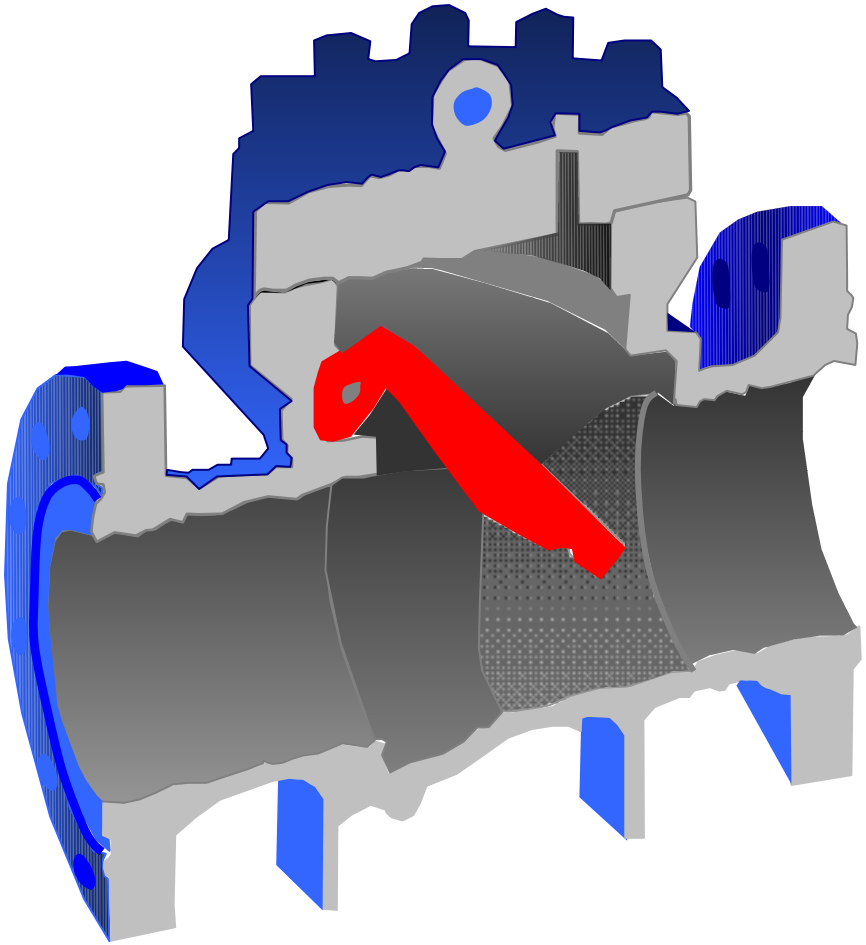
Top Suction
Top Discharge



Suction Nozzle

Discharge Nozzle

Swing Type Check Valve



Valves

- Single disc swing valves
- Double disc or wafer check valves
- Lift-check valves
- Silent or center guide valves
- Ball-check valves
- Cone check valves

Centrifugal Pump Application

- High Flow-rate requirements
- Low Differential Pressure (Lift) requirements
- Low Fluid Viscosity

Centrifugal Pump Operation

- Conversion of rotational driver energy into flow energy
- Work on the fluid is performed by impeller and Volute (higher flow, lower pressure) or Diffuser (lower flow, higher pressure)

Centrifugal Flow

- Centrifugal pumps generate flow by using one of three actions:
  - Radial flow
  - Mixed flow
  - Axial flow

Centrifugal Pump Operation

- Flow Path
- Precautions
  - Prevent Cavitation
  - Avoid Low Flow Conditions

Centrifugal Pump Operation

- Cavitation
  - Formation of and subsequent collapse of bubbles within a pumped fluid.
  - Formation occurs in regions of low pressure and collapse occurs in regions of high pressure.
- Cavitation can result in:
  - Loss of capacity
  - Lowered Discharge Pressure
  - Lower Efficiency
  - Noise, Vibration, and Damage to Pump components.

Cavitation

- Cavitation is Caused by:
  - Vaporization
  - Air ingestion
  - Internal recirculation
  - Flow turbulence
  - Vane Passing Syndrome

Vaporization

- A fluid vaporizes when its pressure gets too low, or its temperature too high. All centrifugal pumps have a required head (pressure) at the suction side of the pump to prevent this vaporization.

Air Ingestion

- Air gets into a system in several ways that include :
  - Through the stuffing box
  - Leaking flanges
  - Suction inlet pipe is out of fluid

Turbulence

- We would prefer to have liquid flowing through the piping at a constant velocity.
- Corrosion or obstructions can change the velocity of the liquid and any time you change the velocity of a liquid you change its pressure.

Vane Passing Syndrome

- You will notice damage to the tip of the impeller caused by its passing too close to the pump cutwater.

Pump Laws

- Velocity is directly proportional to Pump Speed
  - $V_{flow} \alpha N$
- Discharge Head is directly proportional to the square of Pump Speed
  - $H_{pump} \alpha N^2$
- Pump Power consumption is directly proportional to the cube of Pump Speed
  - $P_{pump} \alpha N^3$

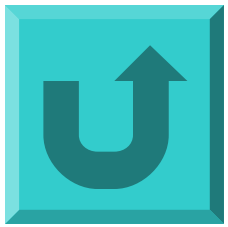
Pump Laws

- Example:
  - N = 1450 RPM
  - V = 400 m3 / hr
  - H = 100 Barg
  - P = 45 kW

Summary

- Review Question and Answer Session

CLICK TO RETURN TO TOPICS



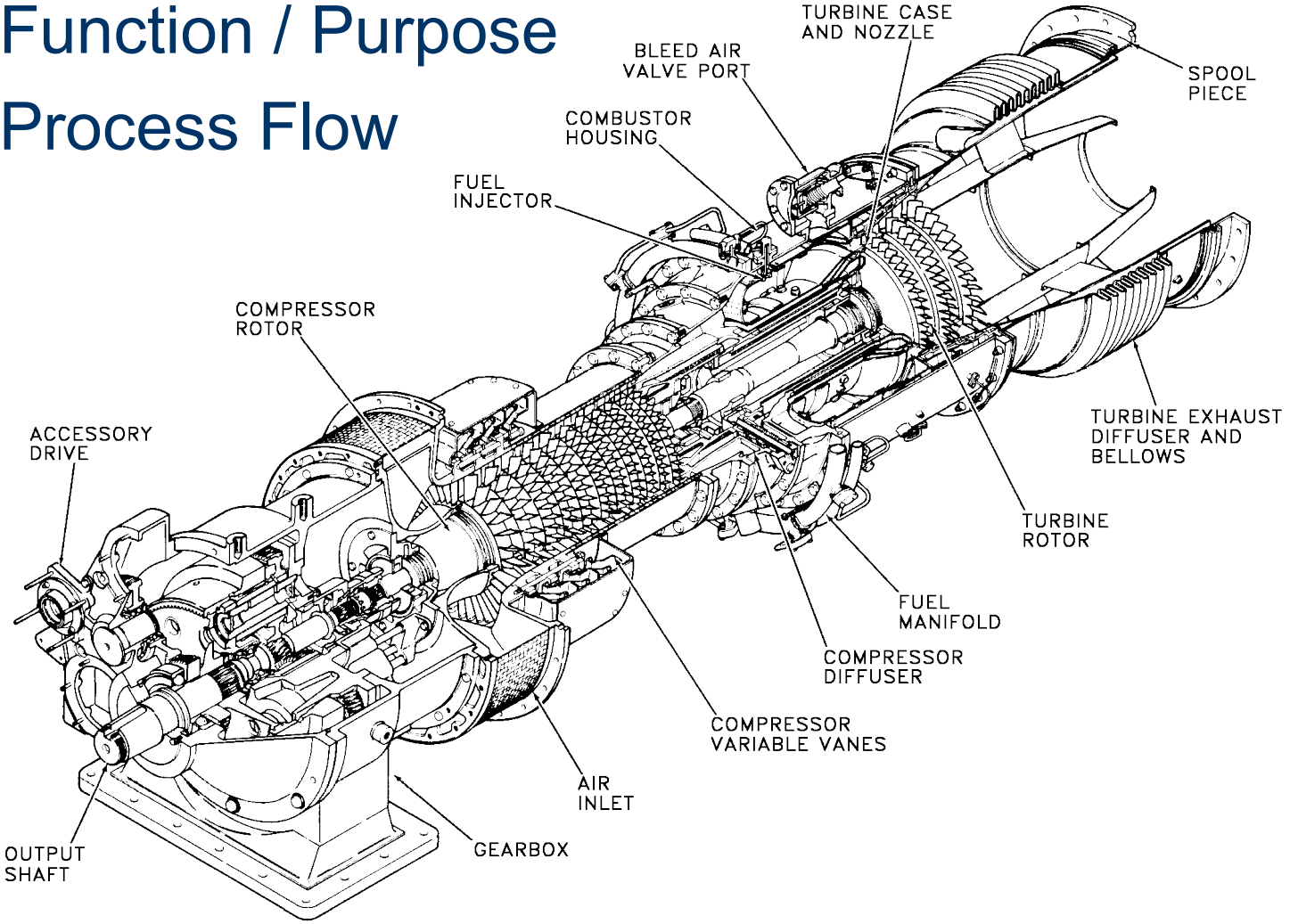
TURBINES

Objectives

- Define Brayton Cycle.
- Turbine Theory of Operation
- Define major components used in a Gas Turbine system.
- Identify Gas Turbine auxiliary systems.
- Define Gas Turbine Maintenance requirements.

Gas Turbine

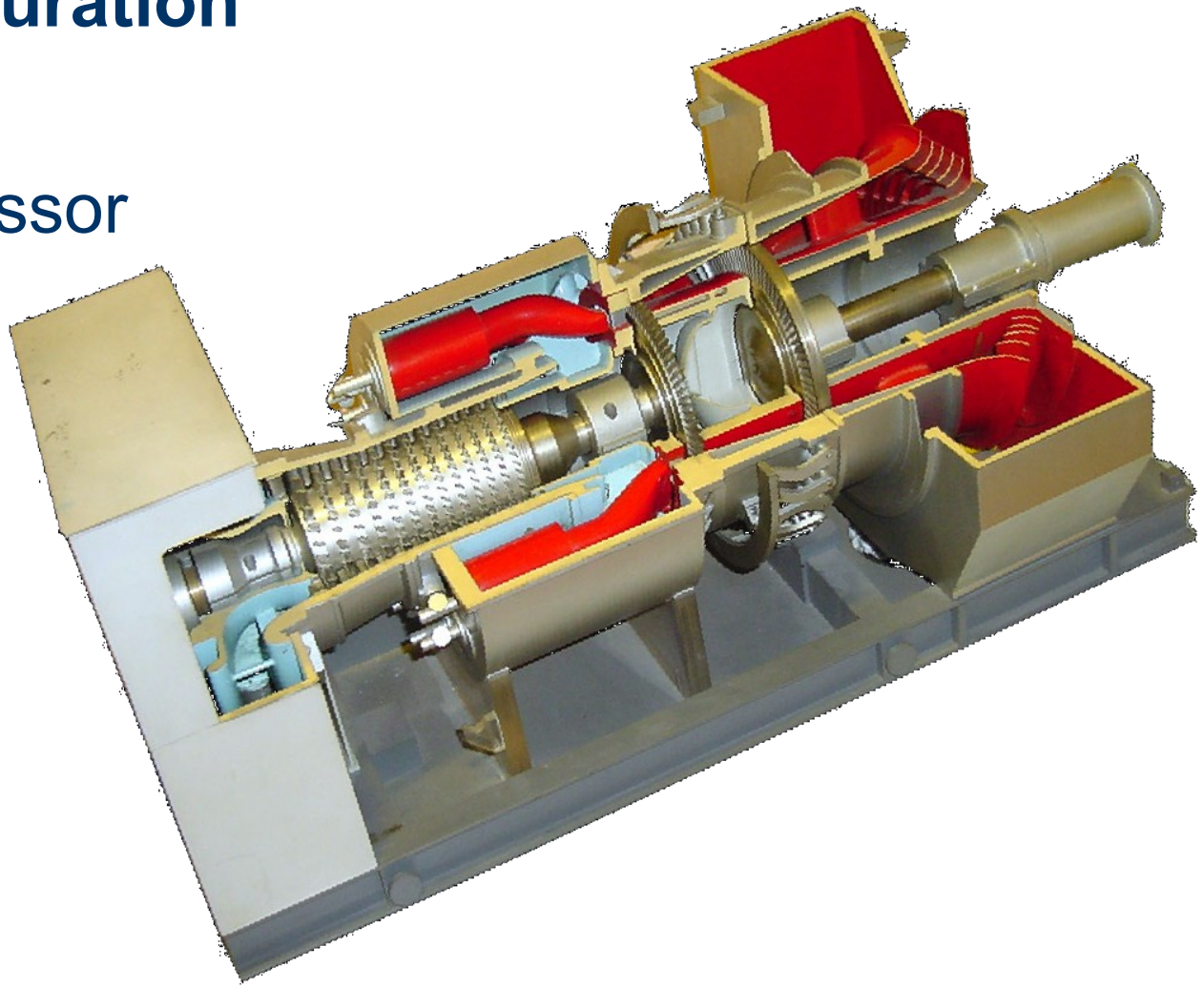
- Function / Purpose
- Process Flow



Gas Turbine

Basic Configuration

- Air Compressor
- Combustor
- Turbine

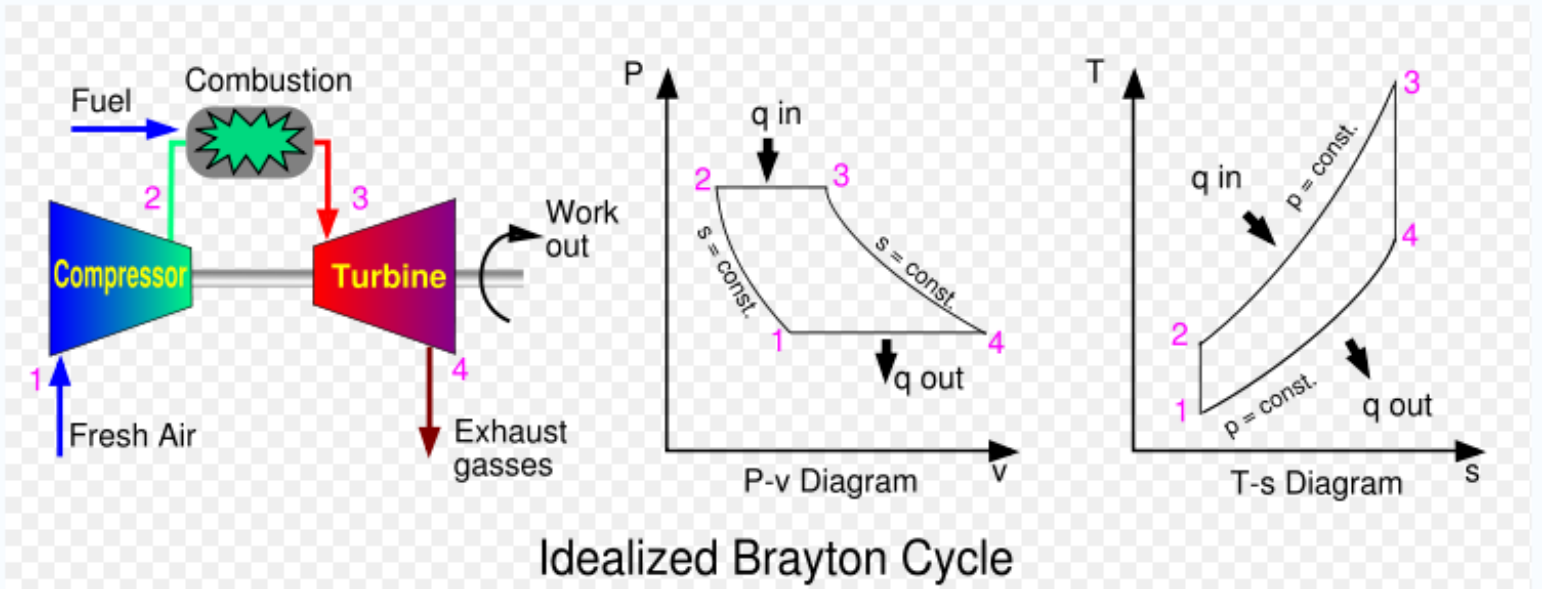


Gas Turbine

- A **gas turbine** extracts energy from a flow of hot gas produced by combustion of gas or fuel oil in a stream of compressed air. It has an upstream air compressor (radial or axial flow) mechanically coupled to a downstream turbine and a combustion chamber in between. "Gas turbine" may also refer to just the turbine element

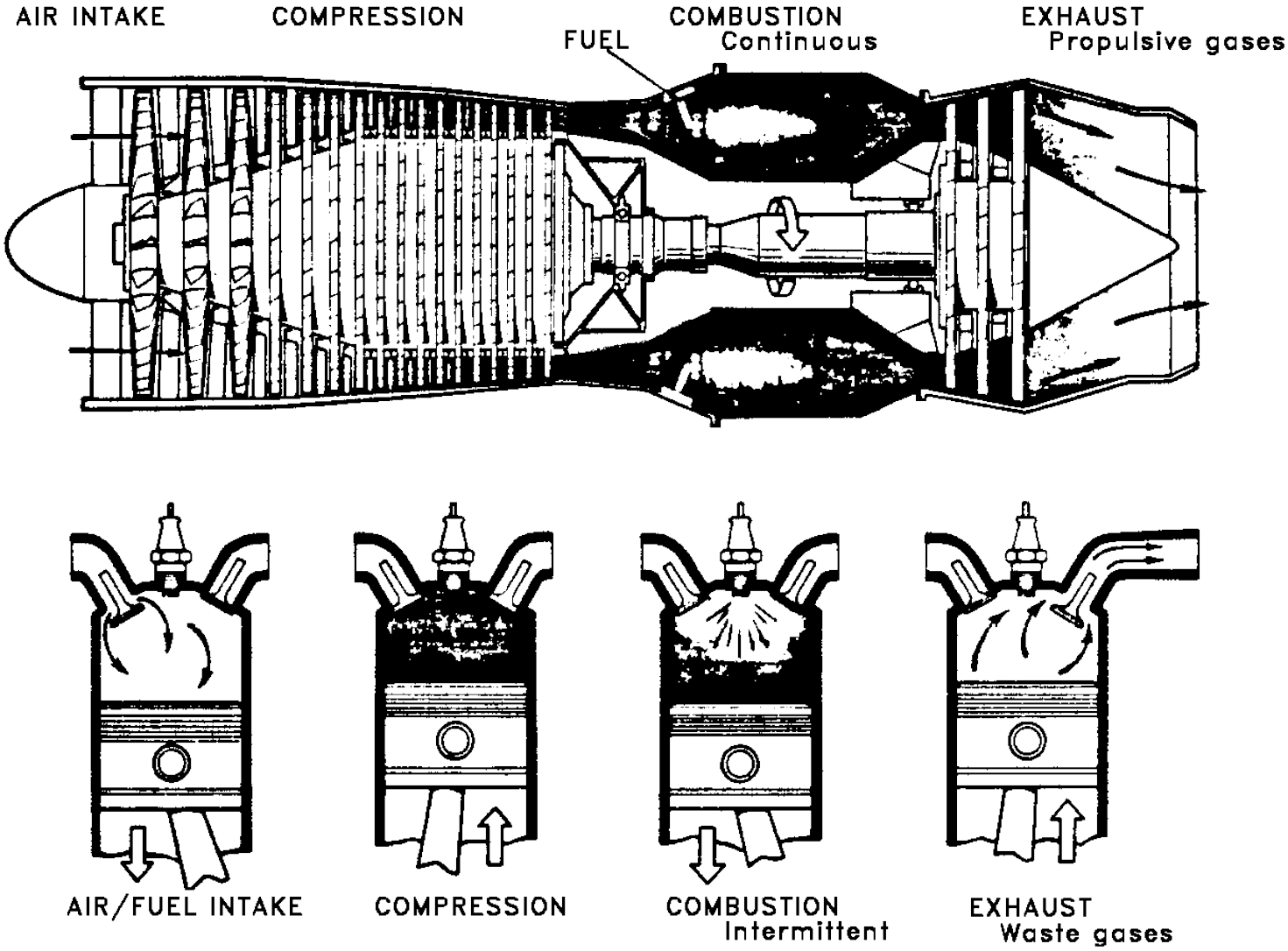
Brayton Cycle

- Gas turbines are described thermodynamically by the Brayton cycle, in which air is compressed isentropically, combustion occurs at constant pressure, and expansion over the turbine occurs isentropically back to the starting pressure.



Idealized Brayton Cycle

Steps of the Brayton Cycle



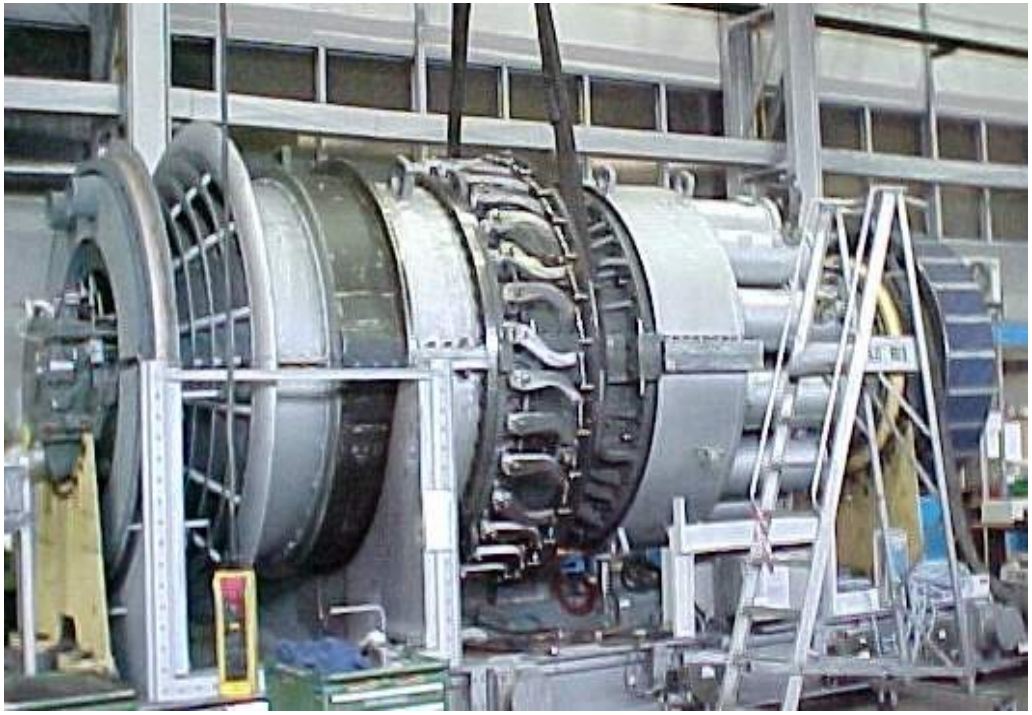
Performance parameters

- Speed of rotation
- Oil Temperature
- Oil Pressure
- Fuel gas pressure
- Rotor axial displacement
- Bearing vibrations
- Exhaust temperature

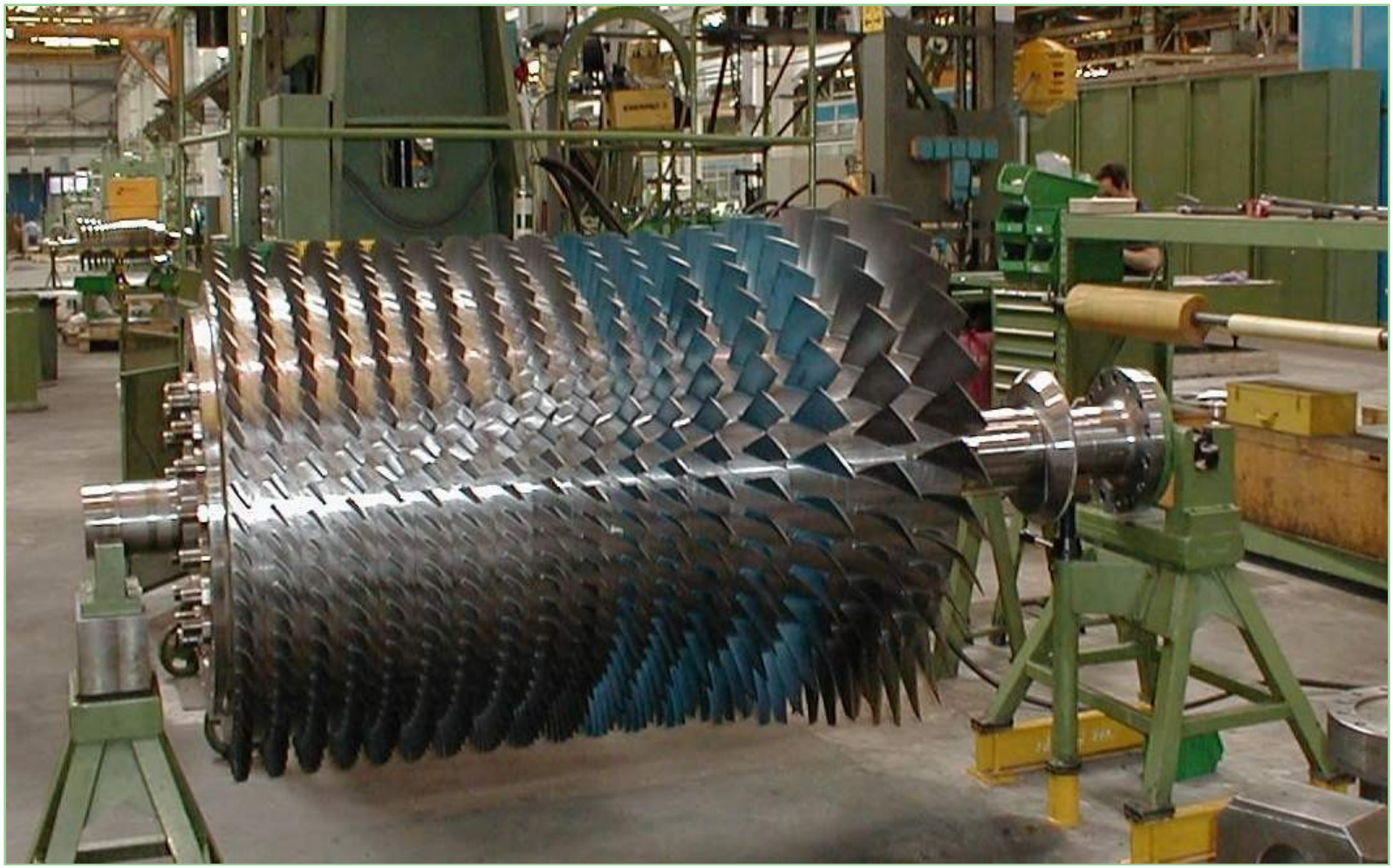
Main Components

- Turbine Casing
- Compressor Section
- Combustion Chamber
- Bearings
- Turbine Rotors
- Auxiliary Systems

Turbine Casing



Compressor Section

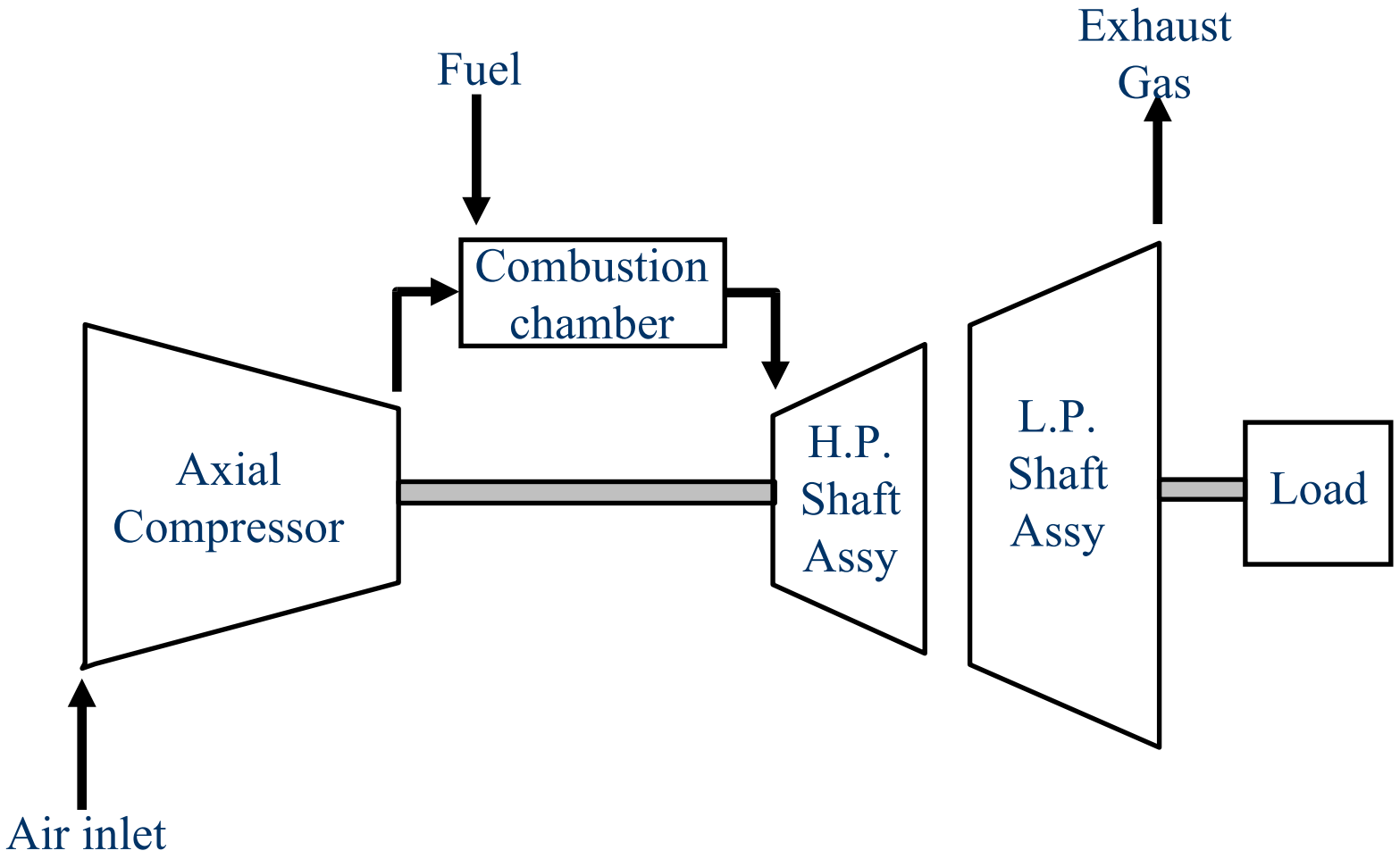


Combustion Chamber



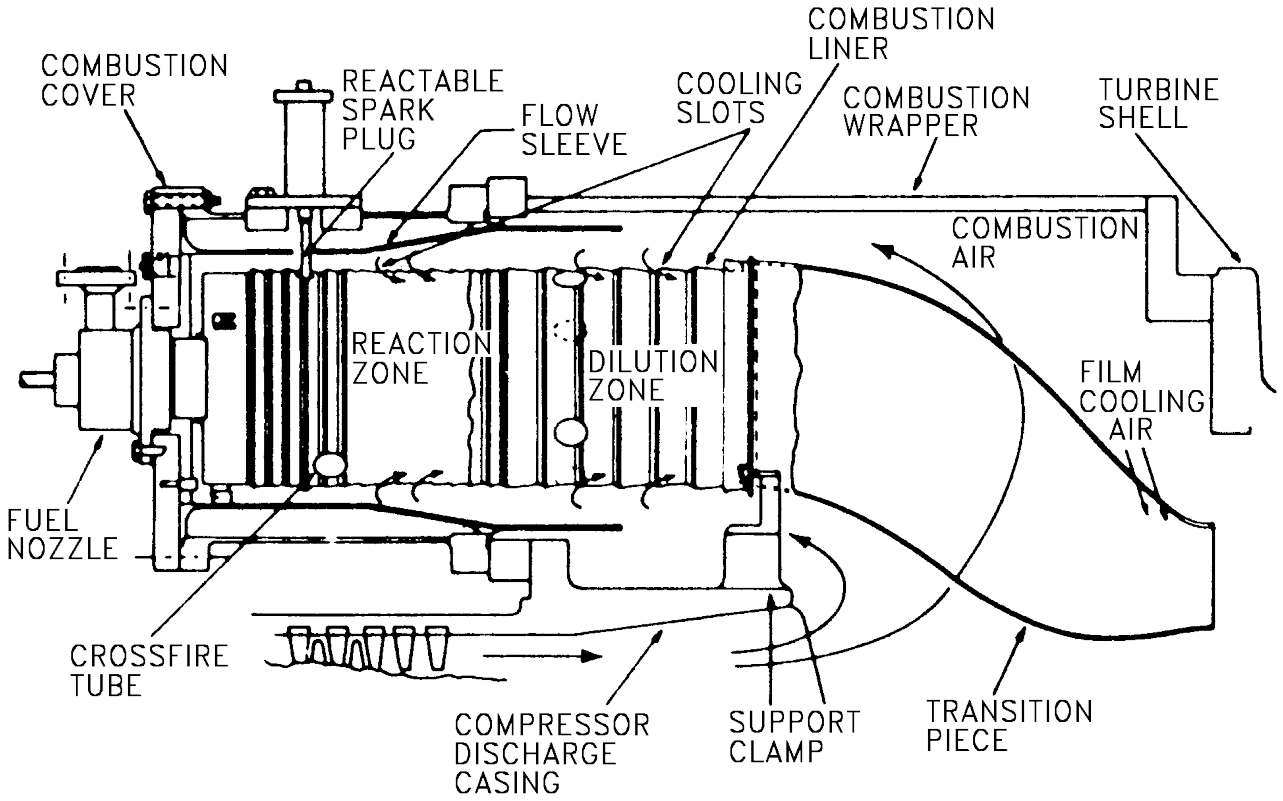


Split Shaft Design

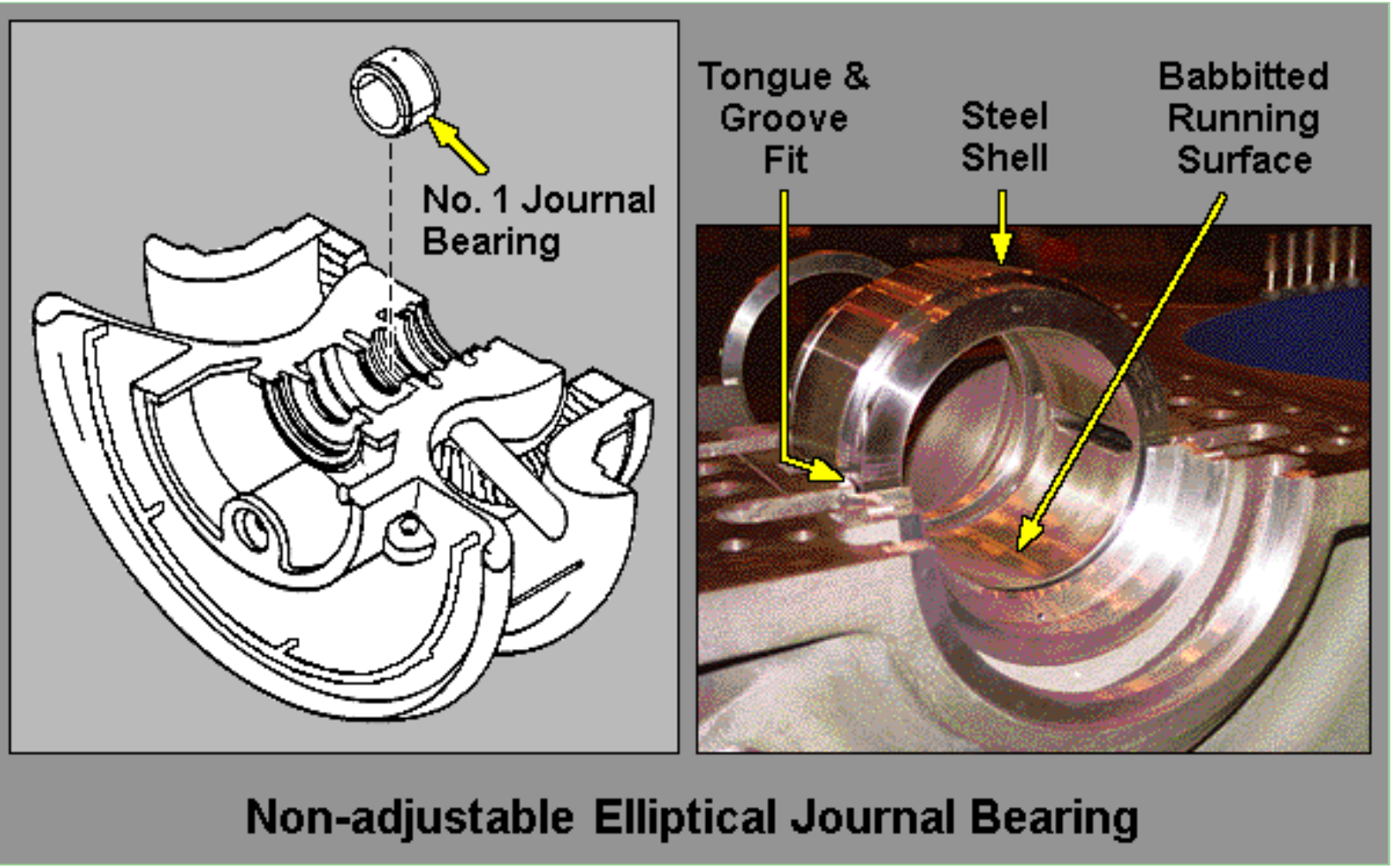


Combustor

- Can-annular Type Combustor Example



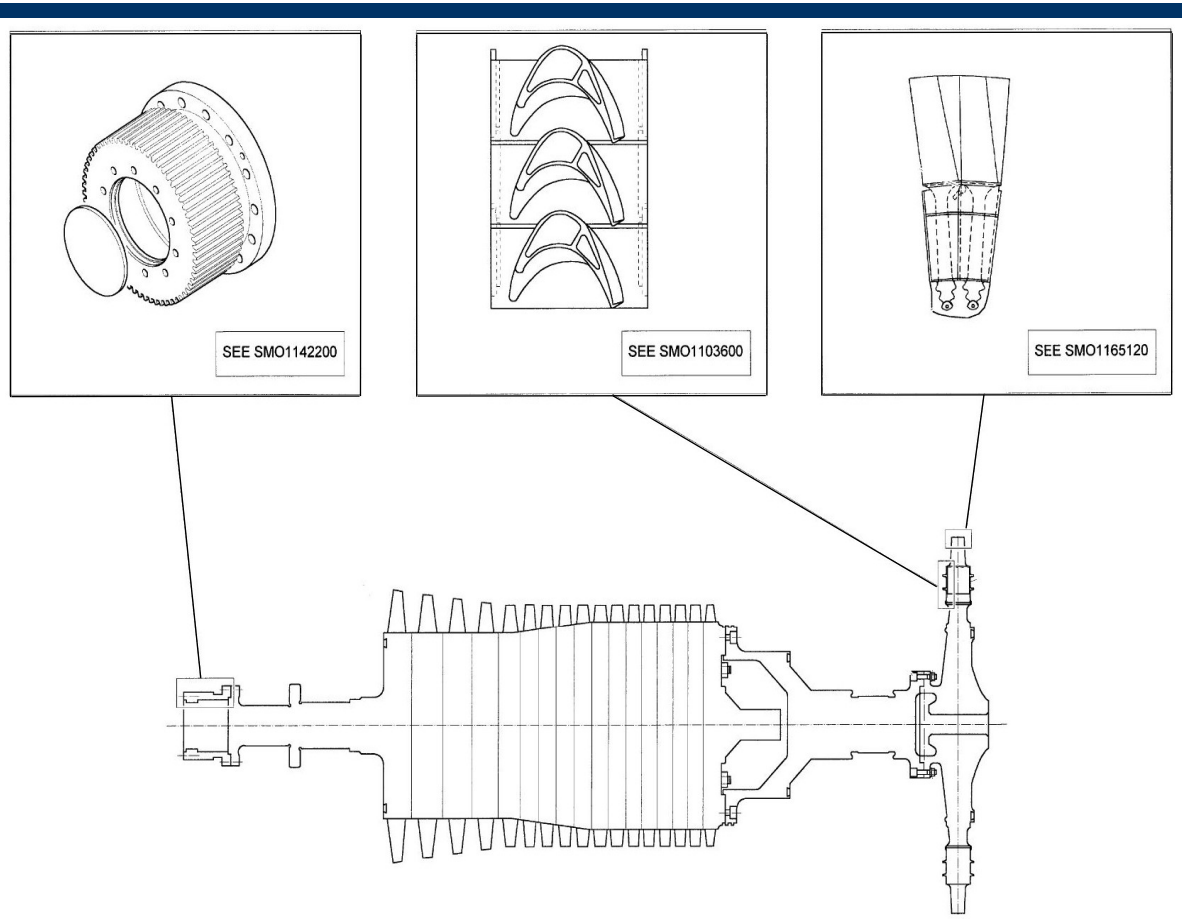
Bearings


No. 1 Journal Bearing
Tongue & Groove Fit
Steel Shell
Babbitted Running Surface
Non-adjustable Elliptical Journal Bearing

Turbine Rotors

- Rotors/Buckets
- Split shaft design
- Variable Nozzle

Rotors/Buckets



Variable Nozzle





Shutdown Sequence

- Normal Shutdown
- Emergency Stop

Normal Shutdown

- Manually initiated, Automatically sequenced
- Turbine is run at idle to reduce thermal stresses
- Turbine may operate on starting system to further reduce stresses
- Unit will be jacked at 1 to 2 rpm for several cool-down hours

Emergency Stop

- Can be manually or automatically initiated
- Automatically sequenced
- Does NOT include a cool-down delay
- When trip is caused by a fire sensor all lube oil flow stops

Filters

- Lube oil filter
- Fuel oil filters
- Hydraulic supply filters
- Air filters

Shutdown Maintenance

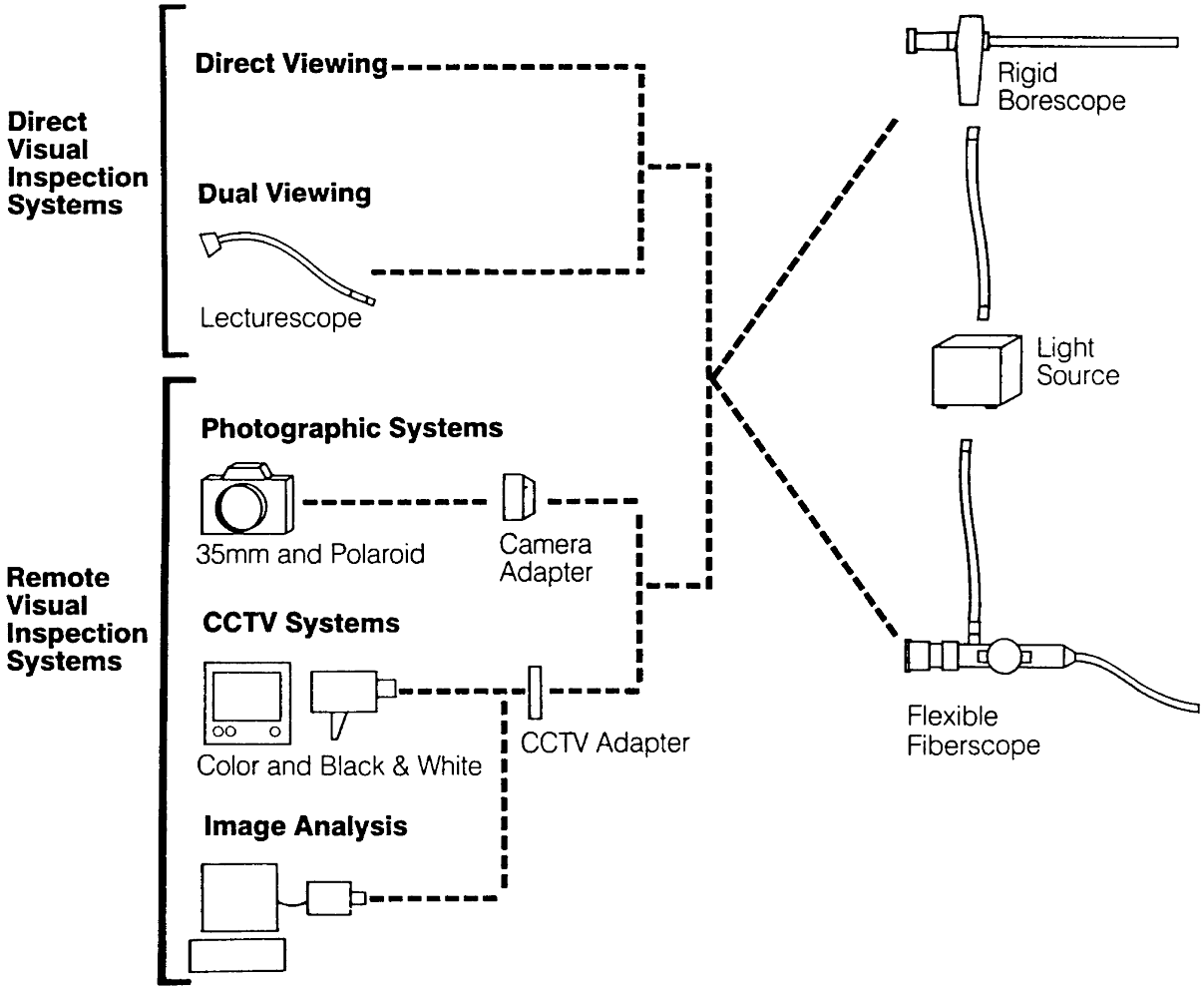
- Major Inspection
- Borescope Inspections
- Combustion Inspection
- Hot Gas Path Inspection

Major Inspection

- Turbine Disassembly
- Initial Alignment Checks
- Component Inspections
- Wear component replacement
- Reassembly
- Final Alignment Checks

Borescope Inspections

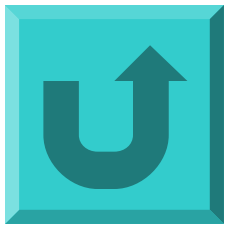
- Overview and Purpose



Summary

- Review Question and Answer Session

CLICK TO RETURN TO TOPICS



FANS AND LOUVERS

Course Objectives

- Define the steps necessary to maintain and replace fan bearings
- Discuss characteristics of Belts
- State the steps necessary to remove, replace and adjust drive belts

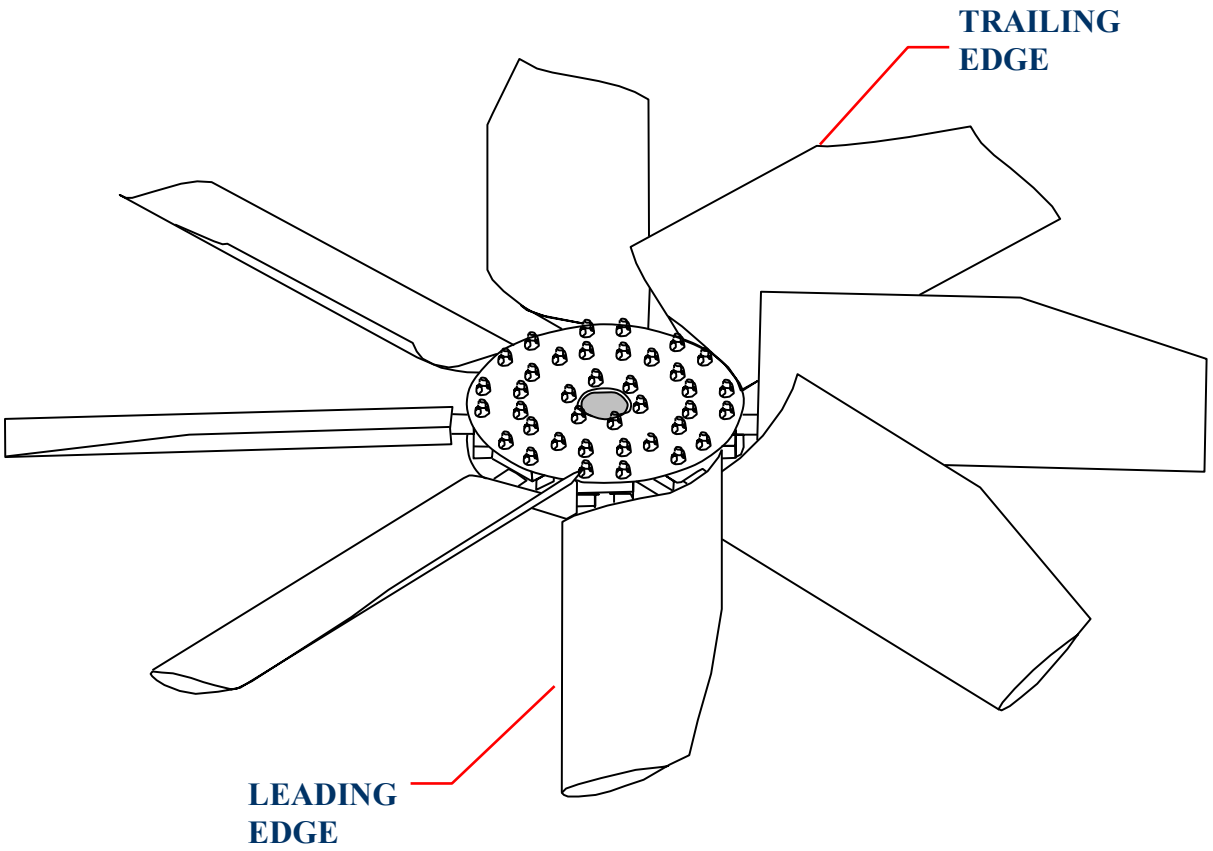
Course Objectives

- Discuss methods of determining cause based upon effect

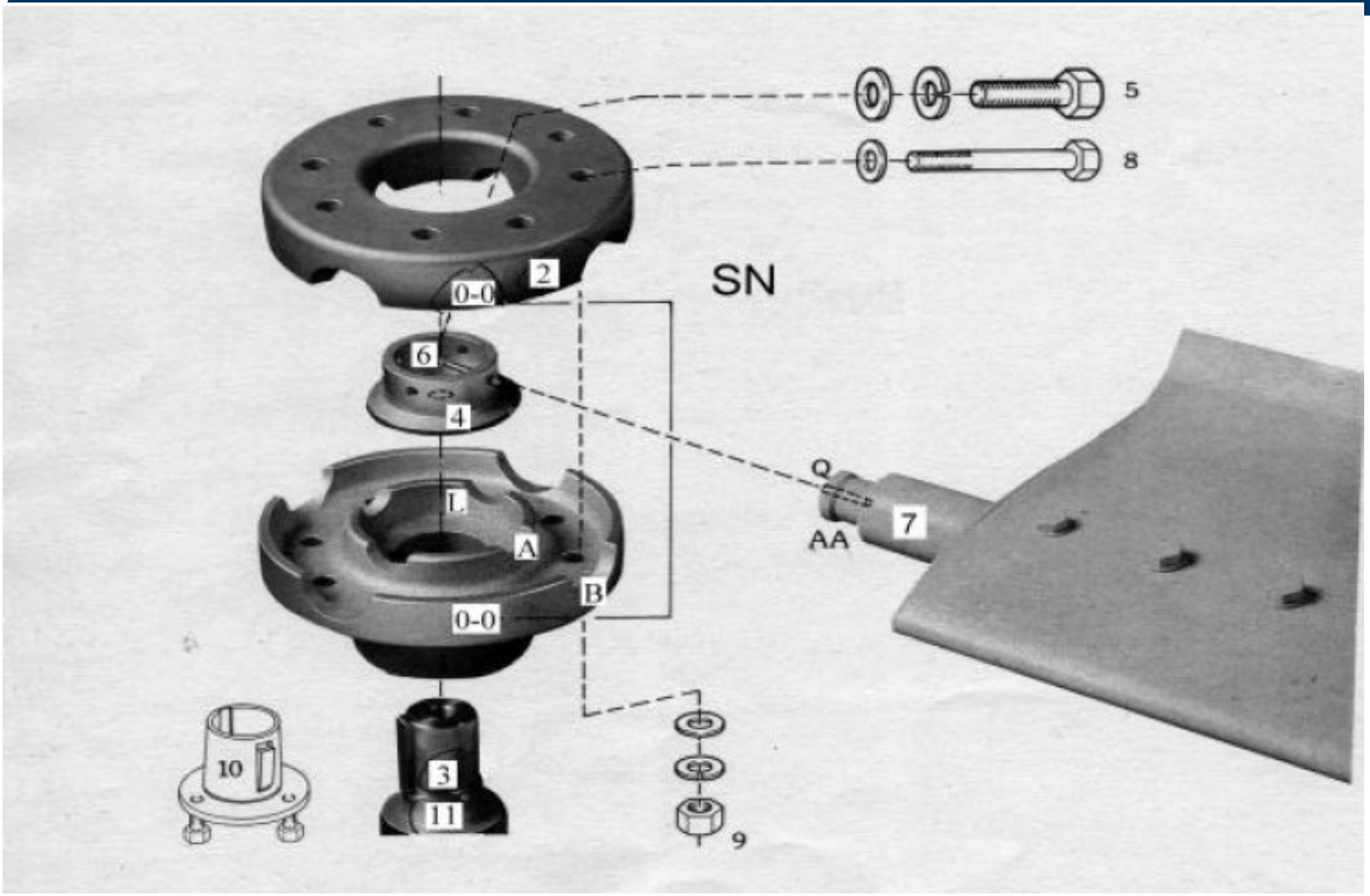
Fan Safety

- Rotating Equipment
- Elevation
- High Temperature
- H2S

Rotor and Hub Assembly Example



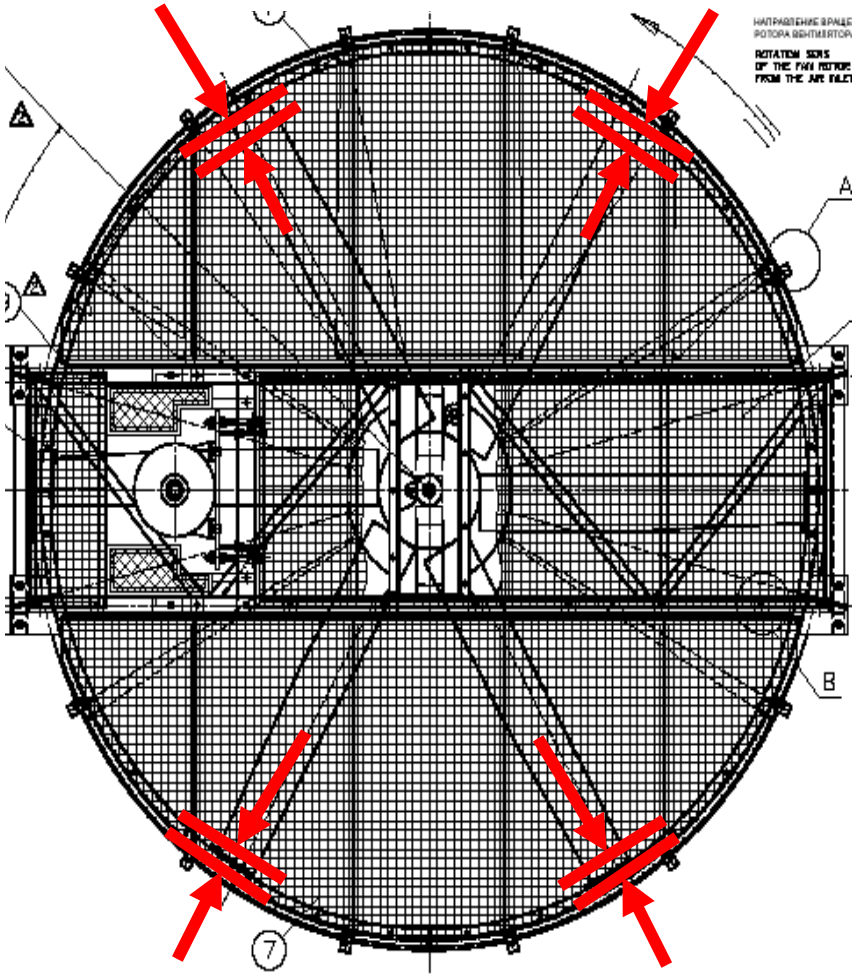
Rotors



Fan Checks

- Adjust the pitch of each blade to the vendor's specified angle
- Verify blades rotate freely
- Verify proper motor rotation

Fin Fan Tip Clearance

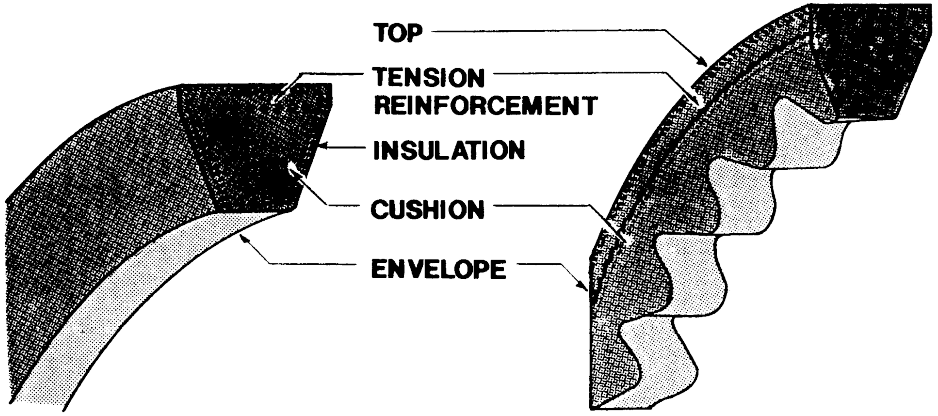


- Blade Tip Clearance
  - Adjust each blade assembly to the vendor's specified tip clearance

Driver

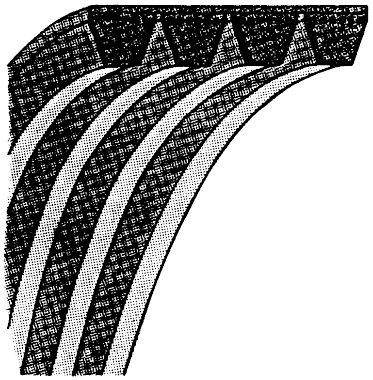
- Variable Speed Drive (VSD) Electric Motor
- Totally Enclosed Fan Cooled (TEFC)
- Explosion Proof

Belts

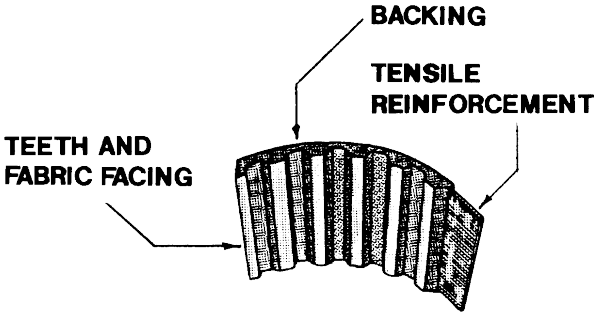


STANDARD V-BELT

COGGED V-BELT

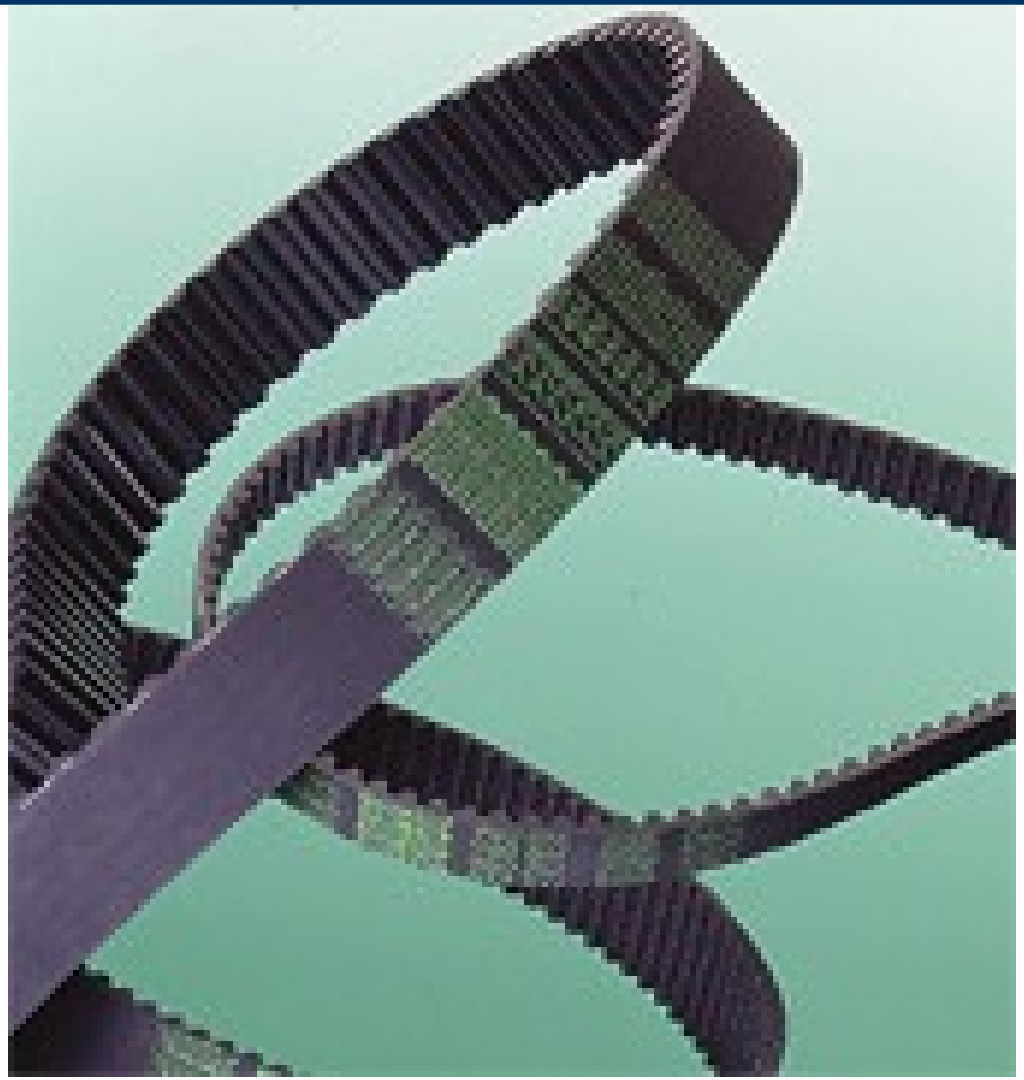


BANDED SET OF STANDARD V-BELTS

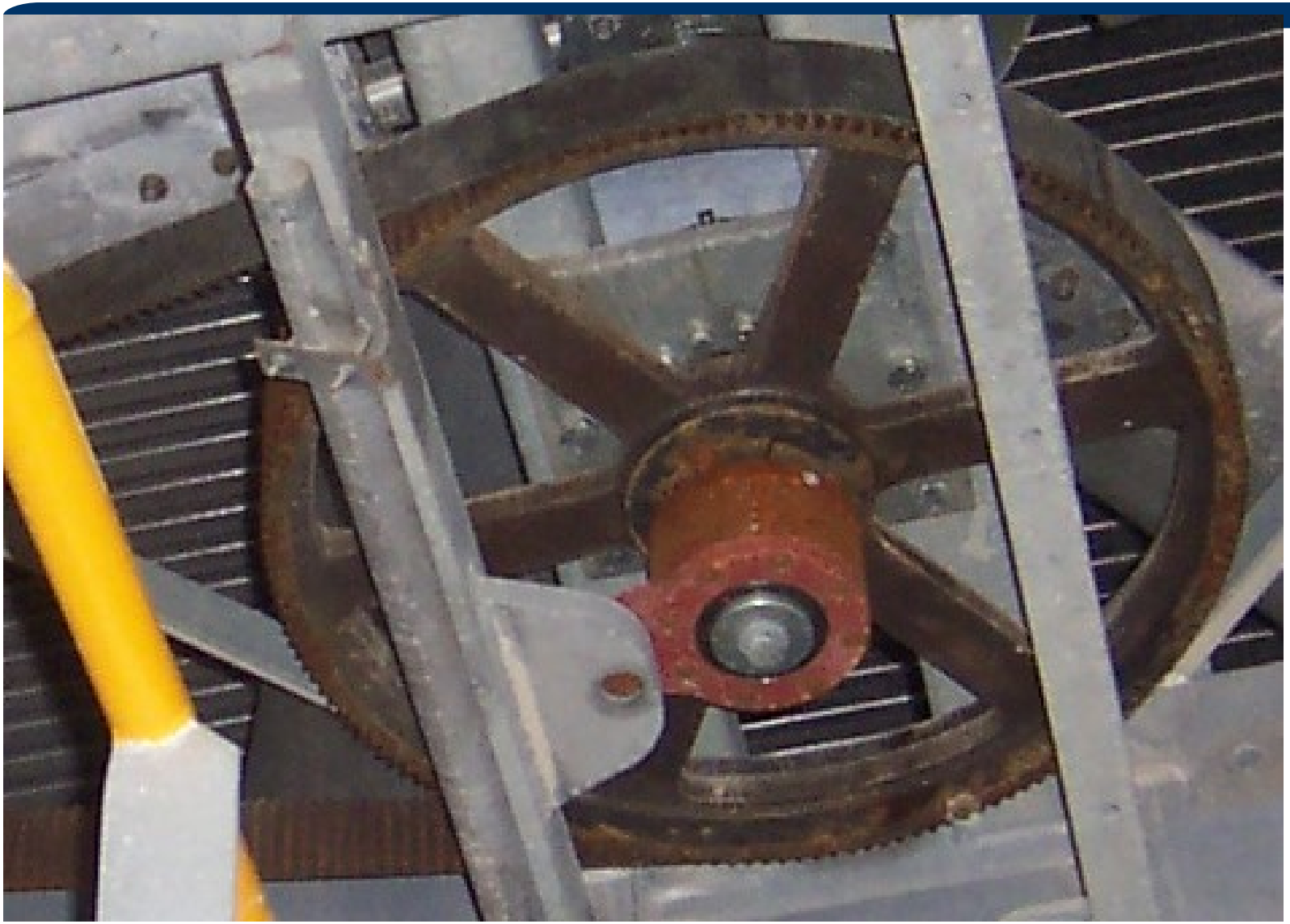


HIGH TORQUE DRIVE (HTD) BELT

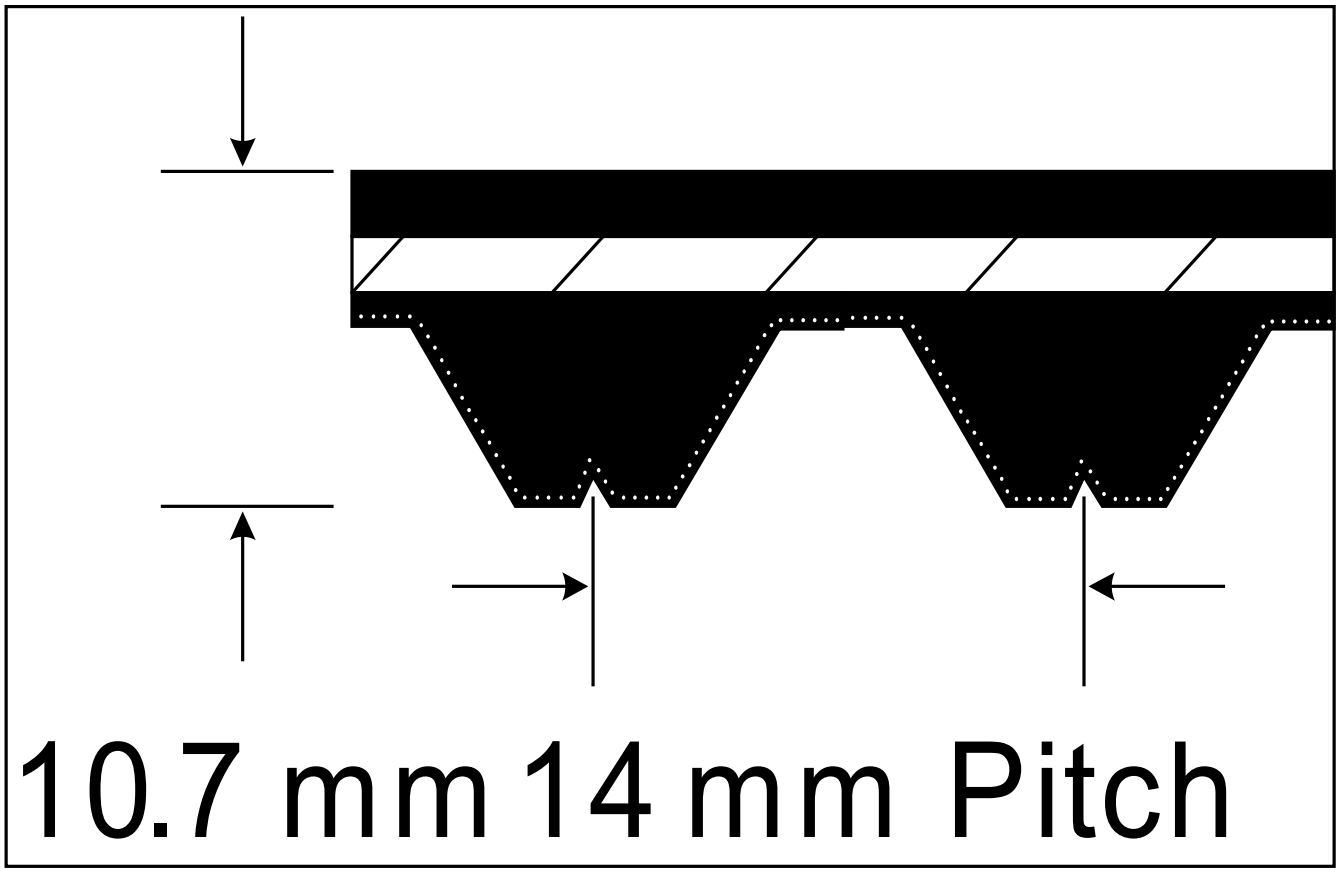
HTD Belts



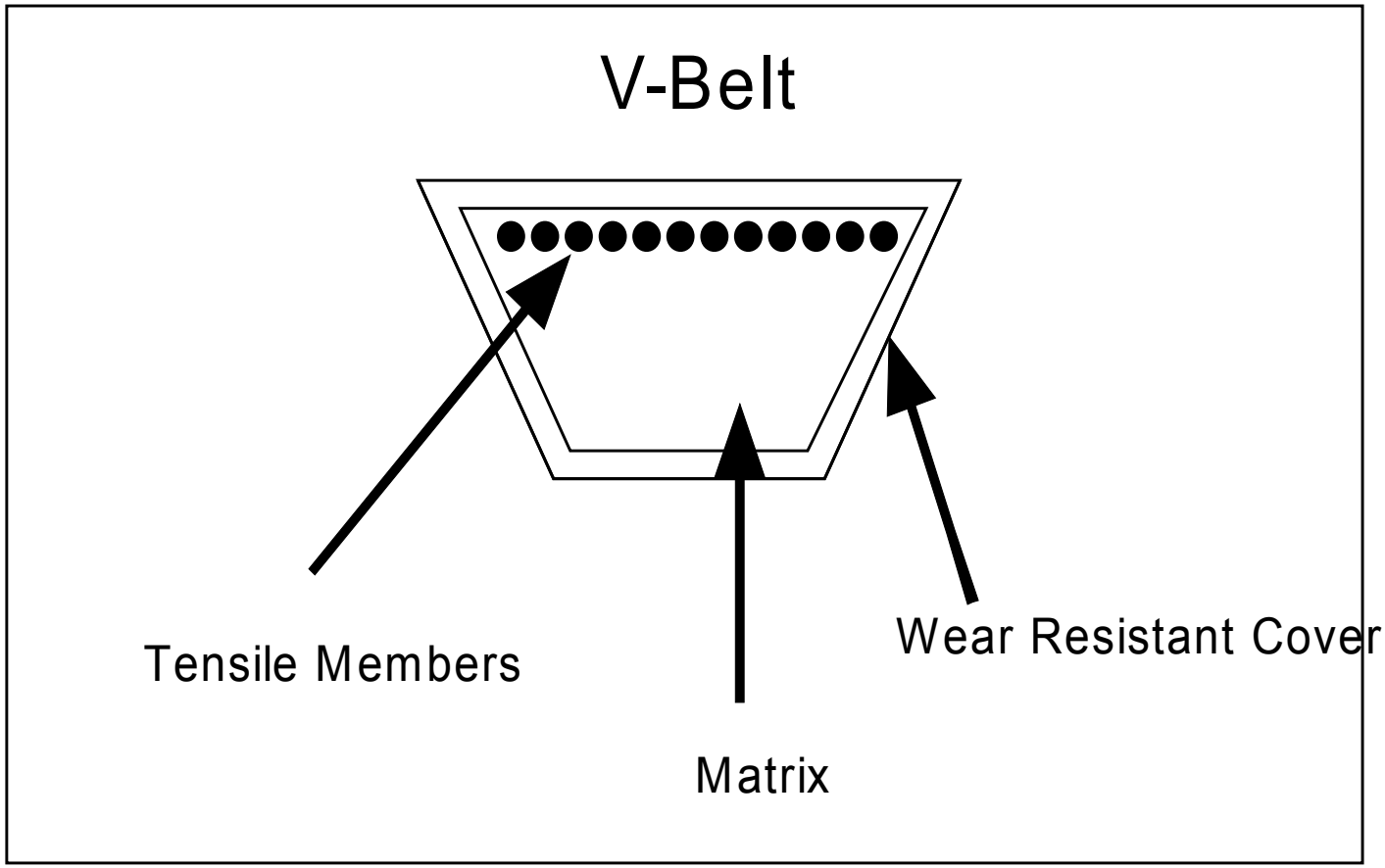
HTD Belts



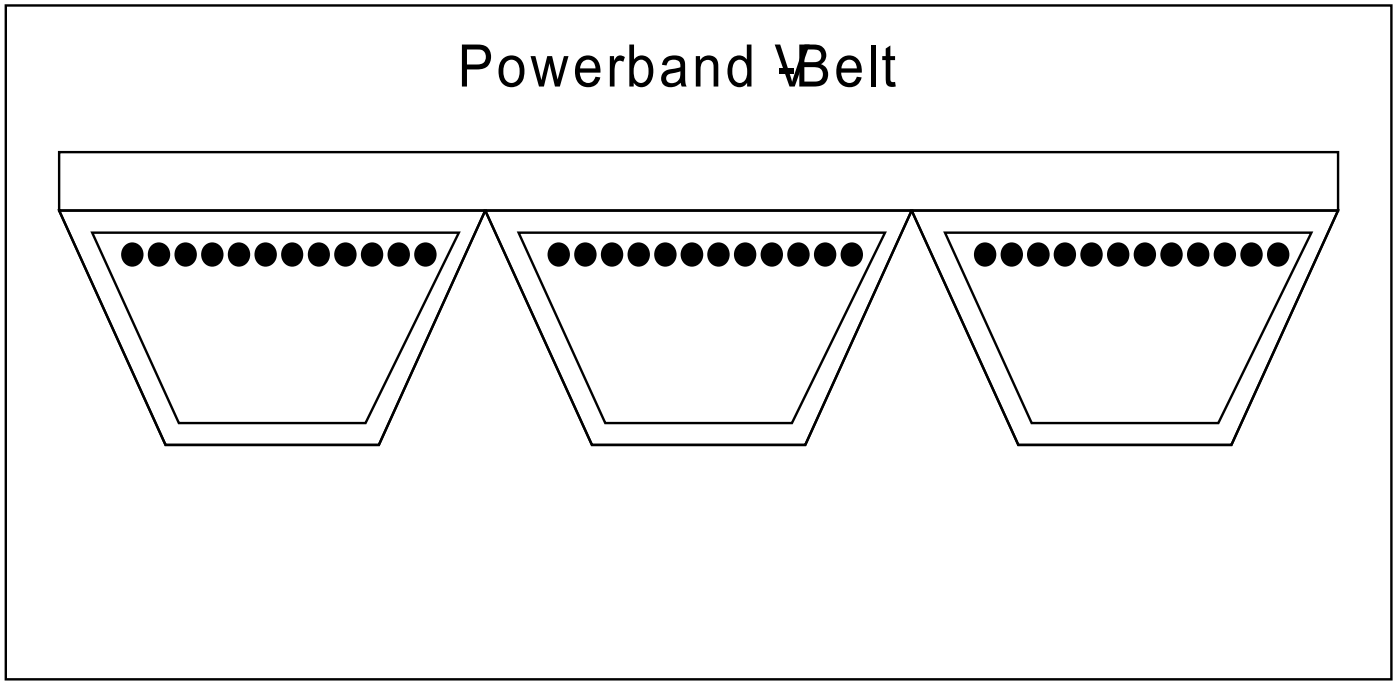
Synchronous Belt



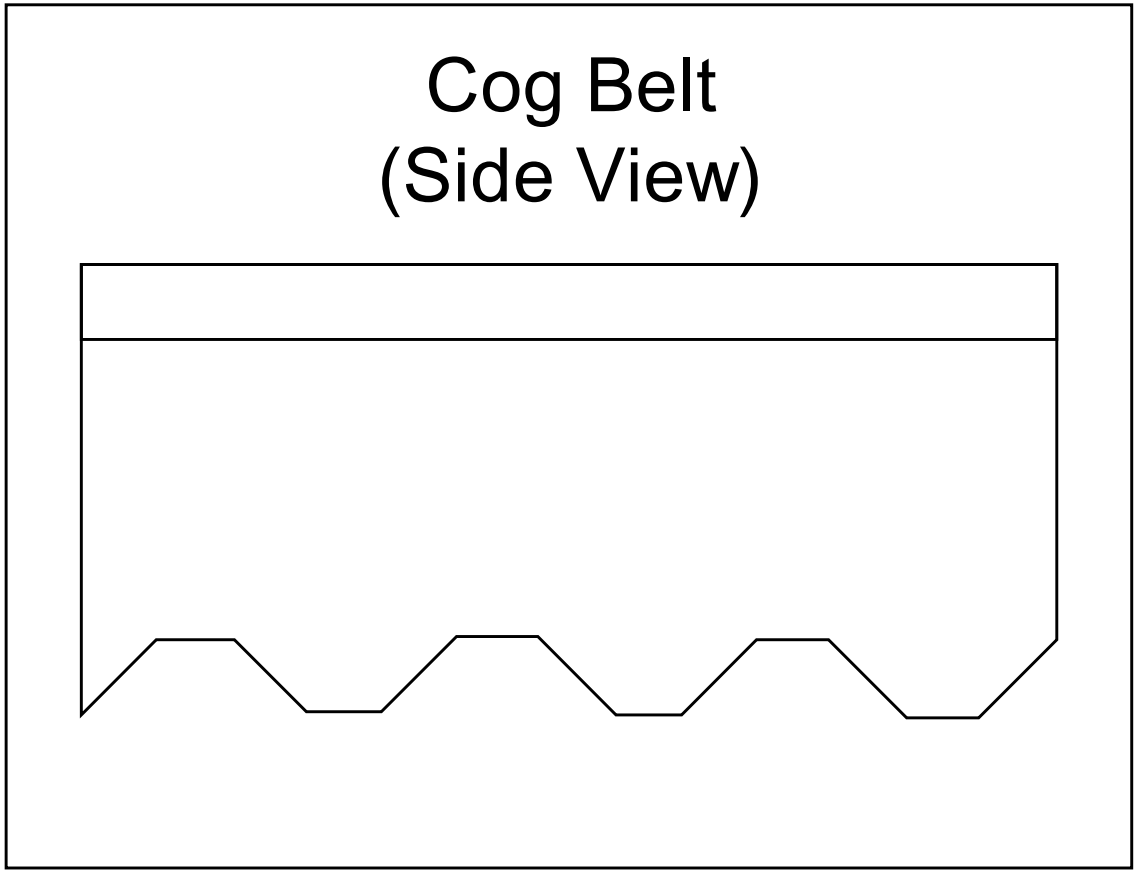
V-belts



Powerband V-belts

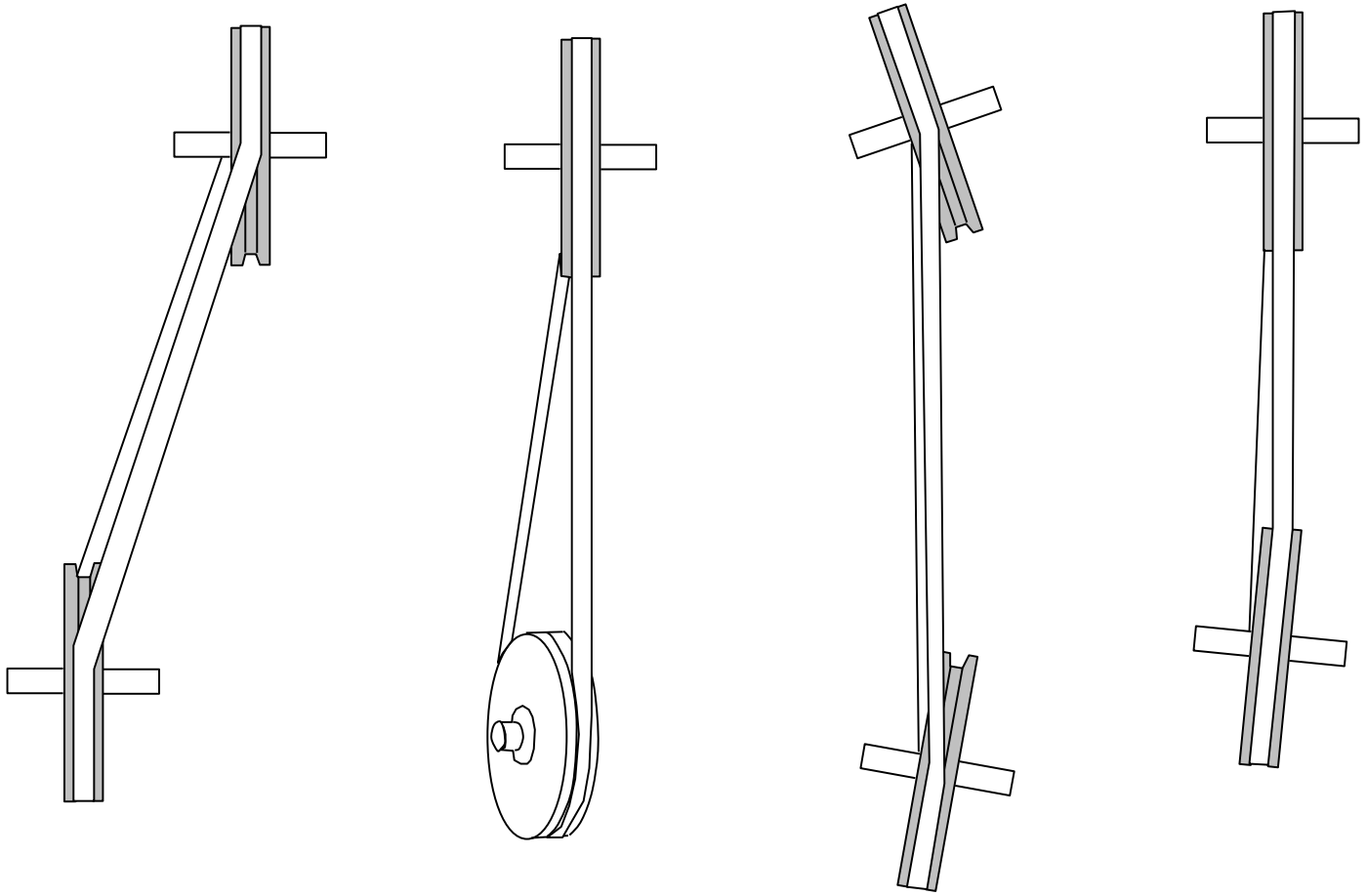


Cog Belts



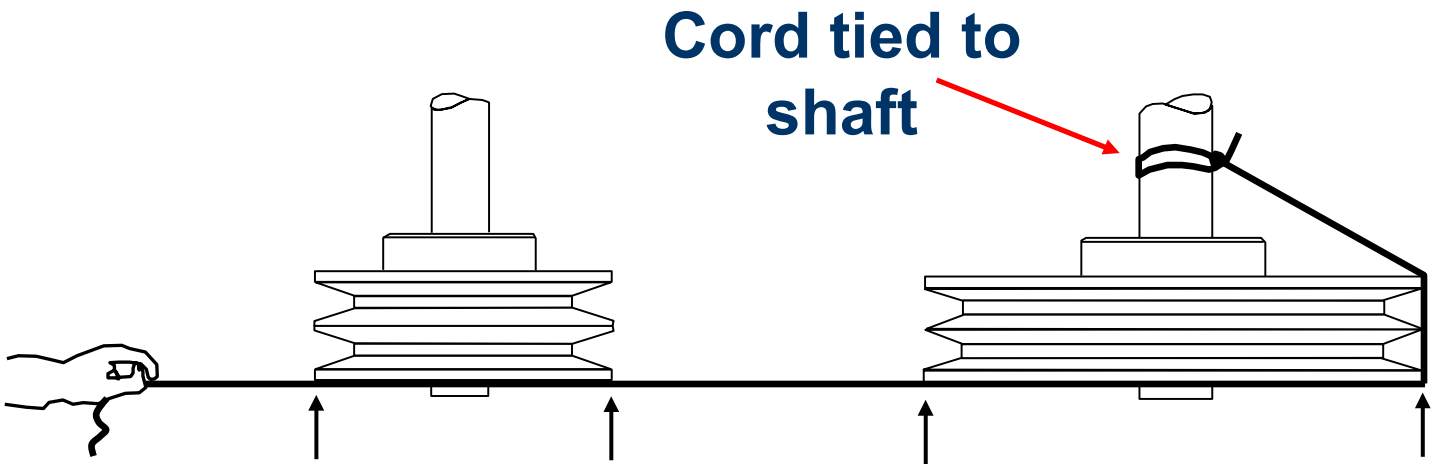
Belt Alignment

- Example mis-alignment of belts

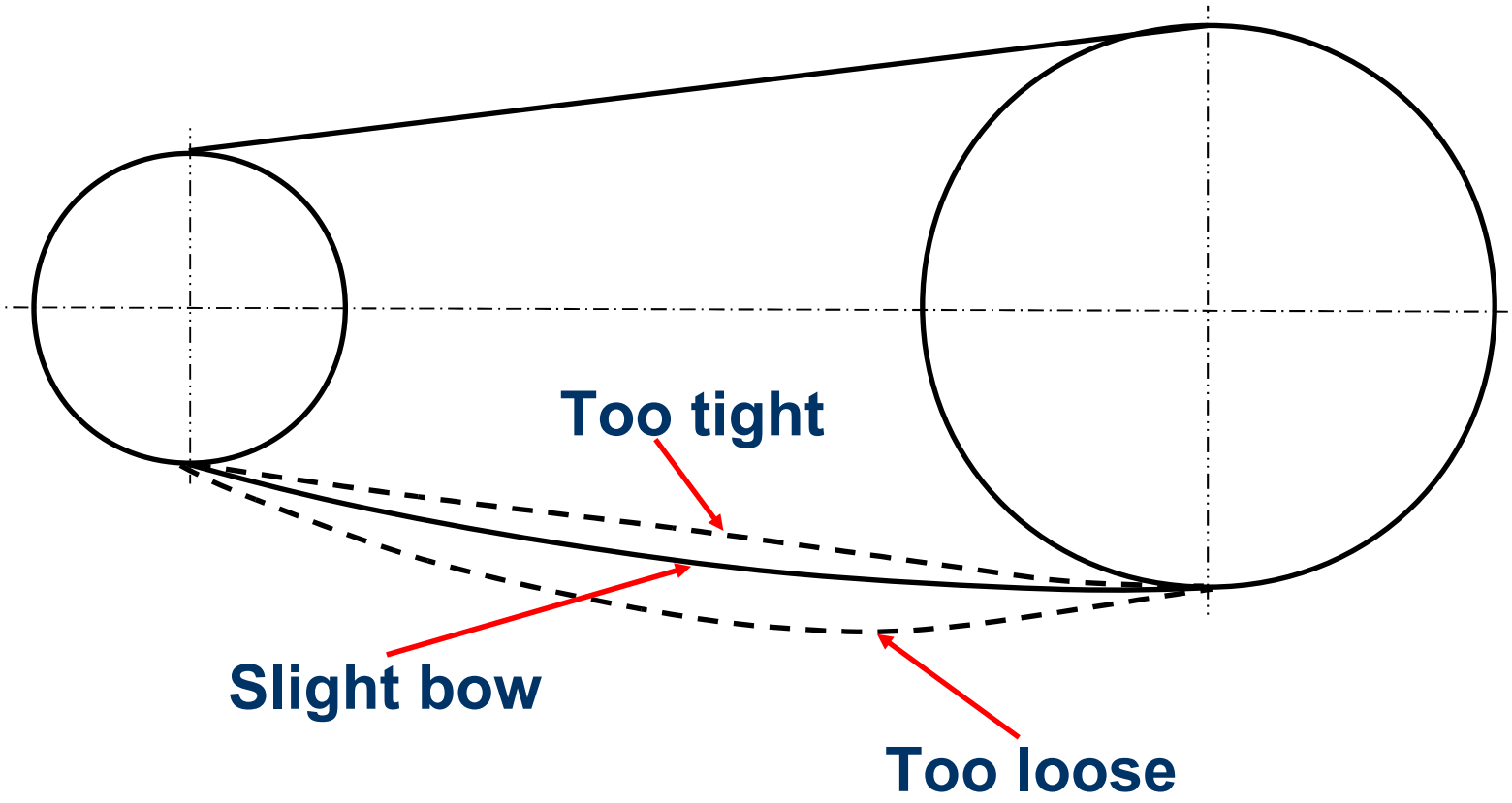


Belt Alignment

- Four Point Touch Alignment



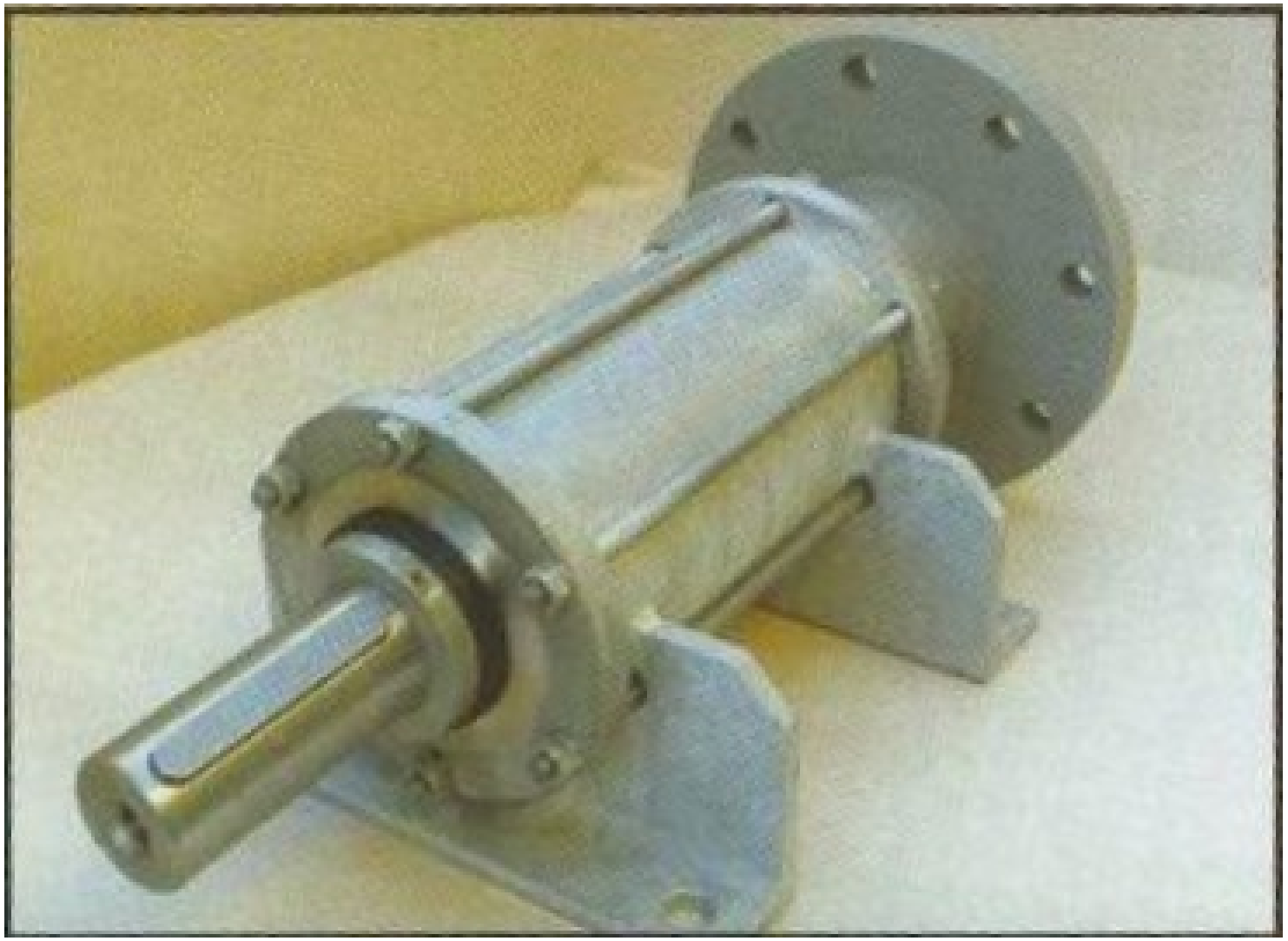
Belt Tensioning



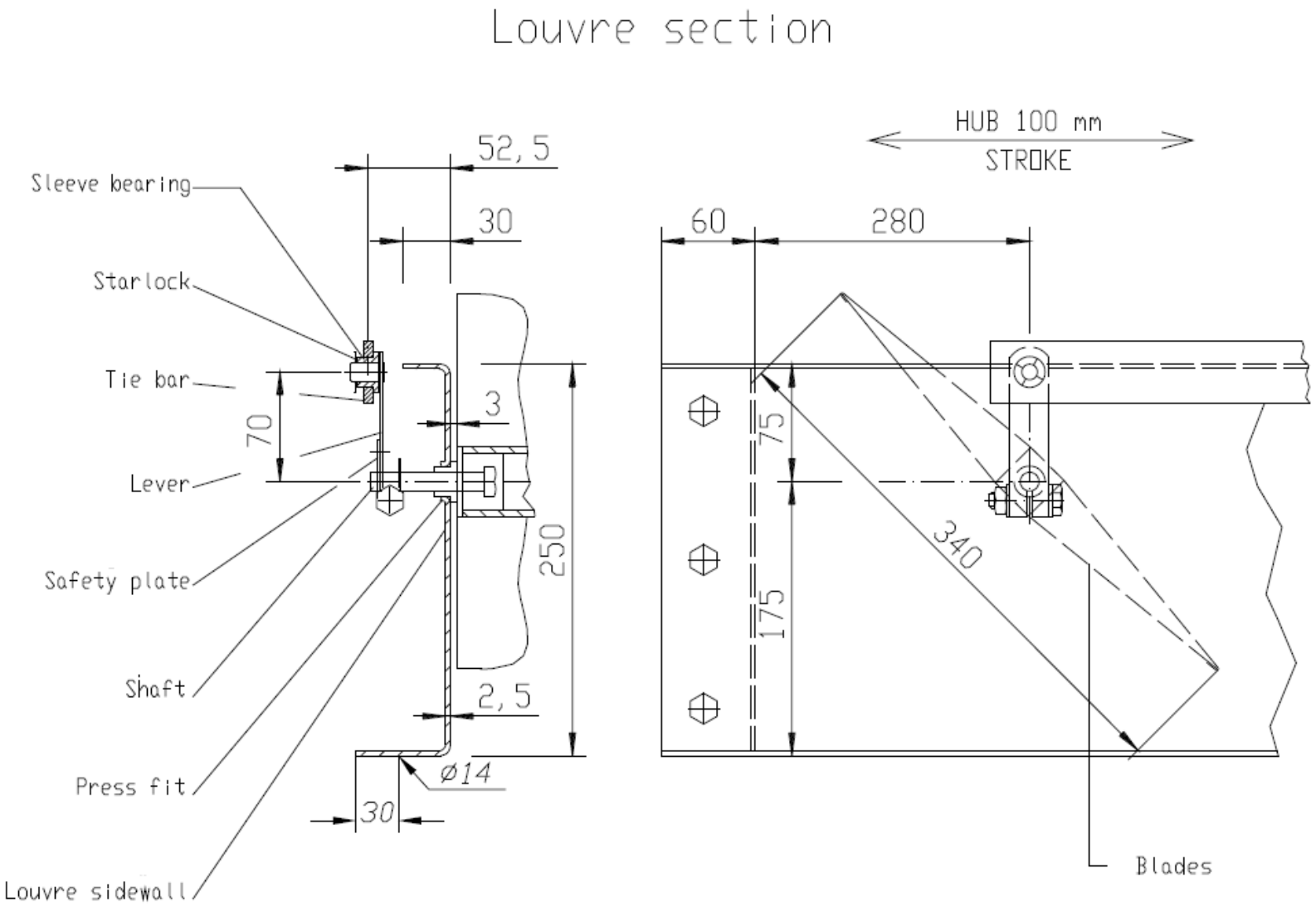
Changing Belts

- Never lever or pry belts onto sheaves or sprockets

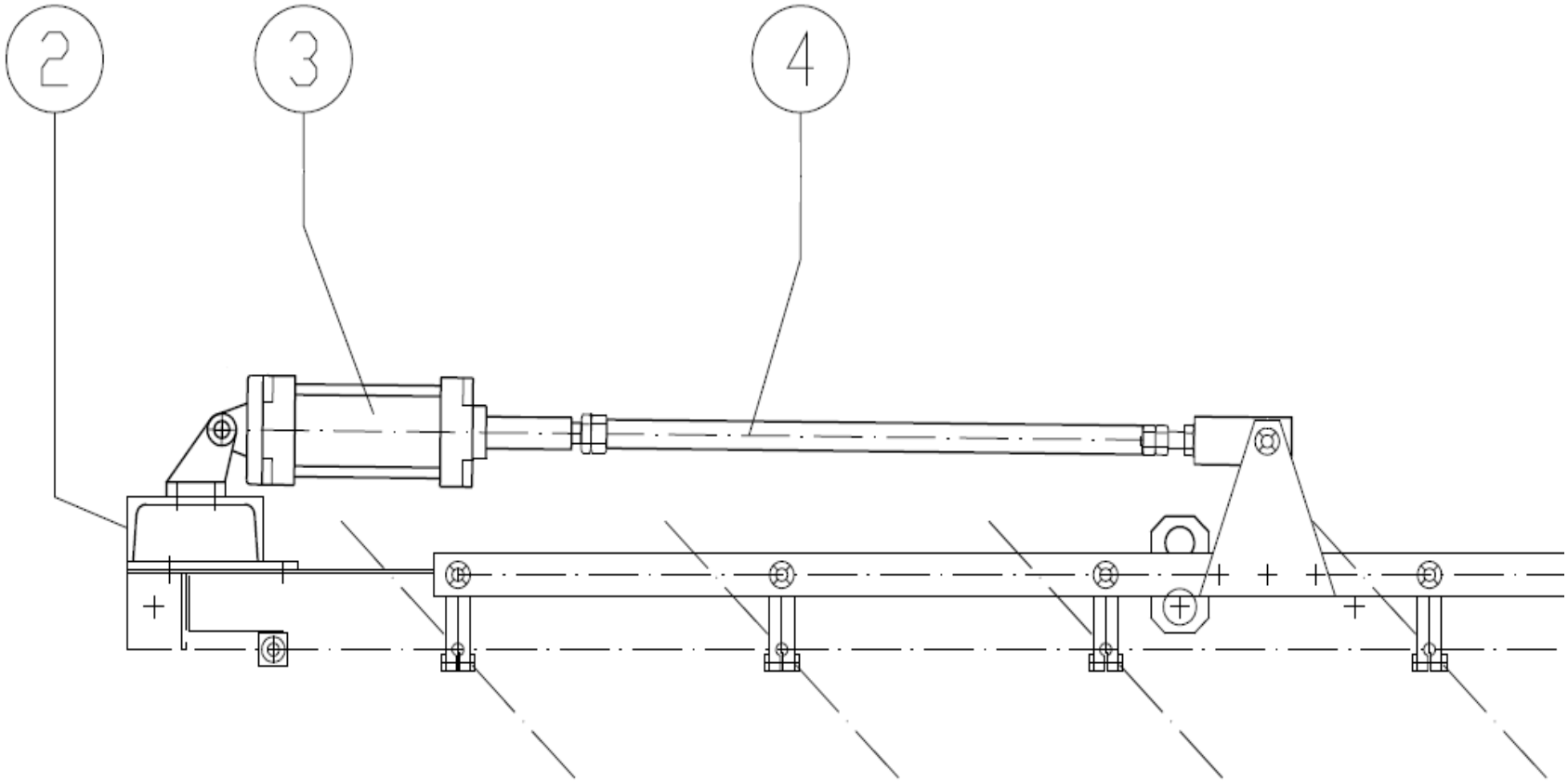
Bearing



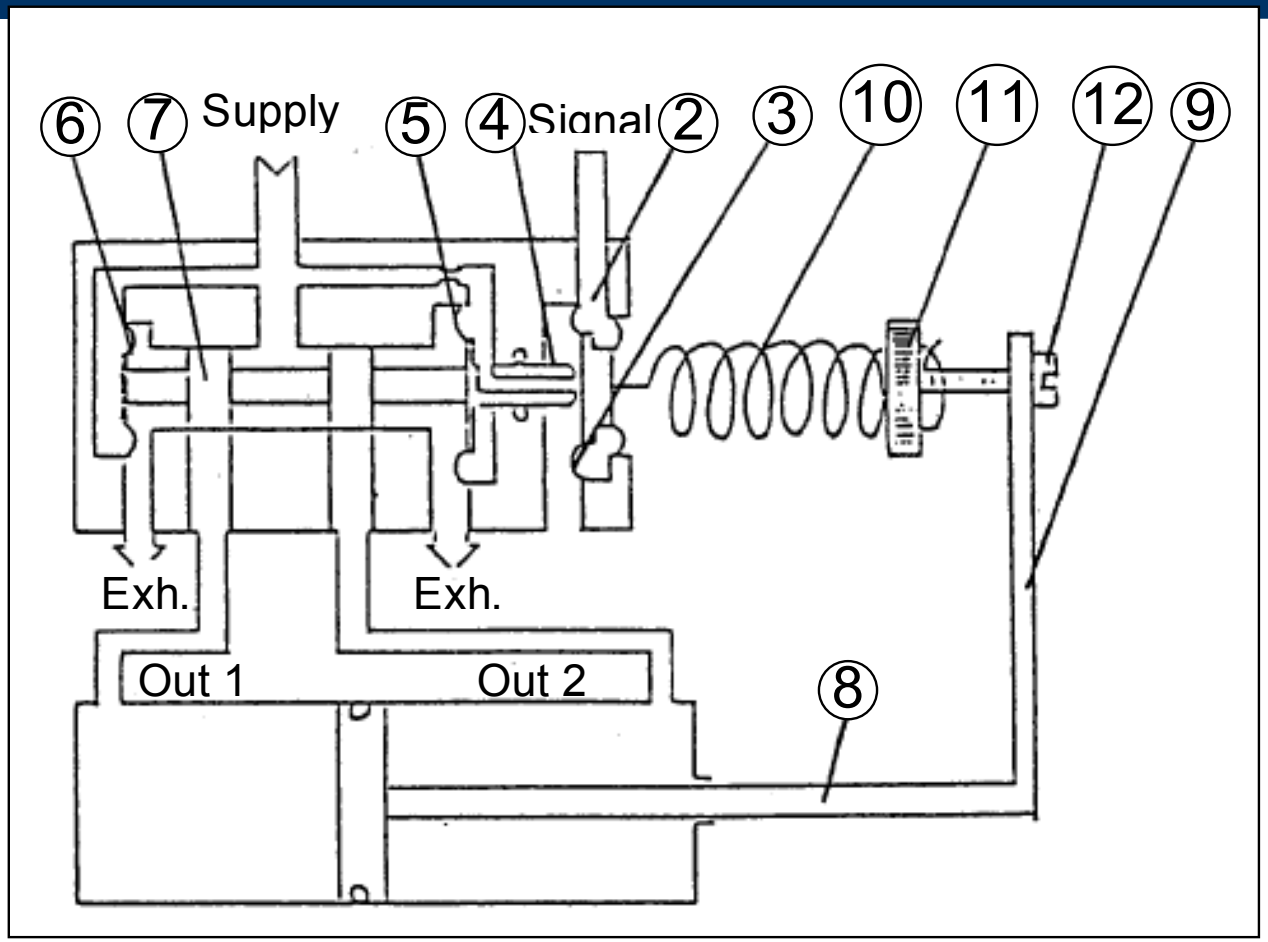
Louvres


Louvre section
HUB 100 mm
STROKE
52,5
30
60
280
Sleeve bearing
Starlock
Tie bar
Lever
Safety plate
Shaft
Press fit
Louvre sidewall
70
3
250
2,5
ø14
30
75
175
340
Blades

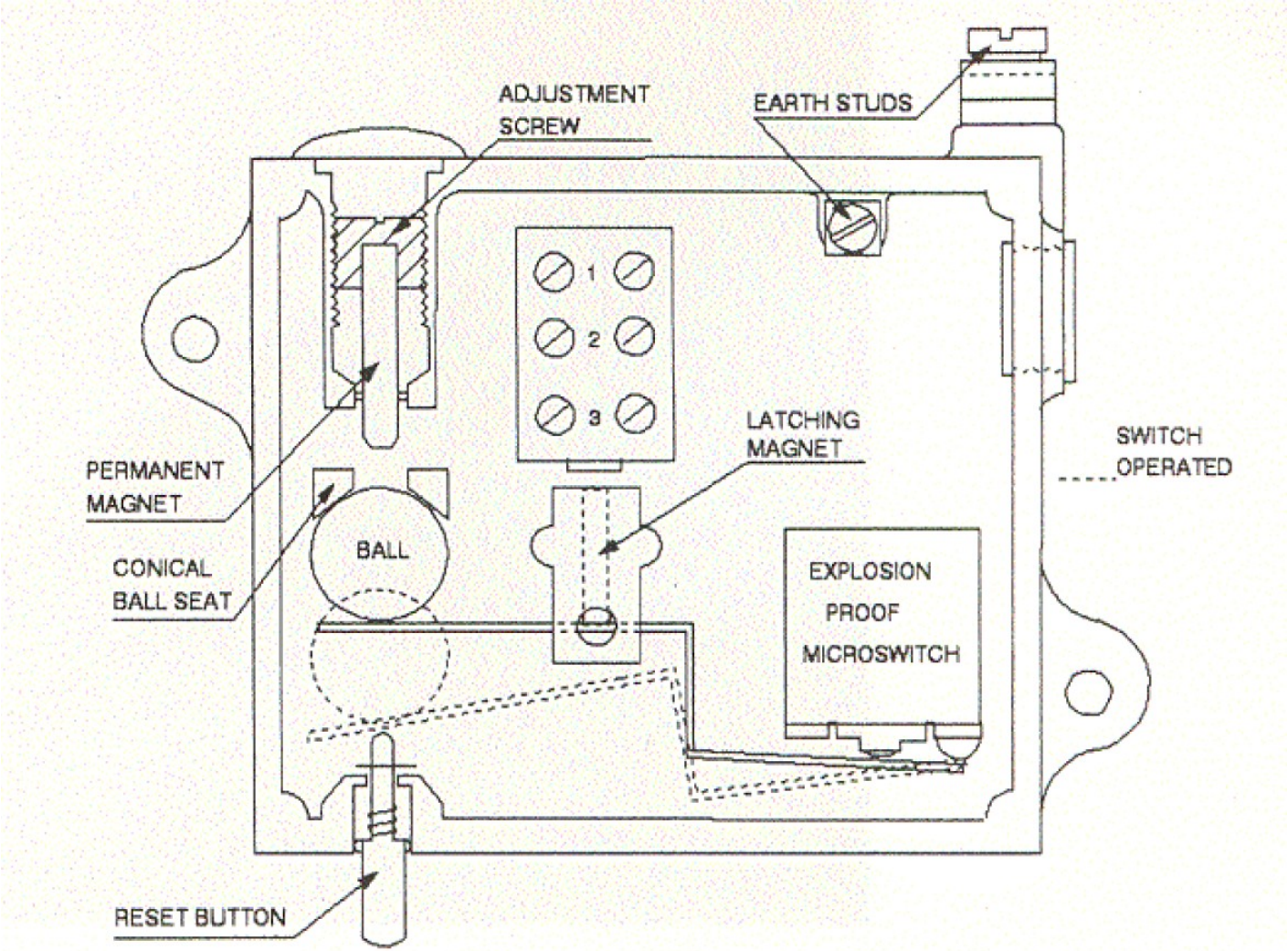
Louvres



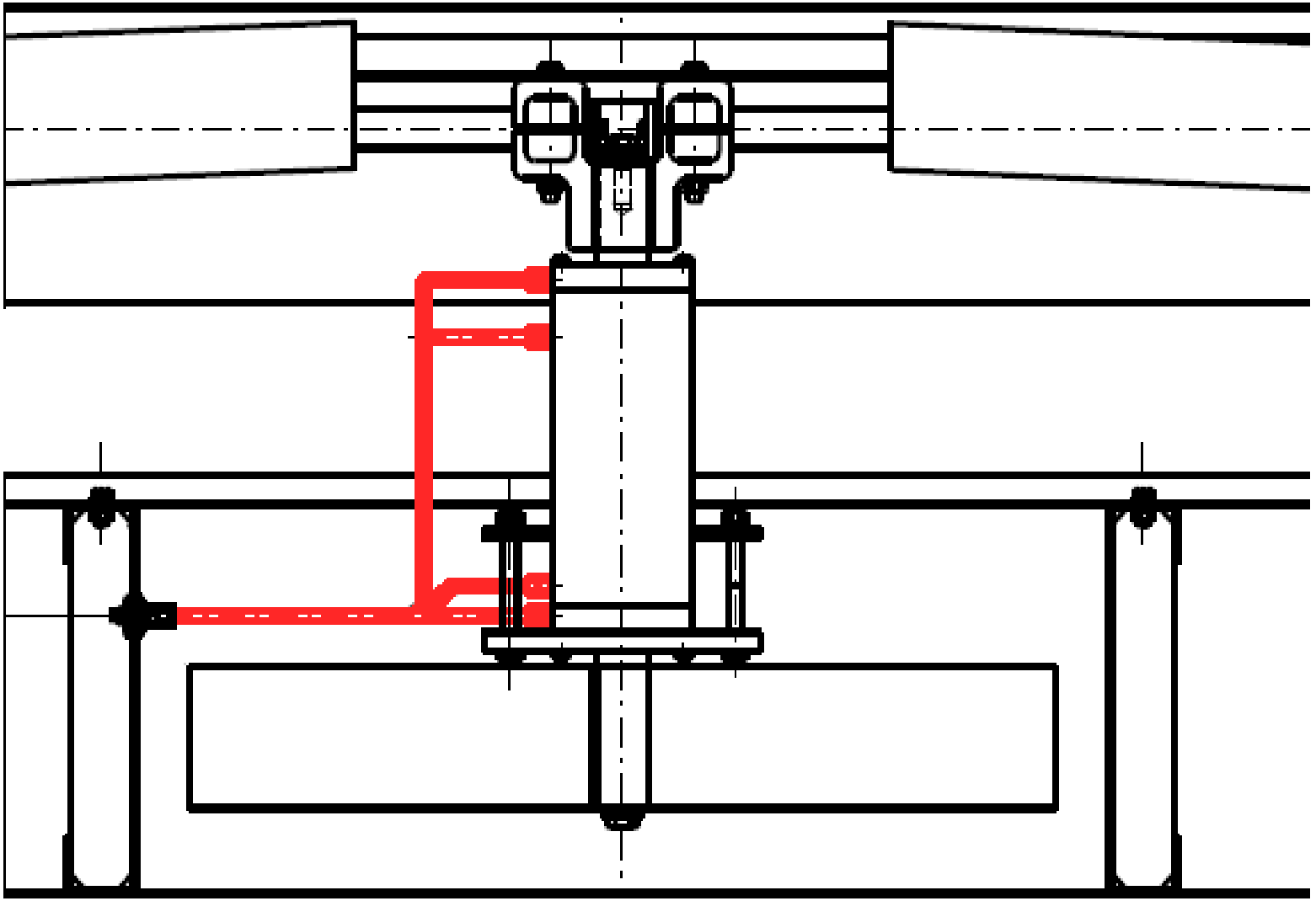
Cylinder Actuator



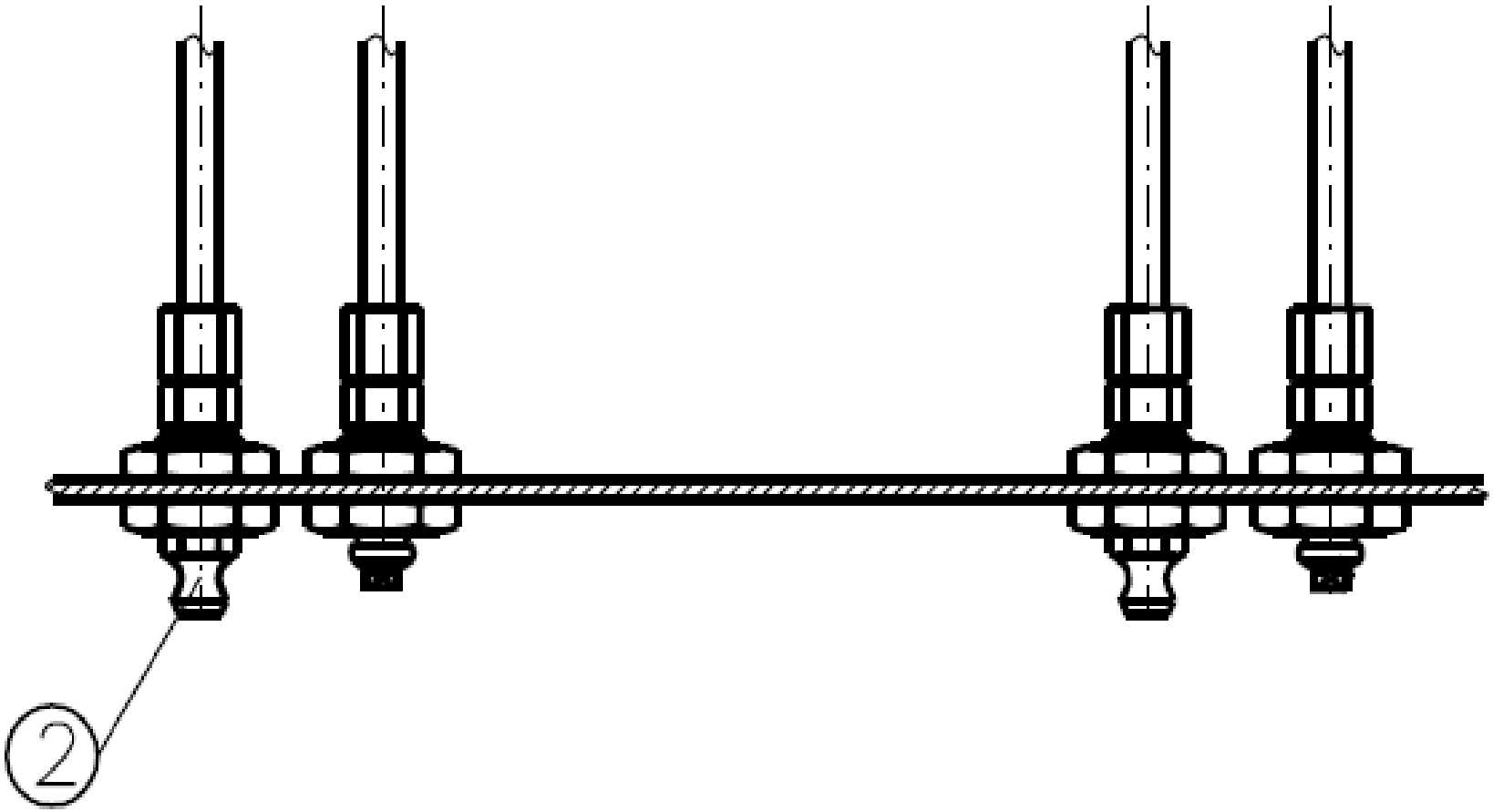
Vibration Switch



Lubrication System



Lubrication System



Maintenance Requirements

- General Inspections
- Blade Angle Adjustment
- Blade Tip Clearance Adjustment
- Bearing Lubrication

Maintenance Requirements

- Vibration Monitoring
- Fan Belt Tensioning
- Fan Belt Alignment

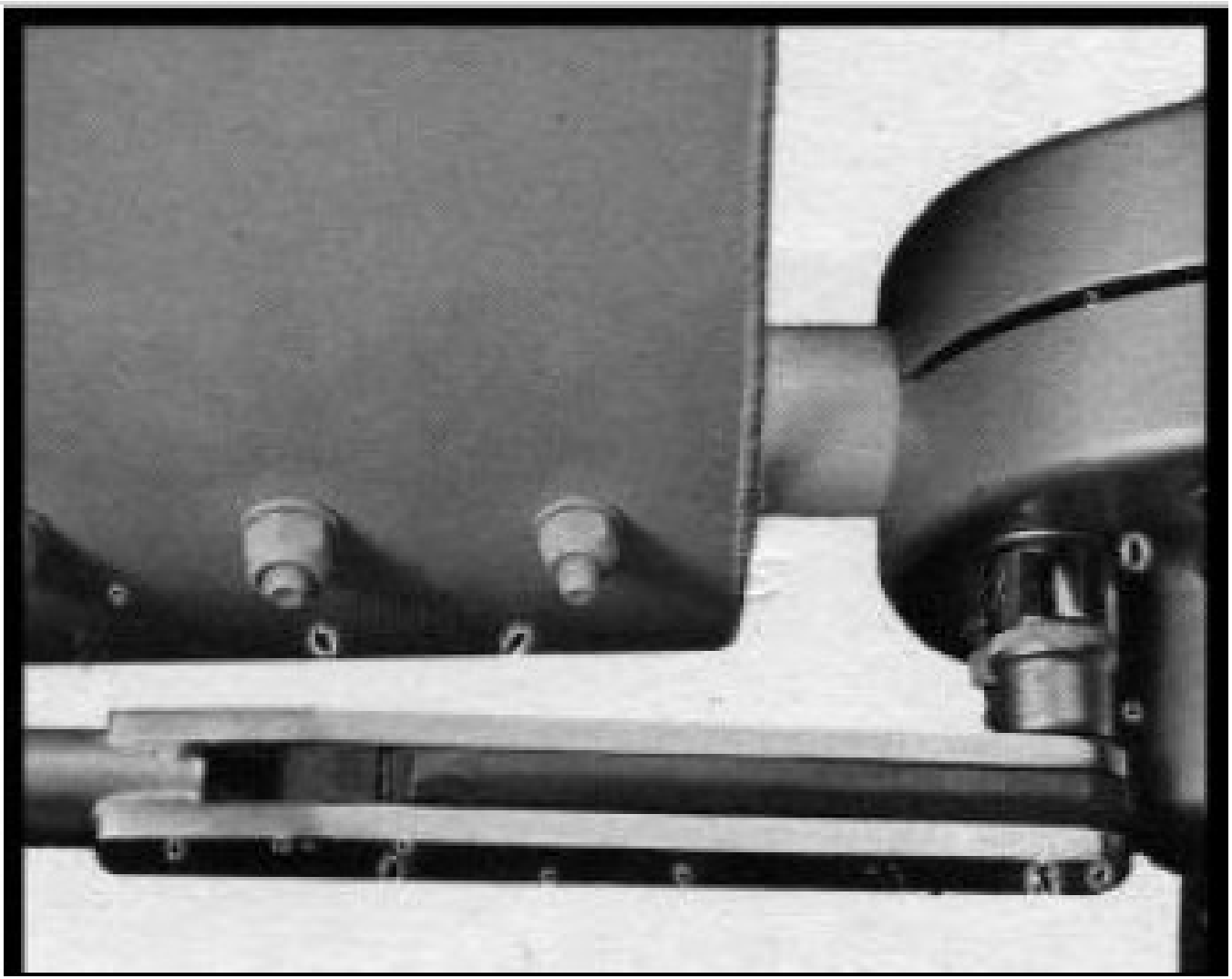
General Inspections

- 24000 Hours - General Inspection and Cleaning
- 90 Days – Vibration Monitoring
- 90 Days – Belt Maintenance

Blade Angle Adjustment

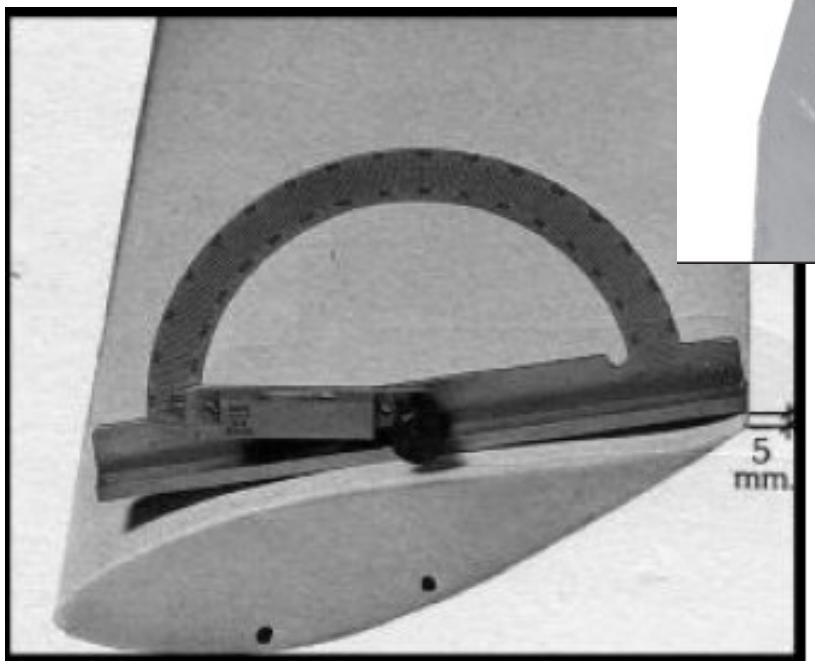
- Position the inclinometer on the least curved part of the blade
- Rotate the blade on its own axis until the desired pitch angle value is obtained
- Repeat operations 1 and 2 for each blade

Blade Angle Adjustment



Blade Angle Adjustment

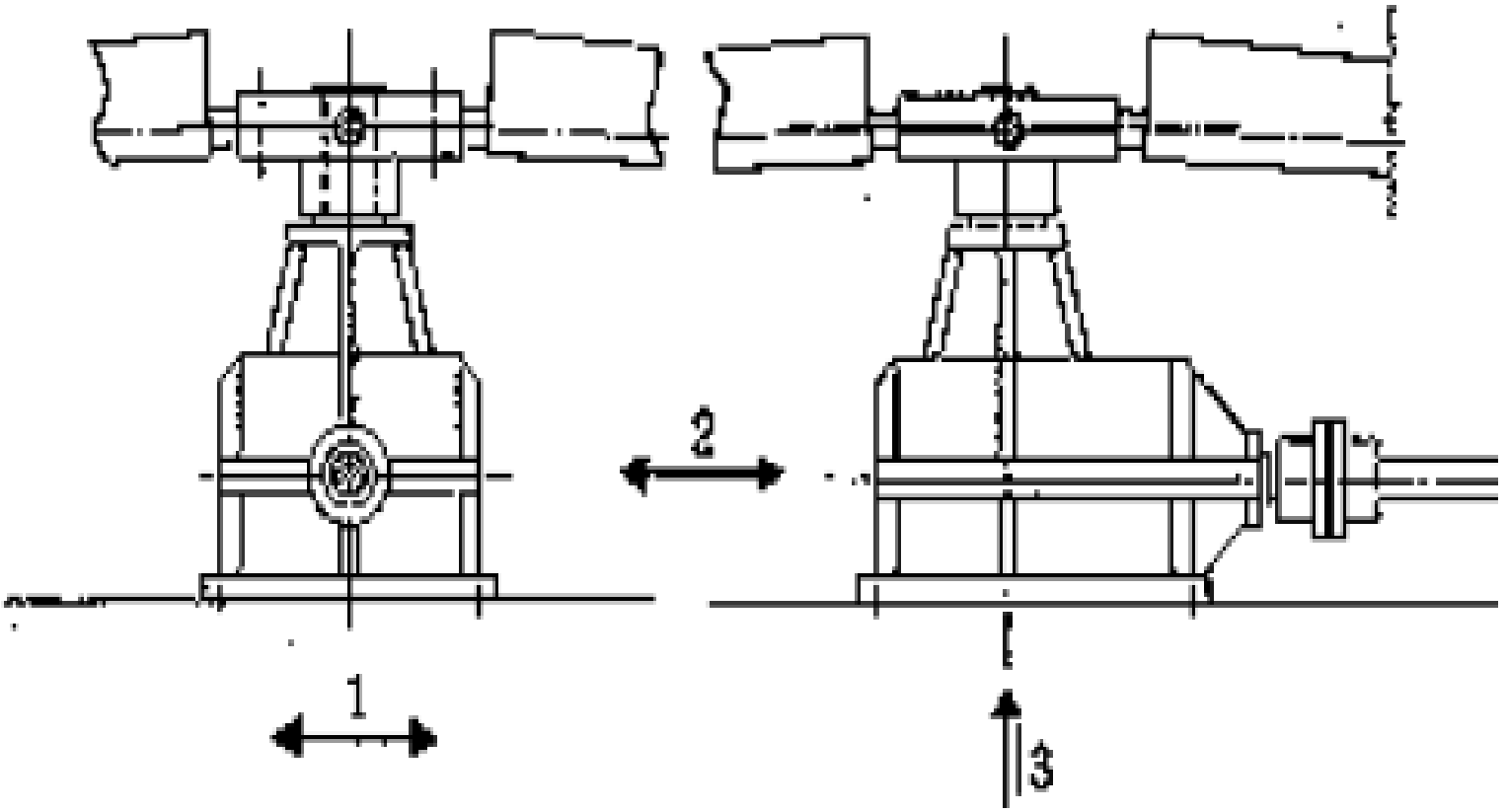




Blade Tip Clearance Adjustment

- Unscrew all the positioning bolts
- Pull each blade out so that the “head” seats firmly against the internal rim of the hub assembly

Vibration Monitoring



Vibration Switch Adjustment

- **Caution:** Isolate power elsewhere before removal of covers
- To set switch, rotate set level screw on top of switch fully clockwise
- Reset switch and check observation window is clear.

Vibration Switch Adjustment

- With machine running normally, rotate set level screw **anti-clockwise** until switch just trips
- Reset carefully; readjust until switch no longer trips
- Adjust clockwise rotation of the set level screw

Vibration Switch Adjustment

- Fill Set Level Screw cavity with Silicone grease and
- Replace cap

Fan Belt Tensioning

- Review Belt drive data sheets
- Belt tensioning is performed by adjusting the motor
- Motor is adjusted until the proper tension is achieved
- Deflection should fall between 9 to 15mm

Fan Belt Alignment

- Axial alignment is performed by moving the motor
- Motor is moved by adjusting 2 nut bolts until proper axial alignment is achieved
- Motor is adjusted until the motor drive pulley and the fan pulley are visually parallel

Troubleshooting

- Excessive Vibration
- Improper Louvre Operation

Fan Vibration

- Imbalanced Blade
- Excessive Blade Pitch Variance
- Misalignment
- Worn Components
- Resonance
- Structural Integrity

Improper Louvre Operation

- Cylinder does not move with rising or falling input signal
  - Cause: Zero adjusting screw is not set properly
  - Solution: Loosen lock-nut and reset the zero adjustment

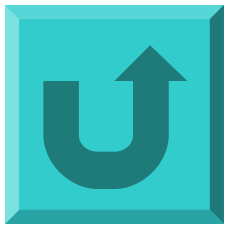
Louvre and Linkage Adjustment

- Cylinder stroke is not in relation to input signal
  - Cause: Adjustment of Span Adjuster is not correct
  - Solution: Remove the set screw of the outer tube and give ideal adjustment while maintaining input signal at 0.6 kg/cm.

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS



LUBRICATION REQUIREMENTS

Objectives

- Define types of lubrication
- Distinguish the difference between grease and oil
- Discuss the hazards of mixing different lubrications
- Describe the proper handling of lubrication
- Describe replacement of Lube Oil filters

Main Topics

- Define types of lubricants
  - Oil
  - Grease
  - ISO and SAE specifications
- Distinguish the difference between grease and oil
- Discuss the hazards of mixing different lubrications

Main Topics

- Describe the proper handling of lubricants
  - Contamination
  - Storage
  - Methods of application
  - Disposal
- Describe replacement of Lube Oil filters.
  - Filter redundancy
  - Flow characteristics,
    DP = Differential Pressure
  - Replace with disposable cartridge

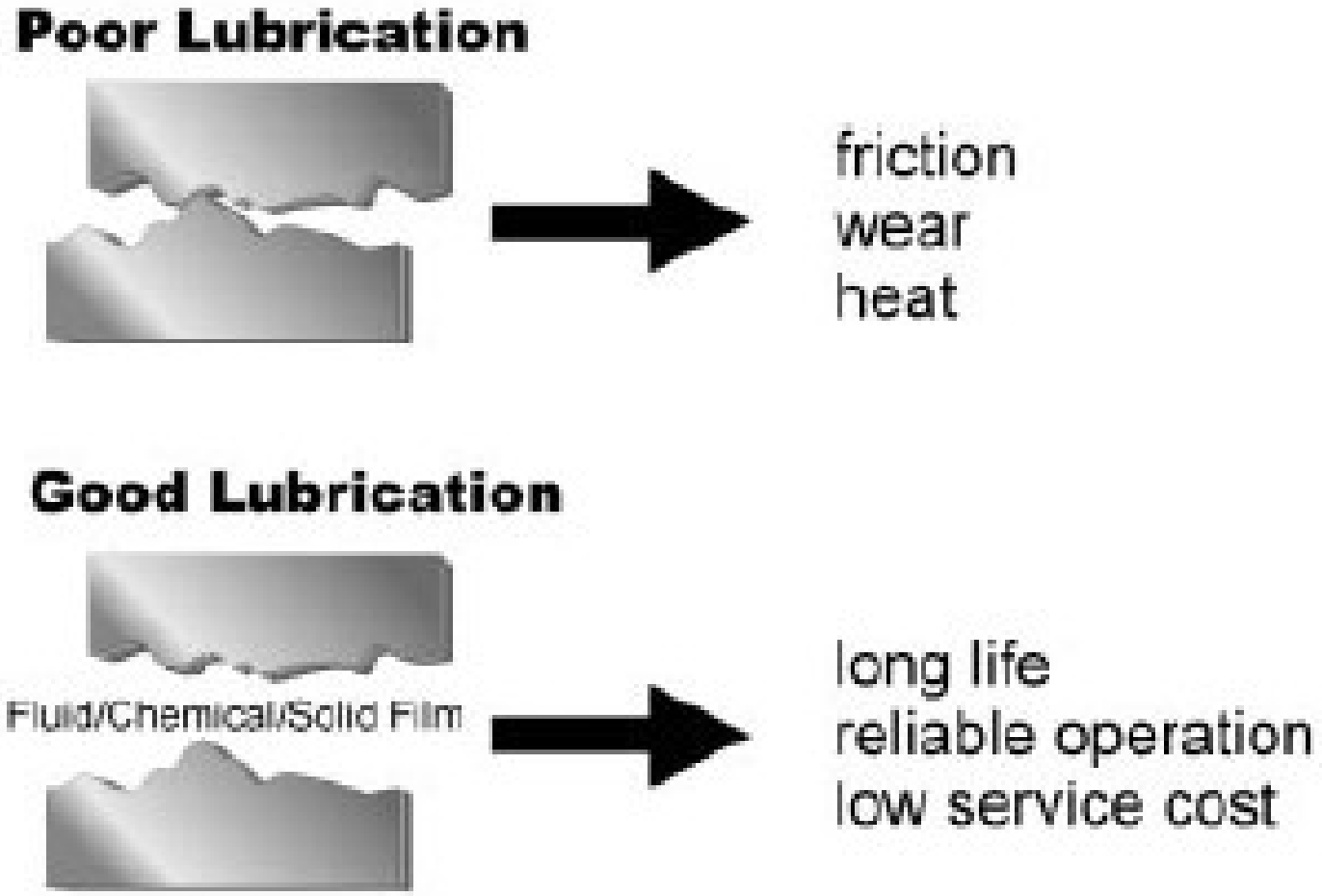
Introduction to Lubrication

- Why use lubricants?
  - Reduce Friction
  - Increase Cooling

Lubrication Functions

- Form a lubricant film between components.
- Reduce the effect of friction
- Protect against corrosion
- Seal against contaminants
- Cool moving parts

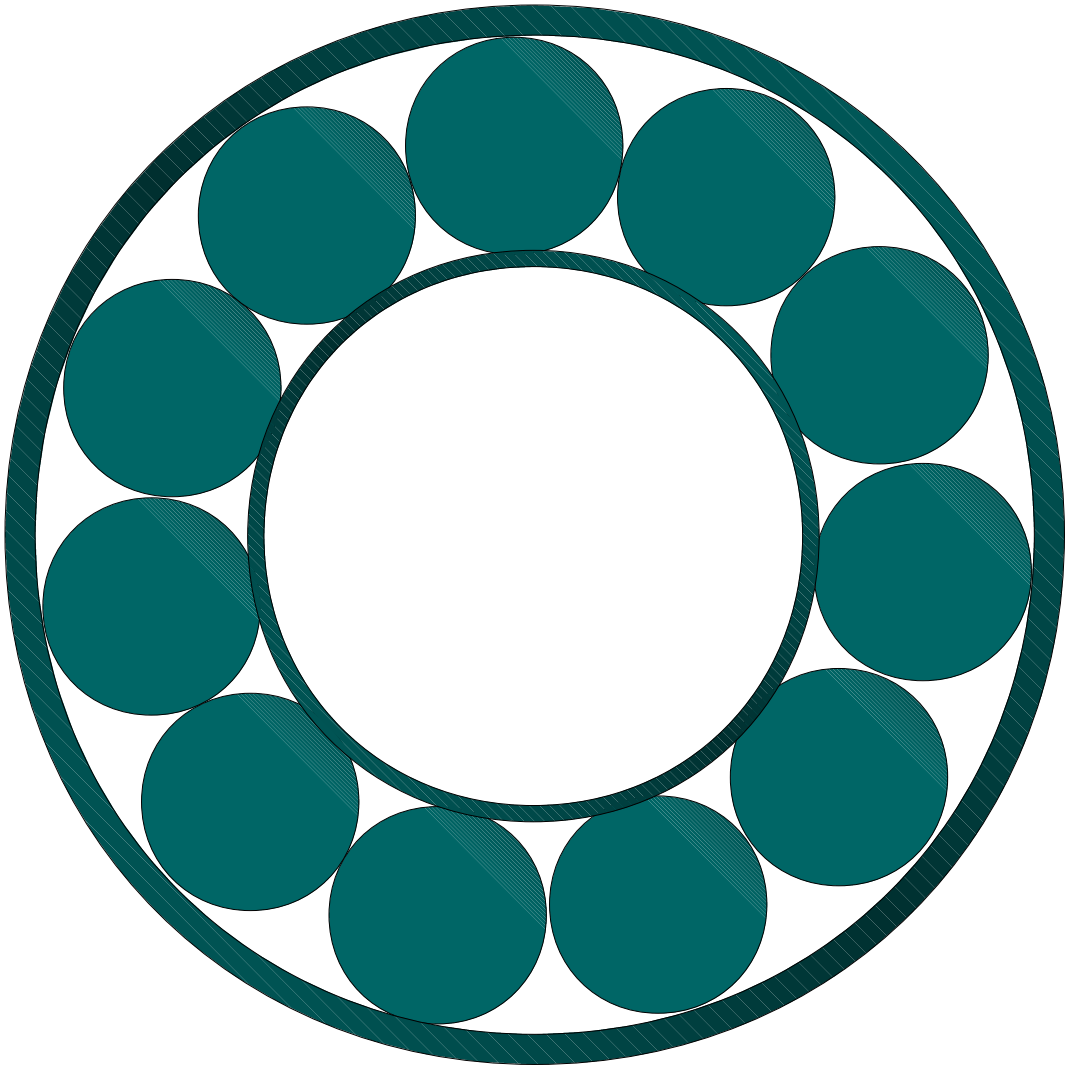
Lubrication



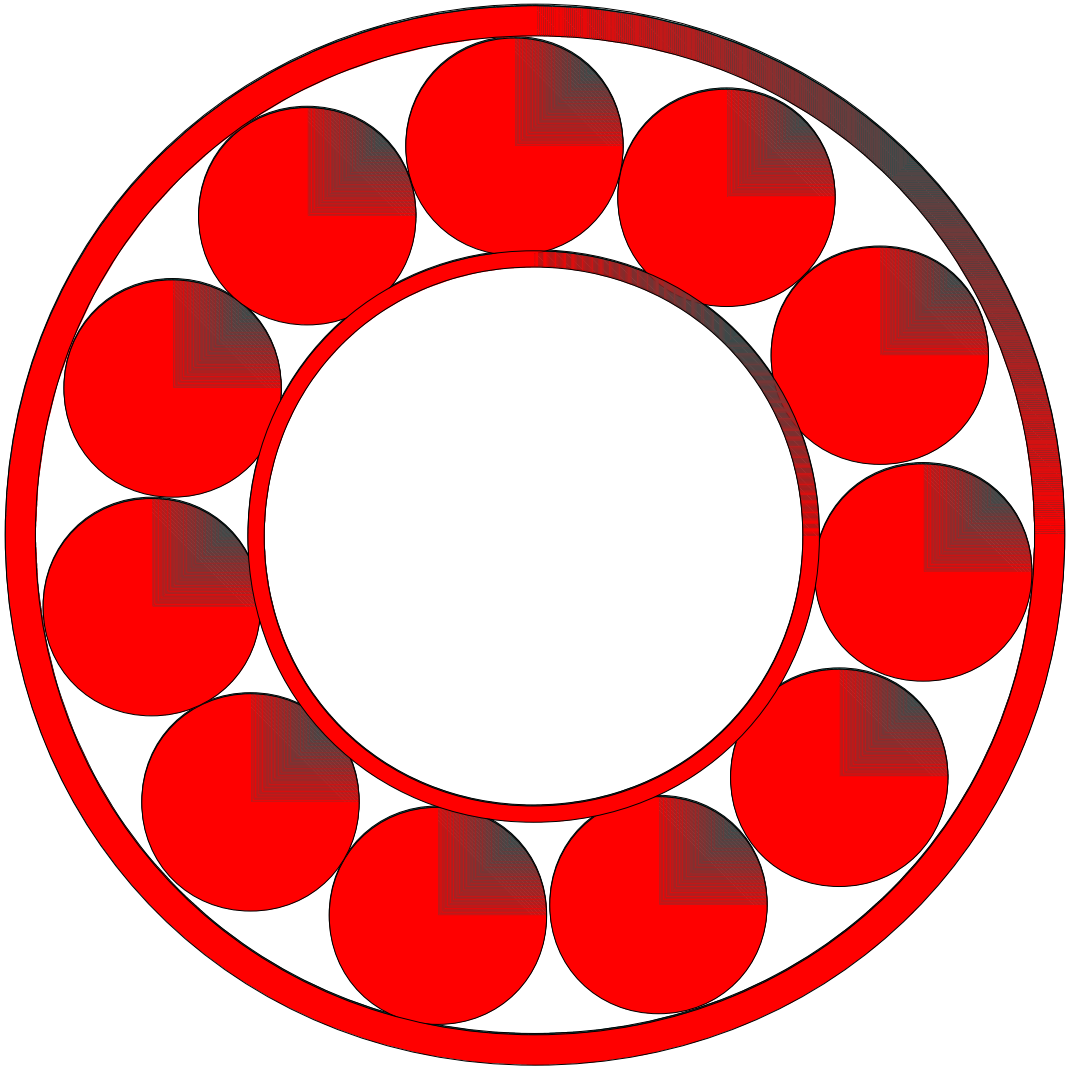
Friction

- Grease and oil lubricate the moving parts of a machine
- Grease and oil reduce friction, heat, and wear of moving machine parts

Oil = Low Friction and Heat



No Oil = High Friction and Heat



Lubrication Prevents Failure of:

- Bearings
- Gears
- Couplings
- Pumps

Lubrication Prevents Failure of:

- Engine components
- Hydraulic pumps
- Gas and Steam Turbines
- Any moving parts

Lubricants prevent failure by:

- Inhibiting rust and corrosion
- Absorbing contaminates
- Displacing moisture
- Flushing away particles

Can lubricants cause damage?

- YES!!
- THE WRONG LUBRICANT CAN CAUSE MACHINE FAILURE!

Lubricant Selection

- Operating temperature
- Load
- Speed
- Environment
- Grease Lubrication
- Oil Lubrication

Grease

- Grease is a heavy, non-liquid lubricant
- Grease can have a mineral, lithium or soap base
- Grease is pasty, thick and sticky
- Some greases remain a paste from below 0°C to above 200°C.
- The flashpoint of most greases is above 200°C
- Grease does not become a mist under pressure

Oil

- Oil can be a heavy or thin liquid lubricant
- Oil can have a natural base (mineral)
- Oil can have a synthetic base (engineered)
- Oil remains liquid from below 0°C to above 200°C.
- The flashpoint of many oils is above 200°C
- The flashpoint is very low for pressurized oil mist. Why?

How are grease and oil different?

- How oil is used:
  - Oil used in closed systems with pumps. An oil sump on a diesel engine pumps liquid oil.
  - Oil is used in gas and steam turbines
  - Oil is used in most machines that need liquid lubricant

How grease is used?

- In areas where a continuous supply of oil cannot be retained, (open bearings, gears chains, hinged joints)
- Factors to be considered when selecting greases are:
  - Type. Depends on operating temperatures, water resistance, oxidation stability etc
  - Characteristics. Viscosity and consistency

Grease or Oil?

- What determines whether a machine needs grease or oil?
- The manufacturer specifies what lubricant is used in their machines, based on the properties of the lubricant. One important property is VISCOSITY.

Viscosity

- Viscosity is a liquid's resistance to flow
- Viscosity affects the thickness of a liquid
- **High viscosity** liquids are hard to pour
- **Low viscosity** liquids are easy to pour

Viscosity Rules of Thumb

- the lower the temperature, the lighter the oil
- the higher the temperature, the heavier the oil
- the heavier the load, the heavier the oil
- the lighter the load, the lighter the oil
- the faster the speed, the lighter the oil
- the slower the speed, the heavier the oil

Viscosity

Temperature affects viscosity.

- Heat decreases viscosity
- Cold increases viscosity
- Viscosity is measured in centistokes (cSt)

Consistency

- Fundamental principle
- Thickener
- Operating temperature
- Mechanical conditions
- Low temperature effect
- High temperature effect

Grease Lubrication

- Thickening agent
- Properties
- Where used

Advantages of Grease Lubrication

- Reduction of dripping and splattering
- Hard to get points
- Reduction of frequency of lubrication
- Helps seal out contaminants and corrosives.
- Ability to cling to part
- Used to suspend other solids

Grease Selection Factors

- Load condition
- Speed range
- Operating conditions
- Temperature conditions
- Sealing efficiency
- External environment

Oil Types

- Two types of lubrication oil are:
- Mineral-based
- Synthetic

Mineral-Based Oil

- Mineral-based oil is refined from crude oil hydrocarbons
- Mineral-based oil has 2 types of base:
  - Naphtha Base
    - A naphtha base is solvent-like
  - Paraffin Base
    - A paraffin base is waxy

Mineral-Based Oil

- Naphtha Base
  - Lower viscosity index (40-80 cs)
  - Lower pour point
  - Less resistant to oxidation and changes in viscosity index
  - Good performance at higher temperatures

Mineral-Based Oil

- Paraffinic Base
  - Higher viscosity index (>95cs)
  - Higher pour point
  - Very resistant to changes in viscosity index and oxidation
  - Thicken at low temperatures

Mineral-Based Oil

- Mineral-based oils are cheaper to buy than synthetics.
- Mineral-based oils can contain traces of sulfur and nitrogen. These impurities can cause oil to form sludge.

Synthetic Oil

- Synthetic oil is NOT refined from crude oil hydrocarbons
- Synthetic oil is made without a mineral base
- Synthetic oil is made by careful control of a chemical reaction that yields a “pure” substance

Synthetic Oil

- Synthetic oils are chemically engineered to be pure. They do not contain the traces of sulfur or nitrogen present in mineral-based oils.
- Synthetic oils are expensive

Synthetic Oil

- Synthetic oil is less flammable than mineral-based oil at low pressure. (Pressure causes most oils to become more flammable)
- Synthetic oils are generally more expensive than mineral based oils

Lubricant Specifications

- ISO = International Standards Organization
- SAE = Society of Automotive Engineers

ISO Lubricant Specifications

- ISO Grade lubricants are for industrial use. ISO specifications exist for lubricants in extreme industrial environments.

ISO Lubricants

ISO GRADE	32	46	68	100
Viscosity 40°C 100°C	30.4 5.2	43.7 6.6	64.6 8.5	30.4 5.2
Flash Point °C(°F)	222(432)	224(435)	245(473)	262(504)
Pour Point °C(°F)	-36(-33)	-36(-33)	-33(-27)	-30(-22)

Using Different Lubricants

- Why do we use different lubricants?
- What happens if oils are mixed?

Mixing Lubricants

- Consequences of mixing different lubricants are:
- Change of viscosity
- Stripping of machine's internal coatings, damage to seals
- Reduced flash point, risk of fire

Mixing Lubricants

- Loss of corrosion protection
- Poor water separation
- Foaming
- Thermal instability

Booster Compressor Lubes

Equipment	Specified Lubricant	Chevron Equivalent	Consumption Rate	Service Interval
Turbine and Compressor Lube Oil System	ISO VG 32	GST ISO 32	5 Liters per day	Based on oil analysis
Electric Motor (Starter)	Grease	SRI Grease NLGI 2	negligible	1750 Hours
Electric Motor (Ventilation)	Grease	SRI Grease NLGI 2	negligible	11500 Hours
Electric Motor (Aux Lube Oil Pump)	Grease	SRI Grease NLGI 2	negligible	3000 Hours
Electric Motor (Aux Lube Oil Cooler)	Grease	SRI Grease NLGI 2	negligible	1000 Hours

Water Pump Lubes

Equipment	Specified Lubricant	Chevron Equivalent	Consumption Rate l/year	Service Interval
Utility Water Pump	Texaco Ursatex SAE 20/20W	Chevron Delo 400 SAE 20	.5L	Yearly
Utility Water Pump Motor	Esso Unirex N3	Chevron SRI Grease 2	50g	2 years
Demineralised Water Pump Motor	Texaco Ursatex SAE 20/20W	Chevron Delo 400 SAE 20	100L	Yearly
Fire Water Jockey Pump	Texaco Ursatex SAE 20/20W	Chevron Delo 400 SAE 20	.5L	Yearly
Fire Water Jockey Pump Motor	Esso Unirex N3	Chevron SRI Grease 2	50g	2 Years

Water Pump Lubes

Equipment	Specified Lubricant	Chevron Equivalent	Consumption Rate g/year	Service Interval
BS12A Fire Water Pump	Texaco Multi-purpose AP EP2	Chevron Dura-Lith EP #2	200	Yearly
Fire Water Pump Motor (SIEMENS)	Shell Alvania G3	Chevron SRI Grease 2	100	3 Years
Fire Water Pump Motor (Caterpillar)	Texaco Ursa Super LA 15W-40	Chevron Delo 400 15W-40	100	3 Years

Nitrogen Generation Lubes

Equipment	Specified Lubricant	Chevron Equivalent	Service Interval
Screw Compressor 72-F 9269/89	Total Dacnis VS 32	Chevron Hydraulic Oil AW ISO 32	4000 hours
73-MGC-9251 A/B Bearings	Total MultiElf	Chevron SRI Grease 2	4500 hours
73-MEA-9202A/B-01/02 Bearings	Filled for life of bearings		

Propane Compressor Lubes

Equipment	Specified Lubricant	Chevron Equivalent	Service Interval
GC 740 compressor and drive bearings, oil pumps	ISO VG 46	Chevron GST ISO 46	Monitor and service if out of spec
MG 741 A/B oil pump drive and electric motor	Shell Alvania R3	Chevron SRI Grease 2	40000 hours or 4.5 years
MEA-709 A1/2/3 oil cooler drive	Shell Alvania R3	Chevron SRI Grease 2	20000 hours or 2.25 years

Propane Compressor Lubes

Equipment	Specified Lubricant	Chevron Equivalent	Service Interval
GC 701 gas compressor and drive bearings, oil pumps	ISO VG 46	Chevron GST ISO 46	Monitor and service if out of spec
MG 711 A/B oil pump drive and electric motor	Shell Alvania R3	Chevron SRI Grease 2	40000 hours or 4.5 years
MEA-708 A1/2/3 oil cooler drive	Shell Alvania R3	Chevron SRI Grease 2	20000 hours or 2.25 years

Fundamentals of Lubrication

- Equipment lubrication
  - Bearings
  - Gears
  - Couplings
  - Pumps
  - Engine components
  - Hydraulic pumps

Lubricant Delivery Methods

- Force Feed Lubricant
- Oil Mist
- Constant Circulation
- Oil Slinger
- Zerk Fittings
- Surface Application (brush or spray)

Force Feed Lubrication

- A force feed lubricant system is like an automated version of the hand held oil can. An automatic plunger applies pressure to deliver a few drops at predetermined time intervals.

Oil Mist Lubrication

- This method keeps rotating machinery operating effectively for extended time periods.

Oil Mist Lubrication

- Centralized lubrication system that generates, conveys and automatically delivers lubricant.
- The generator utilizes the energy of compressed air to atomize oil into micron sized particles
- The particles can be conveyed considerable distances.

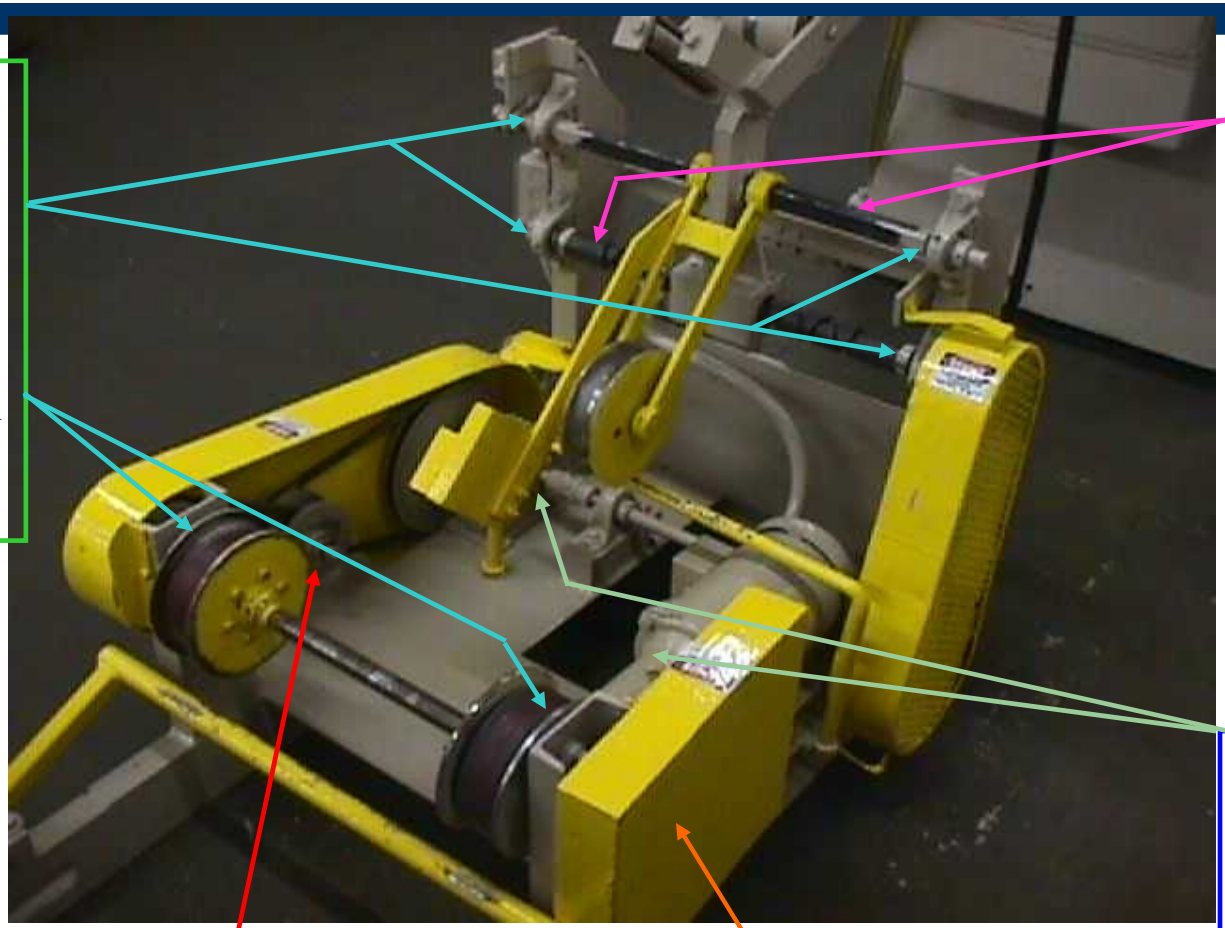
Benefits - Oil Mist Lubrication

- Bearing failures reduced
- Lubricant consumption reduce by 40%
- Equipment runs cooler – Saves energy
- Contaminant's are excluded
- More efficient lubrication

Constant Circulation

- A Constant Circulation system re-circulates oil in a closed system like your heart circulates blood in your body.

Lubrication Check Example



Grease support wheel bearings (Quarterly) 1 to 2 pumps with (Mobil XHP222)

Hand grease square slide shaft and worm shaft (Monthly) 1 to 2 pumps per shaft of (Mobil XHP222)

Check Windup Gear Boxes (Quarterly) Oil type ISO360 (Mobil Gear 636)

Grease Variable Pitch Pulley (Quarterly) 1 to 2 Pumps of (Mobil XHP222)

Hand Oil Roller Chain, [behind guard] (Quarterly) (LPS) (24810)

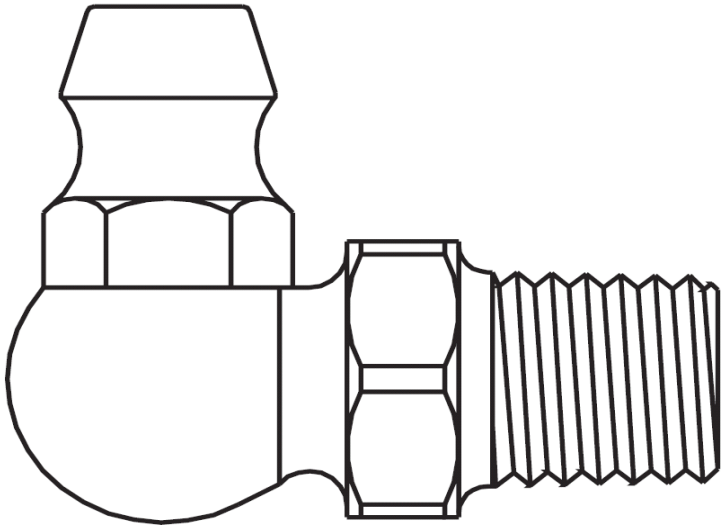
Oil Slinger

- Small disc that loosely rotates on a shaft
- Lubricates moving parts by agitating or splashing oil in the crankcase.
- Allows a thin film of oil to remain on the piston rod.
- The Oil Slinger is installed on the piston rod between the packing case and the wiper case



Zerk Fittings

- Zerk Fittings are grease fill points that have an internal check valve that prevents contaminates from entering the fitting. Always clean the Zerk fitting before applying grease.



Surface Application

- Sometimes lubricants are painted on with a brush, sprayed from an aerosol can, or wiped onto the part.

Pump System

- A Pump System automates lubrication. Grease or oil is fed from a central pump through lines and block valves to the necessary lube points.

Lubricant Storage Factors

- Temperature
- Light
- Water
- Particulate Contamination
- Atmospheric Contamination
- Oil Separation

Storage - Temperature

- High heat (greater than 45°C) and extreme cold (less than 20°C) affect lubricant stability.
- Heat increases oxidation that forms deposits
- Cold can increase sediment and wax formation
- Ideal storage temperature range is 0°C to 25°C

Storage - Light and Water

- Light can change the color and appearance of lubricants. Store lubricants in their original container. Keep out of light.
- Water reacts with additives in the lubricant and forms insoluble matter. Water can cause microbial growth. Keep water out.

Storage - Contamination

- Particles in the air and dust can settle into open containers. Oxygen and carbon dioxide can change the consistency and viscosity of lubricants.
- Always seal lubricant containers tightly.
- Always store and use a clean container.

Storage - Oil Separation

- Oil will naturally separate out of most greases over time.
- Temperature greater than 45°C increase oil separation in grease.

Storage – Shelf Life

Lubricants have a finite shelf life.

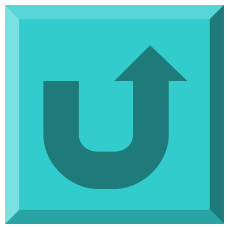
The estimated shelf life for UNOPENED containers in ideal conditions is:

Product	Shelf Life In Years
Base Oils	5+
Lube Oils (Mineral or Synthetic)	5
Greases (Mineral or Synthetic)	5
Rust Preventatives	2
Open Gear Lubes	2

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS

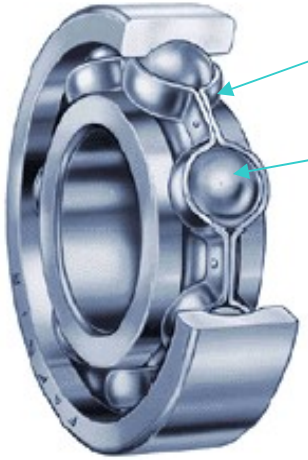


BEARINGS

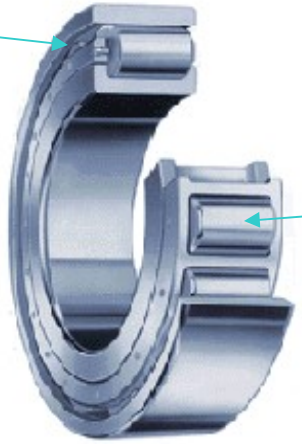
Introduction

- Purpose of a bearing
- Friction bearing
- Antifriction bearing

Bearings



Ball Bearing



Roller Bearing



Sleeve Bearing

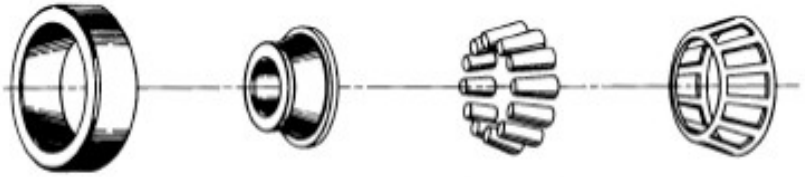
Sleeves and Journals

- Friction bearings
- Journal and Sleeve
- Lubrication
- Rotational Speed
- Highest friction point.

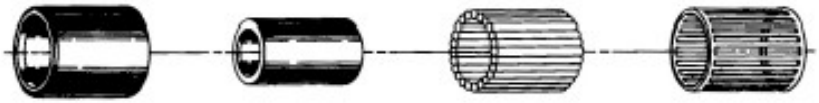
Balls and Rollers

- Rolling contact bearings
- Starting friction
- Cages/Seperators
- Lubrication

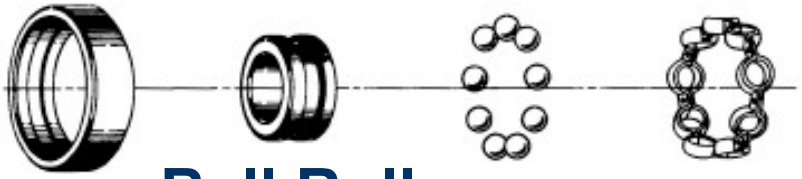
Anti-Friction Bearing Types



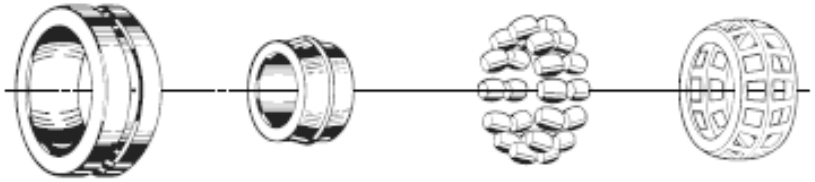
Tapered Rollers



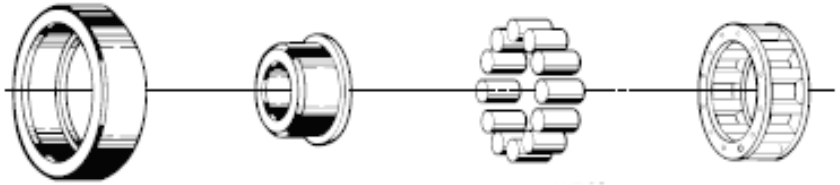
Needle Rollers



Ball Rollers



Spherical Rollers



Cylindrical Rollers

Thrust Bearings



Ball Thrust Bearing



Roller Thrust Bearing



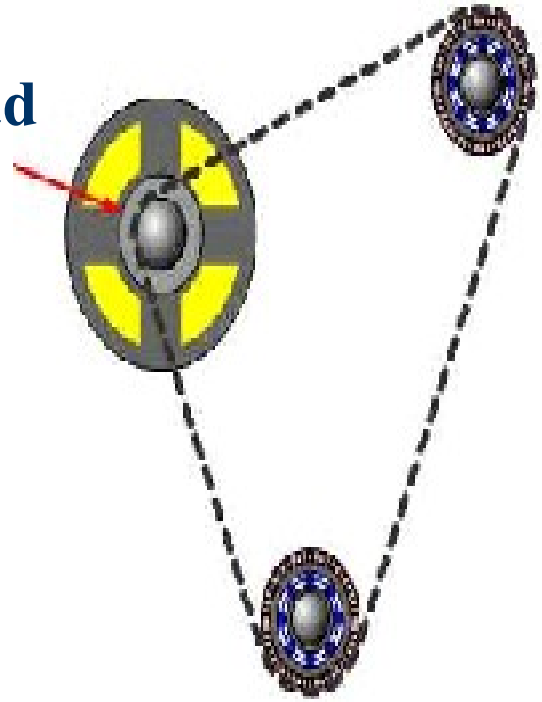
Spherical Roller



Tapered Roller

Bearing Loads

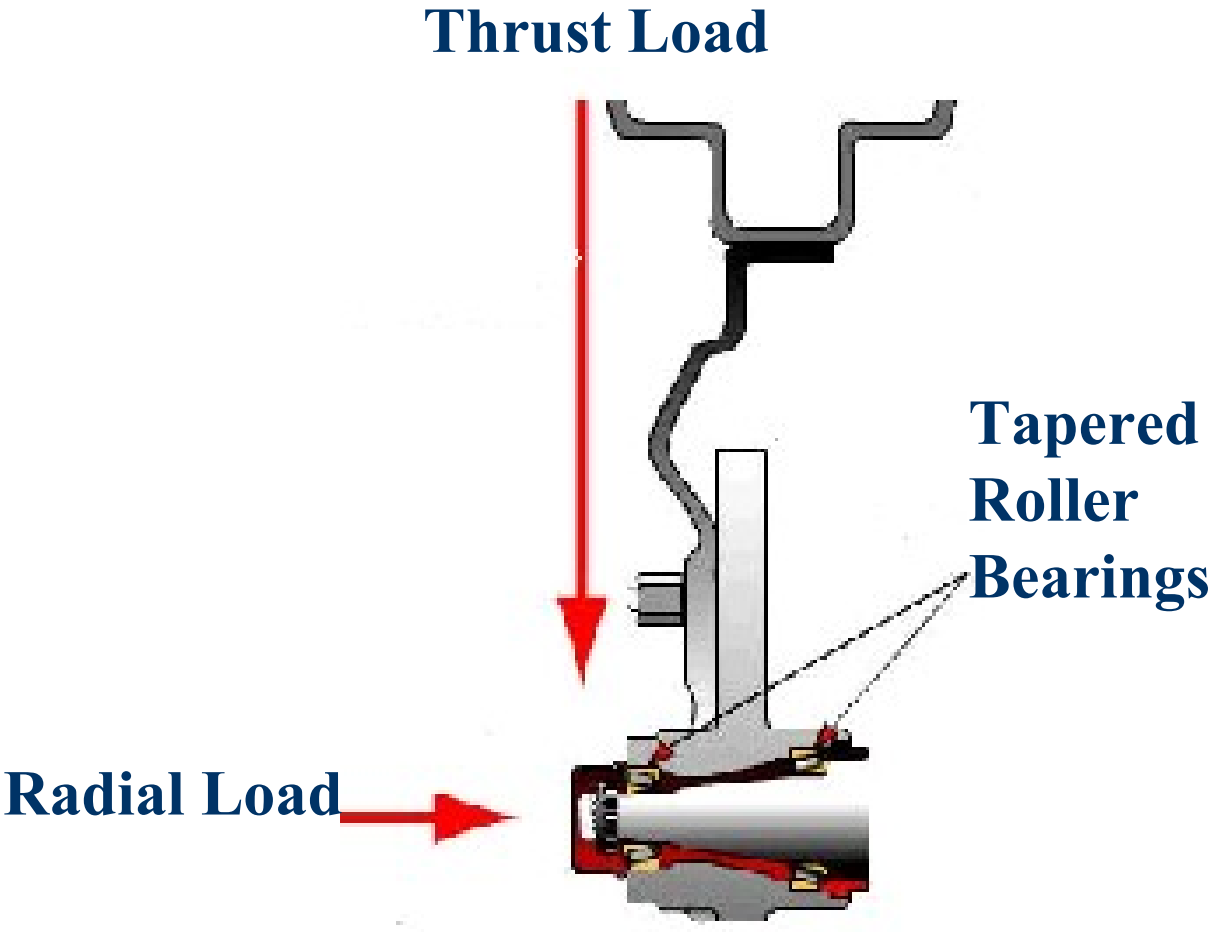
Radial Load



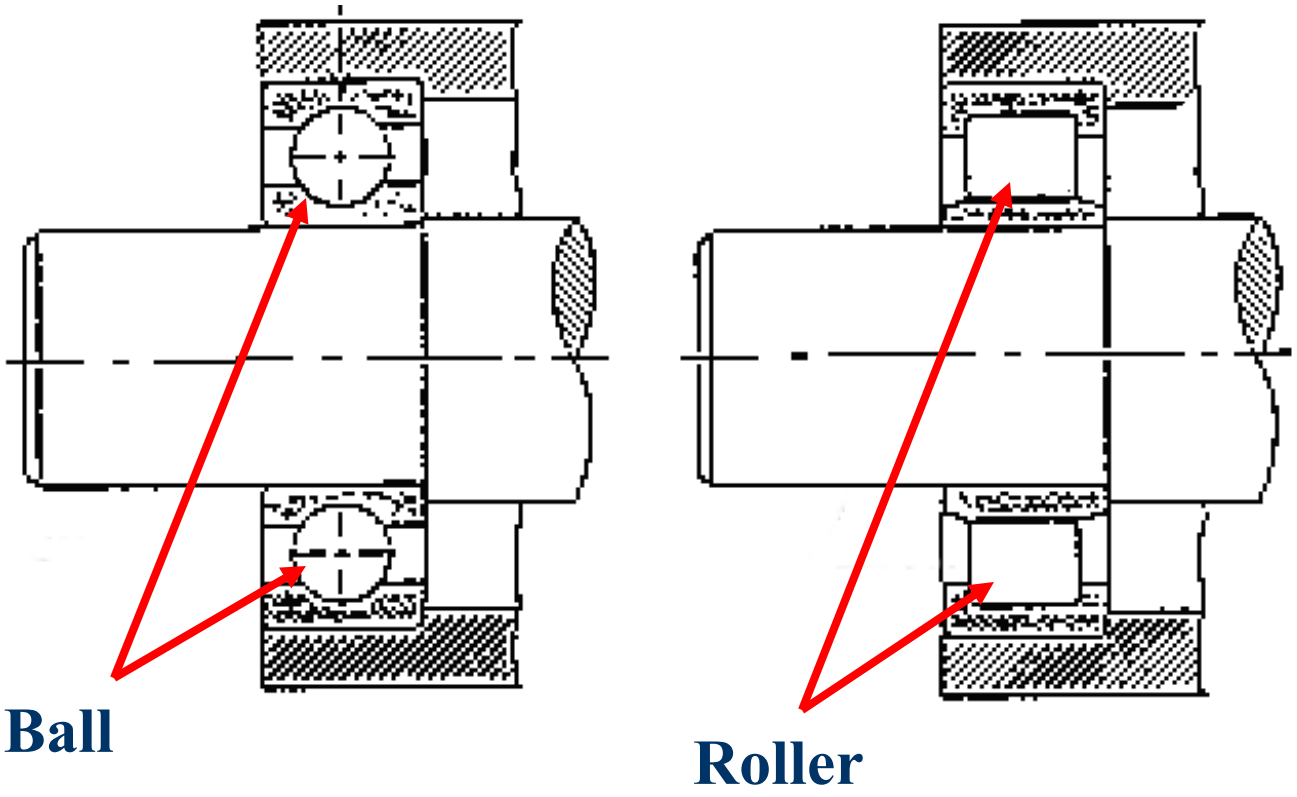
Thrust Load



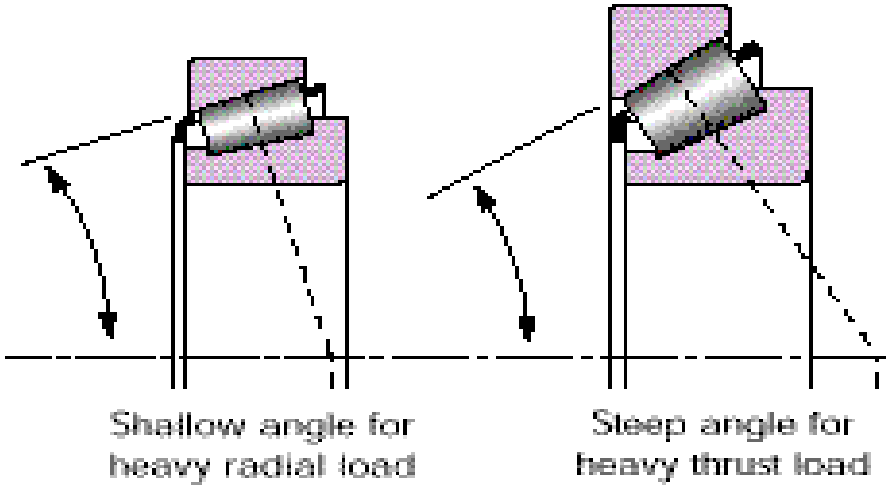
Example of Loads

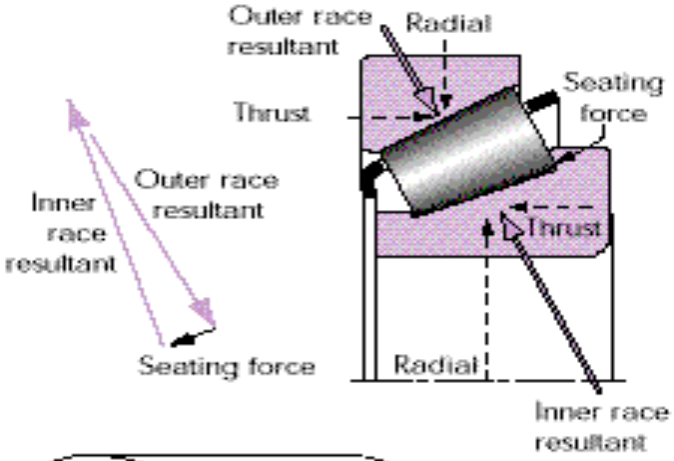


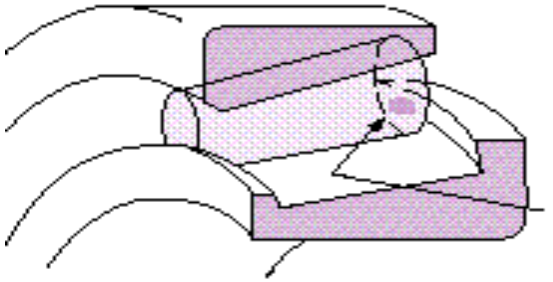
Bearing Contact



Tapered Roller Bearings





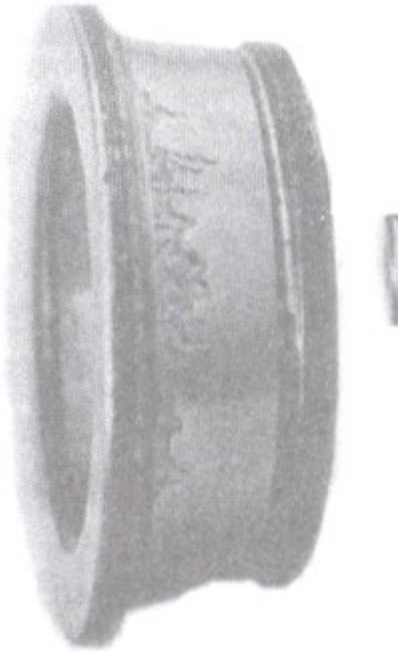


How Do Bearings Fail

- Passage of electric current through the bearing.
- Misalignment.
- Improper mounting.
- Incorrect shaft and housing fits.
- Defective bearing seating on shafts and in housings.
- Ineffective sealing.
- Vibration while bearing is not rotating.
- Inadequate lubrication.

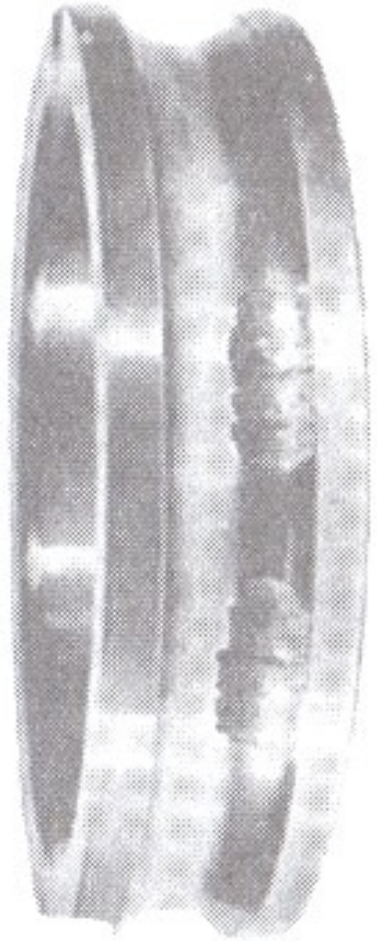
Types of Failure

- Spalling.
- Fretting.



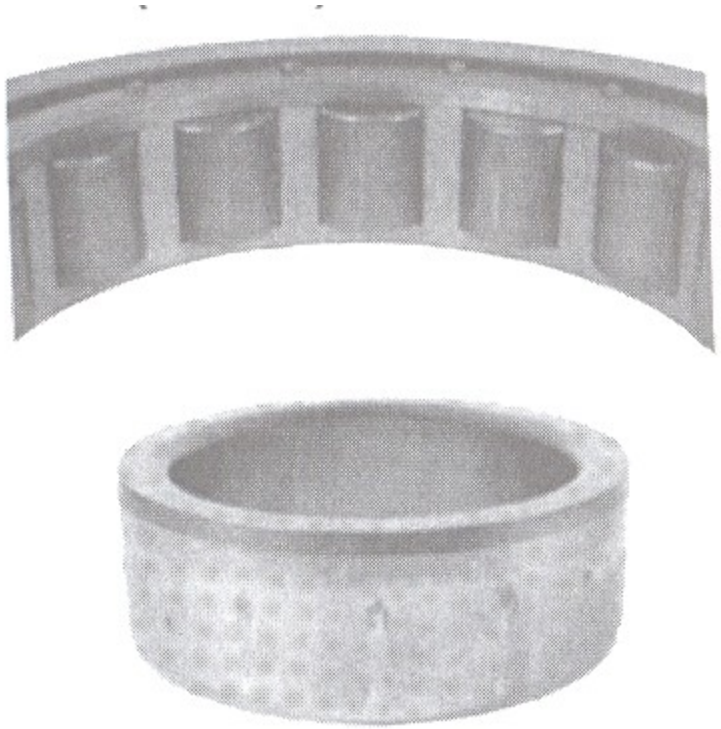
Types of Failure

- Brinelling

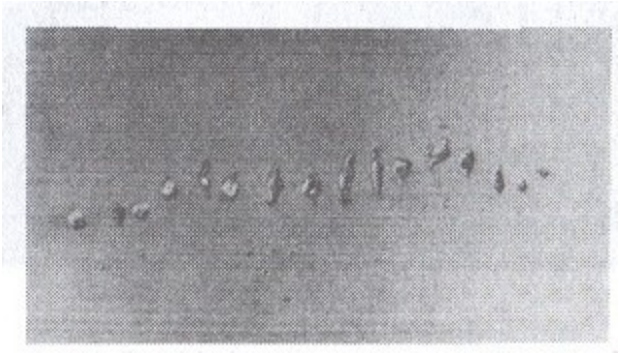


Types of Failure

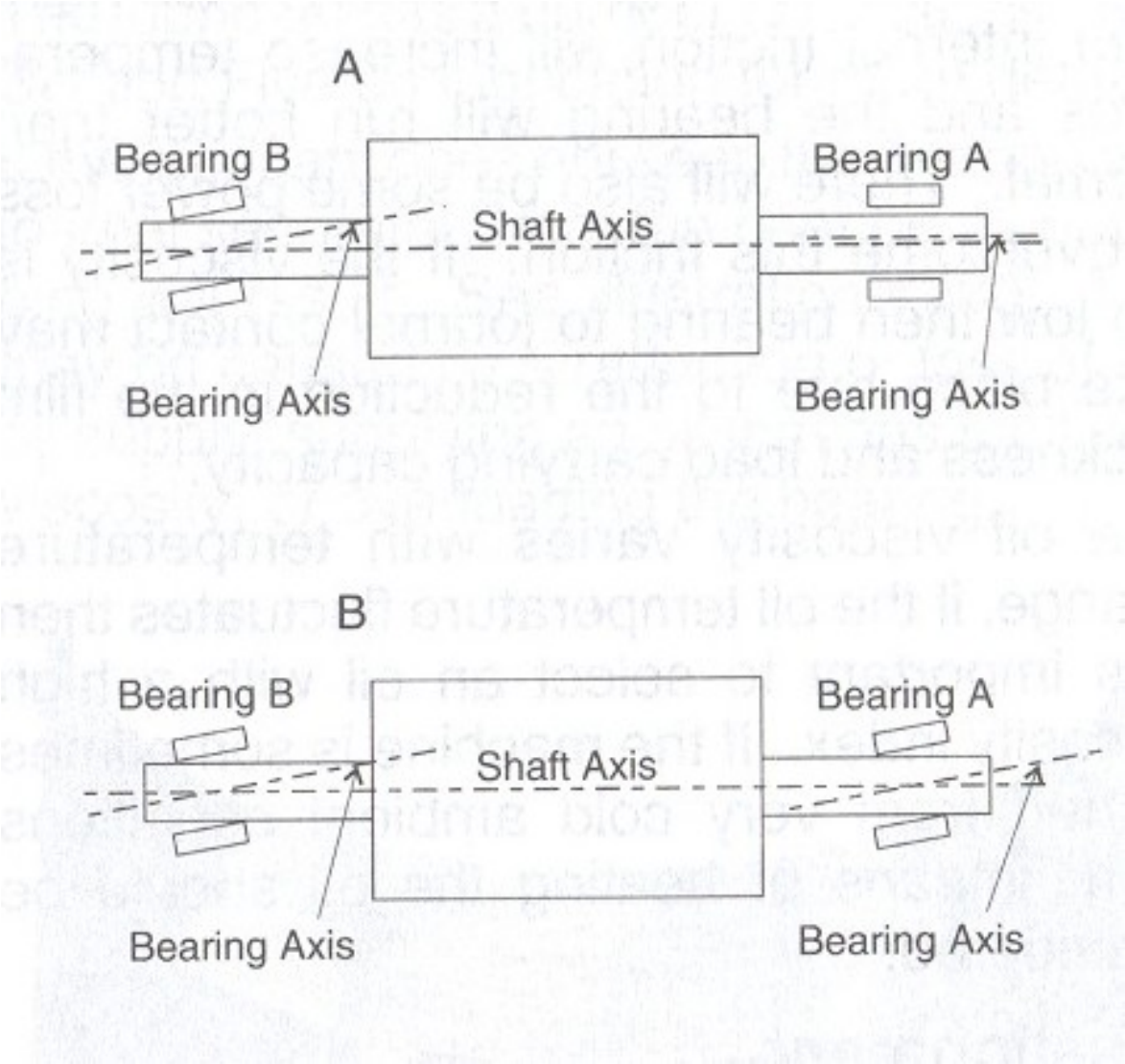
- Vibration
- Electric Currents.



Pitting from large electrical current.



Types of Failure - Misalignment



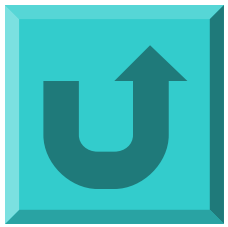
Bearing Lubrication

- All bearings need lubrication to prevent metal-to-metal contact between components.
- Lubrication Practices
- Too Much Lubrication
- Inadequate Lubrication
- Smearing

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS



SEALS

Major Topics

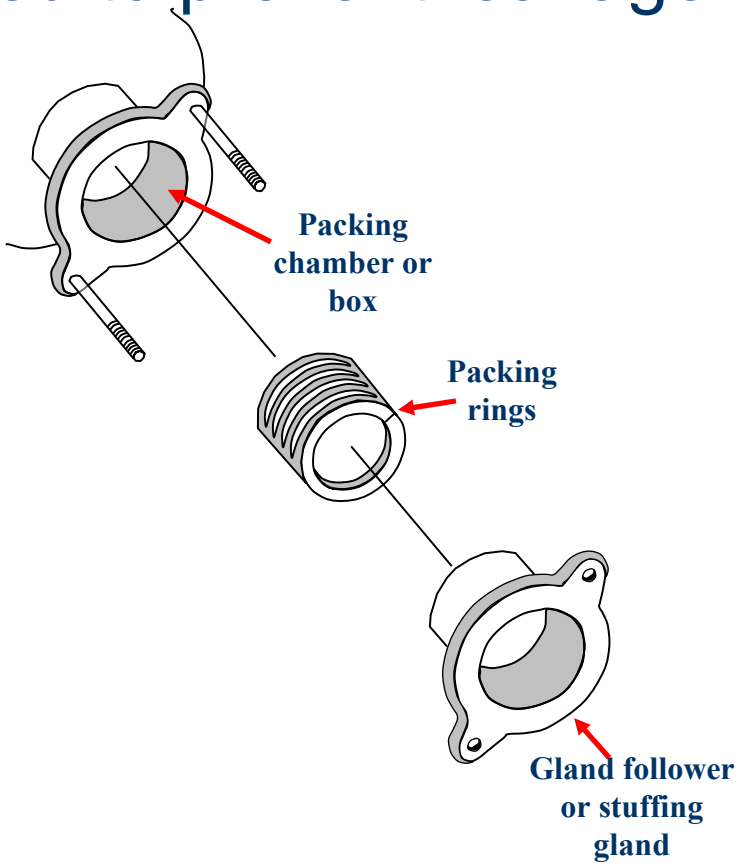
- Seals
- Seal Types
- Dry Gas Seals
- Labyrinth Seals
- Firewater Pump Packing Seals
- Support Systems – Seal Flushing
- Troubleshooting

Purpose

- Shaft Seal Purpose is to prevent leakage into or out of a pump or compressor along its shaft and other moving parts.
- Shaft seals includes two common types.
  - Pack stuffing boxes
  - Simple mechanical seals

Packed Stuffing Box

- A soft pliable material or packing is placed in a box and compressed into rings encircling the drive shaft is used to prevent leakage.



Gland Packing

- Used in Firewater pumps
- Fluid not toxic or flammable
- Leak rate not critical

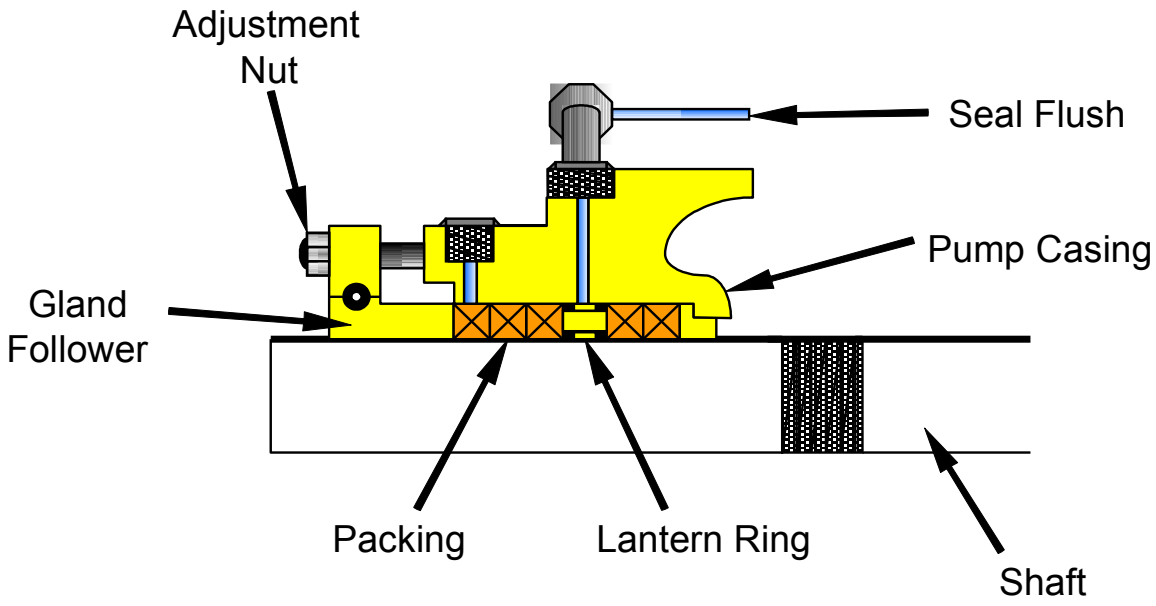
Mechanical Seals

- Fluid is Toxic or Flammable
- Leak rate is critical

Gland Packing

- Description
- Application
- Advantages
- Disadvantages
- Operation

Gland Packing



Gland Packing



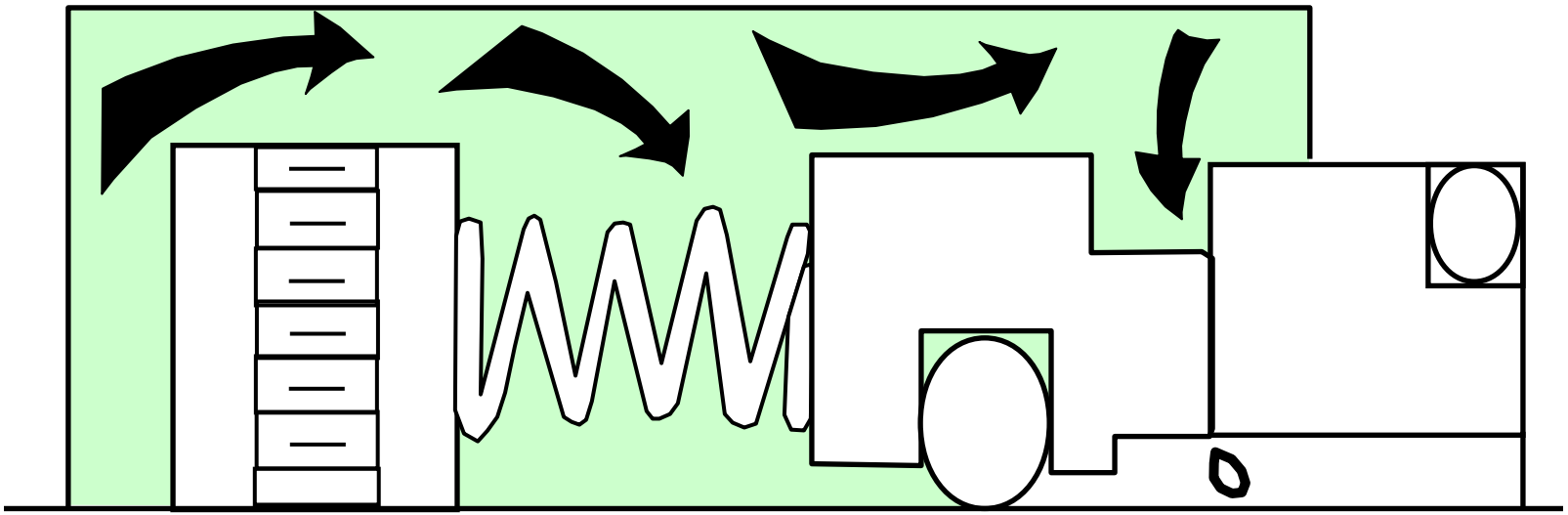
Mechanical Seals

- Pusher Seals
- Bellows Seals
  - Metal
  - Elastomer
- Cartridge Seals

Advantages

- Advantages
  - Extremely low leakage rates can be attained with proper selection and implementation
  - Reduced Preventative Maintenance requirements with proper selection and implementation

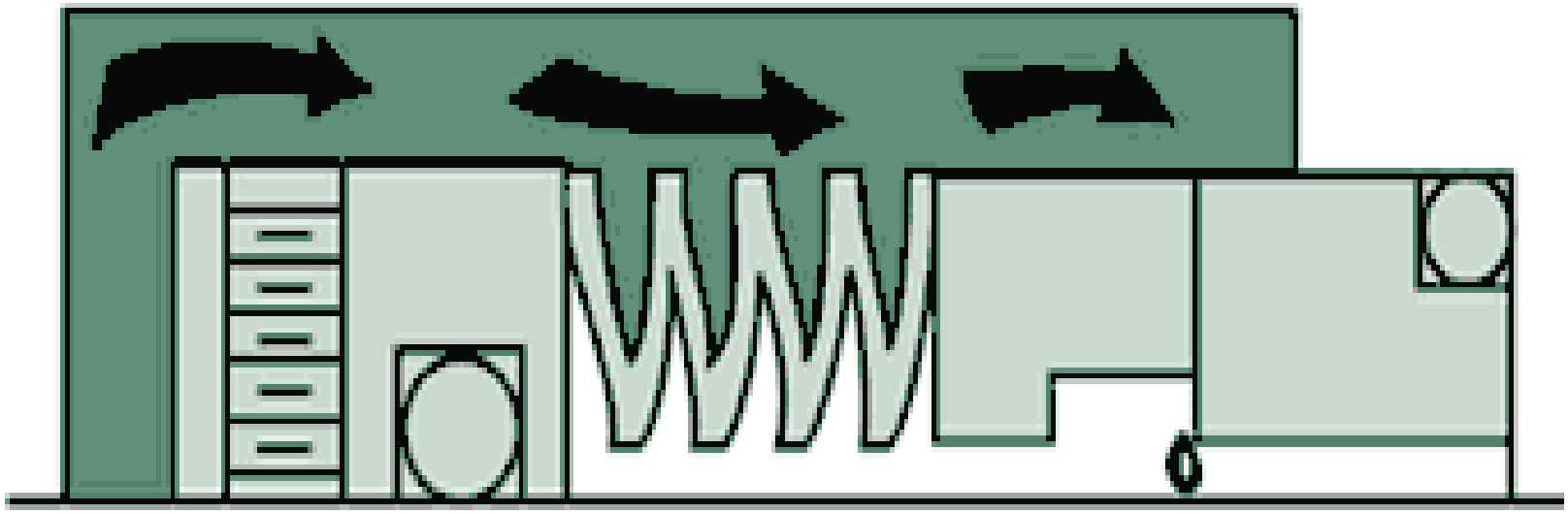
Pusher Seal



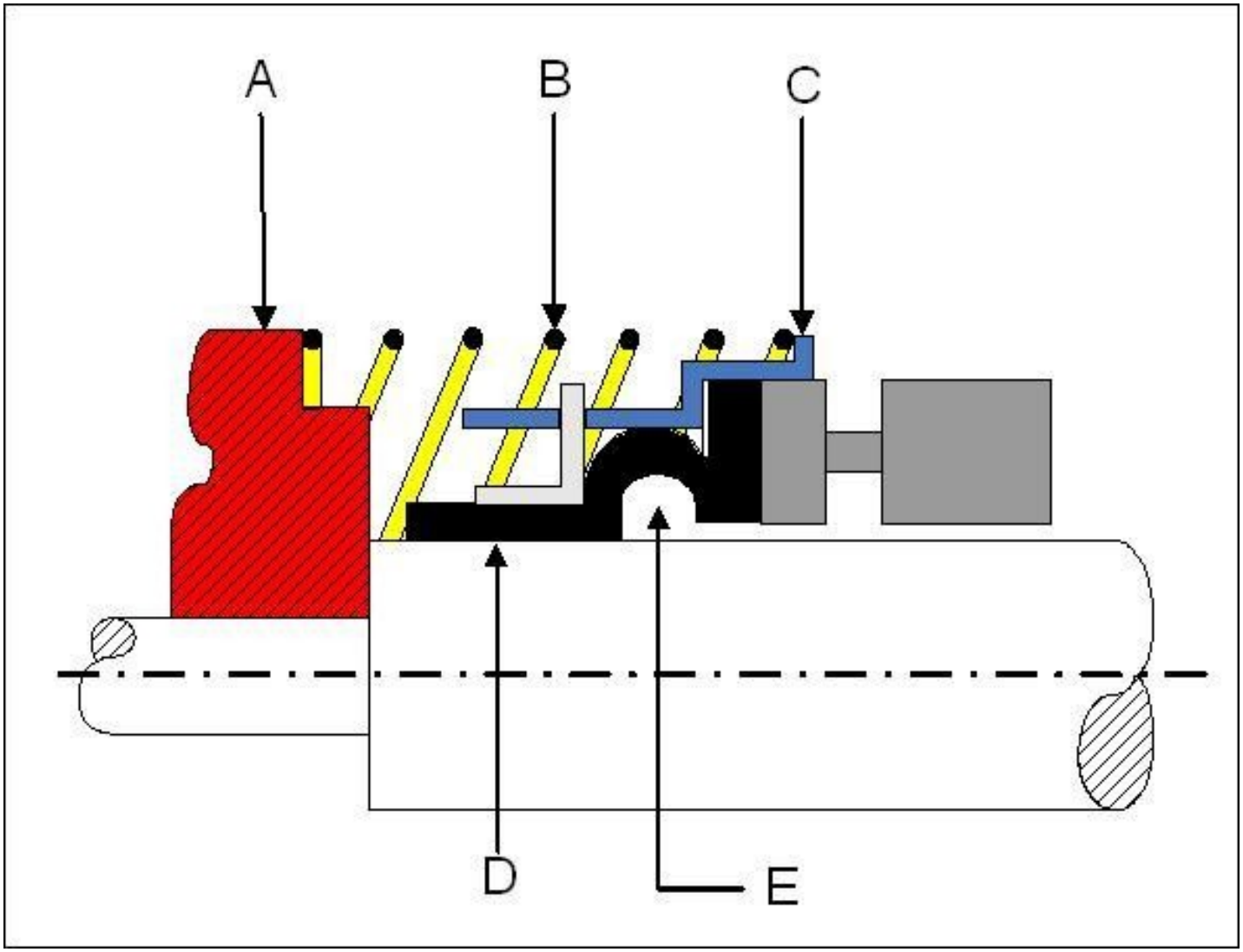
Pusher Seal



Bellows Seals



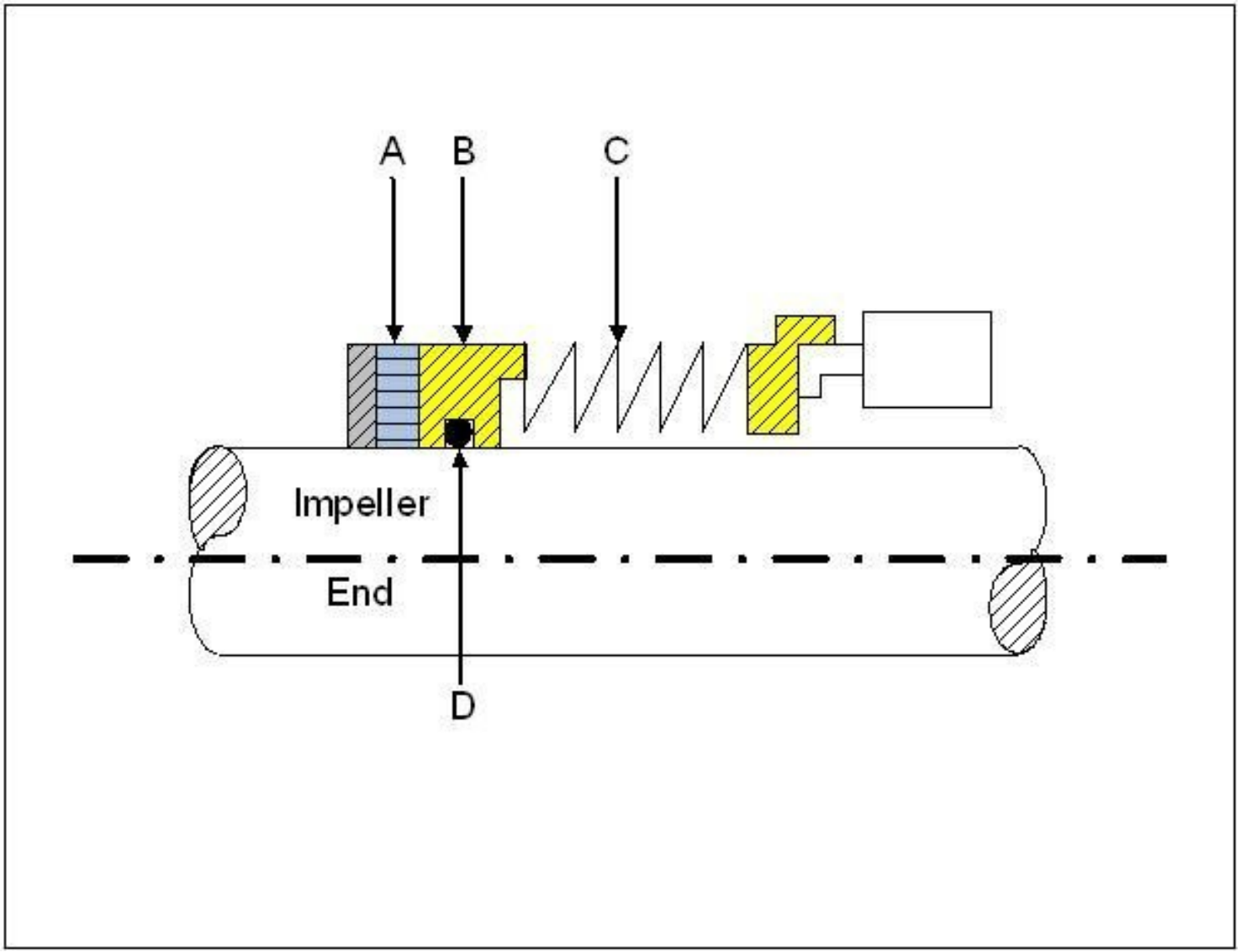
Bellows Seal (Elastomeric)



Bellows Seal (Elastomeric)



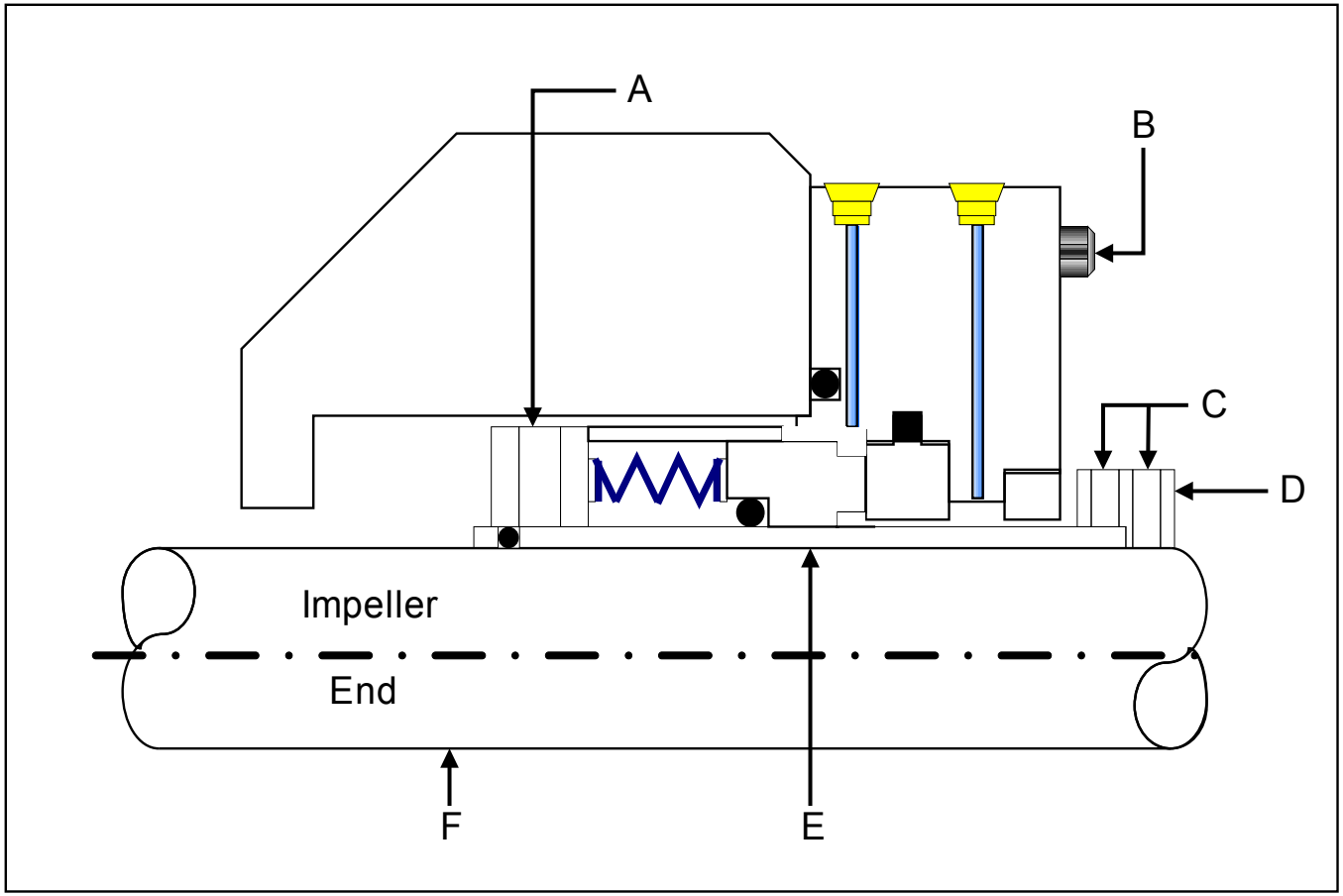
Bellows Seal (Metallic)



Bellows Seal (Metallic)



Cartridge Seals



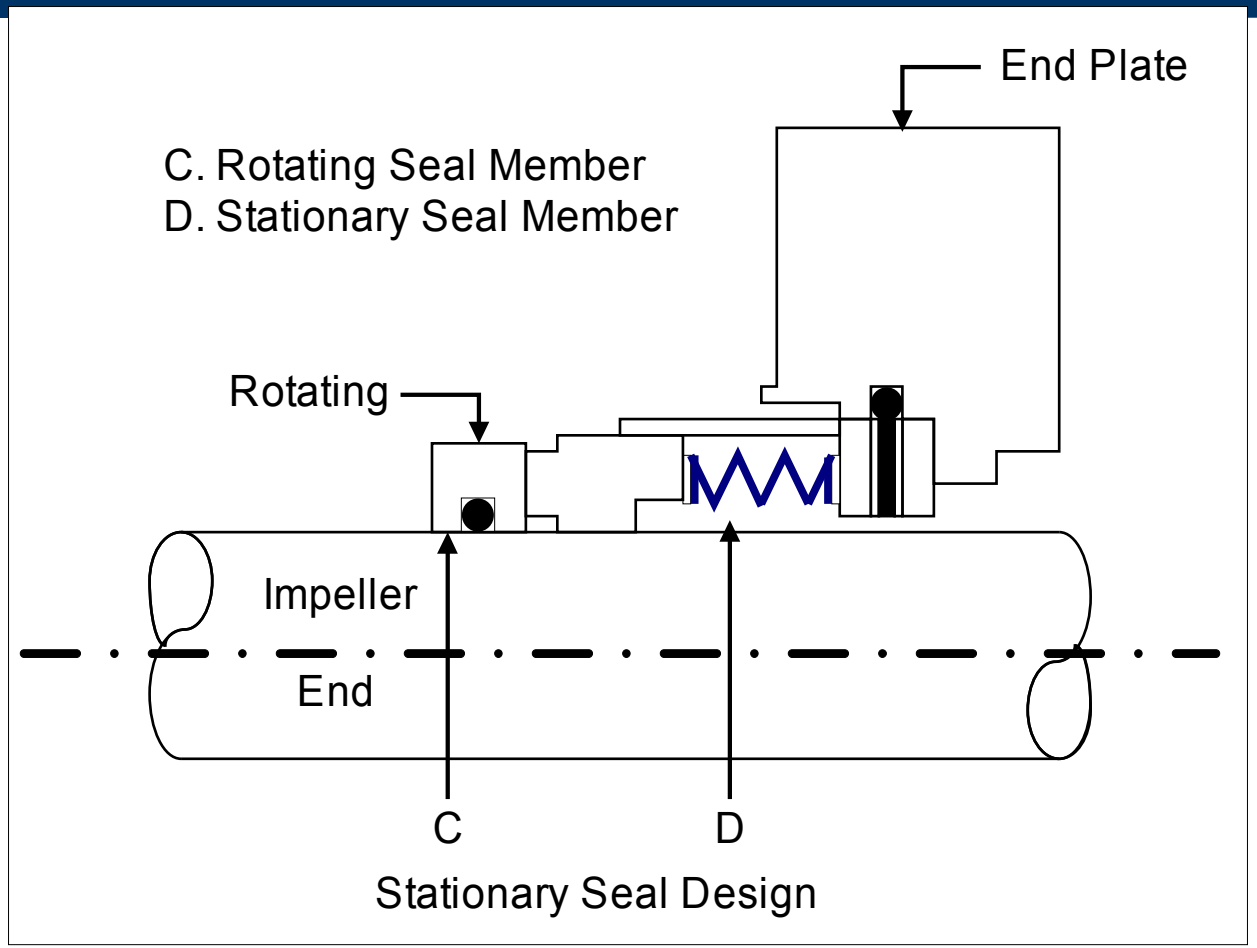
Cartridge Seals



General Terminology

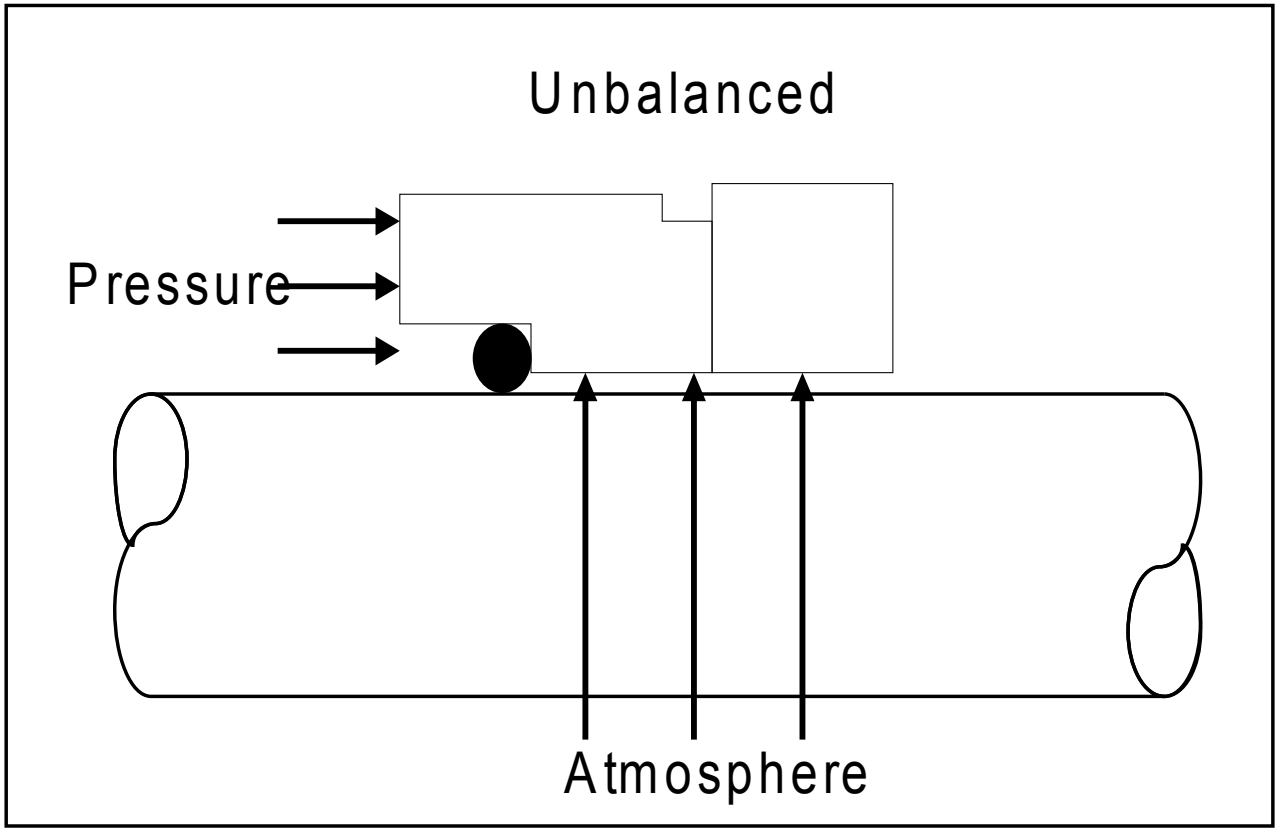
- Rotating Seal
- Stationary Seal
- Balanced Seal
- Unbalanced Seal

Stationary Seal

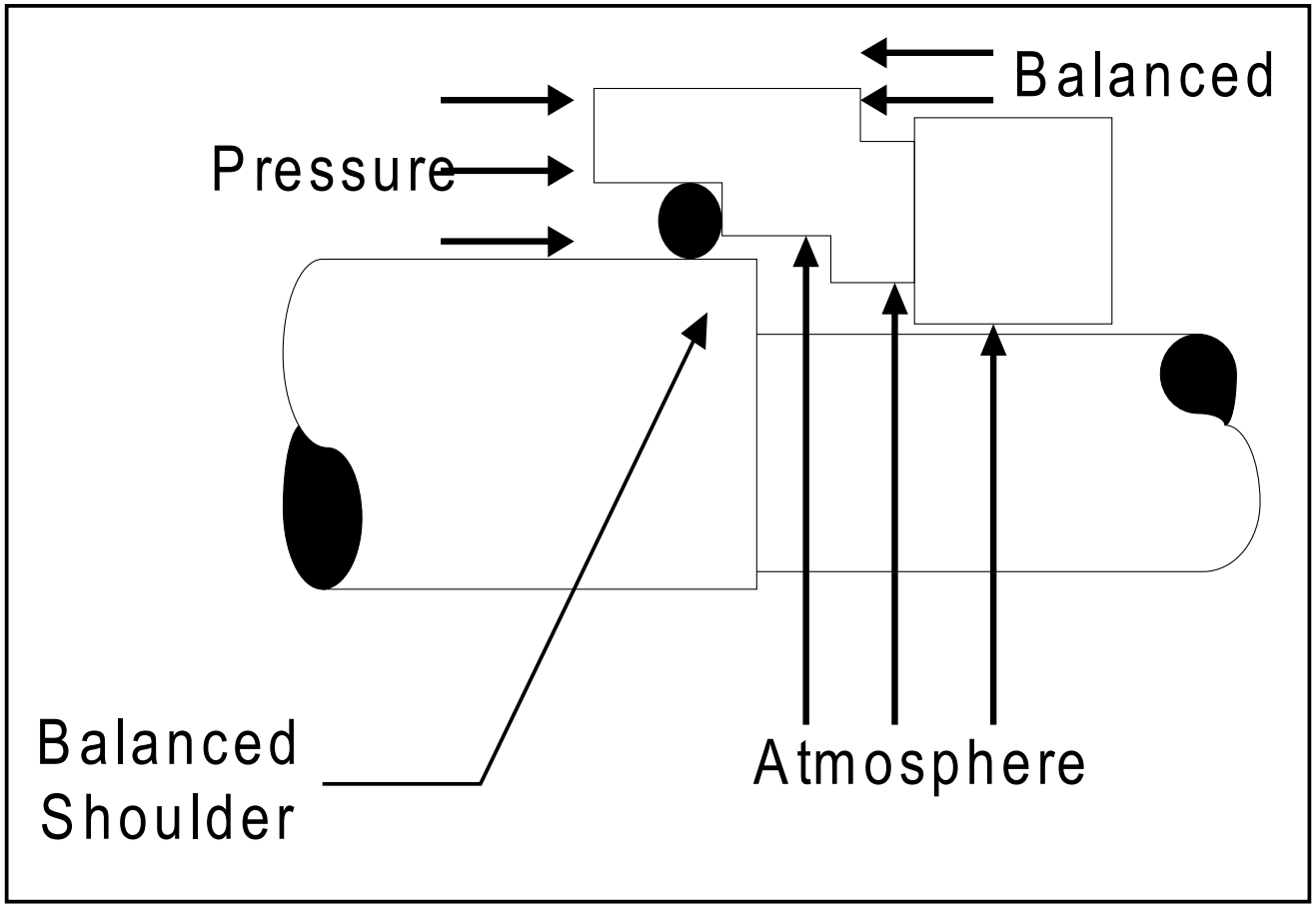


Stationary Seal Design

Unbalanced



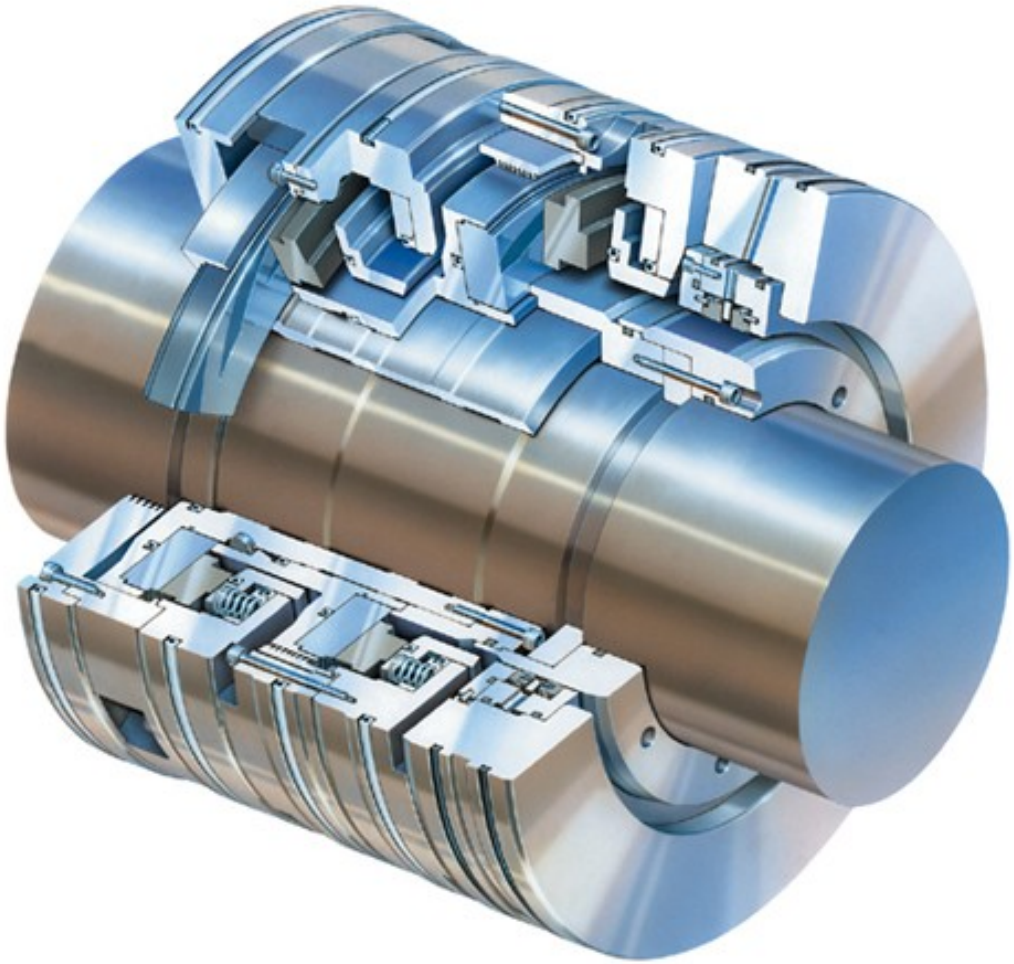
Balanced



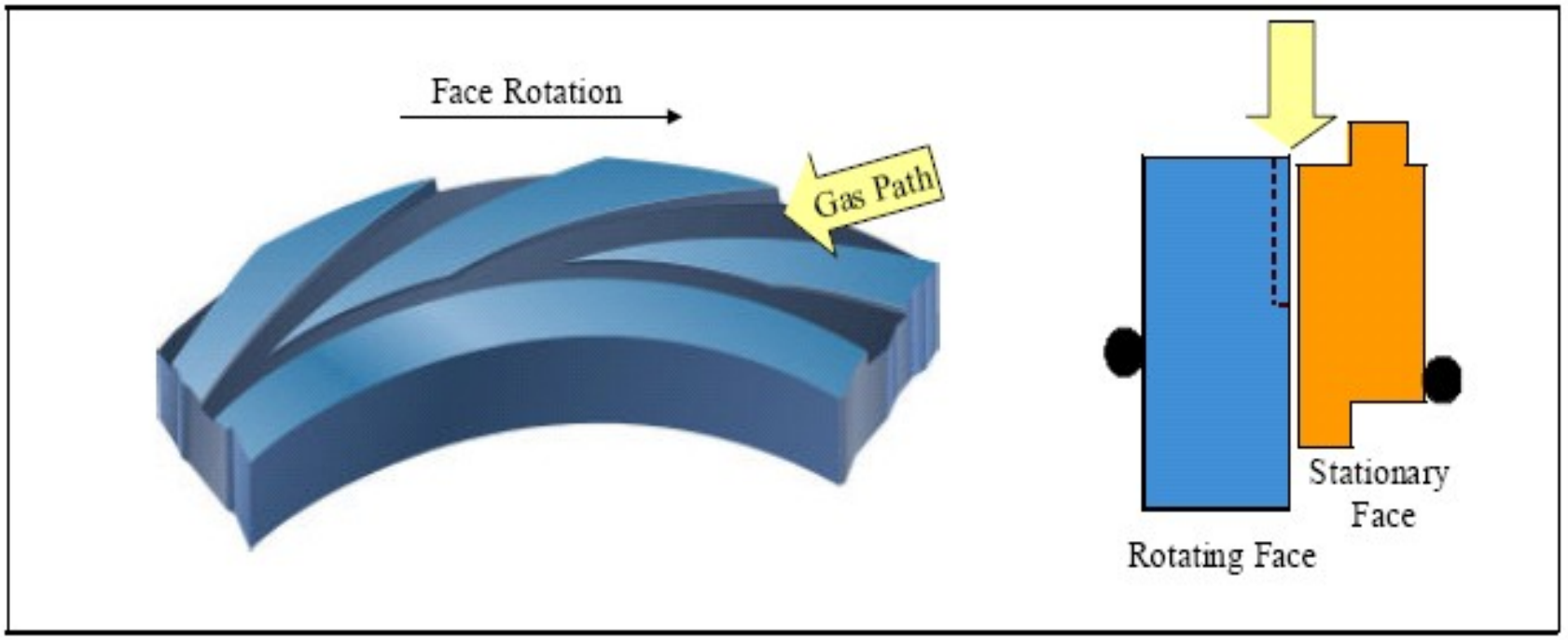
Dry Gas Seals

- Description
- Location
- Maintenance

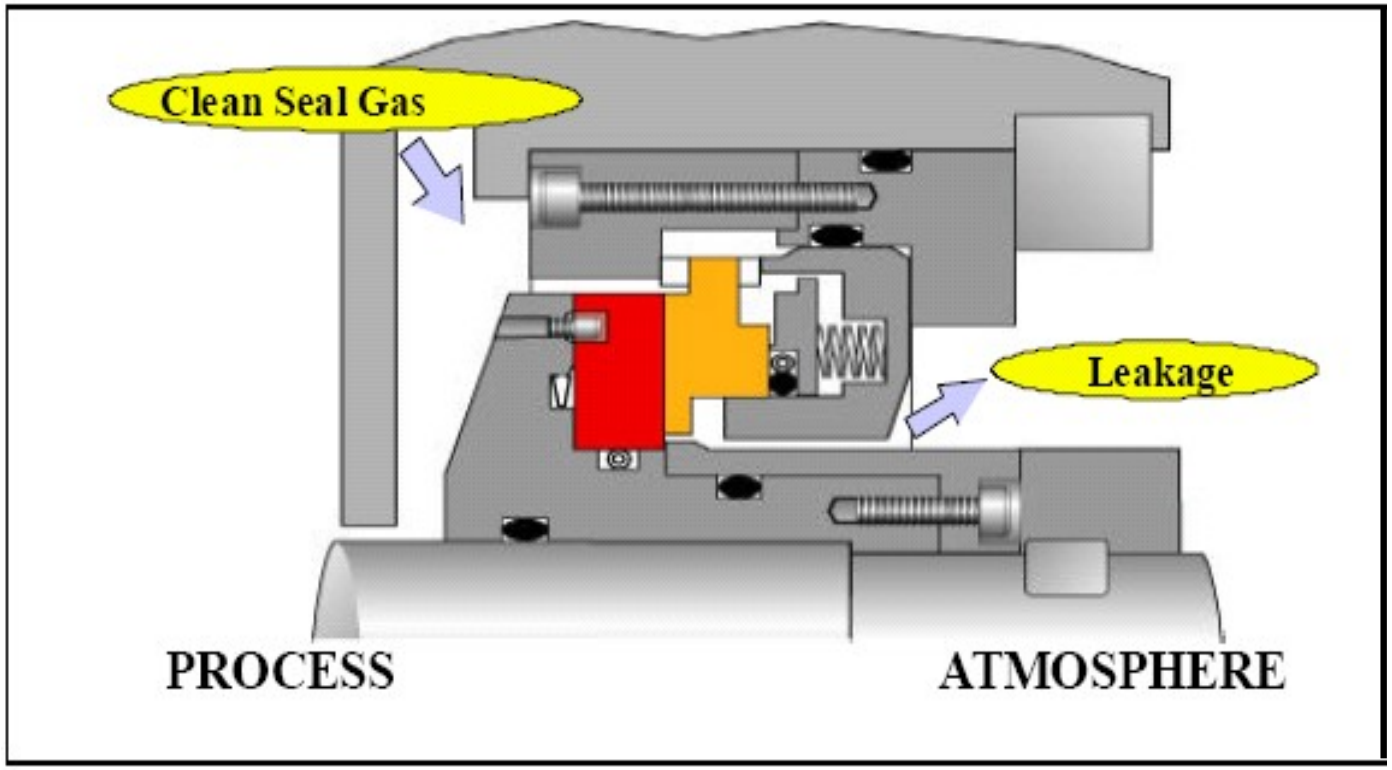
Dry Gas Seal



Description



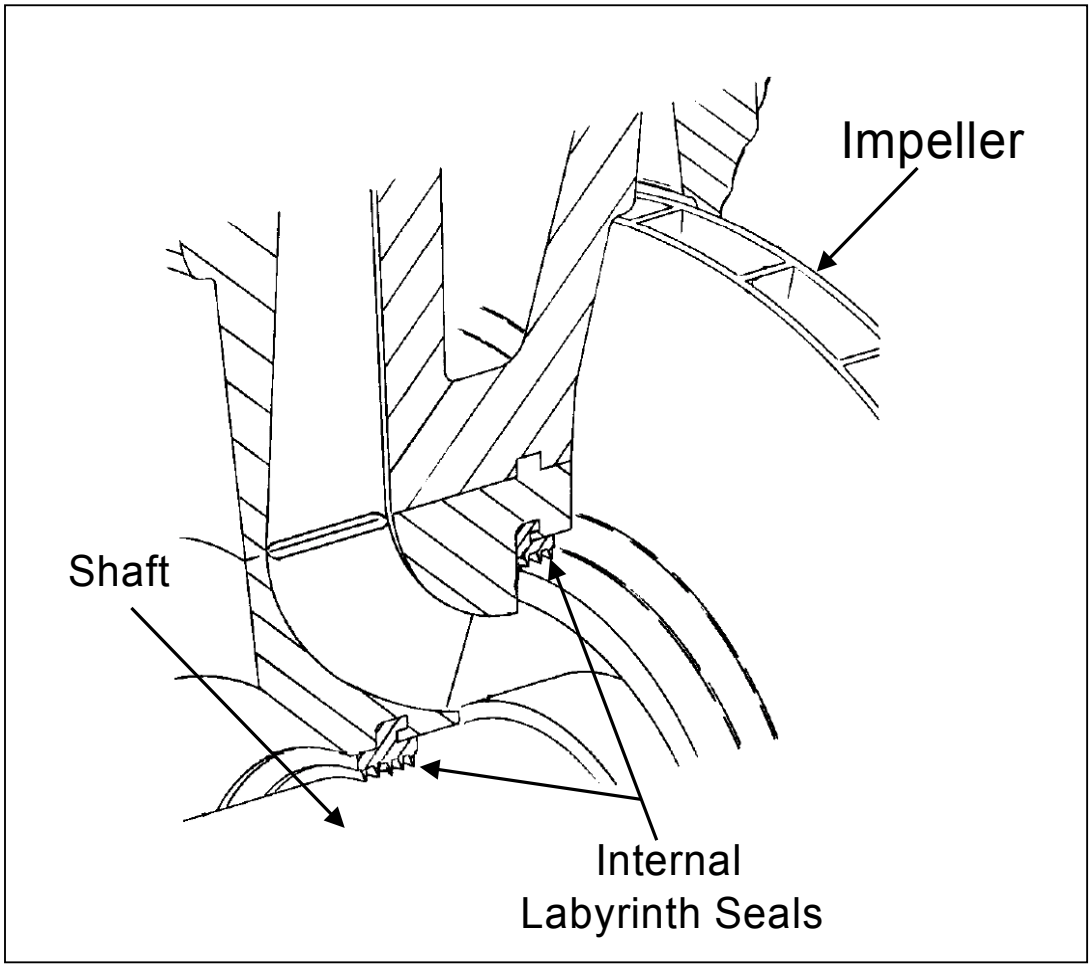
Gas Seal Description



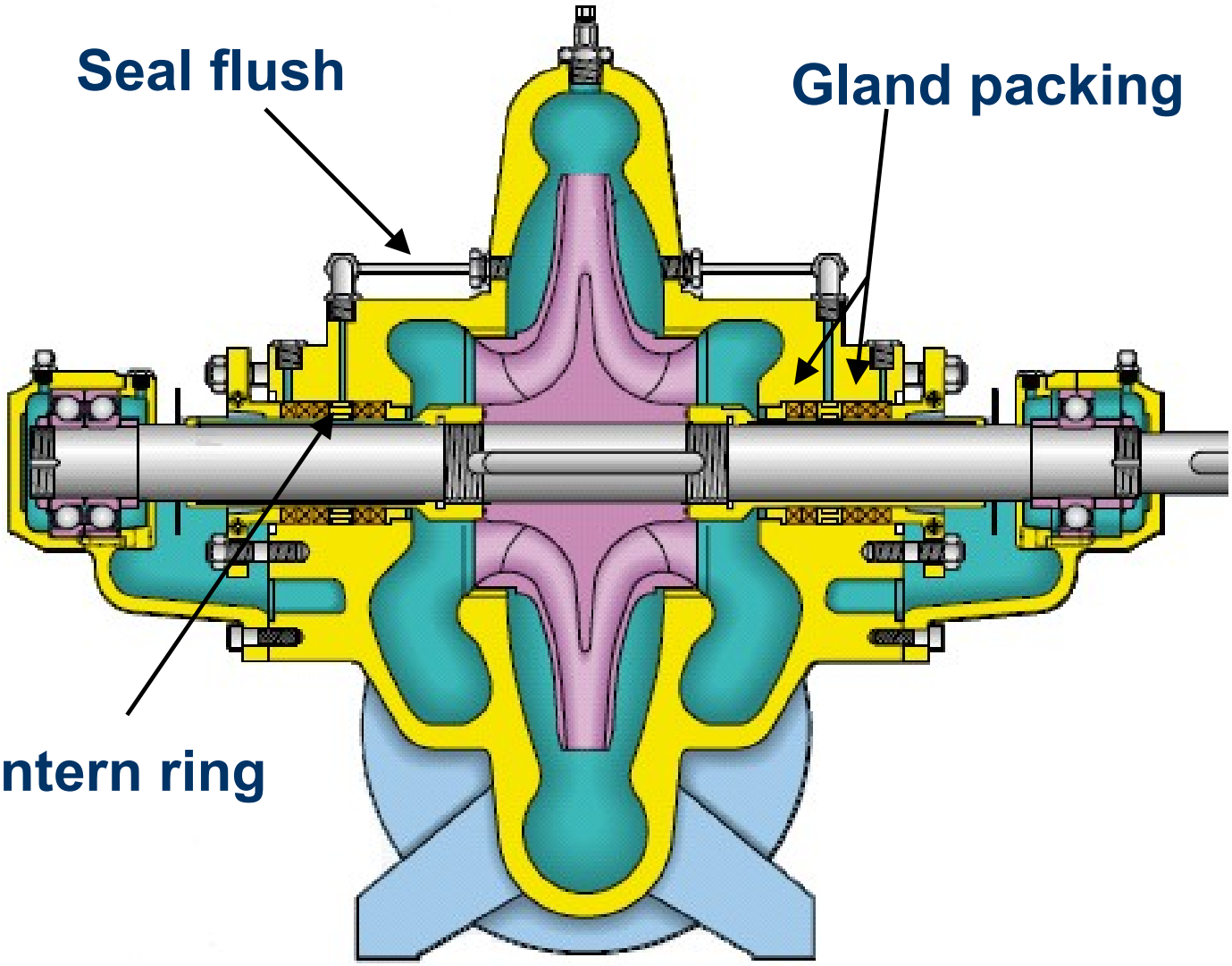
Labyrinth Seals

- Description
- Location
- Maintenance

Description



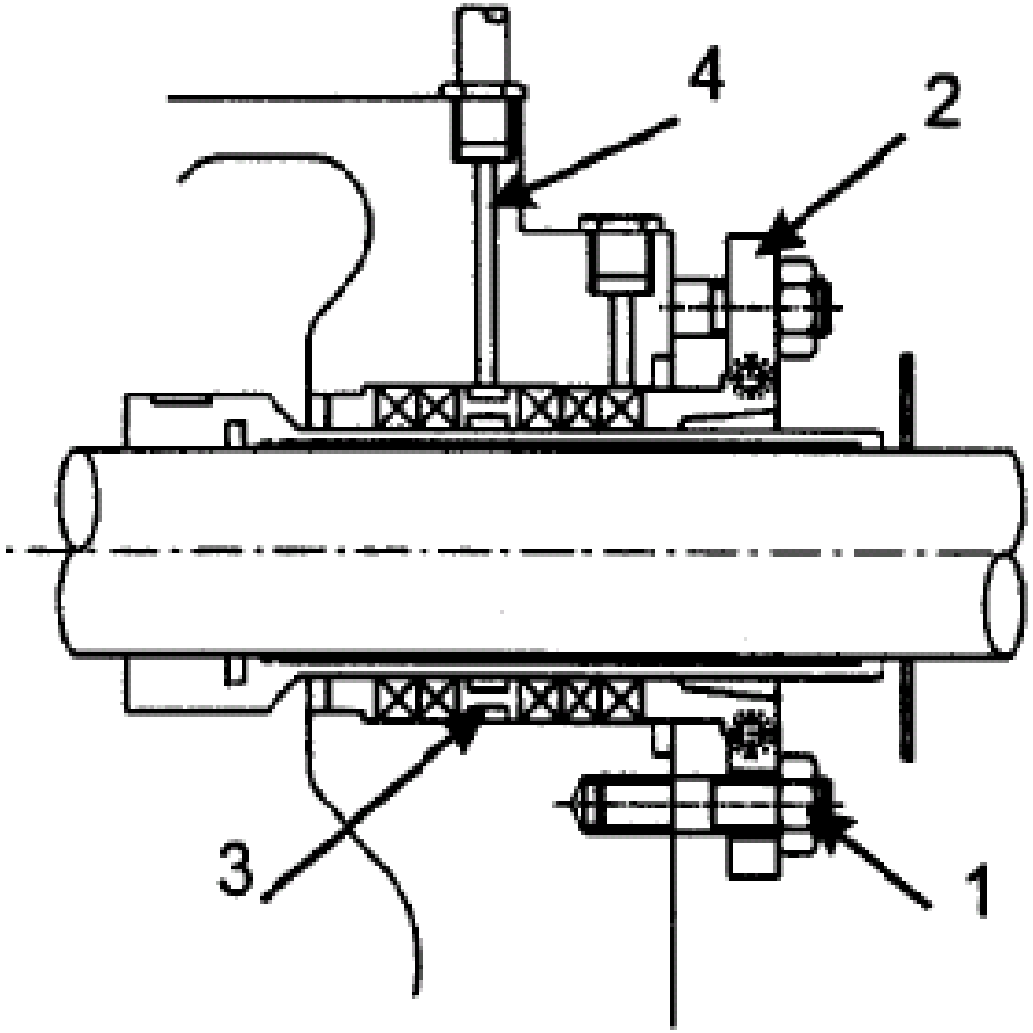
Firewater Pump Diagram



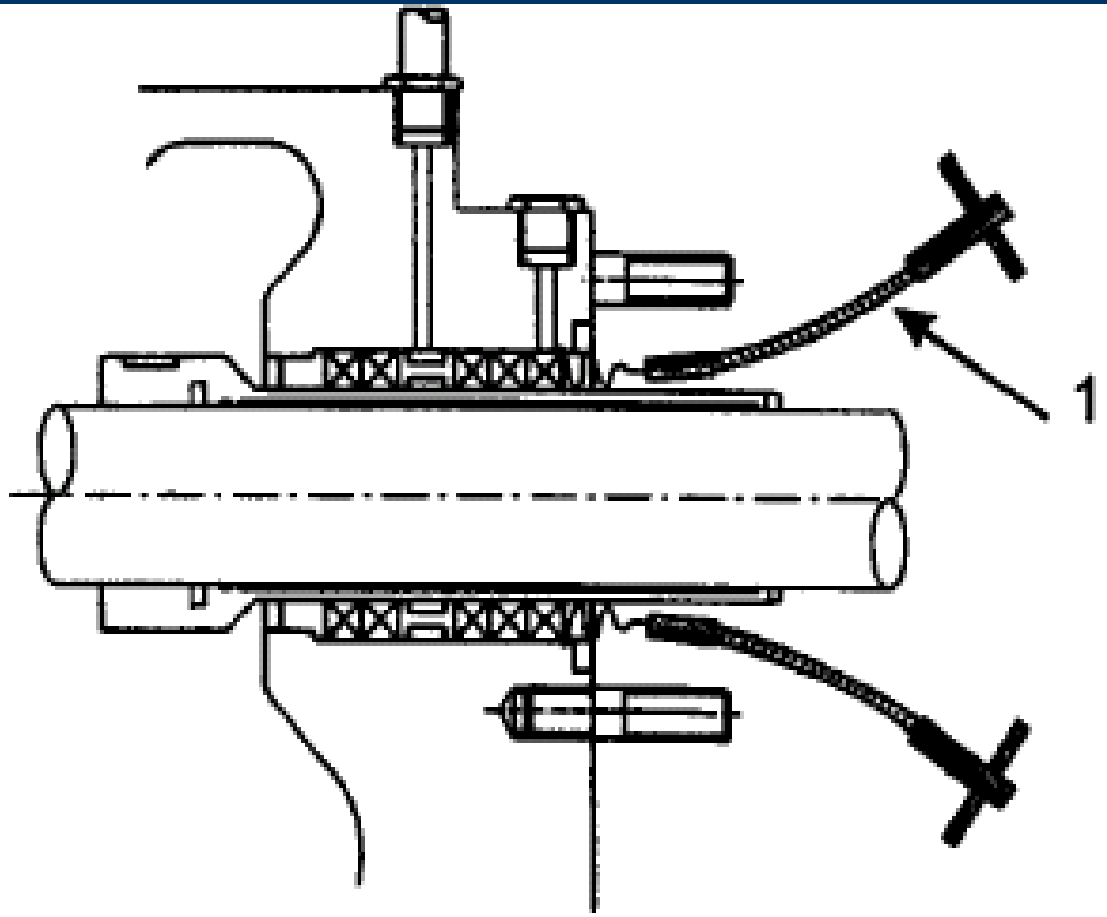
Packing Construction

- Lattyflon 2790AL
  - PTFE Impregnanted
  - Polyacrylic Yarns
  - Silicone Lubricant

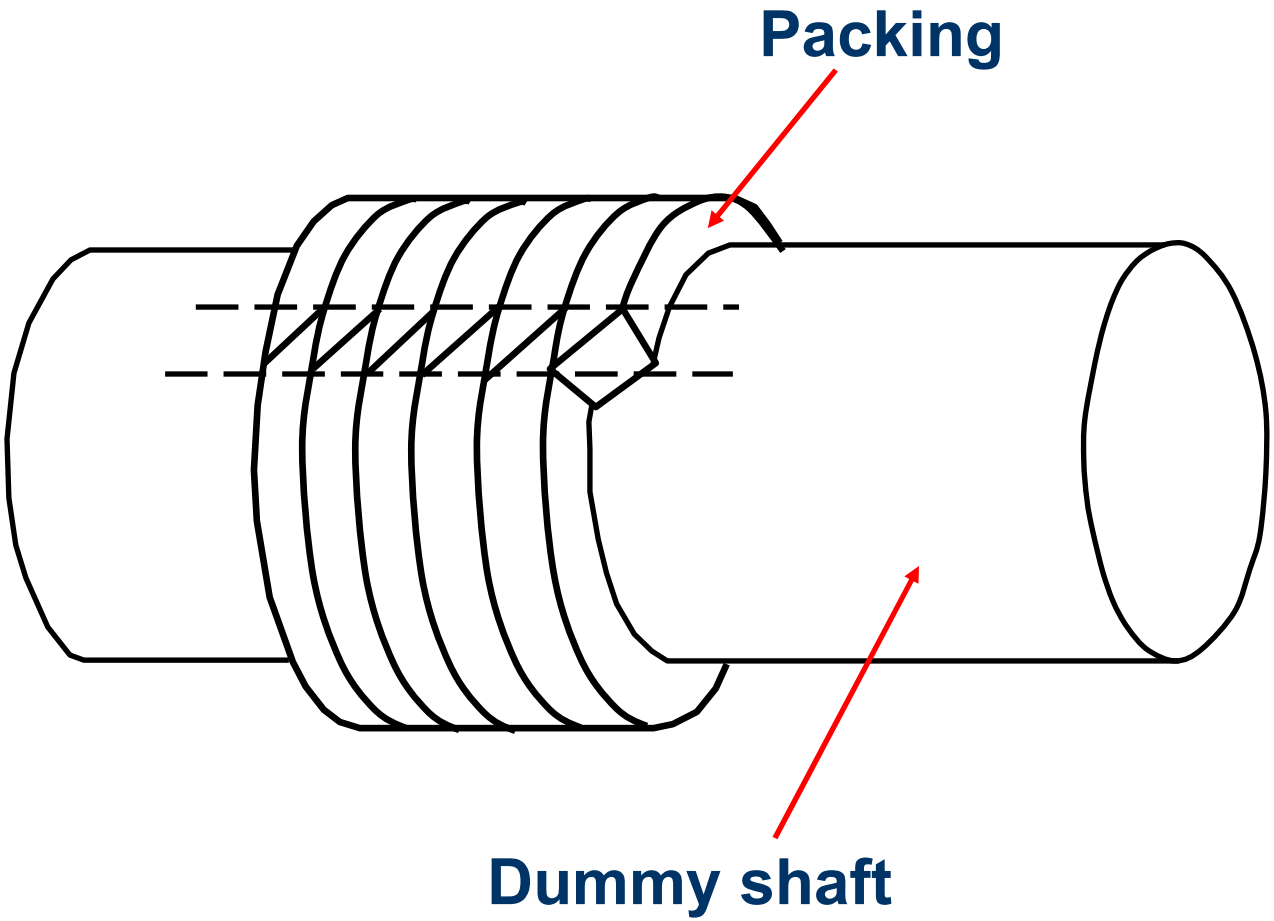
Packing Replacement



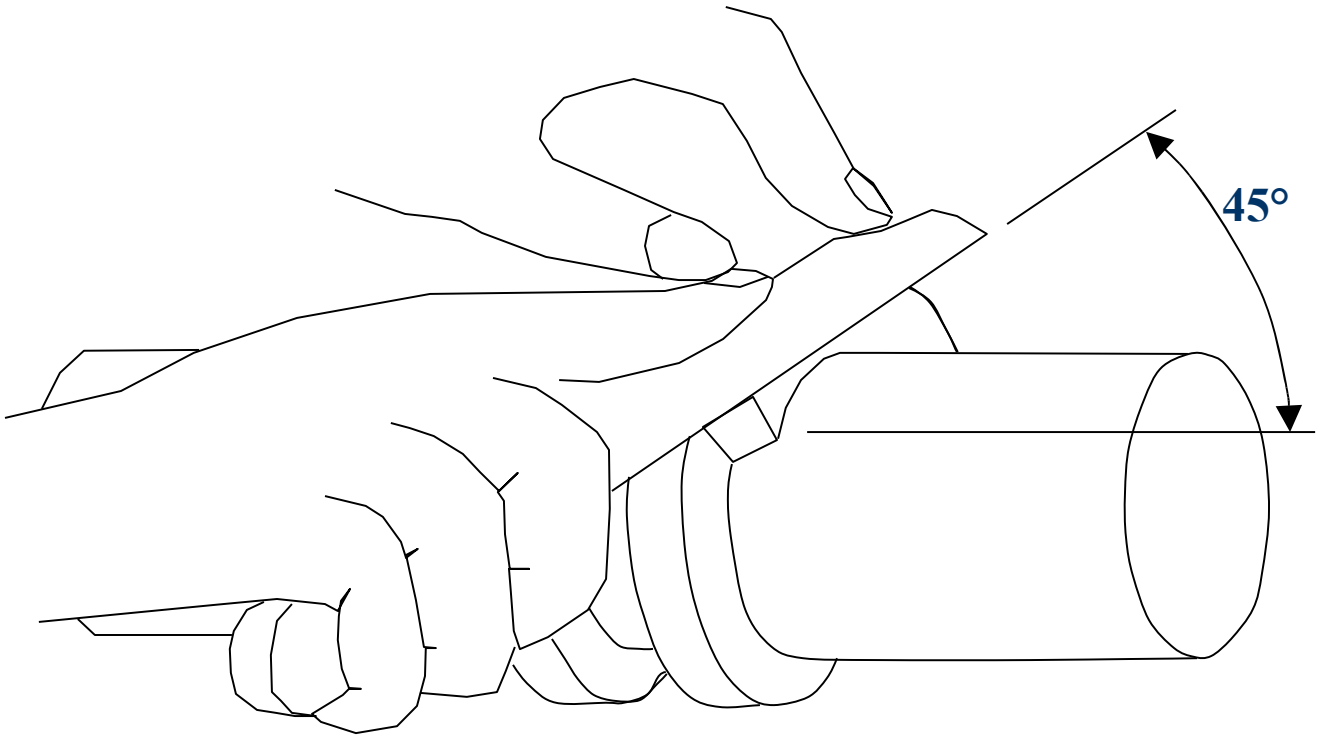
Packing Replacement



Packing Replacement

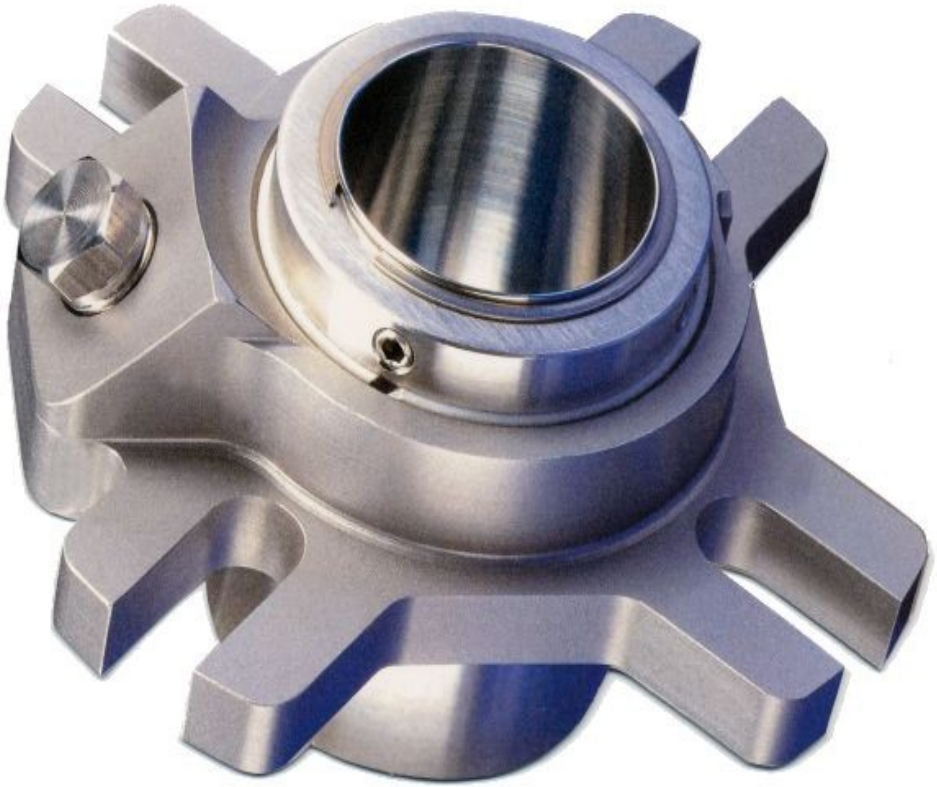


Packing Replacement



Mechanical Seal Service

- Flowserve Single Pusher Cartridge Seal – Type CSCPX



Support Systems - Seal Flush

- Description
- Maintenance

Flushing

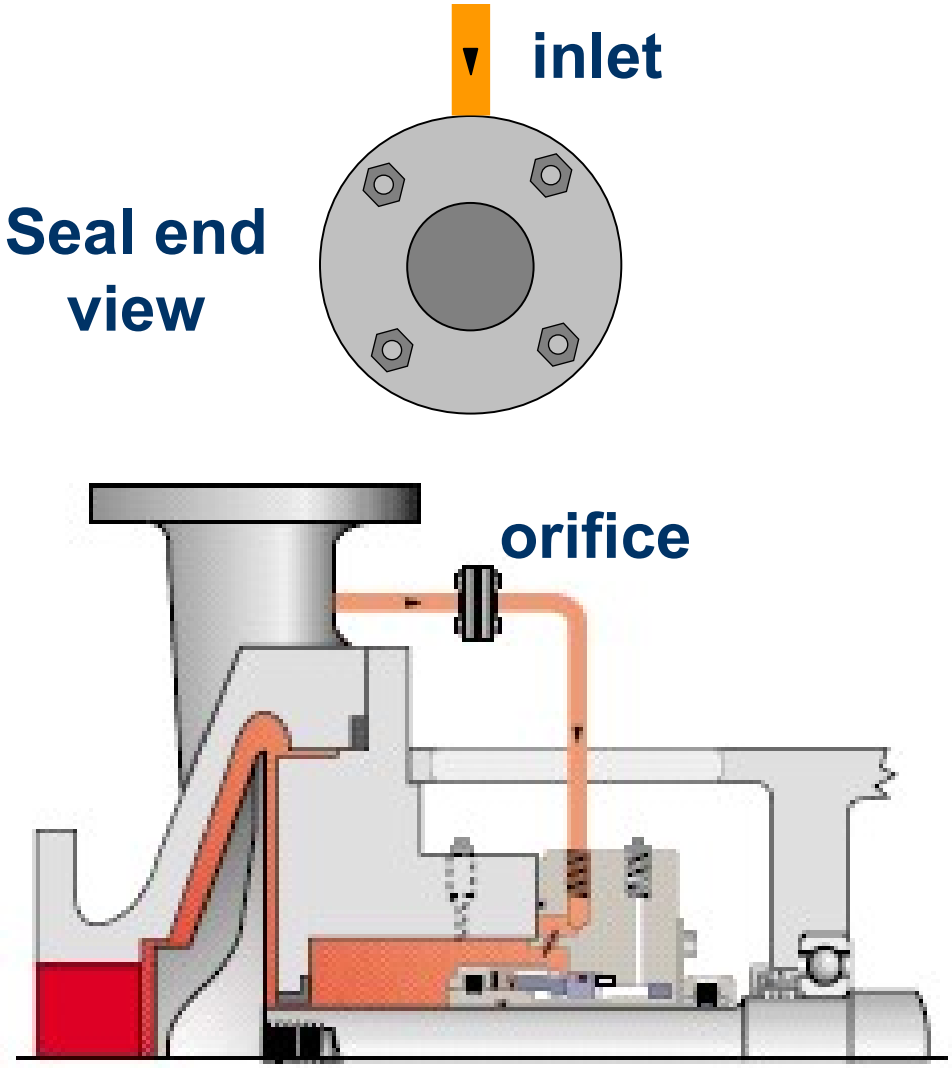
- A small amount of fluid that is introduced into the seal chamber close to the sealing faces
- Improves the fluid conditions near the faces
- Suppress vapor formation at or near the faces by heat removal and pressurization

Seal Flush Piping

- LPG, toxic services, or T> 450°F:
  - Orifice should be provided at the discharge or suction nozzle connection.
  - Flush and quench lines should be Type 316 stainless steel tubing

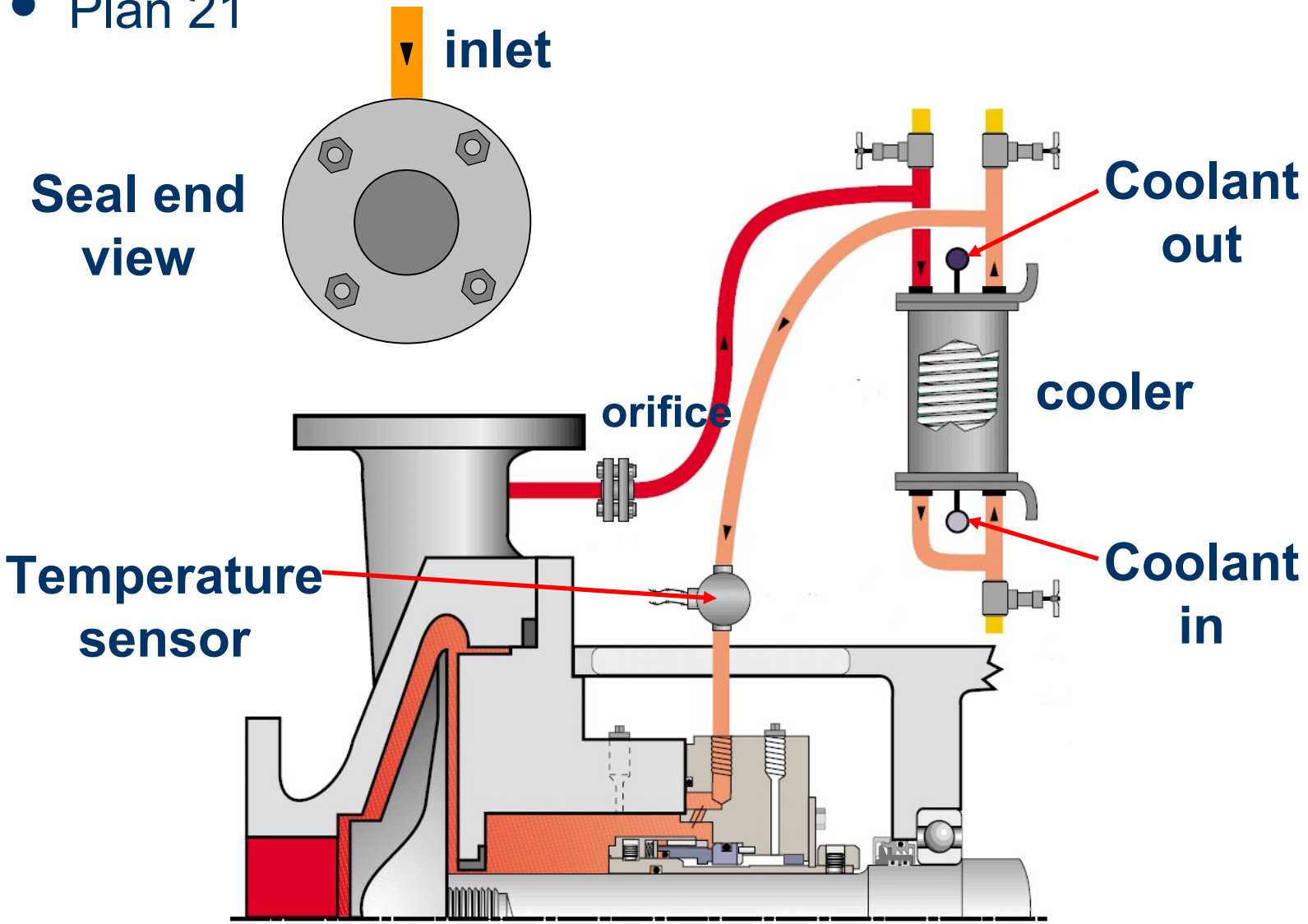
Flush Plans

- Plan 11



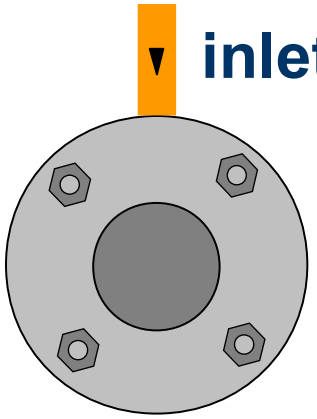
Flush Plans

- Plan 21



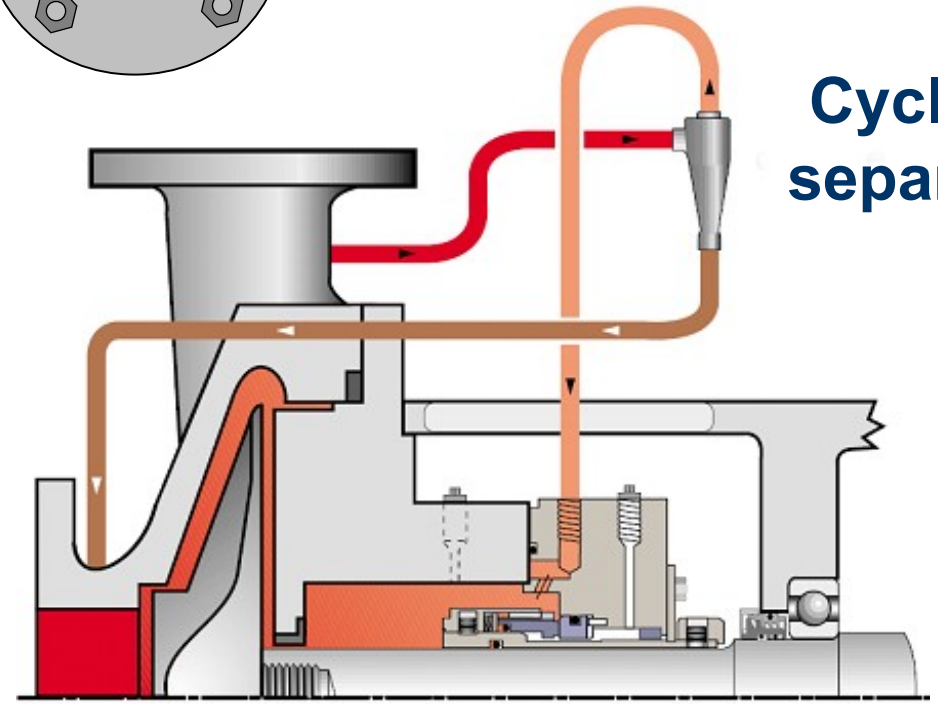
Flush Plans

Plan 31

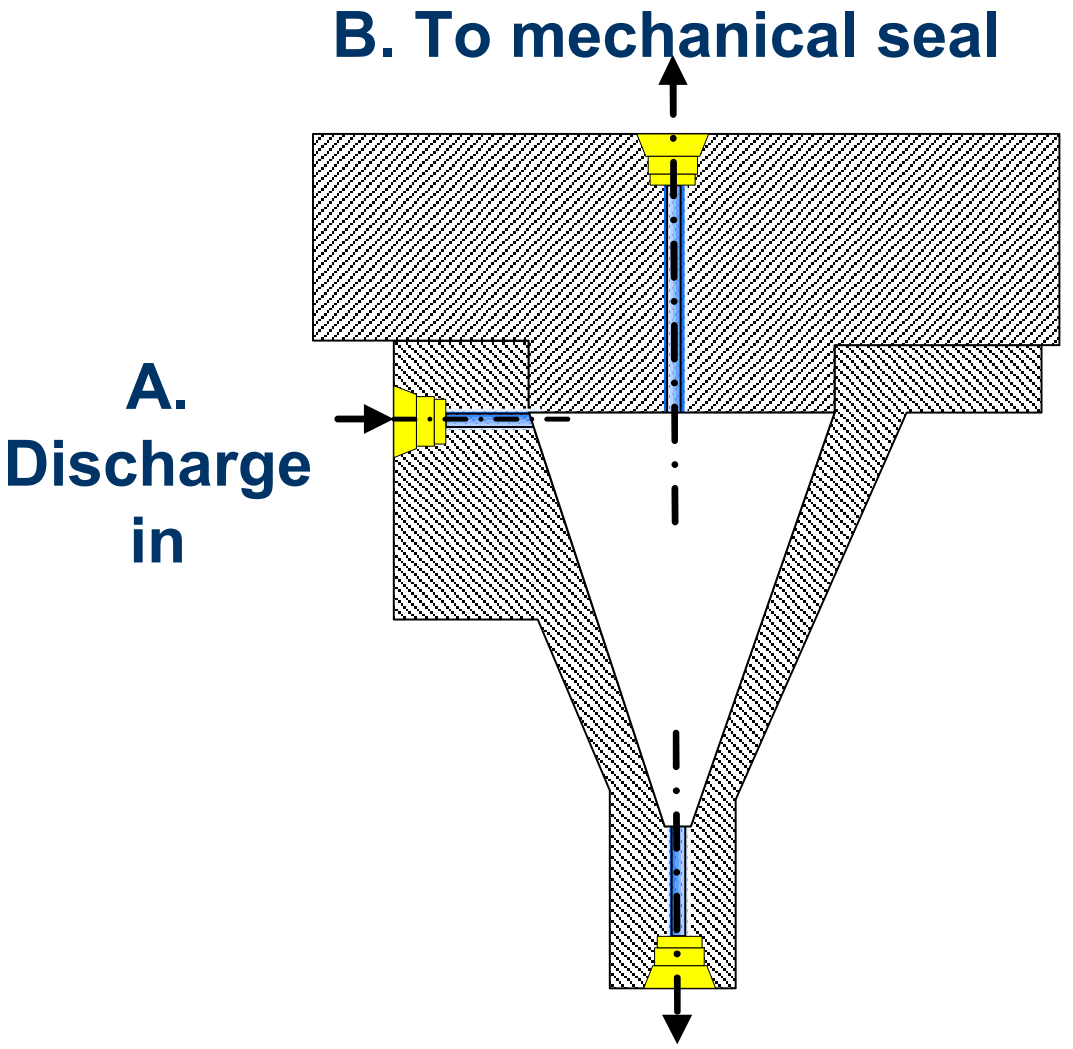


Seal end view

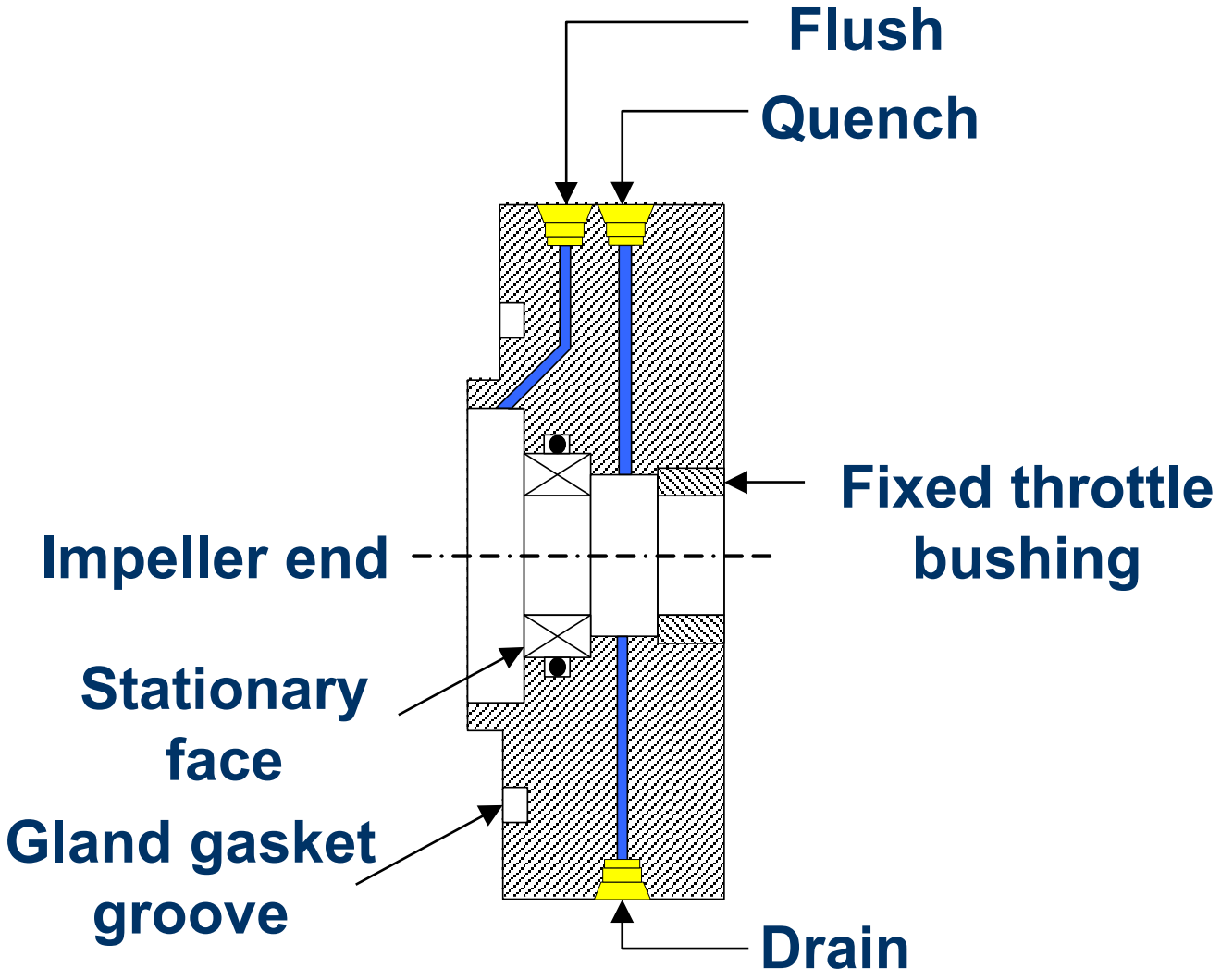
Cyclone separator



Cyclone Separator



Quenching


Flush
Quench
Fixed throttle bushing
Impeller end
Stationary face
Gland gasket groove
Drain

Water Quenching

- When some liquids evaporate, they leave an abrasive crystal deposit behind (caustic, for example)
- Crystals forming on the atmospheric side of the secondary seal stop the secondary and rotating members from moving forward
- This condition is commonly called seal hang-up
- The seal faces are also damaged by abrasives and crystals centrifuged between them
- These problems can be avoided by quenching the seal with water

Other Support Systems

- Cooling
- Pressurization

Pressurization

- Cooling is always preferable to pressurization to suppress vaporization at the seal faces, but cooling is not always feasible
- Often the pressure must be raised in the seal chamber to create the necessary margin between vapor pressure (at seal chamber temperature) and seal chamber pressure

Overview of Seal Failures

- Loss of Face Lubrication
- Bellows cracking
- Corrosion

Overview of Seal Failures

- Corrosion fretting (wear) of the sleeve under the secondary seal
- Coke or crystal build up on the atmosphere side of the seal under the faces

Causes of Seal Failures

- Review Operating Data
- Review Maintenance History

Causes of Seal Failures

- Inspect Mechanical Condition

Causes of Seal Failures

- Inspect Mechanical Seal

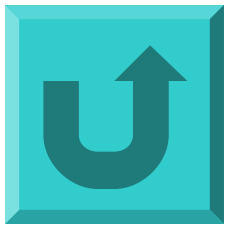
Seal Chamber

- **Causes for a reduction in seal chamber pressure**
- **Causes for a temperature increase at the seal faces**

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS



ALIGNMENT

Major Topics

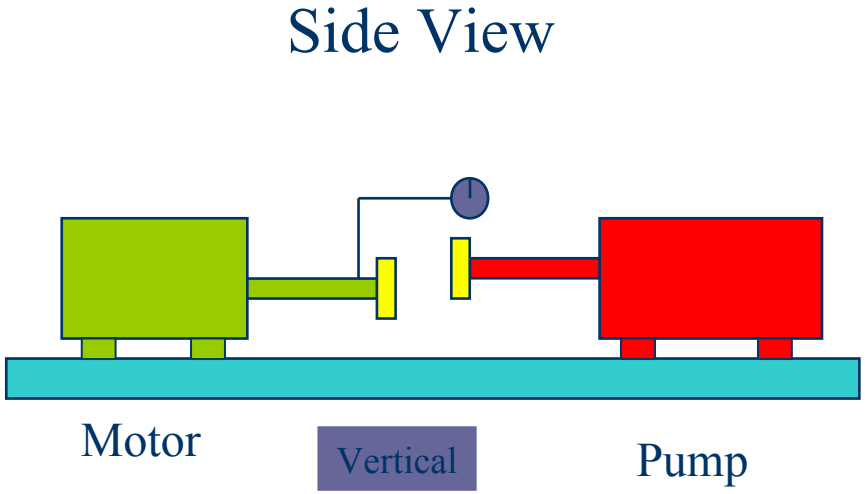
- Alignment Overview
- Methods of Alignment
- Use of the Rotalign® Pro System
- Alignment of Simple Driver/Load Systems
- Soft Foot
- Alignment of Equipment Trains
- Sheave Alignment
- Alignment Troubleshooting
- Thermal Growth

Alignment Overview

- Reasons for Proper Alignment
  - Time
  - Cost
  - Effort

Alignment Terminology

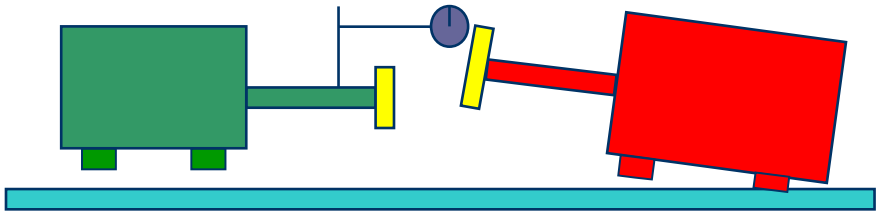
- Offset



Alignment Terminology

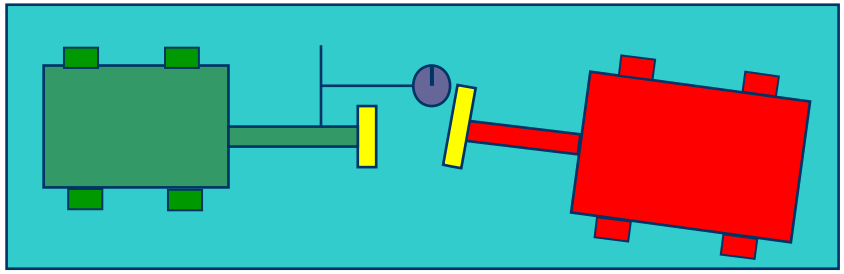
- Angularity

Side View



Motor Vertical Pump

Top View



Motor Horizontal Pump

Methods of Alignment

- Straight Edge



- Dial Indicator



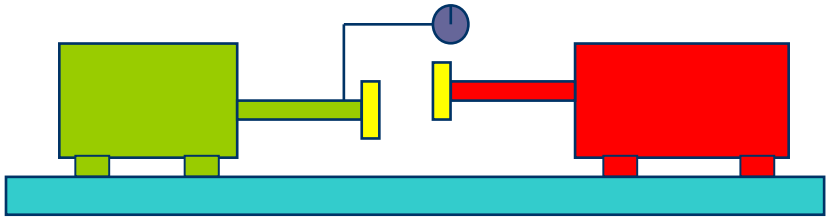
- Laser Alignment



Dial Indicator

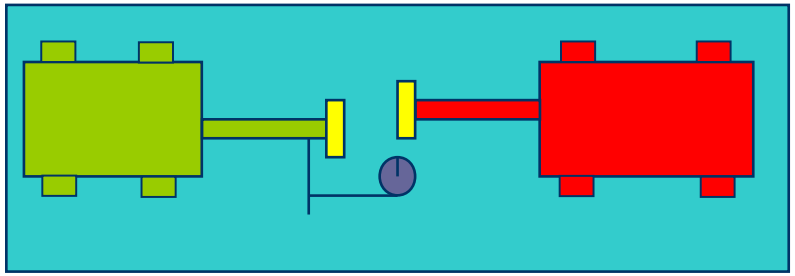
- Rim Alignment

Side View



Motor Vertical Pump

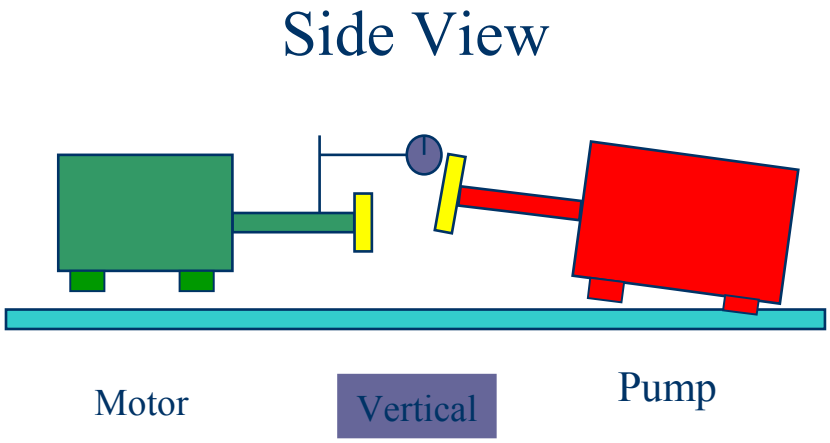
Top View

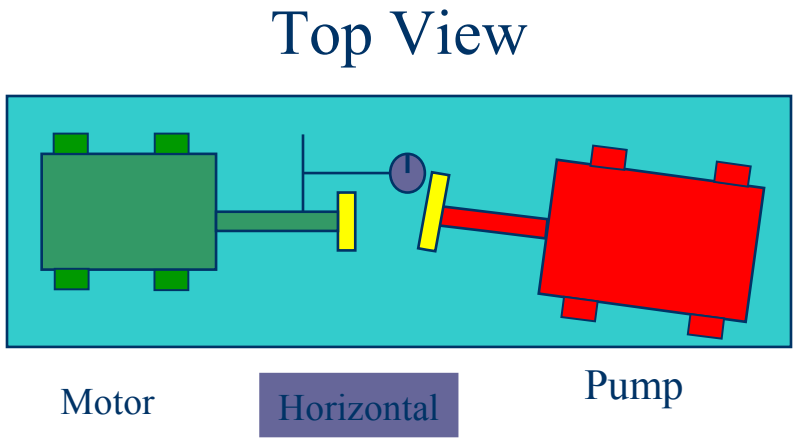


Motor Horizontal Pump

Dial Indicator

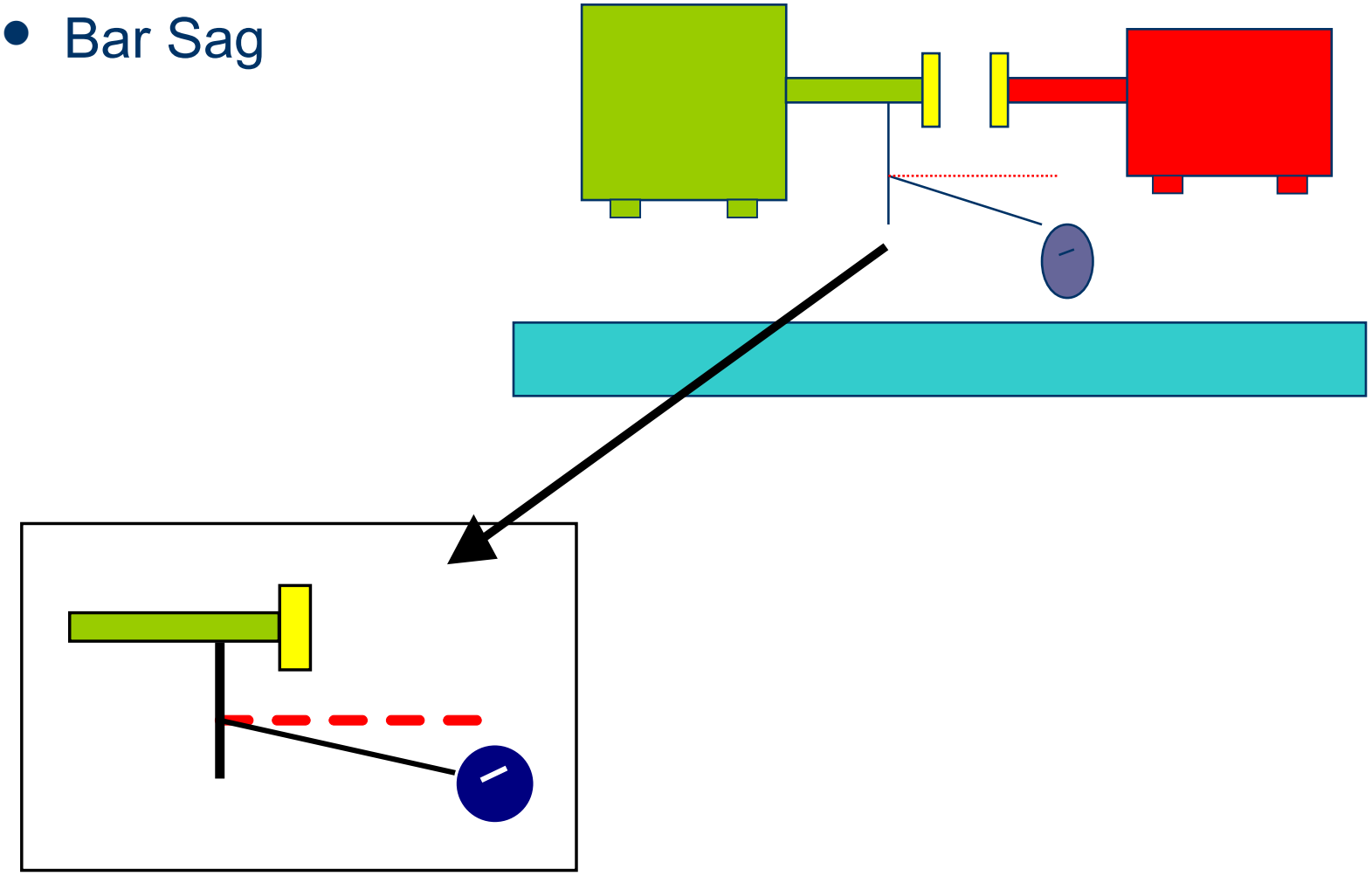
- Face Alignment





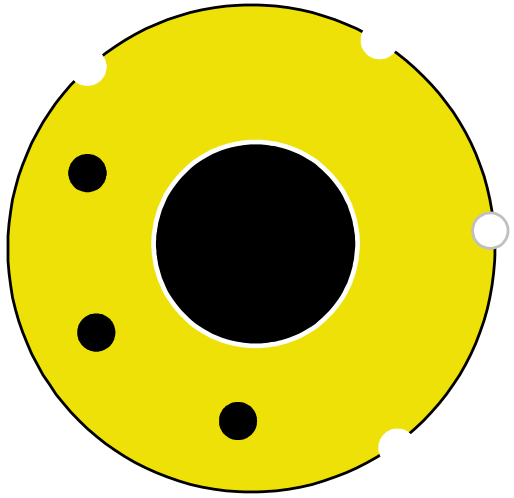
Dial Indicator

- Bar Sag

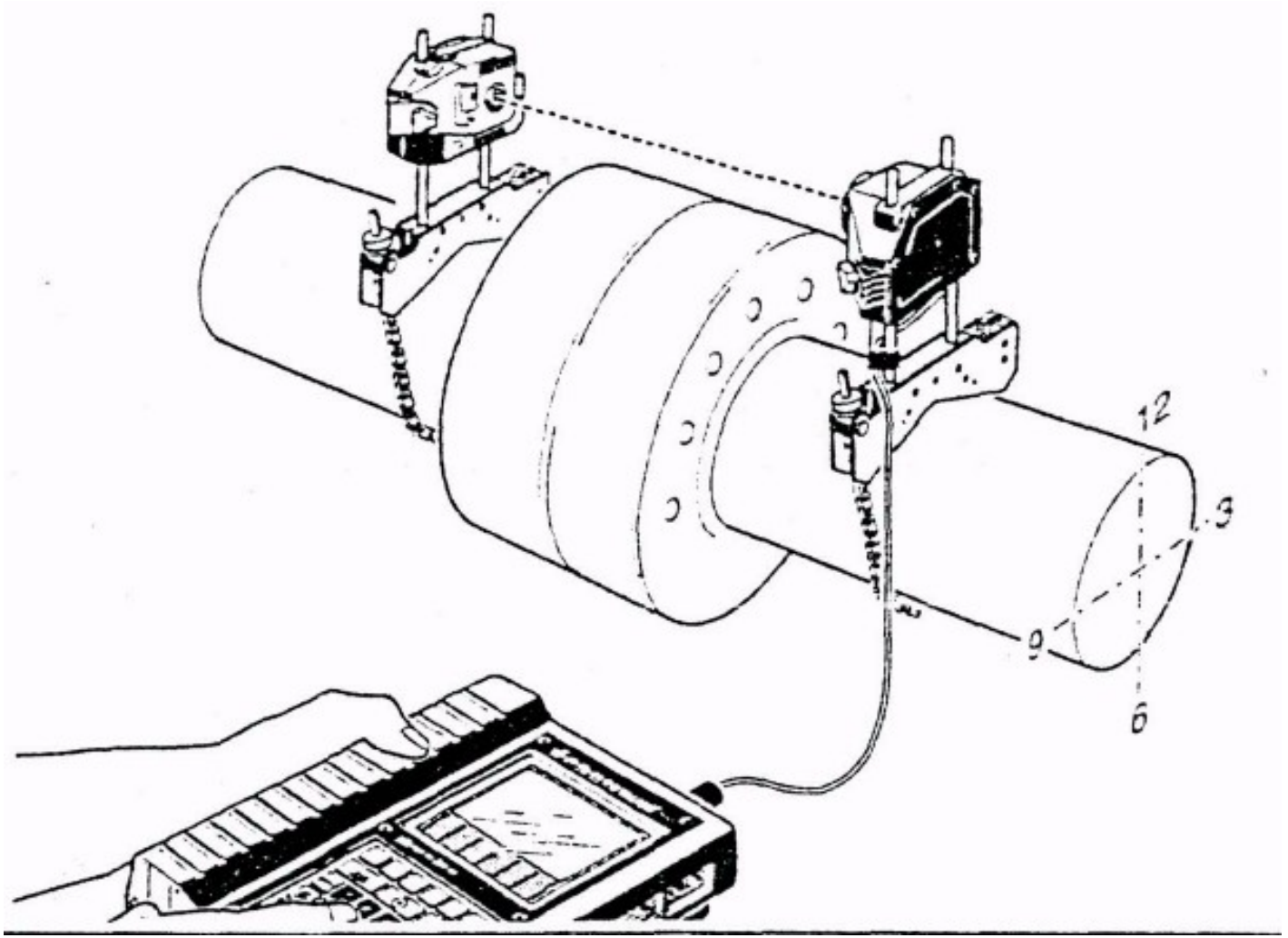


Dial Indicator

- Caution: If the Coupling faces appear as below, it will be necessary to replace



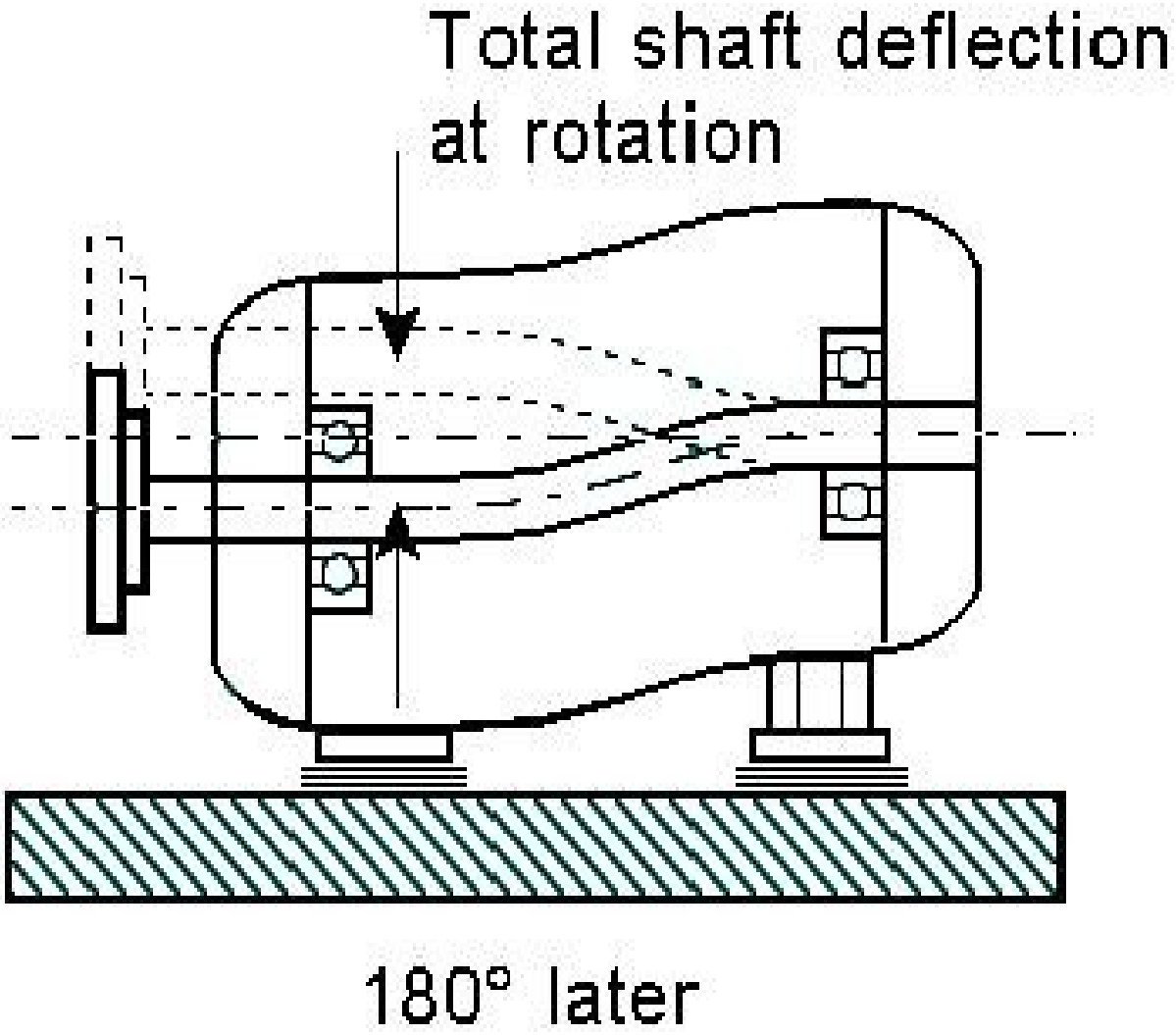
Laser Alignment



Soft Foot

- Any condition where tightening or loosening the bolts of a single foot distorts the machine frame.
- Must be corrected before proper final alignment can be achieved.

Internal Misalignment



Soft Foot

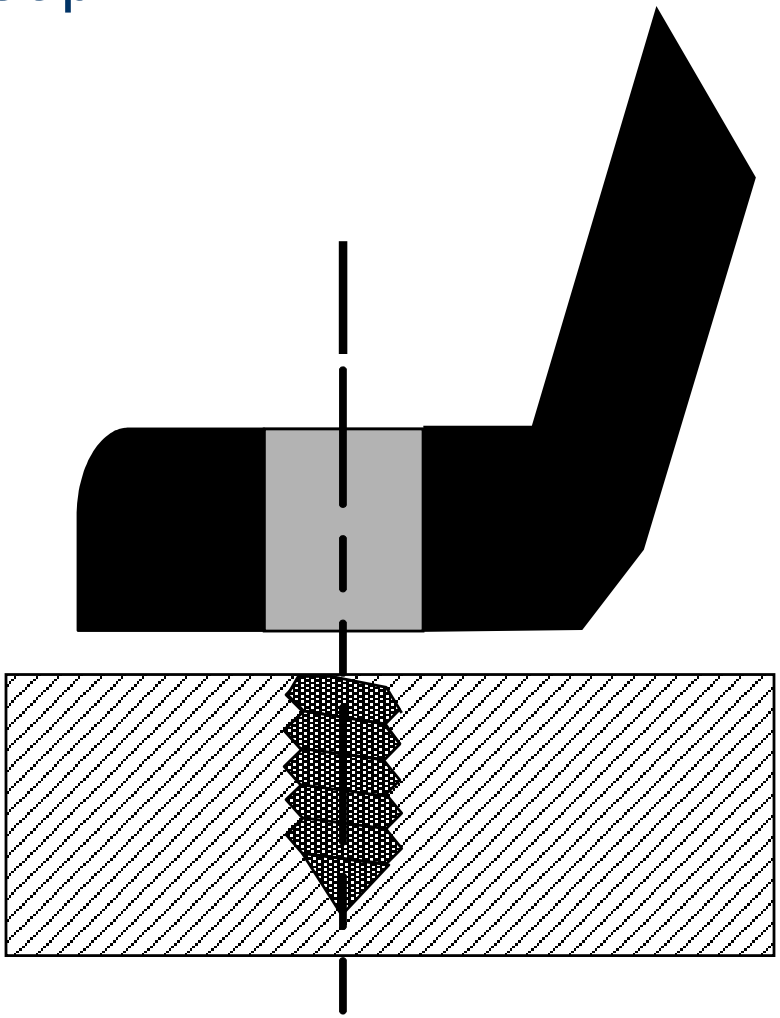
- Causes
  - Bent legs/feet
  - Deformed shims
  - Dirt or debris
  - Strain from attached components
  - Machine frame distortion

Soft Foot

- Effects
  - Vibration
  - Strain and Deformation
  - Bearing Wear/Distortion
  - Premature Equipment Failure

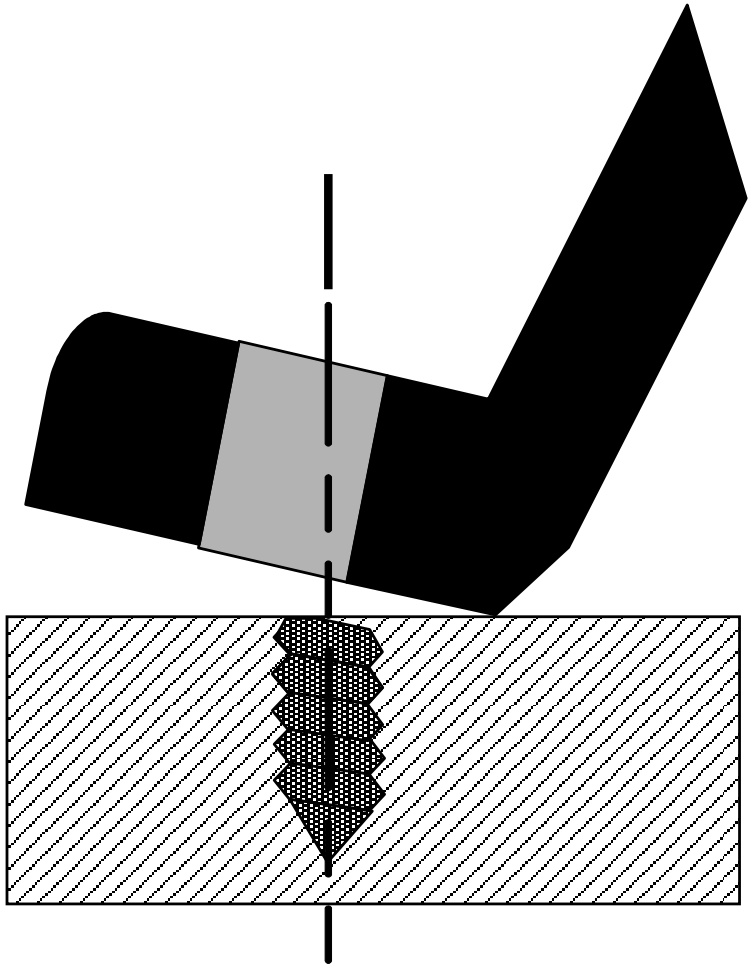
Soft Foot - Types

- Parallel Air Gap



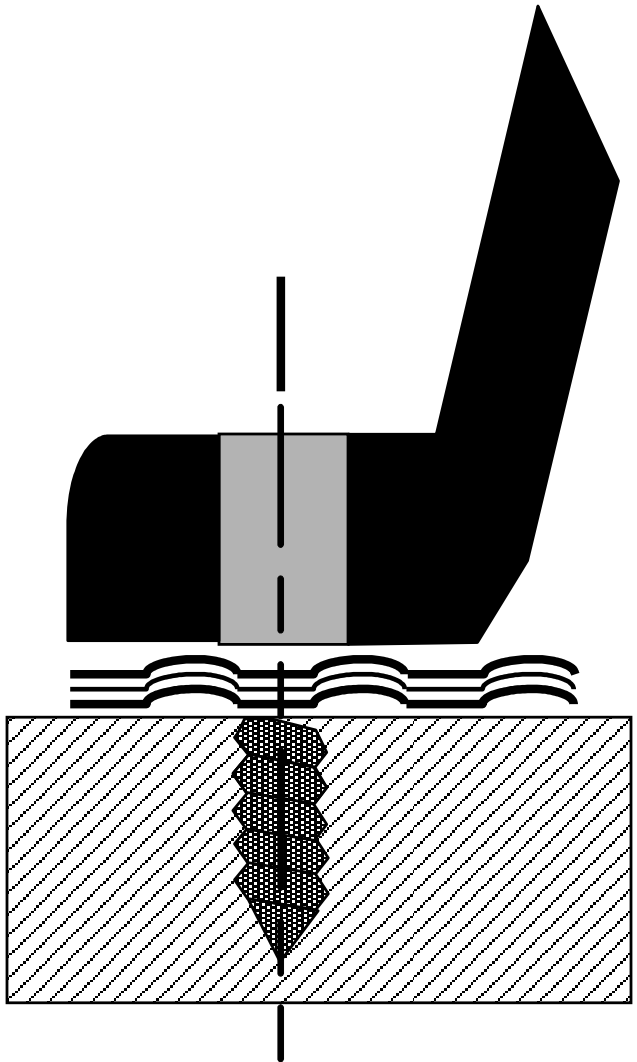
Soft Foot - Types

- Bent



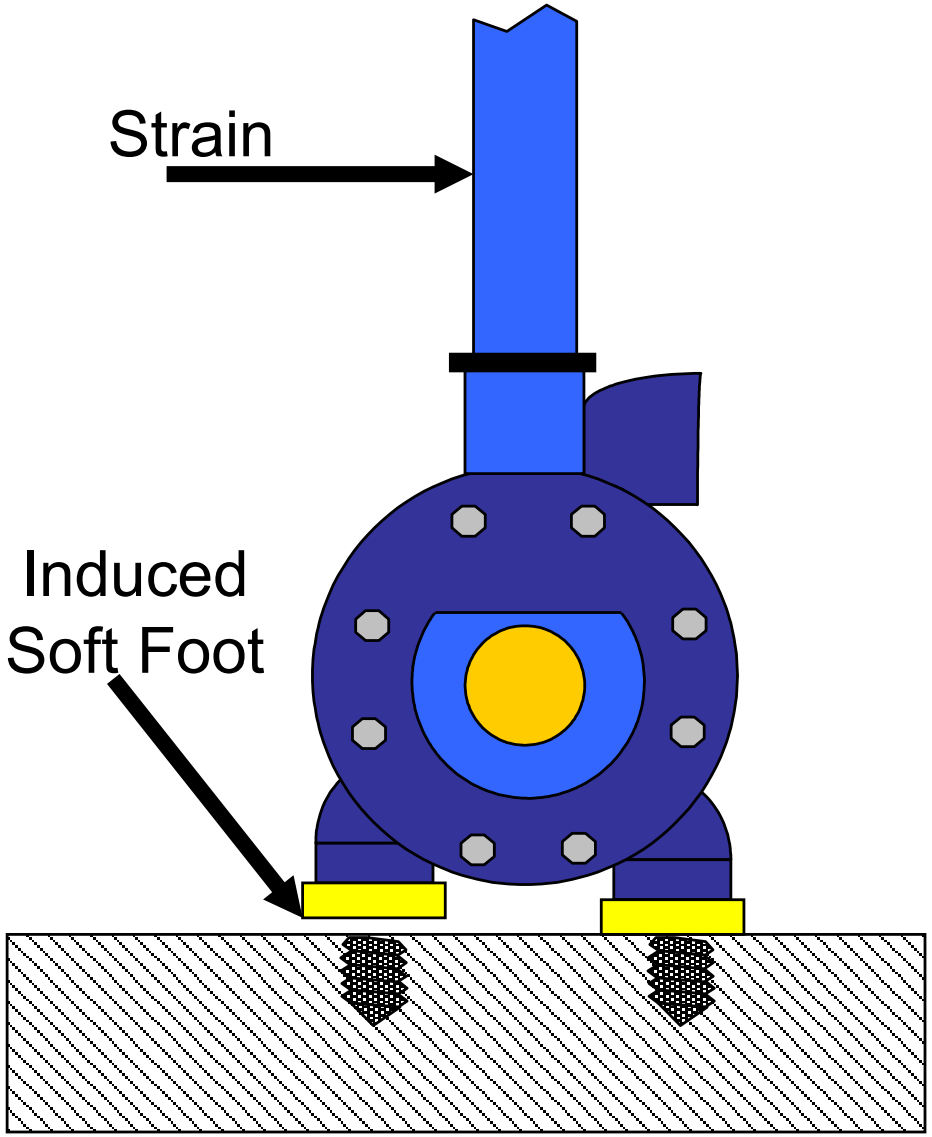
Soft Foot - Types

- Squishy



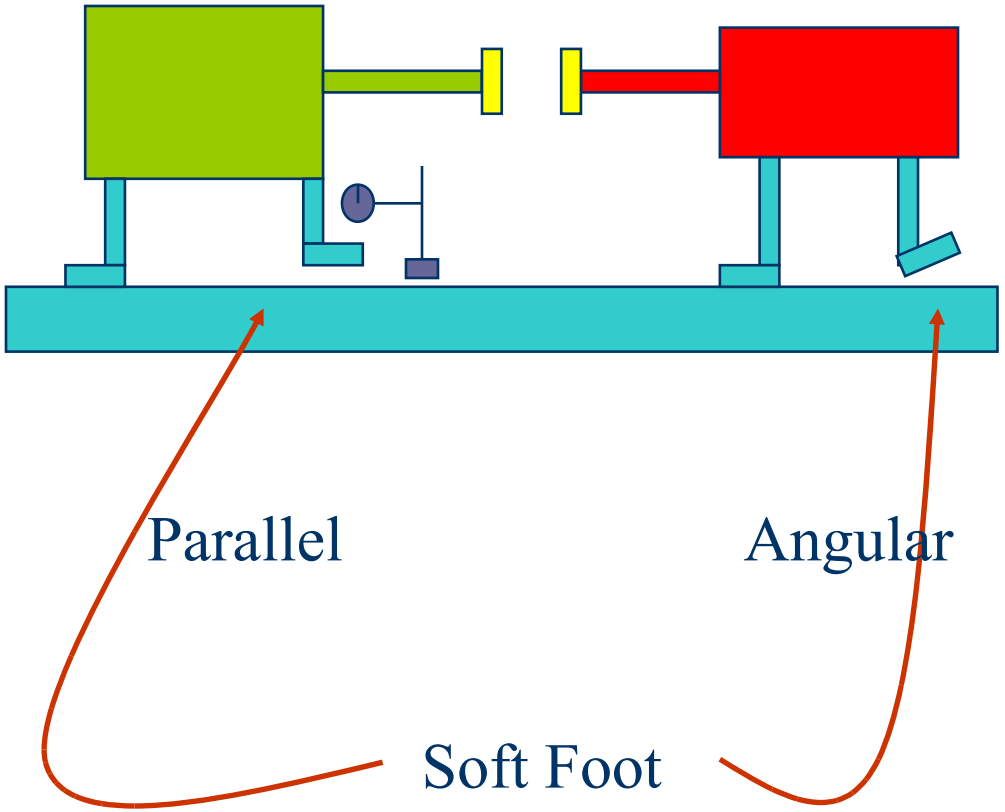
Soft Foot - Types

- Induced



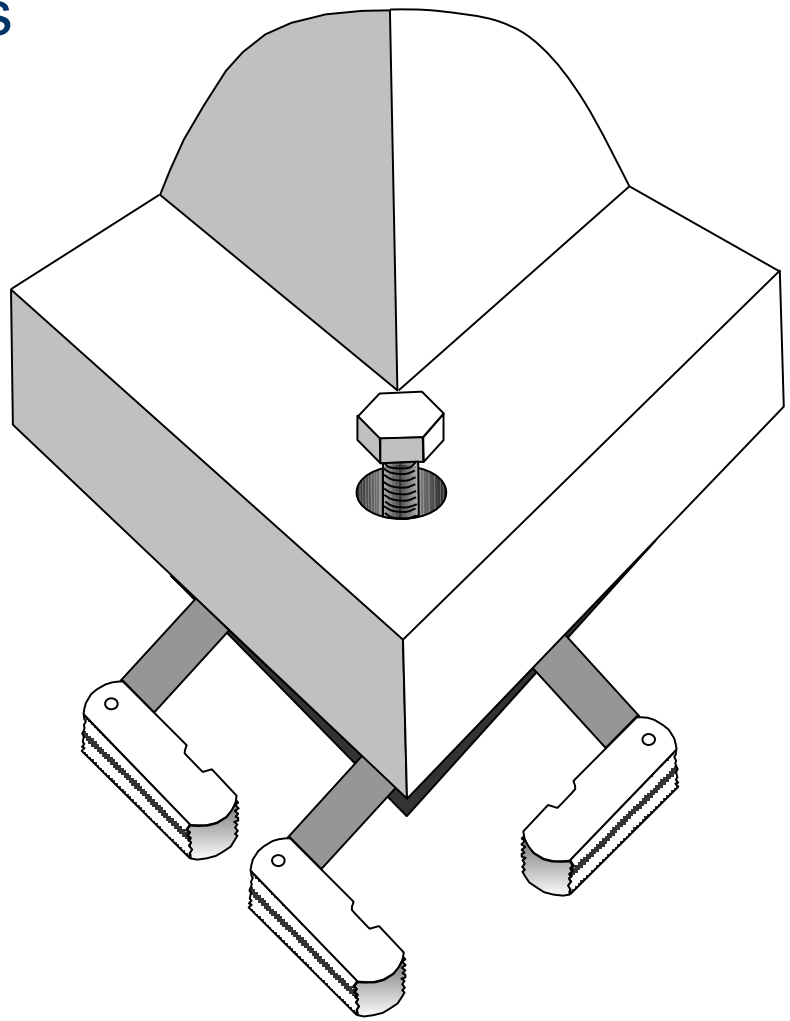
Soft Foot Detection

- Dial Indicator



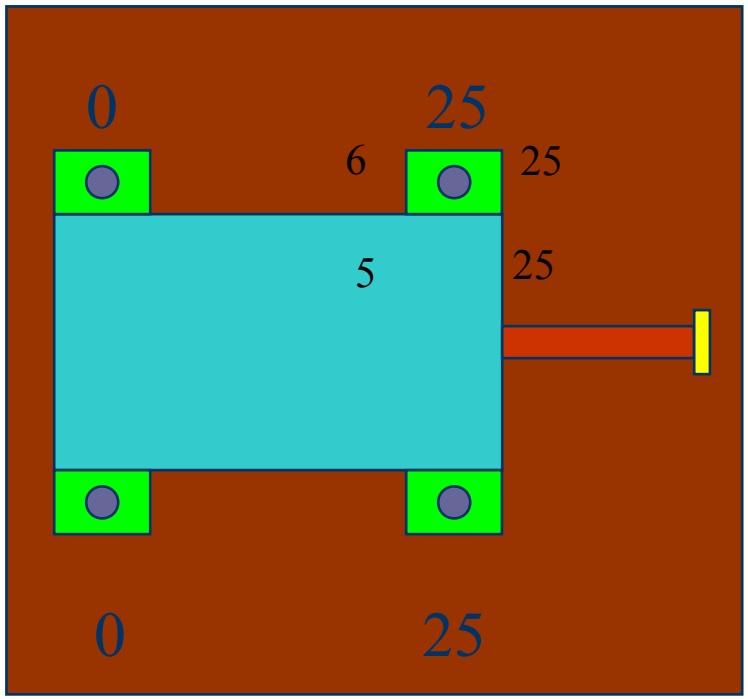
Soft Foot Detection

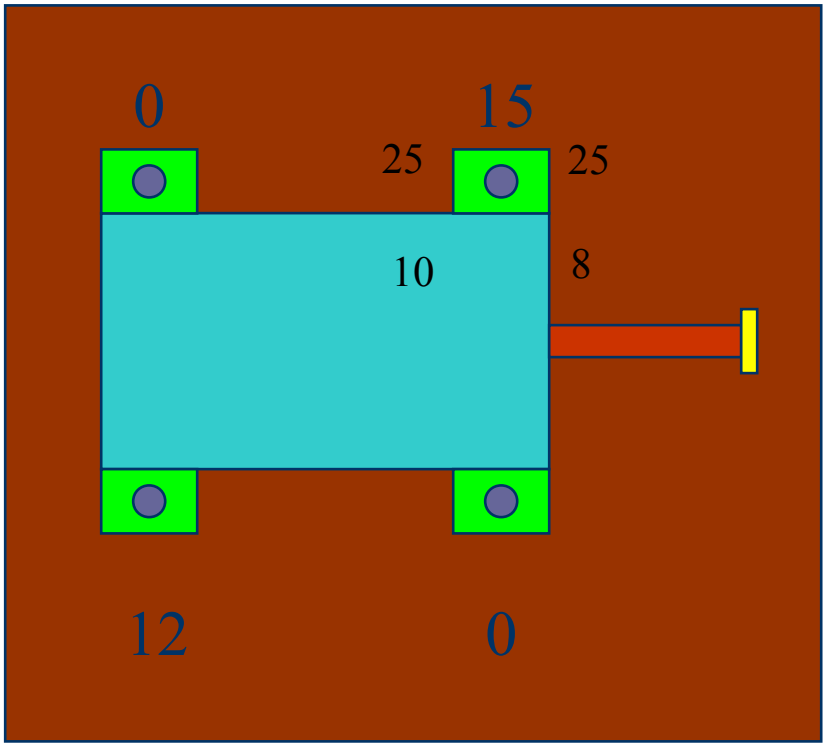
- Feeler Gauges



Soft Foot Detection

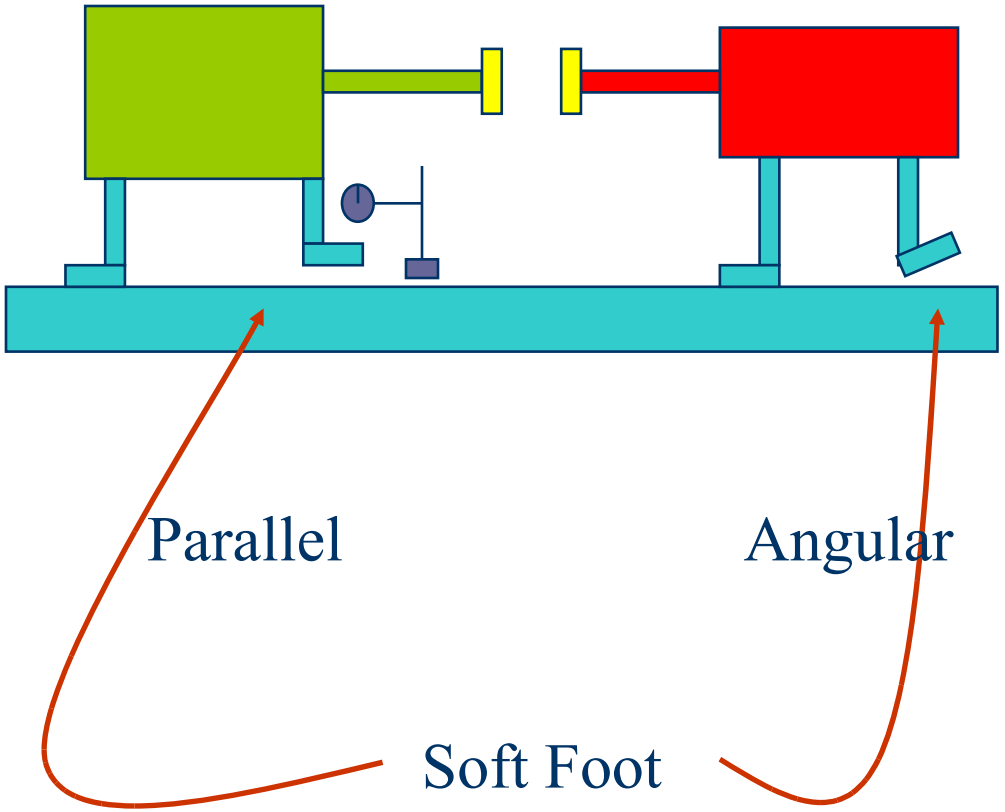
Typical Soft Foot Readings



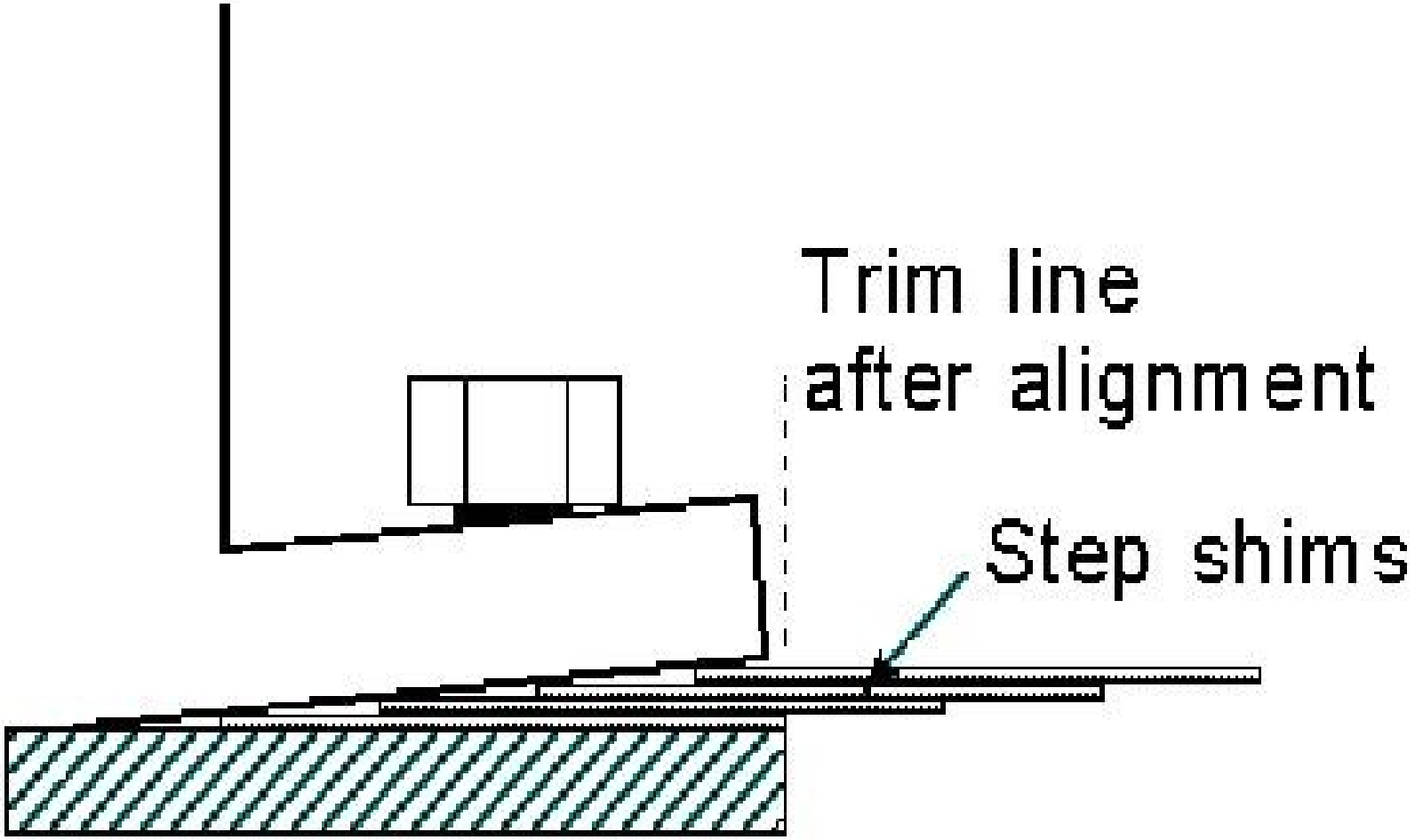


Soft Foot

- Soft Foot Correction



Step Shimming



Sheave Alignment



Alignment Troubleshooting

- Shaft Deflection
  - Cause:
    - Weight of Coupling
    - Shaft Run out
  - Test:
    - Use a dial indicator to measure deflection during 180 degrees of rotation
- Caution:
  - Do Not forget about Bar Sag when performing this test
  - It is better to use two indicators, reverse alignment

Alignment Troubleshooting

- Solution:
  - Replace the coupling with another type of equal Speed (RPM) and Power (HP) rating that is of a lighter weight
  - Remove the coupling and hubs and align machines using just the shafts

Alignment Troubleshooting

- Solution:
  - Replace the machine shaft if necessary
  - Consult the equipment manufacturer

Alignment Troubleshooting

- Shaft Deflection (Continued)
  - Affect on Alignment
    - Alignment readings will be different with and without the coupling
    - No indication what the alignment will be while the machine is in operation

Alignment Troubleshooting

- Bolt Bound
  - Affect on Alignment
    - Motor will not move far enough to bring the motor and pump back into alignment

Alignment Troubleshooting

- Bolt Bound
  - The pump and motor were not aligned properly before the skid was grouted
  - Something, such as a pipe, has moved from its original position
  - The motor or pump is not the same as the original

Alignment Troubleshooting

- Bolt Bound
  - Bolts in improper position
    - Re-position machine on Skid
  - Pipe Strain
    - Correct Piping mis-alignment
  - Wrong Motor / Pump
    - Replace Incorrect Part

Alignment Troubleshooting

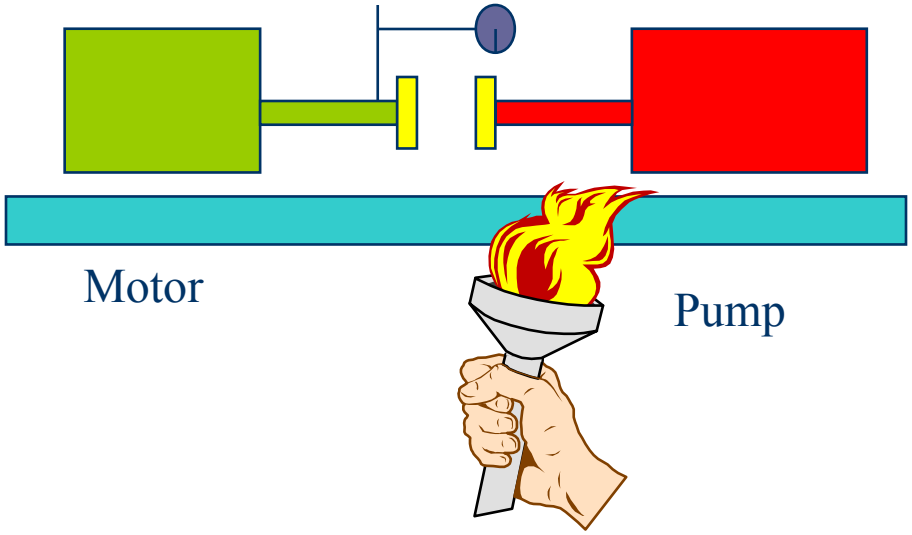
- Coupling Lateral Clearance
  - Cause:
    - Wrong Coupling
    - Improper machine position
    - Excessive Axial Shaft movement

Alignment Troubleshooting

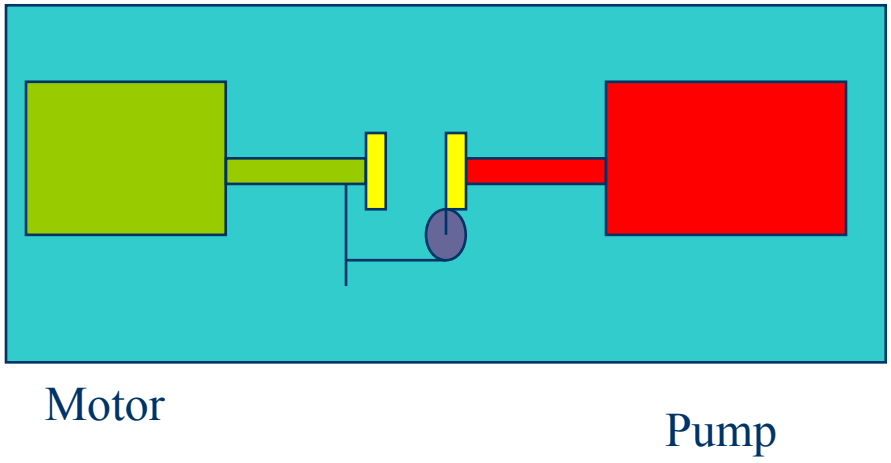
- Solution:
  - Loosen the Shaft grub screws and move the coupling flange(s) as necessary to establish the correct clearance
  - If excessive shaft axial play was present, repair the cause for this play.
  - Consult the equipment manufacturer

Thermal Growth

Side View



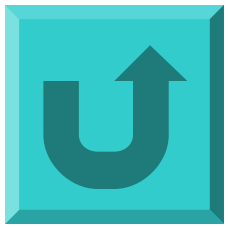
Top View



Summary

- Review Question and Answer Session

CLICK TO RETURN TO TOPICS



VIBRATION ANALYSIS

Course Objectives

- Define the need for analysis
- Define the cause and effects of equipment vibration
- State how vibration is measured

Introduction

- Method to detect and control the mechanical condition of rotating equipment.

What is vibration?

- Motion of a machine from rest.
- Method to detect and control the mechanical condition of rotating equipment.
- Vibration amplitude.
- Vibration facts.

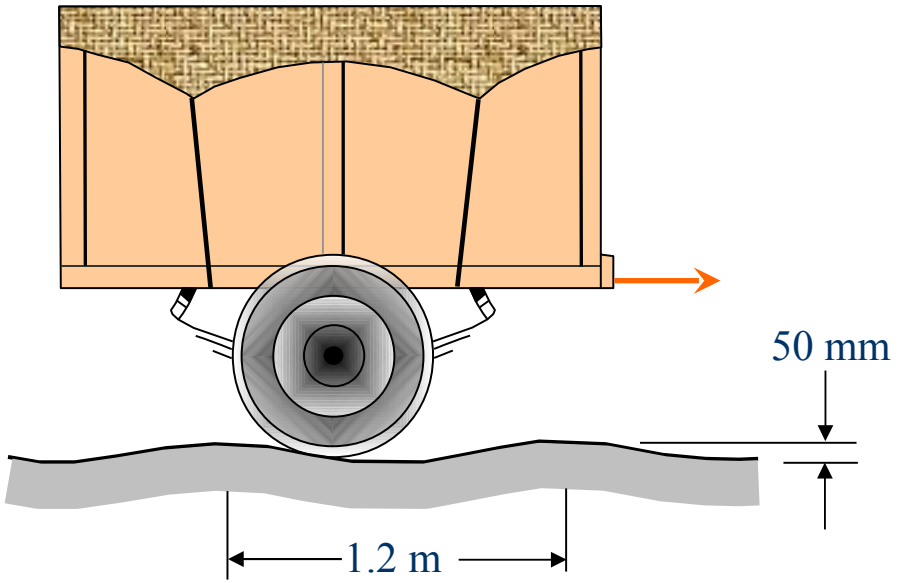
Vibration

Vibration is the mechanical oscillation or motion about a reference point of equilibrium

- Violin string
- Rotating machinery

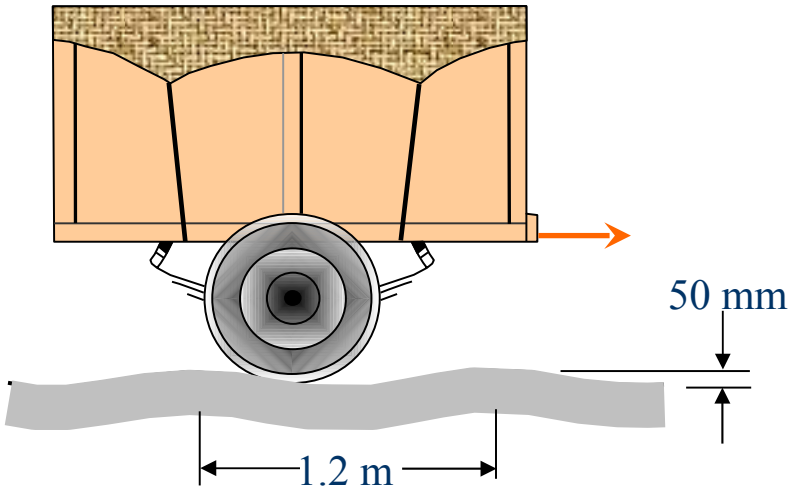
Vibration

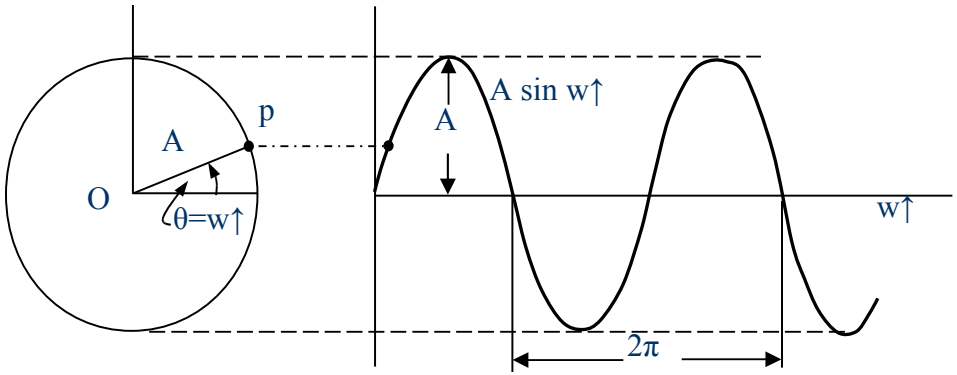
- Vibratory system includes:
  - Spring or Elasticity
  - Mass or Inertia
  - External Force



Oscillatory Motion

External force causes the system to oscillate as the spring stores and releases energy





Vibration

Vibrations may:

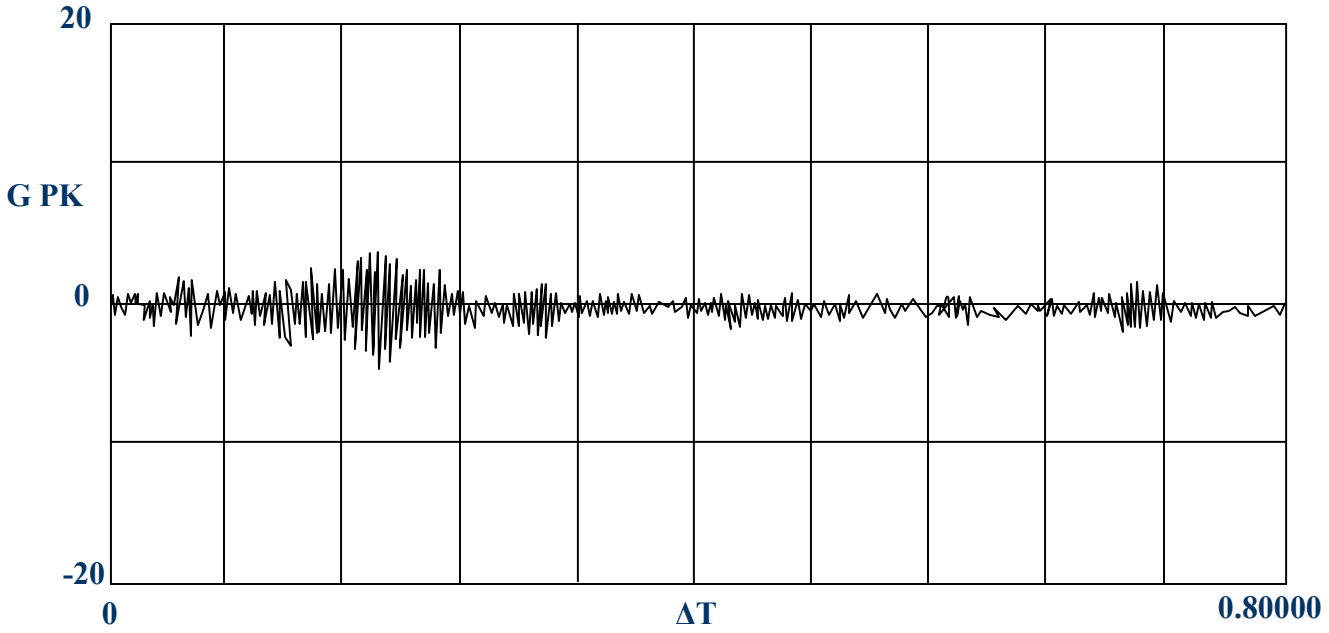
- Repeat (reciprocating machinery)
- Occur at specific times (impact)

Repetitive Vibrations

- The period of repetition may be measured as frequency
- Most equipment vibrations occur between 10 and 2000Hz

Normal Vibrations

- Machines will have a characteristic vibration signature during normal operation



Resonance

- The resonance combines with the natural frequency of the system resulting in an amplified vibration. This can lead to destruction.
  - Example: Bridge resonance

Effects of Machine Vibration

- Efficiency loss
- Wear acceleration
- Machine failure
- Personnel injury

Source of Equipment Vibration

- Normal motion of machine operation
- Unbalanced parts
- Worn bearings
- Loose mounting
- External impact

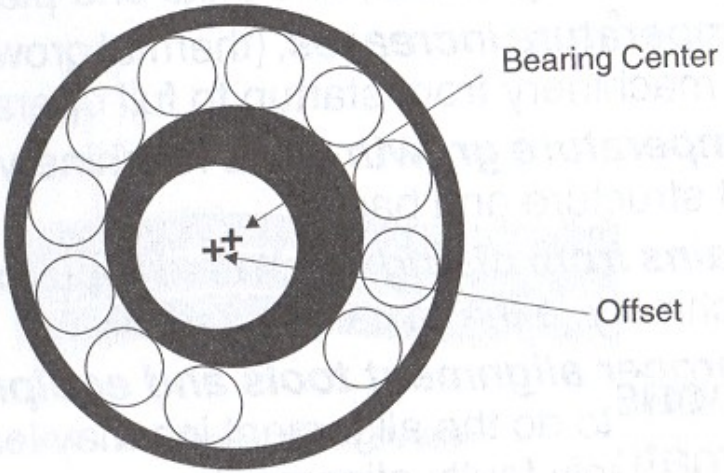
Causes of Unbalance

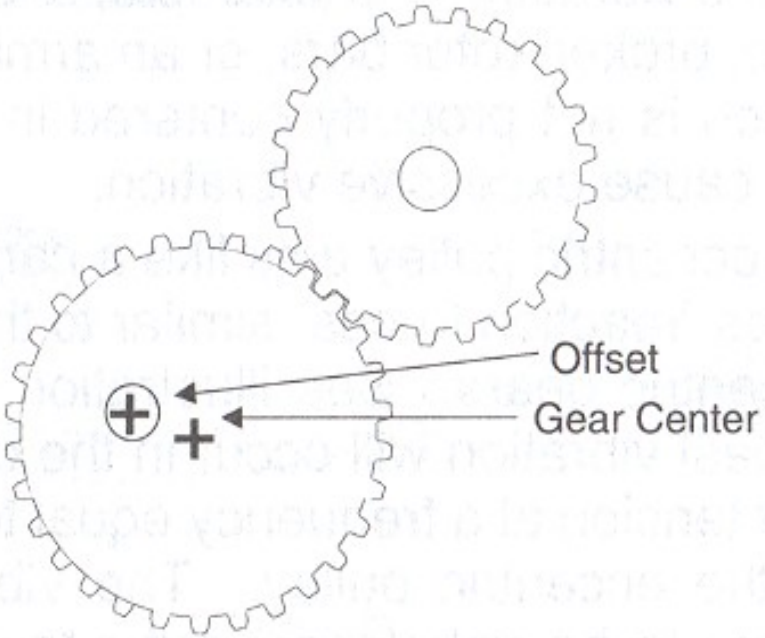
- Deposit and Build-Up
- Corrosion and Wear
- Eccentricity
- Keys And Keyways
- Clearance Tolerances

Misalignment

- Parallel Offset Misalignment
- Angular Misalignment
- Combination
- Tolerances

Eccentricity





Vibration From:

- Bent Shafts
- Faulty Anti-Friction Bearings
- Faulty Journal Bearings
- Belt Drive Problems
- Bad Gears

Vibration Sensors

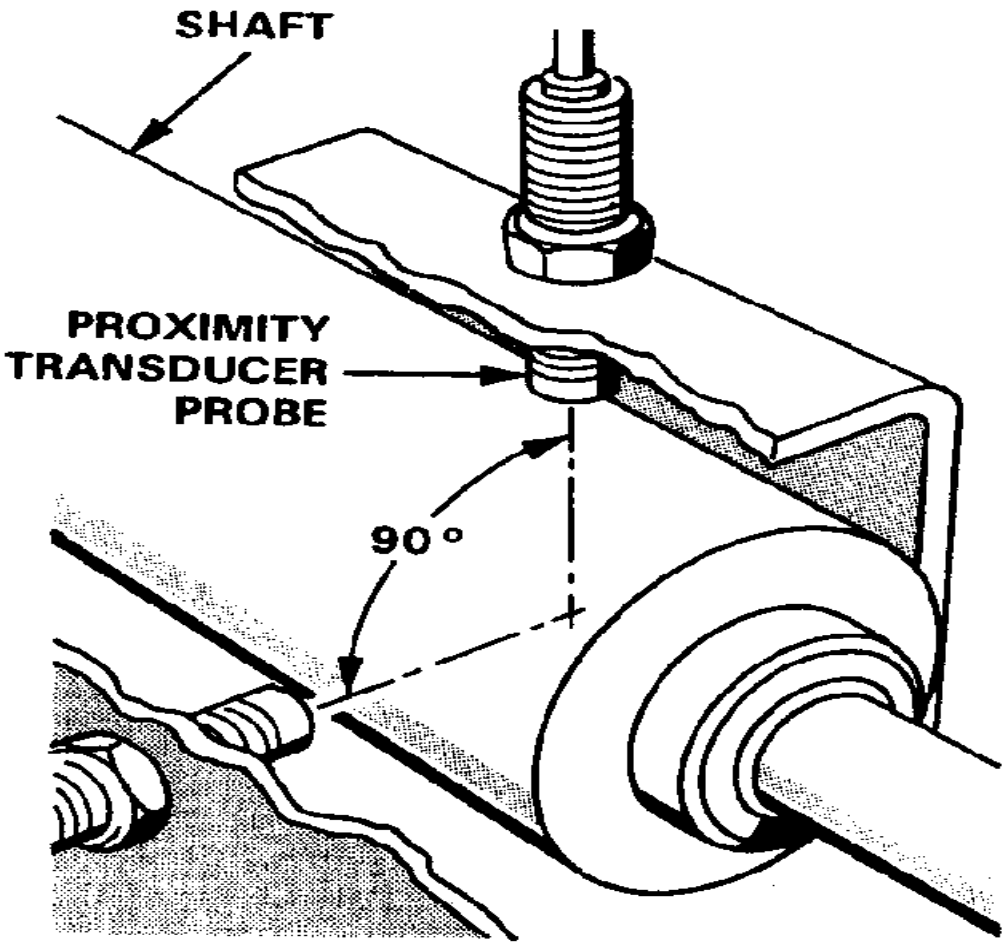
- Sensors convert vibrations into electrical signals
- Two types of sensors
  - Accelerometers
  - Proximity

Velocity Transducer

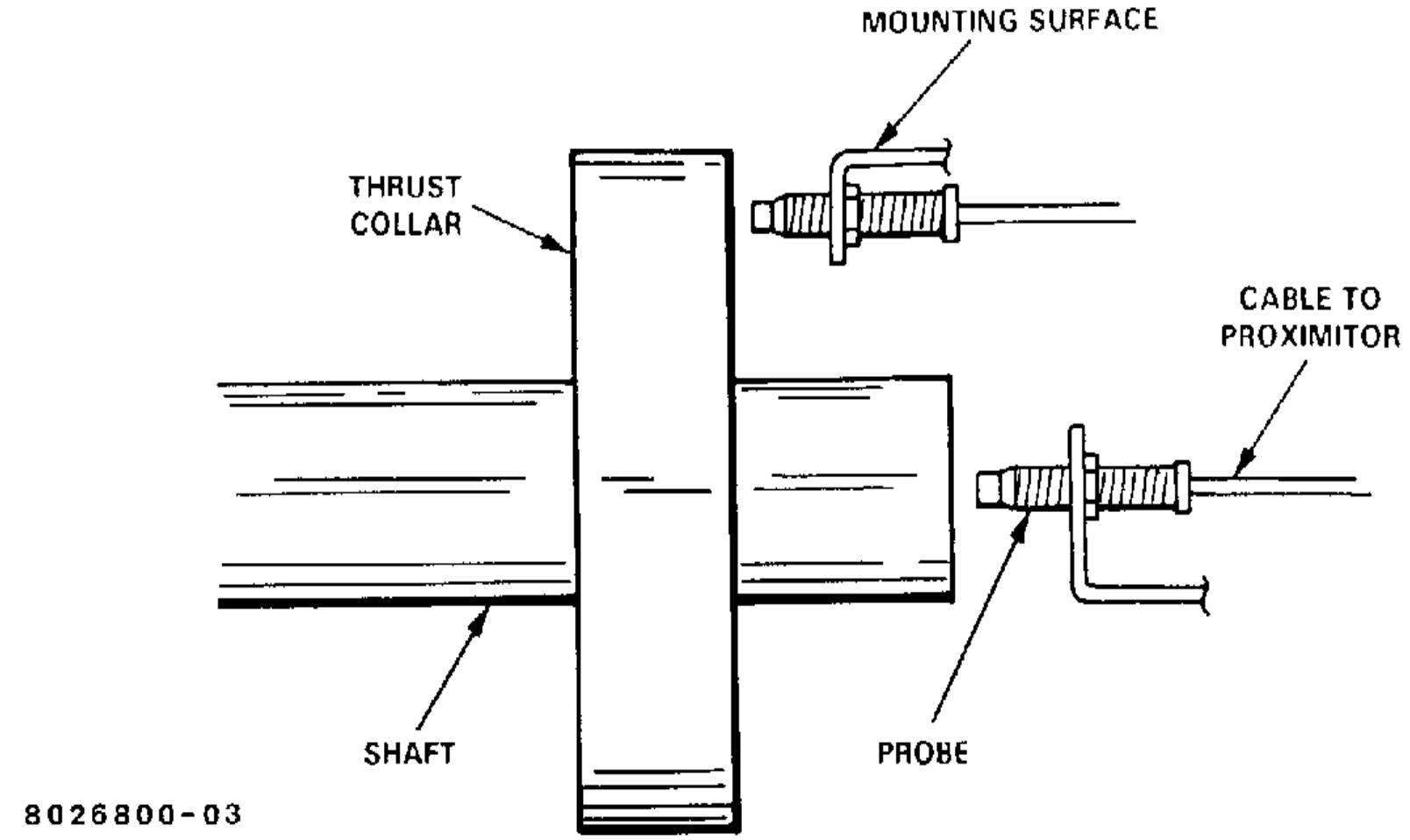




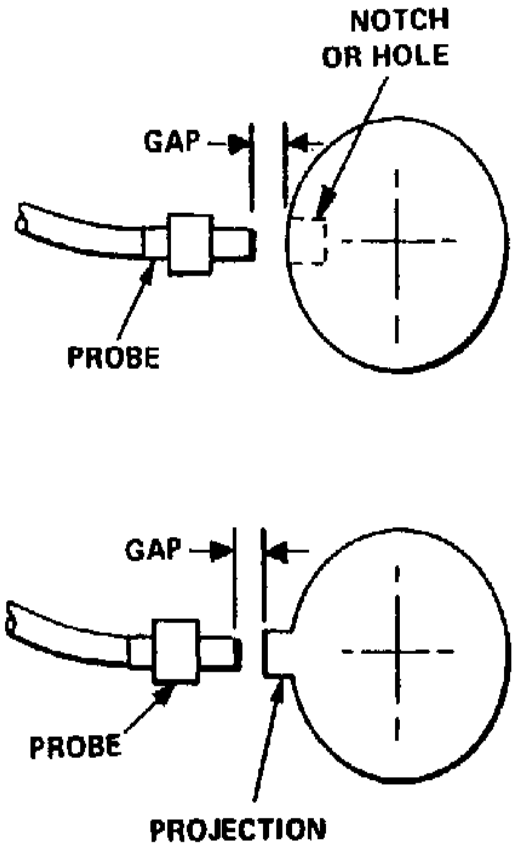
Radial Probe Mounting

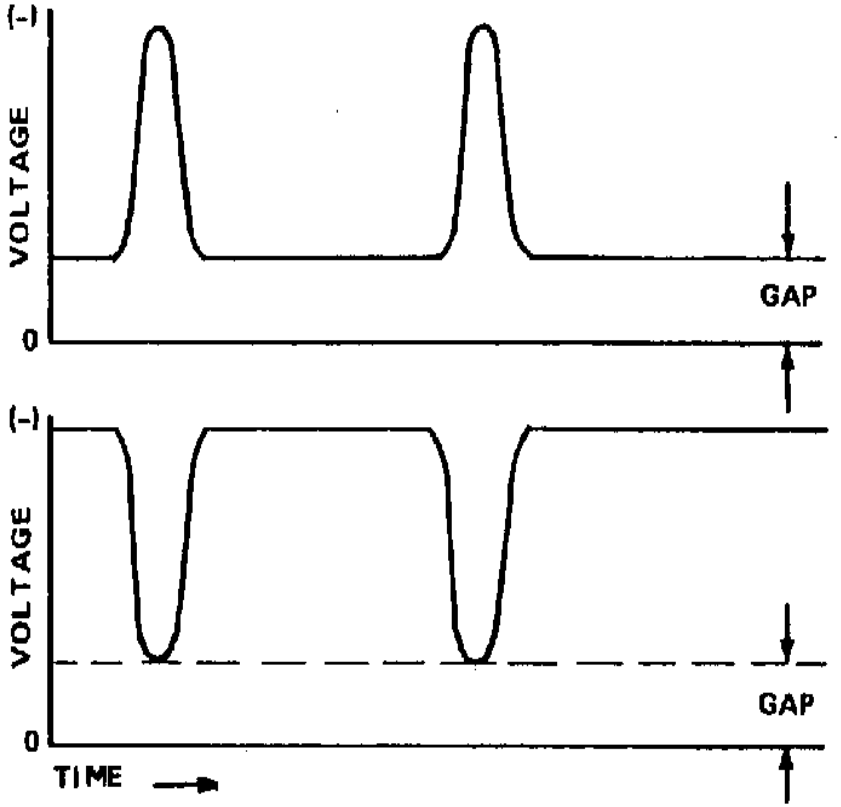


Axial Position



Key Phasor

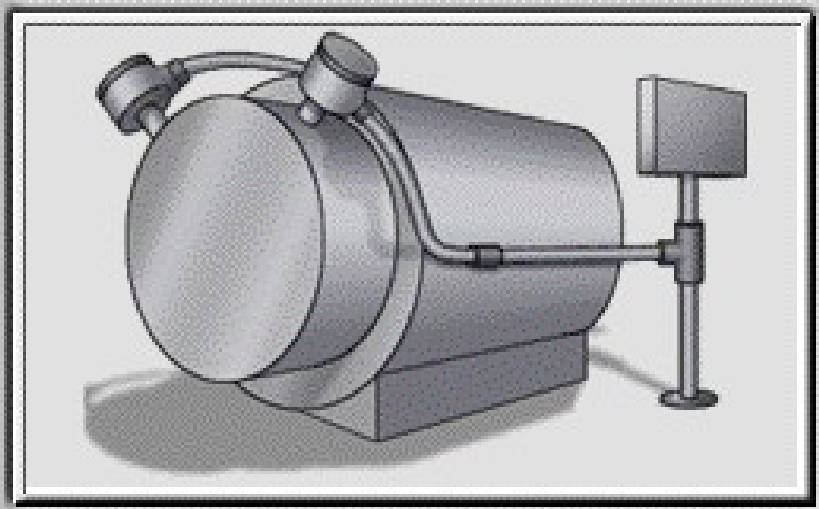

NOTCH
OR HOLE
GAP
PROBE
GAP
PROBE
PROJECTION

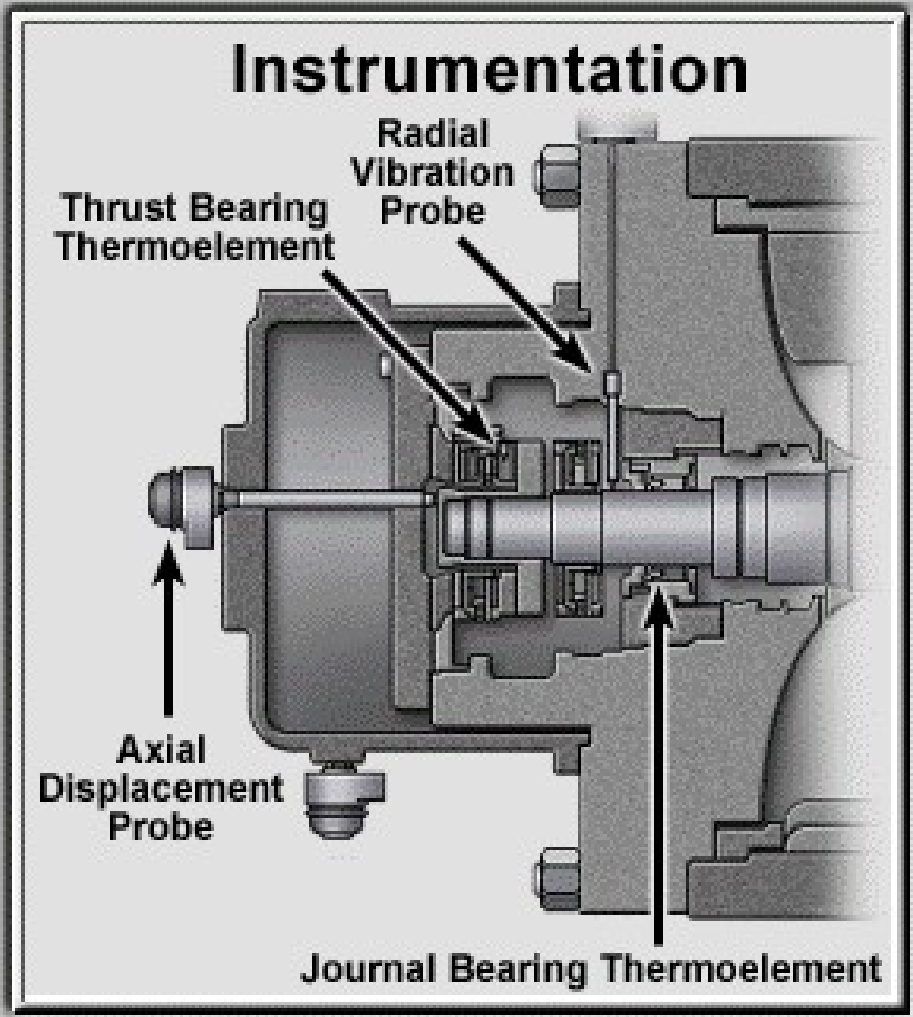

(-)
VOLTAGE
0
GAP
(-)
VOLTAGE
0
GAP
TIME

Proximity Probes

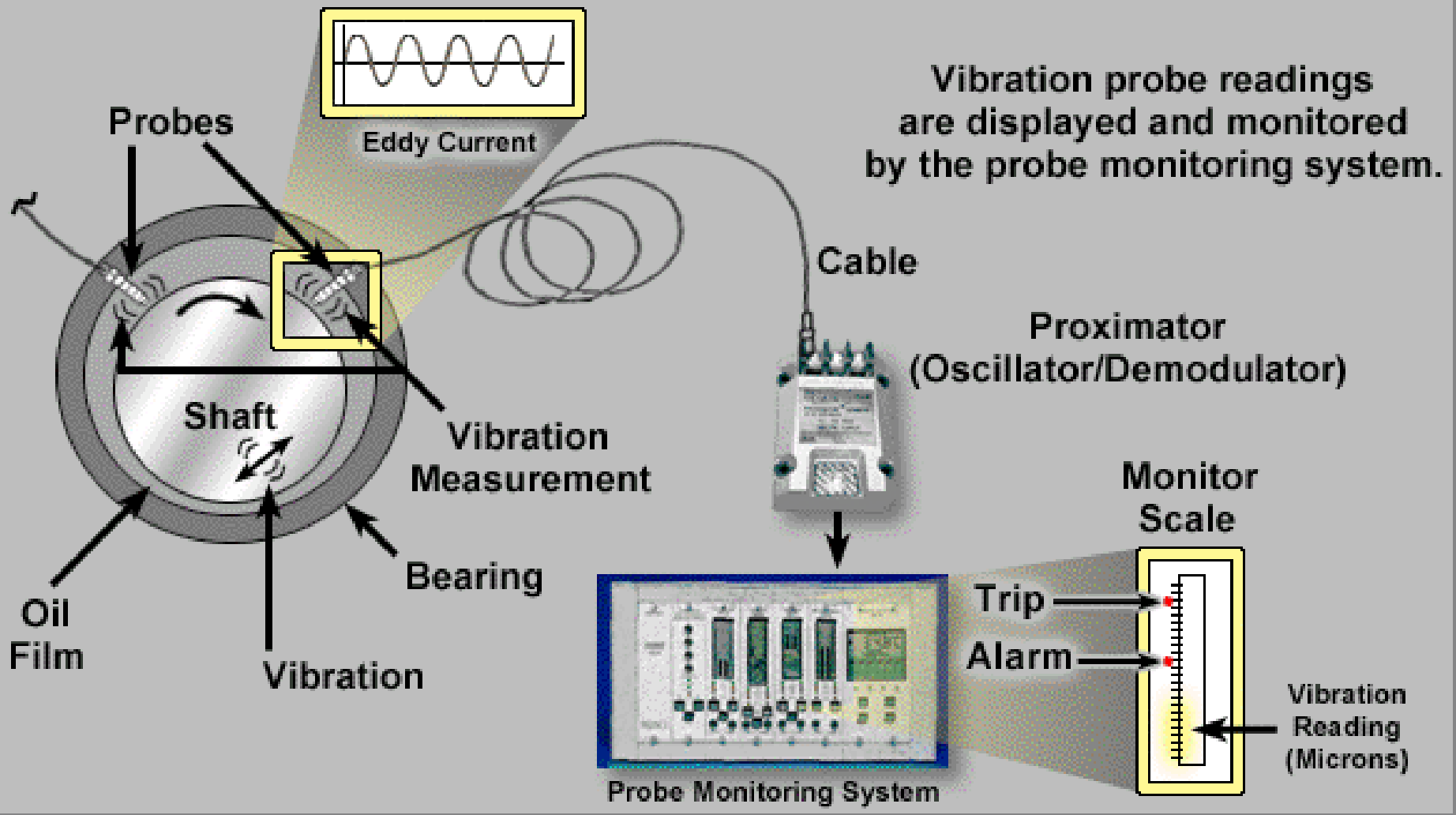
Instrumentation



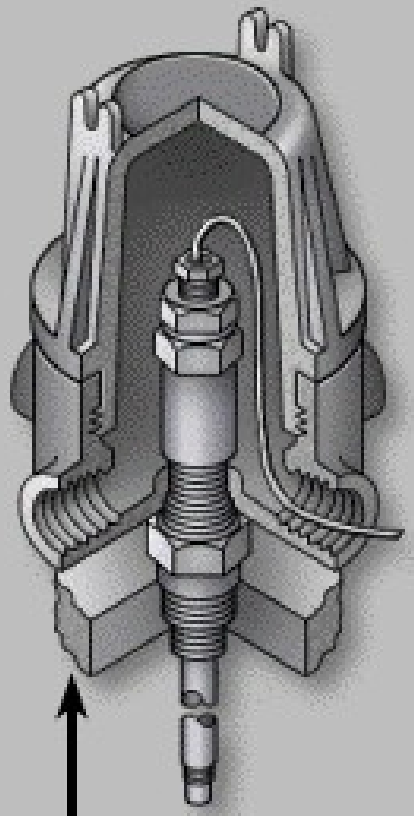


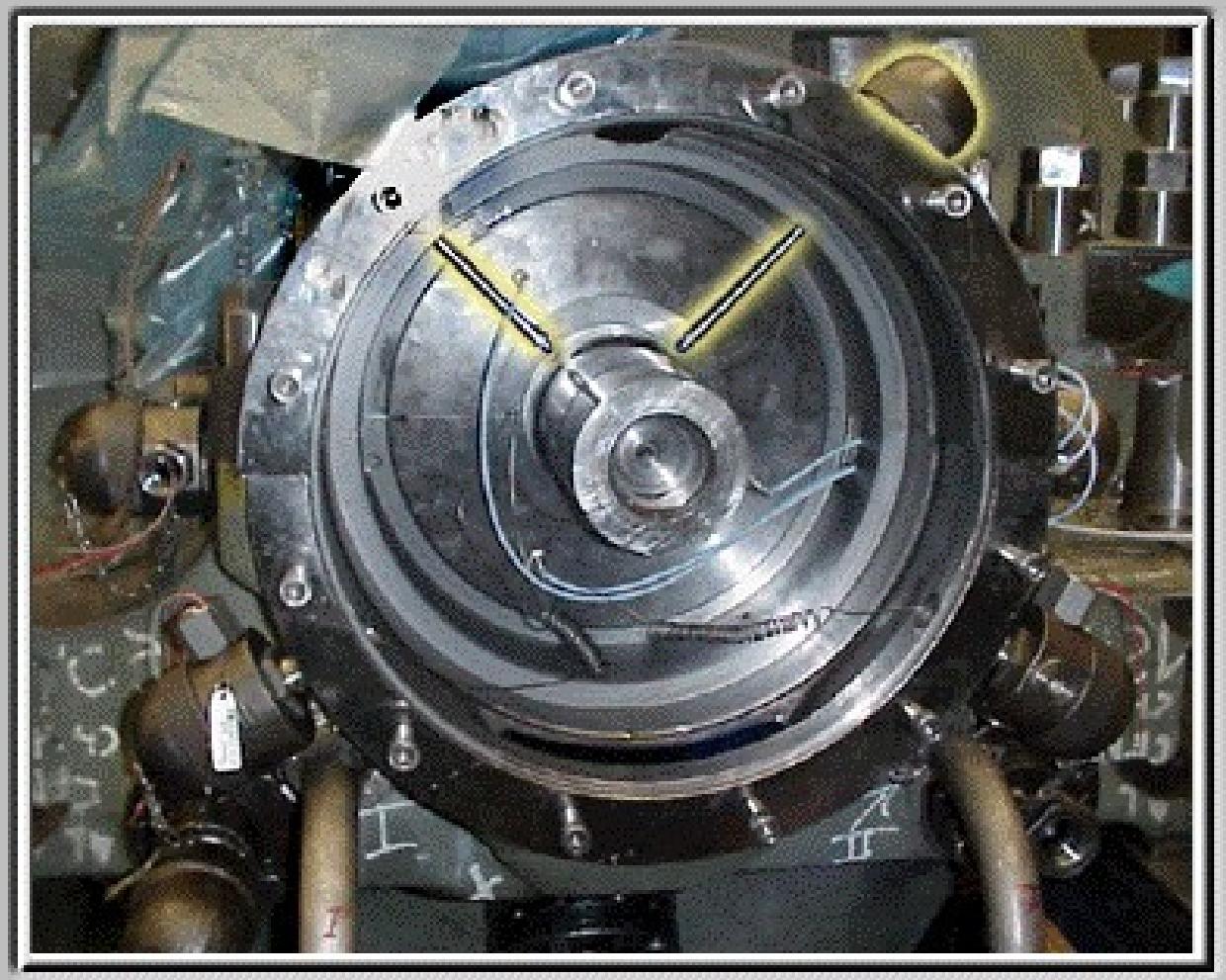


Vibration Monitoring System



Proximity Probes

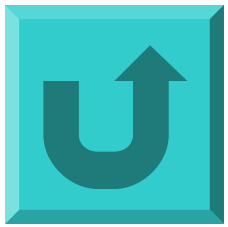




Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS



THERMAL ANALYSIS

Introduction

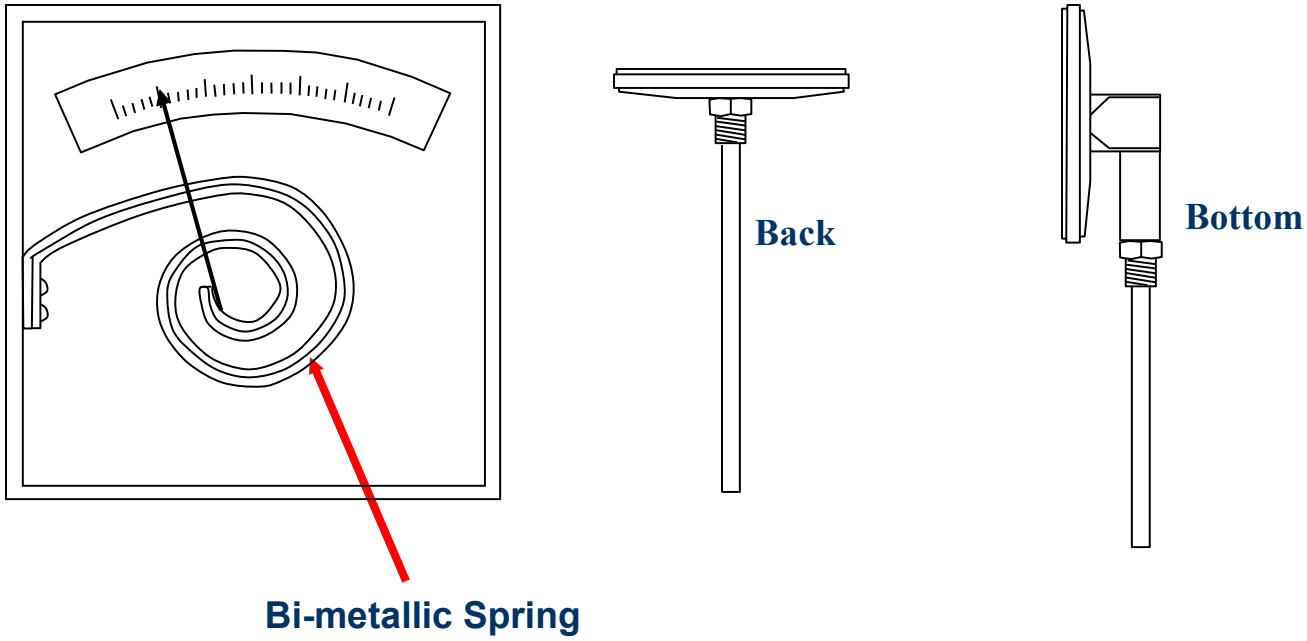
- Purpose of thermal analysis
- Types of equipment used
- Antifriction bearing

Temperature Measurement

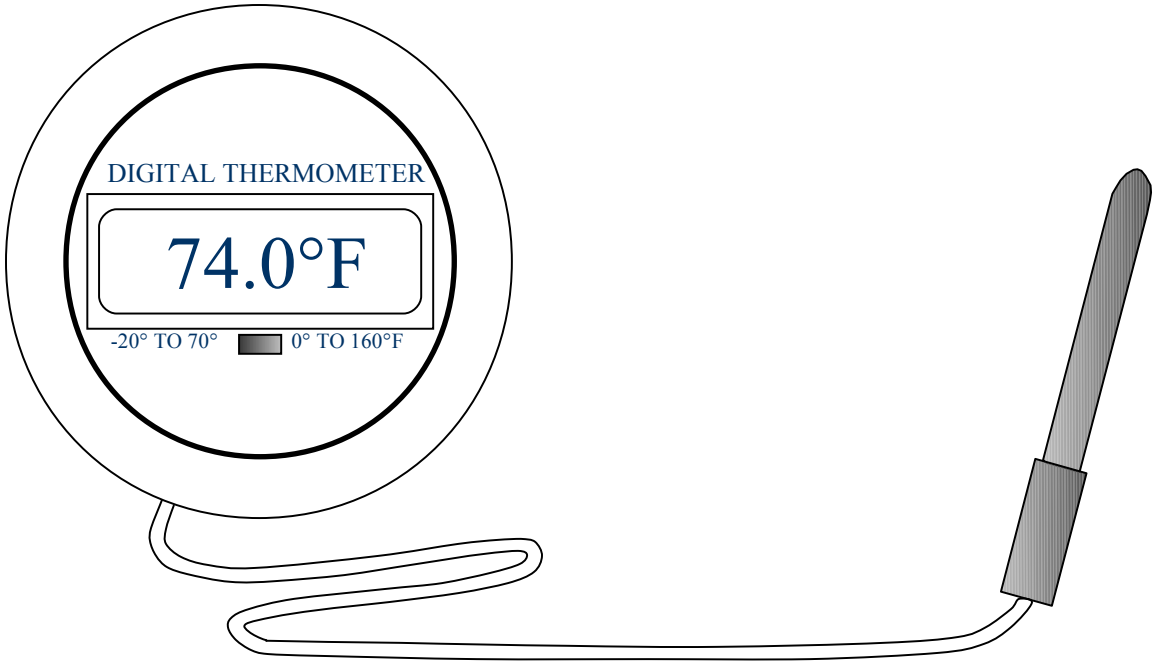
Temperature measurement, just as flow and pressure measurements, is another method for determining both performance and reliability of rotating equipment and hydraulic and lubrication systems.

This condition will continue until component failure occurs. Fluctuating high loads, vibration, metal fatigue, age, and specific operational environments such as: extreme ambient temperatures, wind, chemicals, or dirt in the atmosphere will increase the speed of degradation and the number of faults in electrical systems.

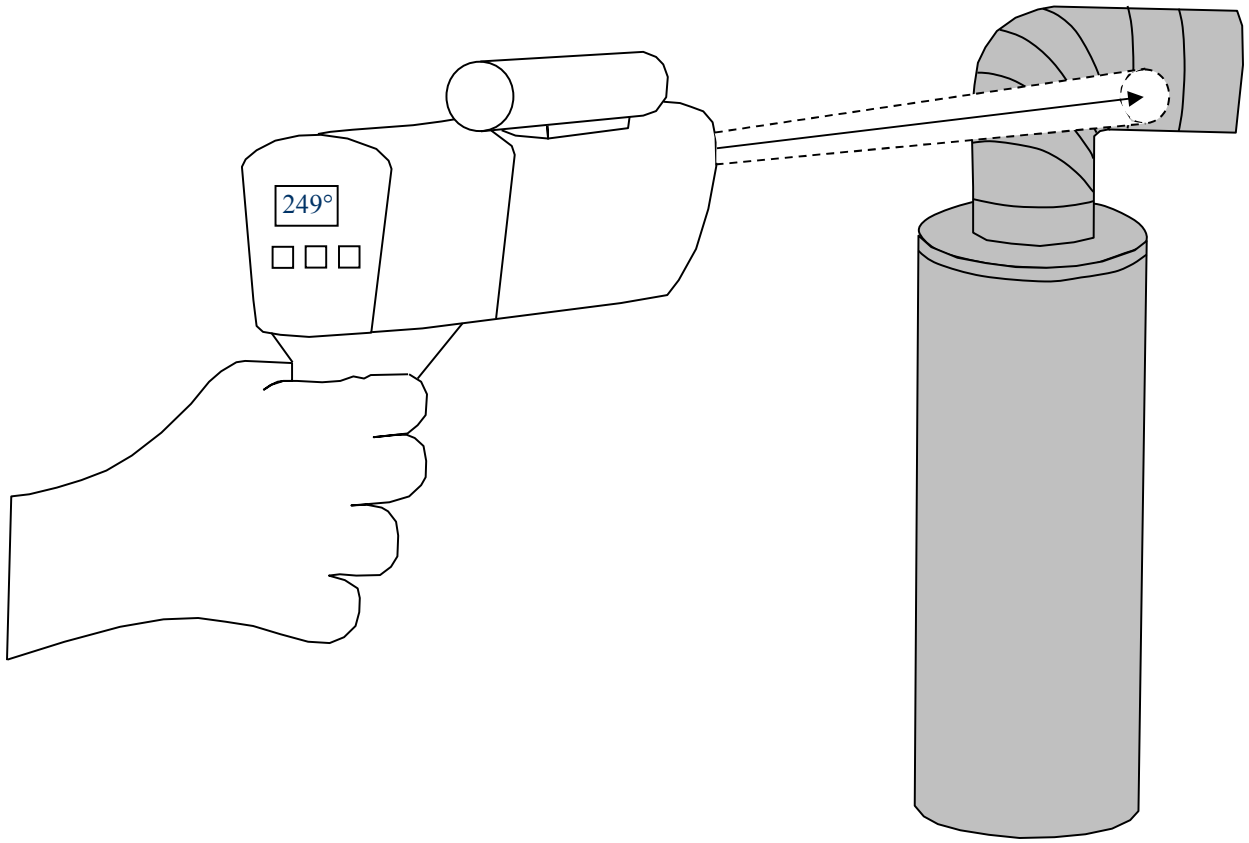
Bimetallic Thermometers



Thermocouples:



Thermographic Instruments:



Evaluating thermal signatures of electrical systems with Infrared Thermography will provide the maintenance department, from point of generation to the end user, with valuable information directly related to operational conditions of virtually every item through which electric current passes through.

To determine an adverse operating temperature of a component, it is necessary to first determine a baseline. For electrical systems the baseline is established when the system is operating under normal load and operating conditions. Once a component or system baseline signature is determined, the thermography technician can identify an anomaly through comparison with the baseline.

Most anomalies in electrical systems are proceeded by a change in its thermal signature. Experienced thermographers are able to identify and analyze problems prior to costly failures. Infrared electrical surveys provide many benefits. Two major advantages of performing infrared thermography surveys are:

Other advantages of an infrared inspection are:

1. **Safety** - Electrical component failure can be catastrophic, injuring personnel or damaging equipment.
2. **Greater System Security** - locate the problems prior to failure greatly reduces unscheduled outages, associated equipment damage and downtime.

Thermal energy generated from an electrical component is directly in proportion to the square of the current passing through it multiplied by the components resistance (I^2R Loss). As the condition of the component deteriorates, its resistance can increase and generate more heat. Then as the component temperature rises the resistance increases further.

When performing an infrared inspection of an electrical system it is important to realize that all of the radiation leaving a surface is not due solely to the temperature of the surface. Unless knowledge, understanding and caution are applied during the analysis portion of the inspection, documentation and interpretation may result in the false conclusion that a fault does or does not exist.

Thermal pattern variations are normally referred to in two ways:

Real Temperature Differences - These are thermal patterns caused only by infrared energy exiting the surface of the object.

Apparent Temperature Differences - they are patterns which are due to factors other than variations of the target surface.

The other three (convection, thermal capacitance, and evaporation) will make a true temperature change at the surface of the component, but it does not provide indication of an electrical fault. In fact, they may actually provide false information by disguising or reducing the amount thermal energy associated with the anomaly, or heat up a component and make it appear to be a fault.

Real	Apparent
I^2R Loss -increased Resistance -load fluctuations	Emittance
Harmonics	Reflectance
Induced heating	Transmittance
Convection	Geometric Variations
Thermal capacitance	

Of the real thermal pattern variations, only three will provide indications of a problem on an electrical system:

1. I^2R Loss
2. Harmonics
3. Induced heating

Remember, the actual component temperature may change or may not change. The thermal variations are not necessarily caused by the electrical components themselves but by outside forces creating the thermal variations, creating or disguising problems.

Many people say it is easy to perform an infrared electrical inspection, be careful - it's easy to be fooled. Beware, IR electrical inspections are one of the most difficult applications if done properly, not just being a "hot spot" finder.

The most common ***loss*** of power in an electric circuit is the heat produced when current flows through a resistance. The exact relationship between the three quantities of heat, current and resistance is given by the equation:

$$P = I^2R$$

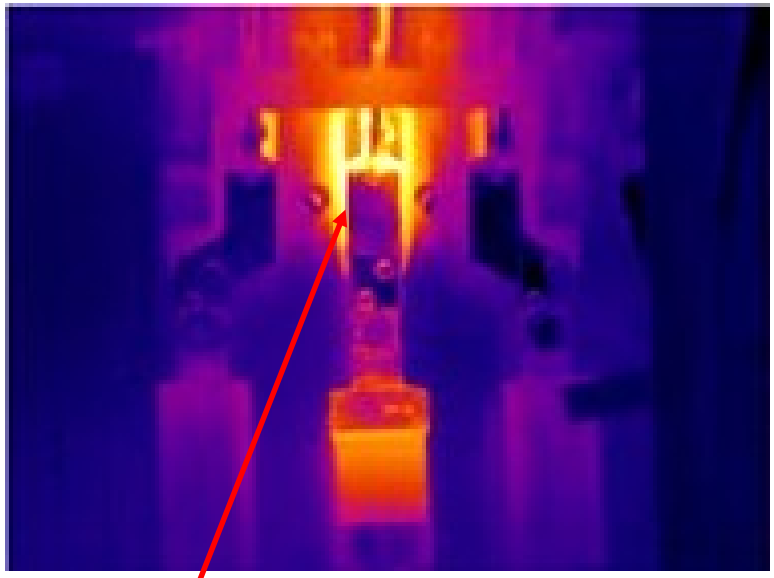
Where **P** = Power and is the rate of doing work or the rate at which heat is produced. It can see from the equation that the amount of thermal energy produced is increased or decreased by increasing or decreasing the current or resistance.

This I^2R heating, as it is often called, takes place in the circuit wires as well as in resistors. The basic unit of Power is the watt, wattage is equal to the voltage (E) across a circuit multiplied by current (I) through the circuit. Below we have divided the effects of power under two headings, since the reason for the power consumption provides an indication as to how the system or components are operating.

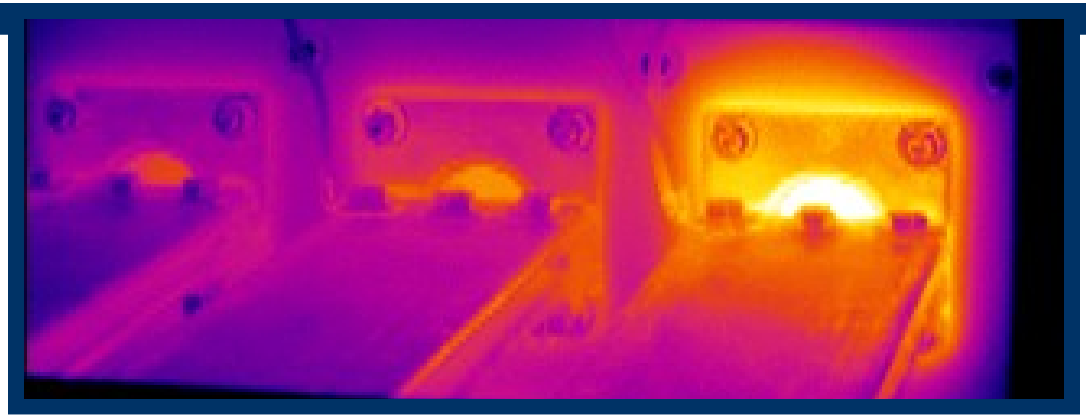
Here we consider a resistor. A resistor in any component in the electric circuit, this can be connections, fuses, switches, breakers, and so on. Under standard operating conditions each component will have a certain "normal" resistance associated with it. It is when the resistance deviates from this norm that the component begins to heat up and must be identified and repaired.

Overheating of components can have several origins. Low contact pressure may occur when assembling a connection or through wear of the material e.g. decreasing spring tension, worn threads or over tightened bolts. Another source could be deteriorated conductors of motor windings. As the component continues to deteriorate the temperature will continue to increase until the melting point of the material is reached and complete failure occurs.

This type of fault can generally be identified because there is a "hottest point" on the thermal image. What this means is, the heat being generated is greatest at the fault point with a tapering off of thermal energy away from the point of highest resistance. Remember, an increase in load will also have a significant effect on increasing the temperature of a high resistance problem (I2R).



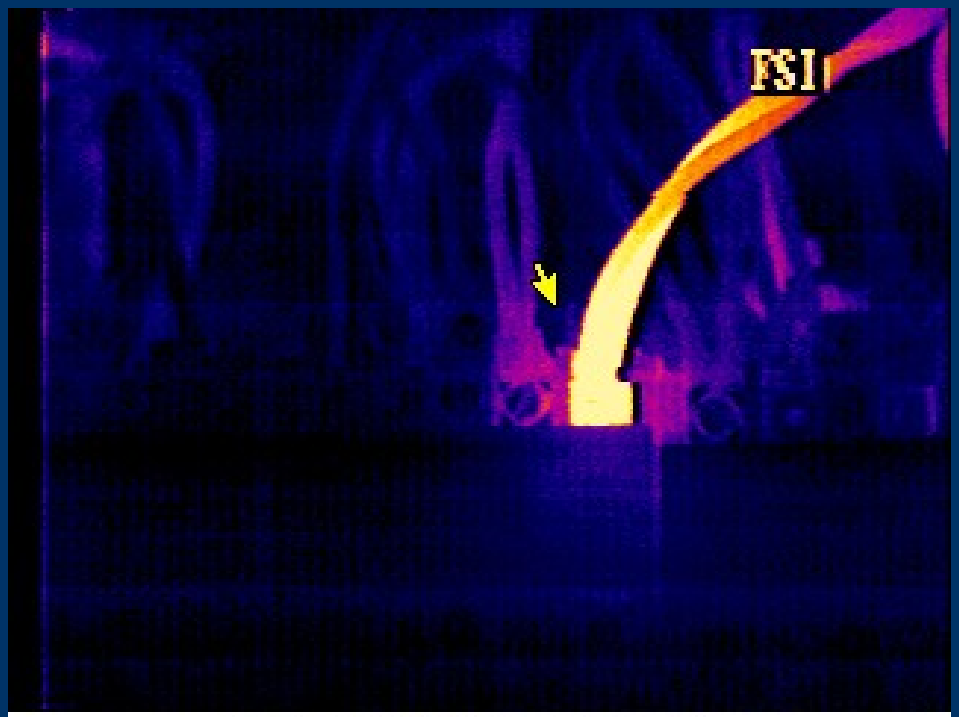
Poor contact B phase breaker

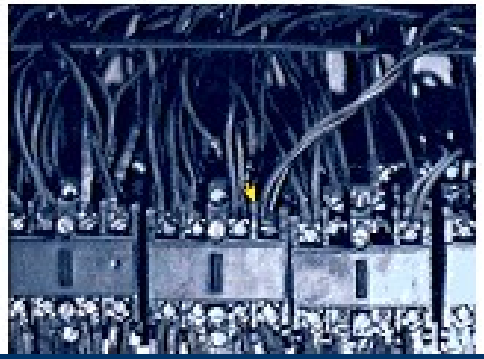


This hot bus stab to the back of the breaker represents an extremely serious problem. Why?
First because of its location in the system. A failure here will typically have significant consequences! Second, the heat appears to be generated inside the breaker. This means the thermal pattern we see is greatly diminished by comparison to the actual point of contact that is inside the breaker. Lastly, the material we are looking at has a very low emissivity, so if it looks at all warm or hot, it is extremely hot! This type of problem should generally be checked and repaired immediately.

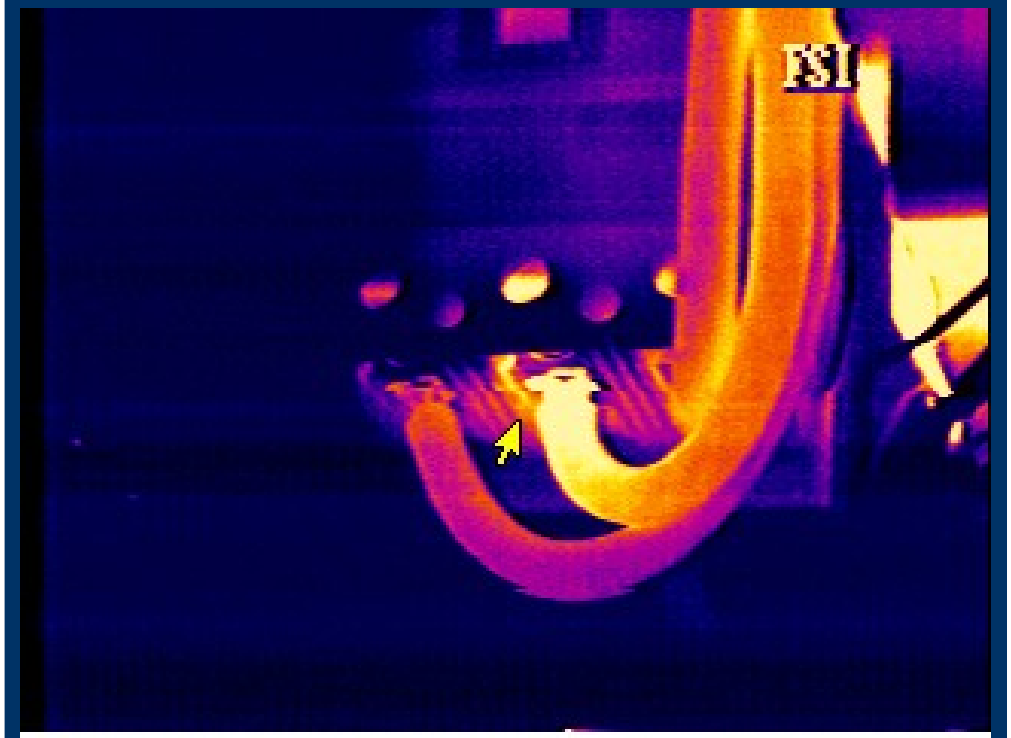
If this is not possible, it should be monitored closely until the next repair opportunity.

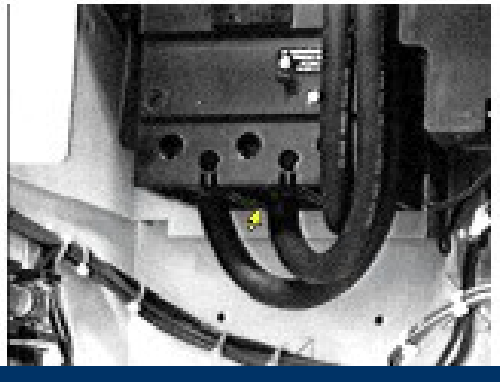
The T2 connection on this starter is approximately 54 degrees F warmer than the T1 connection. When measuring temperatures it is critical to also know the load, since hear output and thus temperatures at this abnormally high resistance connection will increase at the square of the load.



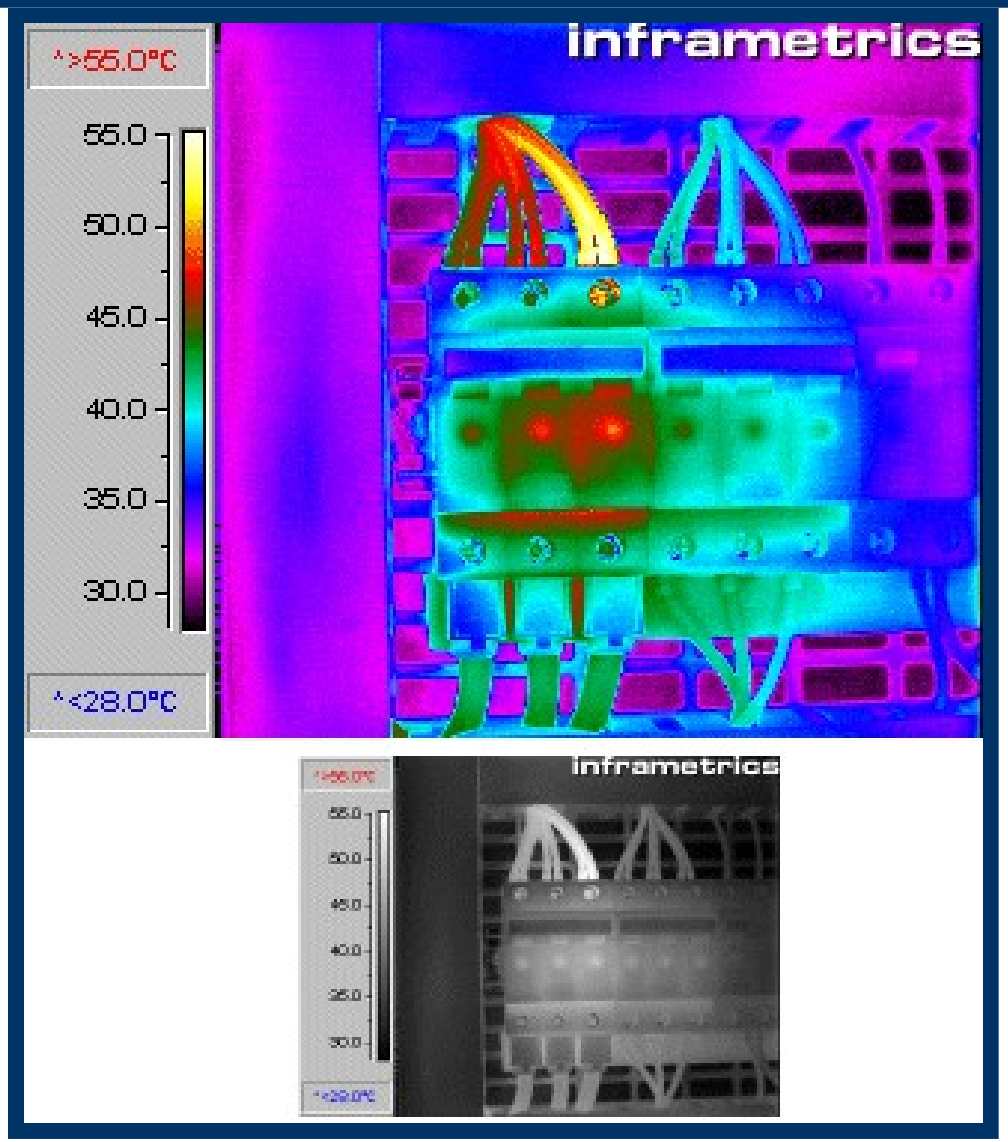


The load-side center phase connection of this primary feed pump breaker is running approximately 21 degrees F over the left phase. Condition of the right phase is unknown, but further investigation is probably warranted.





The right phase of this molded case breaker shows a classic pattern associated with a loose connection. Note how the temperature diminishes further away from the source of the heating, the connection. While loading conditions should be taken into account, this is more than imbalanced load.



Problem Classification	*Phase to Phase Temperature Rise*	*Comments*
Minor	**1º - 10º C**	**Repair in regular maintenance schedule; little probability of physical damage**
Intermediate	**10º - 30º C**	**Repair in the near future (2-4 weeks). Watch load and change accordingly. Inspect for physical damage. There is probability of damage in the component, but not in the surrounding components.**
Serious	**30º - 70º C**	**Repair in immediate future (1-2 days). Replace component and inspect the surrounding components for probable damage.**
Critical	**above 70º C**	**Repair immediately (overtime). Replace component, inspect surrounding components. Repair while IR camera is still available to inspect after.**

**** with wind speed less than 15mph * with load conditions greater than 50%***

***Hint:** Have an electrical contractor use a clamp on ammeter to verify loading.*

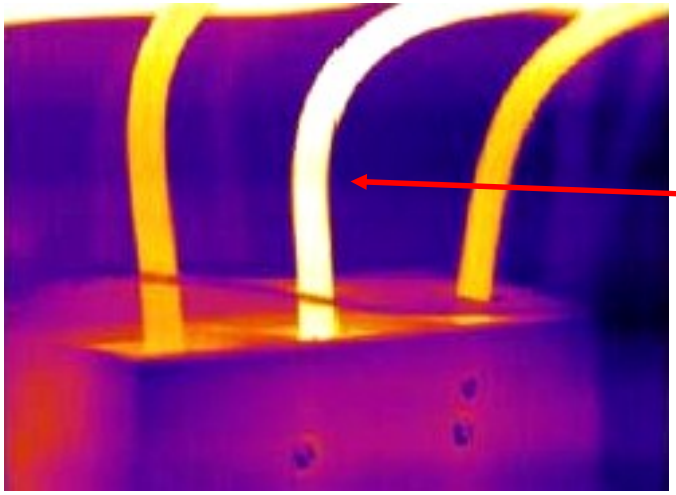
Wind will affect your temperature readings due to convection cooling. This can be compensated in outdoor electrical predictive maintenance applications by multiplying your temp. reading by the correction factors listed below.

Wind Speed (Miles Per Hour)	*Correction Factor*
2	1.00
4	1.30
6	1.60
8	1.68
10	1.96
12	2.10
14	2.25
16	2.42
18	2.60

As the ***load*** increases in a circuit the power output will increase as a square of the load, and the temperature of the entire circuit and components on the circuit will increase. From a thermographic point of view, load is usually looked at as a specific type of problem with specific thermal indications. As the load on an electrical component rises, so does the temperature.

An even load on each phase of a three phase system for example, should result in uniform temperature patterns on all three phases. An anomaly is identified when the overall component and conductor temperature is too high, indicating an overload condition. An unbalanced condition can also be a problem and is identified by the conductors not displaying a balanced or equal thermal pattern and temperature.

Harmonics are currents or voltages that are multiples of the basic incoming 60 HZ frequency serving an electrical distribution system. Possibly the most damaging harmonics are the odd harmonics known as triplens (third harmonics). The triplen harmonics add to the basic frequency and can cause severe over voltage, overcurrent and overheating. Frequency is not the enemy of the electrical system. The real enemy is increased heat caused by higher frequency harmonics.



Harmonics problems on circuit

These triplen harmonics can create drastic overheating and even melting of neutral conductors, connections, contact surfaces, and receptacle strips. Other equipment effected by harmonics are transformers, stand-by generators, motors, telecommunication equipment, electrical panels, circuit breakers, and busbars.

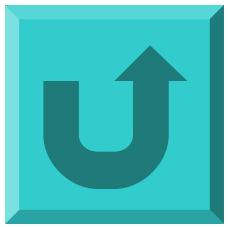
Alternating current in electrical systems naturally induce (***induction***) current flow and magnetic flux into surrounding metallic objects such as conduit, metal enclosures and even structural support steel. This phenomenon will occur in areas of high electromagnetic fields such as high voltage equipment, microwave transmitters, and induction heating equipment. This condition can be induced in ferrous material when an electrically induced electro-magnetic field is present.

Infrared condition monitoring as a part of a total predictive maintenance program can increase reliability and improve operating profit. Infrared thermography will assist in determining equipment and facility maintenance priorities, enhance operational safety and contribute to a stronger bottom line.

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS



PREVENTATIVE MAINTENANCE

Main Topics

- Preventive Maintenance Programs
- Maintenance problems

Maintenance Problems

- Wear and tear
- Careless or untrained personnel
- Improper lubrication
- Excessive loads and speeds
- Incorrect alignment practices
- Vibration

Prevention Troubleshooting

- Troubleshooting is the search for the root cause of a problem
- The need to troubleshoot can be minimized by an effective maintenance programs

Types of Maintenance

- Preventative maintenance
- Condition based maintenance
- Proactive maintenance
- Failure history based maintenance

Preventive Maintenance

- This type of maintenance is performed at set intervals.
- Examples of time-based maintenance include:
  - Monthly calibration checks
  - Weekly lubrication
  - Daily housekeeping

Condition Monitoring

- Temperature
- Vibration
- Changes in noise or sound
- Visually observed changes and problems

Sound/Noise

- Listening
- Sound Measurements

Preventative Maintenance Preparations

- Preparation
- Precautions

Pump Preventative Maintenance

- Observe and record condition of pump
- Listen to pump operation and note unusual sounds.
- Record pressure readings
- Feel for hot spots, take and record any necessary temperatures.
- Feel for unusual vibration. Use vibration meter if necessary.
- Lubricate bearings
- Check mounting bolts
- Check for unusual dirt or corrosion

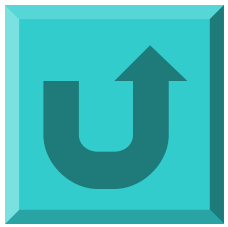
Fan Preventative Maintenance

- Check all fan bolts for tightness
- Check alignment of blades
- Clean blades
- Check fan belts
- Check blades for scale or dirt, clean if required
- Check blade drain holes
- Check clearances

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS



FAULT RECOGNITION

Course Objectives

- Identify types of maintenance problems
- Discuss information gathering for troubleshooting
- Systematically solve equipment problems

Main Topics

- Predictive Maintenance
- Condition Monitoring

Predictive Maintenance

- Systematic method of monitoring equipment.

Predictive Maintenance

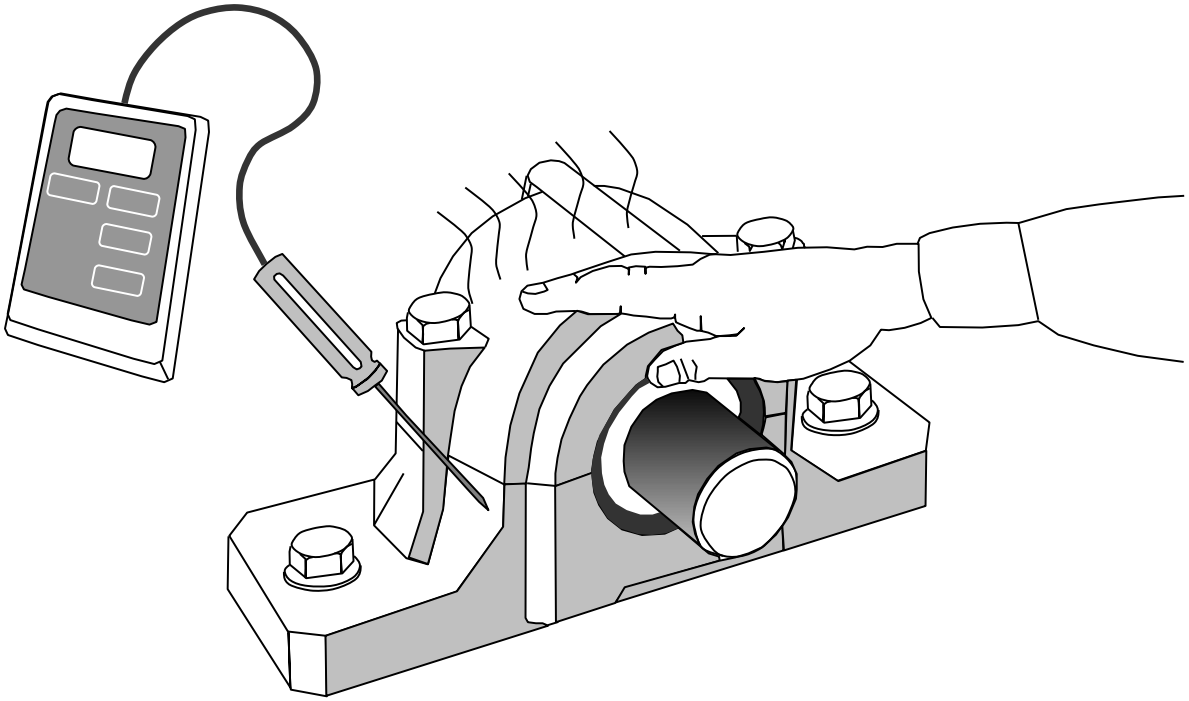
- List the benefits of predictive maintenance

Condition Monitoring

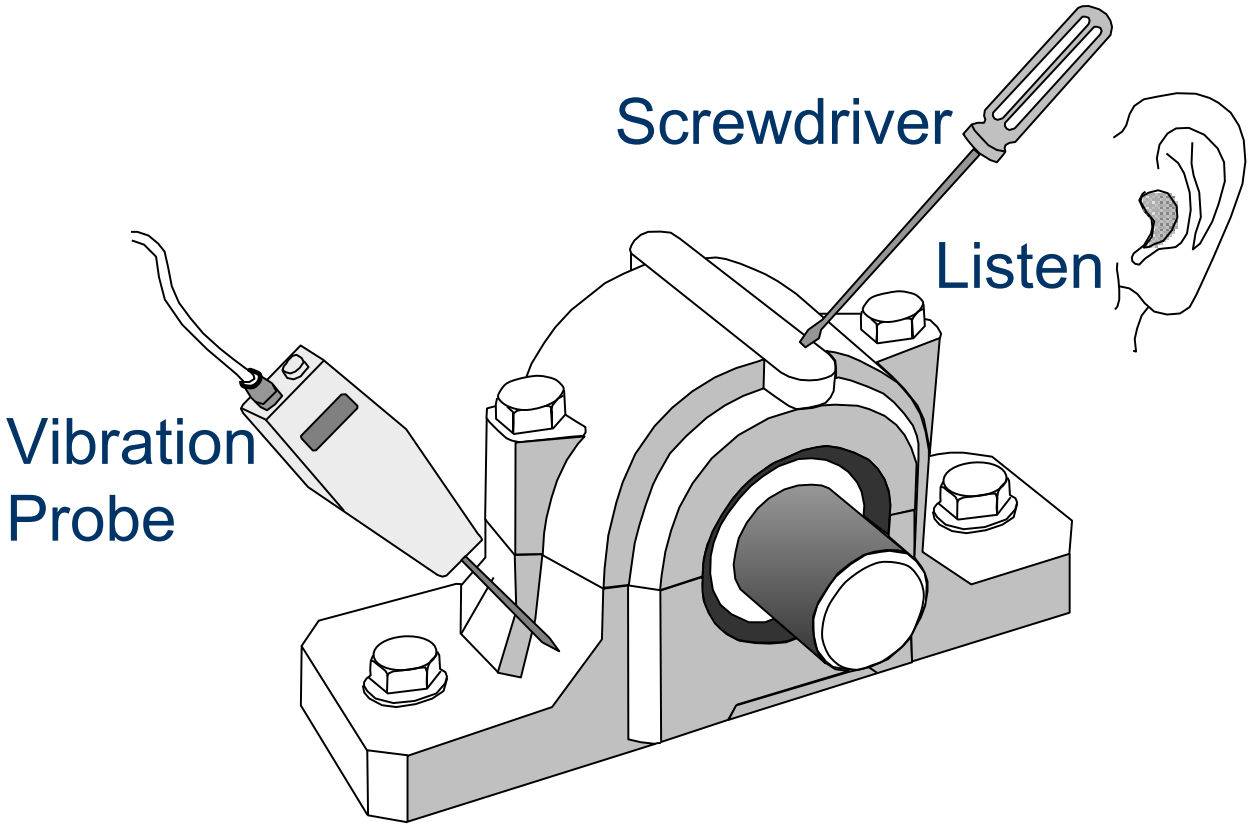
- Temperature
- Vibration
- Changes in noise or sound
- Visually observed changes and problems

Temperature

- Surface Temperature



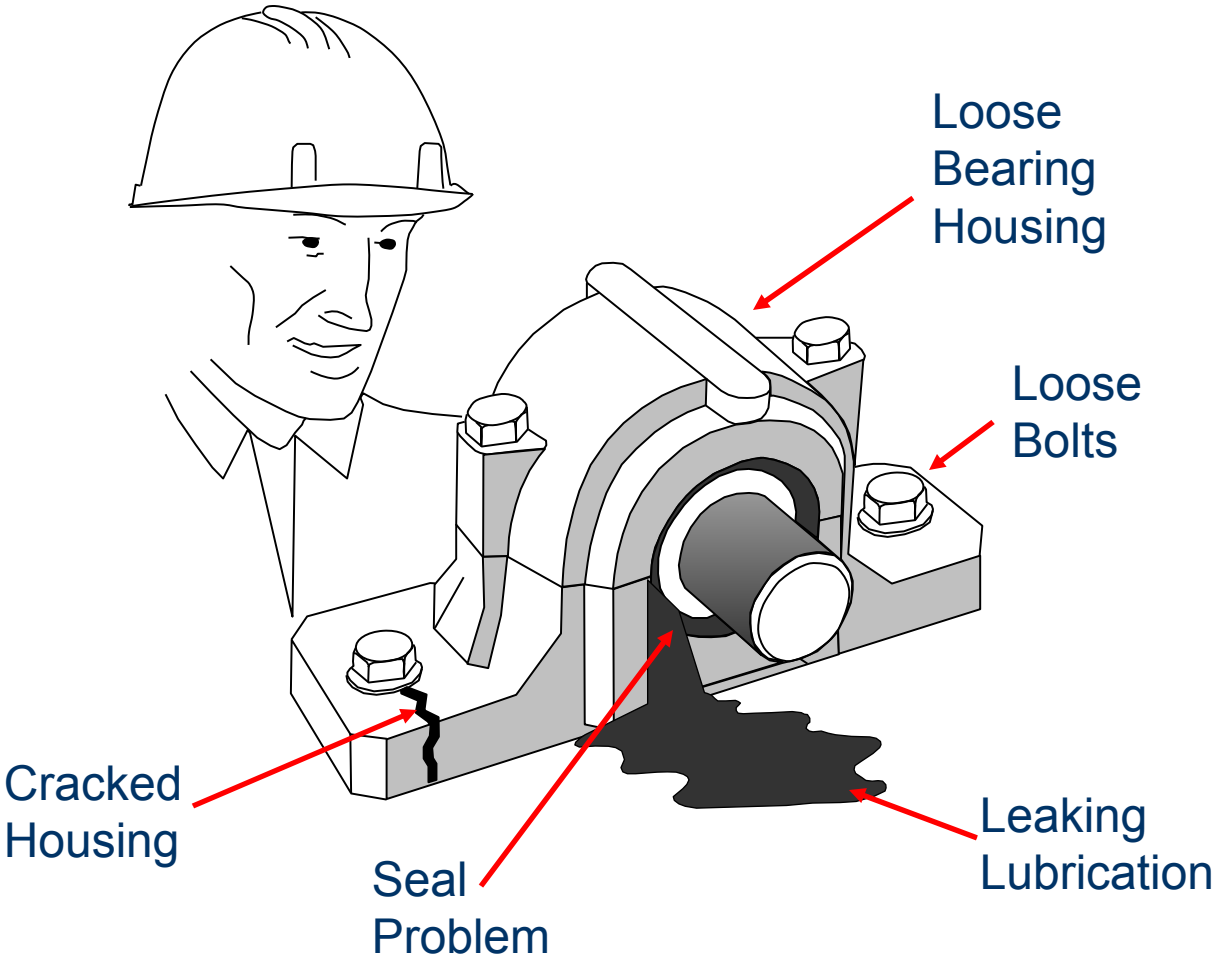
Vibration



Sound/Noise

- Listening
- Sound Measurements

Sight


Loose
Bearing
Housing
Loose
Bolts
Cracked
Housing
Seal
Problem
Leaking
Lubrication

Pump – Steps in Troubleshooting

- Talk to operators
- Ensure other system components are working properly
- Timing of symptoms

  -Sudden symptoms indicate complete failure of parts

  -Gradual symptoms indicate gradual wearing out of parts
- Changes in pump's operating characteristics

Pumps -Symptoms You Can Here

- Loud rattling or clanging noise
- Growling or howling sound
- High-pitched screeching
- Pinpointing Sources

  Use stethoscope, brass sounding rod, or short Length of pipe
- Amplify sound from point of contact with pump

Pumps - Symptoms You Can See

- Abnormal pressure Readings
- Leakage from stuffing box
- Leakage from casing flange
- Lubricant leak from bearing housing

Some Pump Problems/Symptoms

- Bearing Lubrication Leak
- Bearings Damaged
- Bearings Worn
- Casing Flange Bolts Loose
- Casing Flange Gasket Worn
- Casing Wearing Ring Damaged
- Casing Wearing Rings Worn
- Cavitation
- Discharge Strainer Clogged

Pumps – Symptoms You Can Feel

- Excessive Vibration
- Overheating

Summary

- Review
- Question and Answer Session

CLICK TO RETURN TO TOPICS

